Extreme Series

EXAM BOOK

Natural Science

University Entrance Examination

Telegram: https://t.me/Extremeseries

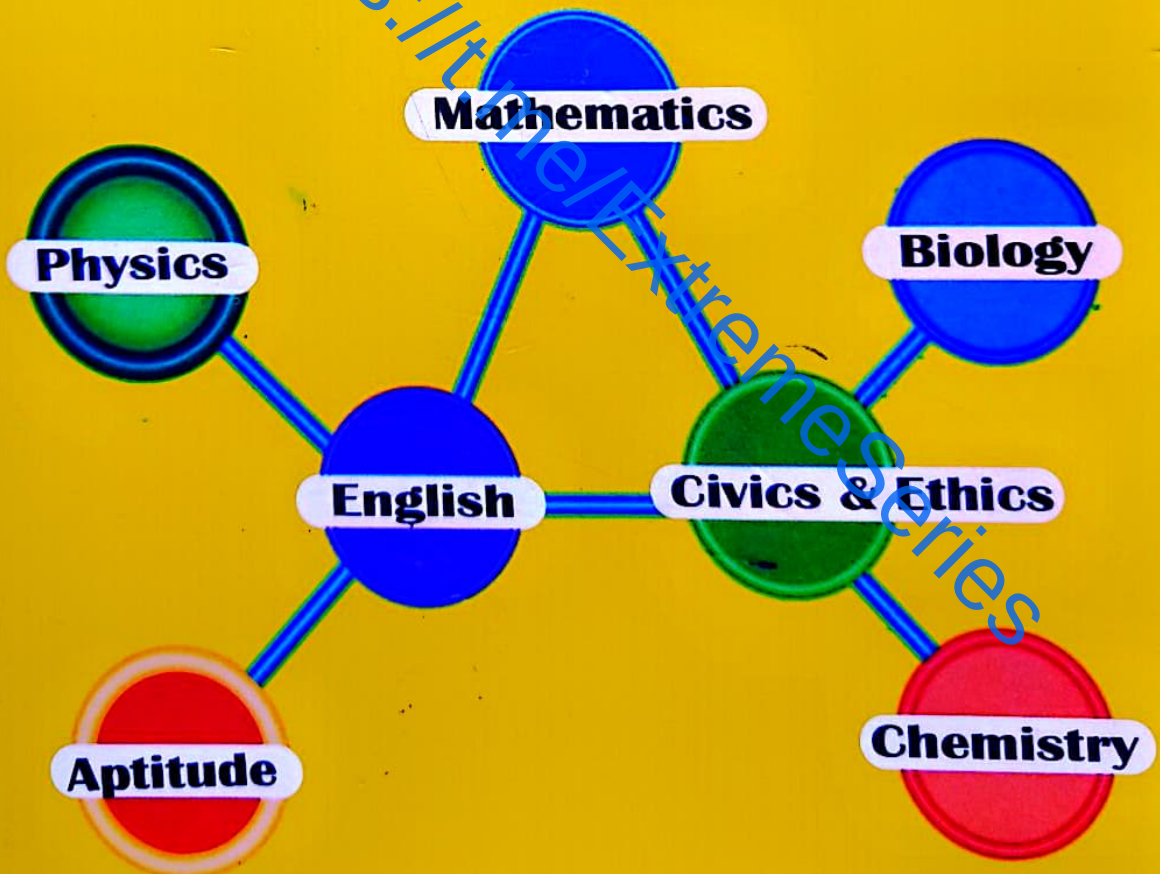


Table of Content

PART ONE

SECTION 1: MATHEMATICS

EUEE 2002.....1
EUEE 2003.....23
EUEE 2004.....40
EUEE 2005.....63
EUEE 2006.....87
EUEE 2007.....113
EUEE 2008.....141
EUEE 2009.....167
EUEE 2010.....189

SECTION 2: CHEMISTRY

EUEE 2002.....217
EUEE 2003.....245
EUEE 2004.....272
EUEE 2005.....306
EUEE 2006.....340
EUEE 2007.....368
EUEE 2008.....396
EUEE 2009.....420
EUEE 2010.....443

SECTION 3: PHYSICS

EUEE 2005.....463
EUEE 2006.....487
EUEE 2007.....510
EUEE 2008.....527
EUEE 2009.....543
EUEE 2010.....559

SECTION 4: BIOLOGY

EUEE 2005.....574
EUEE 2006.....613
EUEE 2007.....638
EUEE 2008.....661
EUEE 2009.....679
EUEE 2010.....699

Telegram: https://t.me/ExtremeSeries

PART TWO

Section 5: English
EUEE 2005 722
EUEE 2006 758
EUEE 2007 792
EUEE 2008 821
EUEE 2009 848
EUEE 2010 877

Section 6: Civics and Ethical Education
EUEE 2005 906
EUEE 2006 929
EUEE 2007 954
EUEE 2008 981
EUEE 2009 1008
EUEE 2010 1038

Section 7: Scholastic Aptitude
How to Workout it out 1066
EUEE 2007 1073
EUEE 2008 1091
EUEE 2009 1106
EUEE 2010 1124
Quantitative Analysis 1140

Telegram: https://t.me/ExtremeSeries

SECTION 1 – MATHEMATICS

1. Which of the following is rational function:

A. $f(x)=\dfrac{\sqrt{2x}-x+1092}{x^4+x^2+\pi}$ C. $h(x)=\dfrac{x^2-2^x+x}{x^4+x+1}$

B. $g(x)=3x^4-\sqrt{x^2+1}+2x$ D. $r(x)=\dfrac{x^2+\sin x+2}{x+1}$

Solution: Rational function $=\dfrac{\text{polynomial}}{\text{polynomial}}$ **Answer: A**

2. If $P(x)=x-4$, $q(x)=x^2+8$ and $r(x)=x^2-2x-8$ then which of the following is the most simplified expression of $f(x)=\dfrac{x}{P(x)}-\dfrac{q(x)}{r(x)}$:

A. $\dfrac{x}{x-4}-\dfrac{q(x)}{r(x)}$ C. $\dfrac{2}{x+2}$

B. $\dfrac{8x}{x-4}$ D. $\dfrac{2x+8}{x^2-2x-8}$

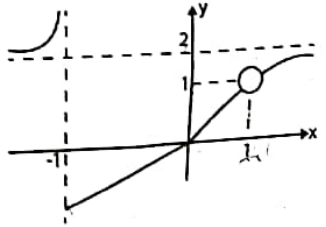
Solution: $f(x)=\dfrac{x}{x-4}-\dfrac{x^2+8}{x^2-2x-8}\Rightarrow f(x)=\dfrac{x}{x-4}-\dfrac{x^2+8}{(x-4)(x+2)}$

$\Rightarrow f(x)=\dfrac{x(x+2)-(x^2+8)}{(x-4)(x+2)}$ ← multiply $\dfrac{x}{x-4}$ by $\dfrac{x+2}{x+2}$

$\therefore f(x)=\dfrac{x^2+2x-x^2-8}{(x-4)(x+2)}=\dfrac{2(x-4)}{(x-4)(x+2)}=\dfrac{2}{x+2}$,

Answer: C

3. Which of the following functions called most likely be sketched as in the figure?



A. $f(x)=\dfrac{2x}{x+1}$

B. $f(x)=\dfrac{2x(x-1)}{x^2-1}$

C. $f(x)=\dfrac{2}{x+1}$

Solution: The graph of $f(x)=\frac{2x(x-1)}{x^2-1}=\frac{2x}{x+1}$ has:

- Vertical asymptote $x = -2$
- Horizontal asymptote, $y = 2$
- Hole at $x = 1$ and $x \to 1, y \to 1$
- x and y intercept at (0, 0)

Answer: B

4. Let ℓ be the line containing (1, 5) and its angle of inclination be $\theta = 135°$. Which of the following is the distance form ℓ to (-1, 1)?

A. $2\sqrt{2}$ B. $3\sqrt{2}$ C. $4\sqrt{2}$ D. $6\sqrt{2}$

Solution:

Suppose slope of $\ell = \tan\theta = \tan(135°) = -1$.

Equation of ℓ: $y-5=(-1)(x-1) \Leftrightarrow \ell: x+y-6=0$.

The distance from ℓ to $(-1, 1)$ is given by:

$$\frac{|Ax+By+C|}{\sqrt{A^2+B^2}}=\frac{|-1+1-6|}{\sqrt{(-1)^2+1^2}}=\frac{6}{\sqrt{2}}=3\sqrt{2}$$

Answer: B

5. The equation of circle with center at (2, −1) and tangent to the y−axis.

A. $(x+2)^2+(y-1)^2=1$ C. $(x-2)^2+(y+1)^2=4$
B. $(x-2)^2+(y+1)^2=1$ D. $(x+2)^2+(y-1)=4$

Solution:

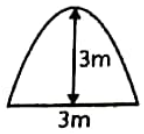
If the circle tangent to y-axis with center (h, k) then the radius will be $r = h$. ∴ The equn. of circle: $(x-h)^2+(y-k)^2=h^2$
$\Rightarrow (x-2)^2+(y+1)^2=4$

Answer: C

Note: If tangent to the x-axis use $(x-h)^2+(y-k)^2=k^2$.

6. Consider a gate whose shape is a parabolic arch of height 3m and base width 3m. If the origin of the coordinate system is at the vertex, the equation that represents the edge of the gate is:

A. $3x^2+3y=0$
B. $4x^2+3y=0$
C. $y=-3x^2+1$
D. $3y^2-4x=0$



Solution:

Since the vertex (0, 0) and the point $\left(\frac{3}{2},-3\right)$ on parabola

$\Rightarrow x^2=4py \Rightarrow \left(\frac{3}{2}\right)^2=4P(-3) \Rightarrow P=-\frac{3}{16}$

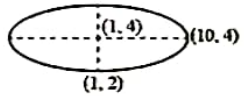
$x^2=4\left(-\frac{3}{16}\right)y \Rightarrow x^2=-\frac{3}{4}y \Leftrightarrow 4x^2+3y=0$

Answer: B

7. Let the center of an ellipse be at (1, 4) and two of its vertices be at (10, 4) and (1, 2). What is the equation of the ellipse:

A. $4(x-1)^2+81(y-4)^2=324$
B. $9(x-1)^2+4(y-4)^2=4$
C. $(x-1)^2+9(y-4)^2=4$
D. $2(x-1)^2+9(y-4)^2=4$

Solution: Distance between centers:
and major vertices, $a = 10 - 1 = 9$
and minor vertices, $b = 4 - 2 = 2$
Since it is x ellipse



$$\therefore \frac{(x-1)^2}{81}+\frac{(y-4)^2}{4}=1 \Leftrightarrow 4(x-1)^2+81(y-4)=324$$

Answer: A

8. What is the length of the minor axis of $x^2+9y^2-2x=0$.

A. $\frac{1}{9}$ B. $\frac{2}{3}$ C. 2 D. 6

Solution: $x^2-2x+9y^2=0 \Rightarrow x^2-2x+1+9y^2=1$

$\Rightarrow (x-1)^2+9y^2=1 \Leftrightarrow \frac{(x-1)^2}{1}+\frac{(y-0)^2}{\frac{1}{9}}=1$, so that $a^2 = 1$ and

$b^2=\frac{1}{9} \Rightarrow b=\frac{1}{3} \Rightarrow 2b=\frac{2}{3} \leftarrow$ lengths of minor axis.

Answer: B

9. The equation $x^2-4y^2=-1$ represents:

A. A hyperbola with one of its foci, at the origin.
B. A parabola with vertex at the origin.

Telegram: https://t.me/ExtremeSeries

C. A hyperbola with vertices at $\left(-\frac{1}{2},0\right)$ and $\left(\frac{1}{2},0\right)$

D. A hyperbola with foci at $\left(0,\frac{-\sqrt{5}}{2}\right)$ and $\left(0,\frac{\sqrt{5}}{2}\right)$

Solution: $x^2-4y^2=-1 \Rightarrow 4y^2-x^2=1$, dividing by (-1).

$\Rightarrow \frac{y^2}{\frac{1}{4}}-\frac{x^2}{1}=1 \Rightarrow a^2=\frac{1}{4} \Rightarrow a=\pm\frac{1}{2}$ and $b^2=1 \Rightarrow b=\pm 1$ and c^2

$=a^2+b^2 \Rightarrow c^2=\frac{1}{4}+1=\frac{5}{4} \Rightarrow c=\pm\frac{\sqrt{5}}{2}$

Since y^2 term comes first, thus it is y – hyperbola, with vertices $\left(0,\frac{-1}{2}\right)$ and $\left(0,\frac{1}{2}\right)$, foci $\left(0,-\frac{\sqrt{5}}{2}\right)$ and $\left(0,\frac{\sqrt{5}}{2}\right)$.

Answer: D

10. Let P (x): x^2+x is positive, then $\neg(\exists x)$ P (x) equal to:

A. $(\exists x)\neg(x^2+x>0)$ C. $(\exists x)(x^2+x\le 0)$

B. $(\forall x)(x^2+x<0)$ D. $(\forall x)(x^2+x\le 0)$

Solution: $\neg(\exists x)(Px) \equiv (\forall x)\neg(\forall x)\neg P(x)$

$\equiv (\forall x)(x^2+x\le 0)$

Answer: D

11. If $(P \wedge Q) \Rightarrow (\neg S \vee R)$ is false, which of the following is true?

A. $S \Rightarrow R$ C. $(P \wedge \neg Q) \Leftrightarrow R$

B. $\neg P \vee \neg S$ D. $P \wedge R$

Solution:

$a \Rightarrow b \equiv F$ if and only if $b = F$, and $a \equiv T$

- $P \wedge Q \Rightarrow (\neg S \vee R) \equiv F$ if and only $\neg S \vee R \equiv F$ and $P \wedge Q \equiv T$
- $\neg S \vee R = F$ if and only if both $\neg S \equiv F$ and $R \equiv F$
- $P \wedge Q \equiv T$ when both $P \equiv T$ and $Q \equiv T$

A. $T \Rightarrow F \equiv F$, B. $F \vee F \equiv F$ C. $F \Leftrightarrow F \equiv T$ D. $T \wedge F \equiv F$

Answer: C

12. Let $A=\begin{pmatrix}2 & 3 & 5\\ 4 & 5 & 7\\ 8 & -3 & 3\end{pmatrix}$ and $B=\begin{pmatrix}-4 & 3 & -5\\ 2 & 1 & 7\\ 6 & 5 & 0\end{pmatrix}$.

If $C = AB + A^T$, then the entry of C_{23} is:

A. 12 B. 18 C. 22 D. 18

Solution: $C = AB + A^T = \begin{pmatrix}2 & 3 & 5\\ 4 & 5 & 7\\ 8 & -3 & 3\end{pmatrix}\begin{pmatrix}-4 & 3 & -5\\ 2 & 1 & 7\\ 6 & 5 & 0\end{pmatrix} + (-3)$

$\Rightarrow C_{23} = (457)\begin{pmatrix}-5\\ 7\\ 0\end{pmatrix} + (-3) = -20+35+(-3) = 12$

Answer: A

13. Given matrix $B=\begin{pmatrix}2 & -1\\ 3 & 0\end{pmatrix}$, what is a matrix A if $AB = A + B$:

A. $\frac{1}{2}\begin{pmatrix}1 & 1\\ 3 & -3\end{pmatrix}$ C. $\frac{1}{2}\begin{pmatrix}1 & 1\\ -3 & 3\end{pmatrix}$

B. $\frac{1}{2}\begin{pmatrix}1 & 2\\ -3 & 3\end{pmatrix}$ D. $\frac{1}{2}\begin{pmatrix}1 & -1\\ 3 & -3\end{pmatrix}$

Solution: Let $A=\begin{pmatrix}a & b\\ c & d\end{pmatrix}$ so that $AB = A + B$

$\Rightarrow \begin{pmatrix}a & b\\ c & d\end{pmatrix}\begin{pmatrix}2 & -1\\ 3 & 0\end{pmatrix} = \begin{pmatrix}a & b\\ c & d\end{pmatrix} + \begin{pmatrix}2 & -1\\ 3 & 0\end{pmatrix}$

$\Rightarrow \begin{pmatrix}2a+3b & -a+0\\ 2c+3d & -c+0\end{pmatrix} = \begin{pmatrix}a+2 & b-1\\ c+3 & d\end{pmatrix}$

$\Rightarrow \begin{cases}2a+3b=a+2 \Rightarrow a+3b-2=0 \ldots\ldots\ldots\ldots(*)\\ 2c+3d=c+3 \Rightarrow c+3d-3=0 \ldots\ldots\ldots\ldots(**)\end{cases}$

and $-a = b - 1 \Rightarrow a = -b + 1$

$-c = d$

Thus, put $a = -b + 1$ in $(*) \Rightarrow -b + 1 + 3b - 2 = 0$

Telegram: https://t.me/ExtremeSeries

$\Rightarrow 2b-1=0 \Rightarrow b=\frac{1}{2}$ and put $d=-c$ in (**)

$\Rightarrow c+3(-c)-3=0 \Rightarrow c=-\frac{3}{2}$

Now, $a=-b+1=-\frac{1}{2}+1=\frac{1}{2}$ and $d=\frac{3}{2}$

$$\therefore A=\begin{pmatrix} a & b \\ c & d \end{pmatrix}=\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{3}{2} & \frac{3}{2} \end{pmatrix}=\frac{1}{2}\begin{pmatrix} 1 & 1 \\ -3 & 3 \end{pmatrix}$$

Answer: C

14. For what value of x is $\begin{vmatrix} 1 & -2 \\ x & x-2 \end{vmatrix}=x+1$:

A. 1 B. $\frac{3}{2}$ C. $\frac{1}{2}$ D. -1

Solution: $(1)(x-2)-(-2)(x)=x+1$

$\Rightarrow 3x-2=x+1 \Rightarrow 2x=3 \Rightarrow x=\frac{3}{2}$

Answer: B

15. Let $A=\begin{pmatrix} a & a & ac \\ b & -b & bc \\ c & c & -c^2 \end{pmatrix}$ then det A is equal to:

A. $2abc^2$ B. $4abc^2$ C. $-4abc^3$ D. 0

Solution:

$$\det(A=(-1)^2 a\begin{vmatrix} -b & bc \\ c & -c^2 \end{vmatrix}+(-1)^3 a\begin{vmatrix} b & bc \\ c & -c^2 \end{vmatrix}+(-1)^4 ac\begin{vmatrix} b & -b \\ c & c \end{vmatrix}$$

$\Rightarrow |A|=a(bc^2-bc^2)-a(-bc^2-bc^2)+ac(bc+bc)$

$|A|=0+2abc^2+2abc^2=4abc^2$

Answer: B

16. Suppose A is 3×3 matrix and $B=\begin{pmatrix} 0 & 4 & 1 \\ 5 & 0 & 6 \\ 0 & 4 & 2 \end{pmatrix}$ If det (A) = 2, the det $(A^{-1}B^t)$ is equal to:

A. -10 B. 40 C. -40 D. 10

Solution: $|A^{-1}B^t|=|A^{-1}||B^t|$ and $|A^{-1}|=\frac{1}{|A|}$

$$\Rightarrow |B^t|=|B|=0\begin{vmatrix} 0 & 6 \\ 4 & 2 \end{vmatrix}-4\begin{vmatrix} 5 & 6 \\ 0 & 2 \end{vmatrix}+1\begin{vmatrix} 5 & 0 \\ 0 & 4 \end{vmatrix}=-20 \text{ and}$$

$$|A^{-1}|=\frac{1}{|A|}=\frac{1}{2}$$

$$\therefore |A^{-1}B^t|=|A^{-1}||B^t|=\left(\frac{1}{2}\right)(-20)=-10$$

Answer: A

17. Let $A=\begin{pmatrix} k & 2 \\ 5 & -1 \end{pmatrix}$ which of the following is necessarily true?

A. For any number k, A is invertible.

B. If $k\neq 10$, then $A^{-1}=\frac{1}{-k-10}\begin{pmatrix} -1 & -2 \\ -5 & k \end{pmatrix}$

C. If $k\neq -10$, then $A^{-1}=\frac{1}{-k+10}\begin{pmatrix} 1 & 2 \\ 5 & -k \end{pmatrix}$

D. If $k\neq -10$, then A is invertible.

Solution: $|A|=-k-10$

Matrix A is invertible if $|A|\neq 0 \Rightarrow k\neq -10$ and

$$A^{-1}=\frac{1}{|A|}\text{adj.}A=\frac{1}{-k-10}\begin{pmatrix} -1 & -5 \\ -2 & k \end{pmatrix}$$

Answer: D

18. For what value k does the system $\begin{cases} kx+y=k \\ x+ky=k \end{cases}$ hav nique solution?

A. $k=1$ C. $k=0$

B. $k\in(-\infty,\infty)\setminus\{-1,1\}$ D. $k\in(-\infty,\infty)\setminus\{1\}$

Telegram: https://t.me/ExtremeSeries

Solution: First $D = \begin{vmatrix} k & 1 \\ 1 & k \end{vmatrix} = k^2 - 1$, so that the system have unique solution if $k^2 - 1 \neq 0 \Rightarrow k \neq 1$, and $k \neq -1 \Rightarrow k \in (-\infty, \infty) \setminus \{-1, 1\}$.

Answer: B

19. The system of equations $\begin{cases} x + y - z = 6 \\ x - 2y = -4 \\ x + 4y - az = b \end{cases}$ has no solution if:

A. $a = 2, b = 16$ C. $a = -2, b = 16$

B. $a = -2, b \neq 16$ D. $a = 2, b \neq 16$

Solution: First, $D = \begin{vmatrix} 1 & 1 & -1 \\ 1 & -2 & 0 \\ 1 & 4 & a \end{vmatrix}, D_Z = \begin{vmatrix} 1 & 1 & 6 \\ 1 & -2 & -4 \\ 1 & 4 & b \end{vmatrix}$

- $D = (1)\begin{vmatrix} -2 & 0 \\ 4 & a \end{vmatrix} - 1\begin{vmatrix} 1 & 0 \\ 1 & a \end{vmatrix} + (-1)^4(-1)\begin{vmatrix} 1 & -2 \\ 1 & 4 \end{vmatrix}$

$= -2a - a - 6 = -3a - 6$

- $D_Z = (-1)^2(1)\begin{vmatrix} -2 & -4 \\ 4 & b \end{vmatrix} + (-1)^3(1)\begin{vmatrix} 1 & -4 \\ 1 & b \end{vmatrix} + (-1)^4 6 \begin{vmatrix} 1 & -2 \\ 1 & 4 \end{vmatrix}$

$\Rightarrow D_Z = -2b + 16 - b - 4 + 24 + 12 = -3b + 48$

They system has no solution if $D = 0, D_Z \neq 0, D_x \neq 0$.

Therefore $D = 0 \Rightarrow -3a - 6 = 0 \Rightarrow a = -2$

$D_Z \neq o \Rightarrow -3b + 48 \neq 0 \neq b \neq 16$

Answer: B

Note: unique solution if $D \neq 0, D_X \neq 0, D_Y \neq 0$ and $D_Z \neq 0$ i.e., $a \neq -2$ and $b \neq 16$.

- Infinite solution if $D = 0, D_X = D_Y = D_Z = 0$ i.e., $a = -2$ and $b = 16$.

20. Let ABCD be a parallelogram and $\overline{AC}$ and $\overline{BD}$ be its diagonals. Which of the following is not true?

A. $\overline{AB} = \overline{AC} - \overline{BC}$ C. $\overline{AC} + \overline{BD} = 2\overline{BC}$

B. $\overline{AB} = \overline{BC} - \overline{BD}$ D. $\overline{CD} + \overline{BD} = \overline{AD}$

Solution:

A. $\overline{AB} = \overline{AC} - \overline{BC} \Rightarrow \overline{AC} = \overline{AB} + \overline{BC}$ from triangle law.

B. $\overline{AB} = \overline{BC} - \overline{BD} \Rightarrow \overline{BC} = \overline{AB} + \overline{BD} = \overline{AD}$

$\Rightarrow \overline{BC} = \overline{AD} \leftarrow$ opposite side of // gram.

$\therefore \overline{AB} = \overline{BC} - \overline{BD}$

C. $\overline{AC} + \overline{BD} = 2\overline{BC} \Rightarrow \overline{BC} = \frac{\overline{AC}}{2} + \frac{\overline{BD}}{2} = \overline{OC} + \overline{BD}$

$\Rightarrow \overline{BO} + \overline{OC} = \overline{BC} \therefore \overline{AC} + \overline{BD} = 2\overline{BC}$

D. $\overline{CD} + \overline{BD} = \overline{AD} \Rightarrow \overline{CD} - \overline{AD} = -\overline{BD}$

$\Rightarrow \overline{CA} = -\overline{BD} \longleftarrow$ False

Answer: D

21. Let A = (0, 1), B = (2, 7) and C = (a, b) be points in the coordinate plane. What is (a, b) so that $\overline{AC} = \overline{AB} + 2\overline{CA}$

Explanation
- $\overline{AB} - \overline{CB} = \overline{AC}$
- $\overline{DB} = -\overline{BD}$

A. $\left(\frac{1}{3}, 2\right)$ B. (1, 3)

C. (2, 3) D. $\left(\frac{2}{3}; 3\right)$

Solution: $\overline{AC} = (a - 0, b - 1), \overline{AB} = (2 - 0, 7 - 1) = 2, 6$

$\Rightarrow (a - 0, b - 1) = (2, 6) + 2(-a, 1 - b)$

$\Rightarrow (a, b - 1) = (2 - 2a, 6 + 2 - 2b)$

- $a = 2 - 2a$ and $b - 1 = 8 - 2b$
- $3a = 2$ and $3b = 9$

$\Rightarrow a = \frac{2}{3}$ and $b = 3$

$\therefore (a, b) = \left(\frac{2}{3}, 3\right)$ **Answer: D**

22. If $\bar{u} = 2i - j, \|\bar{v}\| = 5$ and $\bar{u}$ and $\bar{v}$ are perpendicular vectors then $\|\bar{u} + 3\bar{v}\|$ is equal to:

A. $\sqrt{55}$ B. $9\sqrt{5}$ C. $\sqrt{230}$ D. 20

Solution: Recall $\|\bar{a} + \bar{b}\| = \sqrt{(\bar{a} + \bar{b}).(\bar{a} + \bar{b})}$

Telegram: https://t.me/ExtremeSeries

$$\therefore \|\vec{u}+3\vec{v}\| = \sqrt{(\vec{u}+3\vec{v}).(\vec{u}+3\vec{v})}$$
$$= \sqrt{|\vec{u}|^2 + 6|\vec{u}||v|\cos(90^o) + 9|v|^2}$$
$$= \sqrt{\sqrt{5}^2 + 0 + 9(5)^2} = \sqrt{230}$$

Answer: C

23. Let ℓ be a line given by $x - 2y = 0$. A parametric vector equation of the line that passes through (5, 0) and perpendicular to ℓ is:

A. $(x, y) = (5, 0) + t(1, 2), t \in (-\infty, \infty)$
B. $(x, y) = (5, 0) + t(1, -2), t \geq 0$
C. $(x, y) = (5 + t, -2t), t \in (-\infty, \infty)$
D. $(x, y) = (5 - t, 2t), t \geq 0$

Solution: $x - 2y = 0 \Rightarrow x = 2y \Rightarrow y = \frac{1}{2}x \Rightarrow \left(x, \frac{1}{2}x\right) = t(2, 1)$. Let

$\vec{v} = (a, b) \leftarrow$ direction vector perpendicular to ℓ

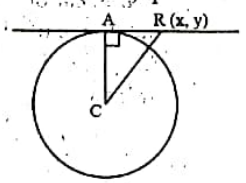
$\Rightarrow (a, b).(2, 1) = 0 \Rightarrow 2a + b = 0 \Rightarrow b = -2a$
$\Rightarrow (a, -2a) = a(1, -2)$
$\Rightarrow \vec{v} = t(1, -2)$,
$\therefore (x, y) = (5, 0) + t(1, -2) \leftarrow$ vector equation.
$\therefore (x, y) = (5 + t, -2t) \leftarrow$ parametric vector equn.

Answer: C

24. Consider circle that passes through A = (2, 3) and centered at C = (1, 1). The equation of the line tangent to the circle at A is:

A. $3x + y = 9$ C. $3x - 2y = 0$
B. $x + 2y = 8$ D. $x + y = 5$

Solution: Let R = (x, y) point on the line A = (2, 3) point of tangency on circle C = (1, 1) center of circle.



$\Rightarrow \overline{RA}.\overline{AC} = 0 \Rightarrow (2 - x, 3 - y).(-1, -2) = 0$

$\Rightarrow x - 2 + 2y - 6 = 0 \Rightarrow x + 2y = 8$

Answer: B

25. If the equation of a line ℓ is given by the dot product: $(x - 1, y + 3).(2, -1) = 0$, which of the following is the equation of the circle with center at (−2, 1) and ℓ is its tangent line?

A. $(x + 2)^2 + (y - 1)^2 = 2\sqrt{5}$
B. $(x + 2)^2 + (y - 1)^2 = 10$
C. $(x + 2)^2 + (y - 1)^2 = 5\sqrt{2}$
D. $(x + 2)^2 + (y - 1)^2 = 20$

Solution: $(x - 1, y + 3).(2, 1) = 2x - 2 - y - 3 = 0 \Rightarrow 2x - y = 5$

The perpendicular distance (radius) from line ℓ: $2x - y - 5 = 0 + 0$ center of circle (−2, 1) is given by

$$r = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}} = \frac{|-4 + (-1)(1) - 5|}{\sqrt{4+1}} = \frac{10}{\sqrt{5}} = 2\sqrt{5}$$

$\therefore$ Equation of circle: $(x + 2)^2 + (y - 1)^2 = 20$, **Answer: D**

26. Suppose anti-clock wise rotation about the origin through 90° is followed by reflection in the x-axis. What is the image of the ellipse given by $\frac{1}{4}(x-1)^2 + (y+2)^2 = 1$ under the two transformations?

A. $4(x - 2)^2 + (y + 1)^2 = 4$ C. $(x - 1)^2 + 4(y + 2)^2 = 4$
B. $4(x - 2)^2 + (y - 1)^2 = 4$ D. $(x + 1)^2 + 4(y - 2)^2 = 4$

Solution: Center $(1, -2) \xrightarrow{R_{90^o}} (2, 1) \leftarrow$ rotation with 90°
$M(2, 1) = (2, -1) \leftarrow$ reflection on x-axis.
$x -$ ellipse become y-ellipse.
$\Rightarrow$ Image will be $4(x - 2)^2 + (y + 1)^2 = 4$.

Answer: A

27. A translation takes $x^2 + (y + 1)^2 = 5$ to $(x - 2)^2 + y^2 = 5$. What is the image of (1, 3) under this transformation?

A. (3, 4) B. (3, 2) C. (−1, 2) D. (−2, 3)

Solution:
Center of $x^2 + (y + 1)^2 = 5$ is (0, −1)
Center of $(x - 2)^2 + y^2 = 5$ is (2, 0)
$T(0, -1) = (2, 0)$, $\vec{v} = (2 - 0, 0 - (-1)) = (2, 1) \leftarrow$ vector
$\therefore T(1, 3) \xrightarrow{\vec{v}=(2,1)} (2 + 1, 3 + 1) = (3, 4)$

Answer: A

Telegram: https://t.me/ExtremeSeries

28. Which of the following is a point of intersection of the sphere $(x-1)^2+(y-2)^2+(z+1)^2=6$ and the y-axis?

A. $(0,-2,0)$ B. $(0,2,0)$ C. $(0,-4,0)$ D. $(0,4,0)$

Solution: $(0, y, 0)\leftarrow$ the set of all point in y-axis.

$\Rightarrow (0-1)^2+(y-2)^2+(0+1)^2=6 \Rightarrow (y-2)^2=4$

$\Rightarrow y-2=2$ or $y-2=-2$

$\Rightarrow y=4$ or $y=0$

$\Rightarrow (0,4,0)$ or $(0,0,0)$ pt. of inter section. **Answer: D**

29. If $\theta = \arccos\left(\frac{1}{2}\right)$, what is the value of $\tan\theta$?

A. $\frac{1}{\sqrt{3}}$ B. $\sqrt{3}$ C. $\frac{\sqrt{3}}{2}$ D. 1

Solution: $\theta=\cos^{-1}\left(\frac{1}{2}\right)\Rightarrow \cos\theta=\frac{1}{2}$, $\sin\theta=\sqrt{1-\frac{1}{4}}=\frac{\sqrt{3}}{2}$

$\tan\theta=\frac{\sin\theta}{\cos\theta}=\frac{\sqrt{3}}{2}.(2)=\sqrt{3}$

Answer: B

30. If $x=\arcsin\left(\frac{3}{4}\right)$, the value of $\cos 2x$ is:

A. $\frac{3}{4}$ B. $\frac{-3}{4}$ C. $\frac{-1}{8}$ D. $\frac{1}{8}$

Solution: $x=\sin^{-1}\left(\frac{3}{4}\right)\Rightarrow \sin x=\frac{3}{4}$, $\cos x=\sqrt{1-\left(\frac{3}{4}\right)^2}=\frac{\sqrt{7}}{4}$

$\therefore \cos 2x=\cos(x+x)=\cos^2 x-\sin^2 x=\frac{7}{16}-\frac{9}{16}=\frac{-2}{16}=\frac{-1}{8}$

Answer: C

31. Let $f(x)=-3\sin\left(\frac{1}{2}x\right)$ which of the following is true?

A. The amplitude o the graph of f is -3.

B. The range of f is $[-1, 1]$.

C. f is decreasing on $[-2\pi, 0]$.

D. f is increasing on $[\pi, 3\pi]$.

Solution:

- Amplitude, $|a|=|-3|=3$
- Range is $[-3, 3]$
- f is increasing on: $\frac{1}{2}x=\left[\frac{\pi}{2},\frac{3\pi}{2}\right]\Rightarrow[\pi, 3\pi]$

Answer: D

32. Which of the following is equal to $\tan\left(x-\frac{\pi}{4}\right)$ if $\cos x\neq 0$.

A. $\frac{\sin x-\cos x}{\sin x+\cos x}$ C. $\frac{\sin x+\cos x}{\sin x-\cos x}$

B. $\frac{\cos x-\sin x}{\cos x+\sin x}$ D. $\frac{\cos x+\sin x}{\cos x-\sin x}$

Solution:

$$\tan\left(x-\frac{\pi}{4}\right)=\frac{\tan x-\tan\frac{\pi}{4}}{1+\tan x\tan\frac{\pi}{4}}=\frac{\tan x-1}{1+\tan x}$$

$$\Rightarrow \frac{\tan x-1}{1+\tan x}=\frac{\frac{\sin x}{\cos}-1}{1+\frac{\sin x}{\cos x}}=\frac{\sin x-\cos x}{\sin x+\cos x}$$

Answer: A

33. In the interval $[0, 2\pi]$, the solution set of $\cos^2 x+\sin x\cos x=1$.

A. $\left\{0,\frac{\pi}{4},\pi\frac{5\pi}{4}\right\}$ C. $\left\{0,\frac{\pi}{4},\frac{3\pi}{4},\pi\right\}$

B. $\{0, \pi\}$ D. $\left\{0,\frac{\pi}{4},\pi\right\}$

Solution: $\Rightarrow \cos^2 x+\sin x\cos x-1=0$

$\Rightarrow -\sin^2 x+\sin x\cos x=0\Rightarrow \sin x(-\sin x+\cos x)=0$

$\Rightarrow \sin x=0$ or $\cos x-\sin x=0\Rightarrow \cos x=\sin x$

$\Rightarrow x=\pi k$ or $\tan x=1\Rightarrow x=\frac{\pi}{4},\frac{\pi}{4}+\pi=\frac{5\pi}{4}$

$\therefore S.S=\left\{0,\frac{\pi}{4},\pi,\frac{5\pi}{4}\right\}$ **Answer: A**

Telegram: https://t.me/ExtremeSeries

34. The expression $\frac{\cos A}{1+\sin A}+\frac{\cos A}{1-\sin A}$ is equal to:

A. 2 csc A C. 2 cos A
B. 2 sec A D. 2 sin A

Solution: $\cos A\left[\frac{1}{1+\sin A}+\frac{1}{1-\sin A}\right]=\cos A\left[\frac{1-\sin A+\sin A}{1-\sin^2 A}\right]$

$\Rightarrow \cos A\left[\frac{2}{\cos^2 A}\right]=2\left[\frac{1}{\cos A}\right]=2\sec A$

Answer: B

35. In the set of complex numbers, the solution set of $x^2 - 4x + 5 = 0$ is:

Solution: use $x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}=\frac{4\pm\sqrt{16-20}}{2}=2\pm 2i$

$\therefore$ S.S = {2 + 2i, 2 – 2i}

36. Let z be complex number, such that $\left|\frac{3z}{-4+3i}\right|=(2-i)\left(\overline{2-i}\right)$, then the modulus of z, is equal to:

A. $\frac{25}{3}$ B. $\frac{10}{3}$ C. $\frac{14}{3}$ D. $\frac{25}{7}$

Solution: $\left|\frac{3z}{-4+3i}\right|=(2-i)(2+i)\Rightarrow\frac{3|z|}{\sqrt{4^2+3^2}}=4+1=5$

$\Rightarrow\frac{3|z|}{5}=5\Rightarrow 3|z|=25\Rightarrow|z|=\frac{25}{3}$ **Answer: A**

37. In the set of complex numbers, find the solution set of $z^3 - 8=0$.

Solution: $z^3-8=0\Rightarrow(z-2)(z^2+2z+4)=0$

$\Rightarrow z-2=0$ or $z^2+2z+4=0$

$\Rightarrow z=2$ or $z=\frac{-2\pm\sqrt{4-16}}{2}=-1\pm\sqrt{3}\,i$

$\therefore$ S.S = {2, $-1+\sqrt{3}\,i$, $-1-\sqrt{3}\,i$} ← in rectangular form

In polar form, $z = x + yi = r(\cos\theta + i\sin\theta)$

$\Rightarrow z_1=-1+\sqrt{3}\,i\Rightarrow|z_1|=r=\sqrt{1^2+\sqrt{3}^2}=2$,

$\theta=\tan^{-1}\left(-\sqrt{3}\right)=\frac{2\pi}{3}\Rightarrow z_2=-1-\sqrt{3}\,i$

$\Rightarrow|z_2|=r=\sqrt{1^2+\sqrt{3}^2}=2,\theta=\tan^{-1}\left(\frac{-\sqrt{3}}{-1}\right)=\frac{4\pi}{3}$

Therefore the polar form:

$S.S=\left\{2\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right),2\left(\cos\frac{4\pi}{3}+i\sin\frac{4\pi}{3}\right)\right\}$

38. Solve, $z^3 + 8z = 0$, where z = x + yi

Solution: $z(z^2+8)=0\Rightarrow z(z-\sqrt{8}i)\left(z+\sqrt{8}i\right)=0$

$\Rightarrow z=0$, $z=\sqrt{8}\,i$ or $z=-\sqrt{8}i$

$\therefore$ S.S = $\left\{0,2\sqrt{2}i,-2\sqrt{2}i\right\}$

39. If k- the term of a sequence is $a_k=\frac{(-1)^k(2-k)}{2k-1}$ what is its next term a_{k+1} when k is even:

A. $\frac{k-1}{2k+1}$ C. $\frac{k-3}{2k+1}$
B. $\frac{3-k}{2k+3}$ D. $\frac{1-k}{2k+1}$

Solution: $a_{k+1}=\frac{(-1)^{k+1}(2-(k+1))}{2(k+1)-1}=\frac{(-1)^k(-1)(1-k)}{2k+1}$

$\Rightarrow a_{k+1}=\frac{-(1-k)}{2k+1}=\frac{k-1}{2k+1}$, if $(-1)^k=1$, for $k=2, 4, 6, \ldots$

Answer: A

40. Consider the sequence $\{a_n\}$ where the first term is $a_1 = 2$ and $a_{n+1} = a_n + 4$ for all $n \geq 1$. Then, the sum $a_3 + a_4 + a_5 + \ldots a_{35}$ is equal to:

A. 2460 B. 2458 C. 2442 D. 2450

Solution: $d=\frac{a_{n+1}-a_n}{n+1-n}=\frac{(a_n+4)-a_n}{1}=4$

$\Rightarrow$ Total term from a_3 to a_{35} is $35 - 3 + 1 = 33$ and

$a_3 = A_1 = a_1 + (n-1)d = 2 + 2(4) = 10$

$\therefore S_{33}=\frac{33}{2}(2A_1+32d)=\frac{33}{2}(2(10)+32(4))=2442$

Answer: C

Telegram: https://t.me/ExtremeSeries

41. Suppose a geometric sequence $\{G_n\}_n^{\infty}$ has terms $G_{10} = 192$ and $G_6 = 12$. Which of the following is equal to G_2?

A. $\frac{3}{2}$ B. $\frac{3}{4}$ C. $\frac{3}{8}$ D. $\frac{3}{16}$

Solution: $\frac{G_{10}}{G_6} = \frac{G_1 r^9}{G_1 r^5} = \frac{192}{12} \Rightarrow r^4 = 16 \Rightarrow r = 2$

From G_6 we have, $G_6 = G_1 r^5 \Rightarrow 12 = G_1 (2)^5 \Rightarrow G_1 = \frac{12}{32} = \frac{3}{8}$

$G_2 = G_1 r \Rightarrow G_2 = \left(\frac{3}{8}\right)(2) = \frac{6}{8} = \frac{3}{4}$

Answer: B

42. The sum of the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{(3^n)^2}$ is equal to:

A. $\frac{9}{8}$ B. $\frac{9}{10}$ C. $\frac{-9}{10}$ D. $\frac{-1}{10}$

Solution: $\sum_{n=1}^{\infty} \frac{(-1)^n}{(3^n)^2} = \frac{-1}{3^2} + \frac{1}{3^4} - \frac{1}{3^6} + \ldots, G_1 = \frac{-1}{9}, r = -\frac{1}{9}$

$\Rightarrow S_{\infty} = \frac{G_1}{1-r} = \frac{\frac{-1}{9}}{1+\frac{1}{9}} = \left(-\frac{1}{9}\right)\left(\frac{9}{10}\right) = \frac{-1}{10}$

Answer: D

43. Which of the following is true about $\left\{2 + \frac{(-1)^n}{n+1}\right\}_{n=1}^{\infty}$?

A. The least upper bounds of the sequence is $\frac{7}{3}$.

B. The sequence is monotonic.

C. 1.6 is its lower bound.

D. $2 - \frac{1}{15}$ is term of sequence.

Solution: If $n = 1, 3, 5, 7, \ldots$

$\Rightarrow \left\{2 + \frac{(-1)^n}{n+1}\right\} = \left\{2 + \frac{(-1)}{n+1}\right\} = \left\{2 - \frac{1}{2}, 2 - \frac{1}{4}, \ldots 2\right\}$

If $n = 2, 4, 6, 8, \ldots$ then $\left\{2 + \frac{1}{n+1}\right\} = \left\{2 + \frac{1}{3}, 2 + \frac{1}{7}, \ldots 2\right\}$

$\therefore \frac{3}{2} \le \left\{2 + \frac{(-1)^n}{n+1}\right\} \le \frac{7}{3}$, so that

glb $= \frac{3}{2}$, and lub $= \frac{7}{3}$

Answer: A

44. Which of the following is a divergent sequence?

A. $\left\{\left(\sqrt{2}\right)^{-n}\right\}$ B. $\left\{5^{\frac{1}{n}}\right\}$ C. $\left\{\left(\frac{\pi}{3}\right)^n\right\}$ D. $\left\{1 + \frac{(-1)^n}{7^n}\right\}$

Solution: $\lim_{n\to\infty}\left(\frac{1}{\sqrt{2}^n}\right) = 0 \leftarrow$ convergent sequence

- $\lim_{n\to\infty} 5^{\frac{1}{n}} = 1 \leftarrow$ convergent
- $\lim_{n\to\infty}\left(1 + \frac{(-1)^n}{7^n}\right) = 1 \leftarrow$ convergent

But $\lim_{n\to\infty}\left(\frac{\pi}{3}\right)^n$ doesn't exist because $\frac{\pi}{3} > 1$

$\therefore \left\{\left(\frac{\pi}{2}\right)^n\right\}$ diverges

Answer: C

45. $\lim_{n\to\infty} \frac{2n^2 - n + 1}{6n^2 + 1}$ is equal to:

A. 2 B. $\frac{1}{3}$ C. $\frac{2}{3}$ D. ∞

Solution: Dividing numerator and denominator by n^2.

$\Rightarrow \lim_{n\to\infty} \frac{2 - \frac{1}{n} + \frac{1}{n^2}}{6 + \frac{1}{n^2}} = \frac{2 - 0 + 0}{6 + 0} = \frac{2}{6} = \frac{1}{3}$

Answer: B

Telegram: https://t.me/ExtremeSeries

46. $\lim_{x\to 1}\frac{x^3+x-2}{x^2-x}$ is equal to:

A. −2 B. 1 C. 4 D. doesn't exist

Solution: $\lim_{x\to 1}\frac{x^3+x-2}{x^2-x}=\lim_{x\to 1}\frac{(x-1)(x^2+x+2)}{(x-1)(x)}=1+1+2=4$

Answer: C

47. What is the left hand limit, $\lim_{x\to 0}\frac{|x|-xe^x}{x}$:

A. 0 B. −2 C. −1 D. Doesn't exist

Solution: Note, $|x|=-x$ if $x<0$

$\Rightarrow \lim_{x\to 0^-}\frac{|x|-xe^x}{x}=\lim_{x\to 0^-}\frac{-x-xe^x}{x}=\lim_{x\to 0^-}\frac{x(-1-e^x)}{x}$

$=-1-1=-2$

Answer: B

48. Let $f(x)=\begin{cases}a-\cos x, x\le 0\\ 3e^x-a, x>0\end{cases}$ for what value of a f is continuous function.

A. 1 B. $\frac{3}{2}$ C. 2 D. 3

Solution: $\lim_{x\to 0} f(x)=f(0)\Rightarrow \lim_{x\to 0^+} f(x)=f(0)$

$\Rightarrow 3-a=a-1\Rightarrow 4=2a\Rightarrow a=2,$

Answer: C

49. If $f(x)=x^3$, then $\lim_{h\to 0}\frac{f(2+h)-f(2)}{h}$ is equal to:

A. 4 B. 1 C. 12 D. ∞

Solution: Note, $f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$

$f(x)=x^3\Rightarrow f'(x)=3x^2\Rightarrow f'(2)=3(2^2)=12$

$\therefore f'(2)=\lim_{h\to 0}\frac{f(2+h)-f(2)}{h}=12$

Answer: C

50. Which of the following is the equation of the line tangent to the graph of $f(x)=\begin{cases}x-\pi, x\le 0\\ 1-2\sin x, x>0\end{cases}$ at $(\pi, f(\pi))$

A. $y=2x-2\pi+1$ C. $y=-2x+2\pi+1$

B. $y=2x-2\pi-1$ D. $y=-2x+2\pi-1$

Solution: $y-f(\pi)=f'(\pi)(x-\pi)\Rightarrow f'(x)=\begin{cases}1, x<0\\ -2\cos x, x>0\end{cases}$

$\Rightarrow f'(\pi)=-2\cos(\pi)=(-2)(-1)=2$

$\therefore y-1=2(x-\pi)\Rightarrow y=2x-2\pi+1$

Answer: A

51. Suppose $f(x)$ is differentiable at all $x\in(-\infty,\infty)$ and the graph of $y=f'(x)$ is as shown in the figure below. Which of the following is true:

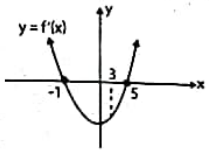
A. $f(x)$ increasing on $[3,\infty\}$

B. $f(x)$ attains its minimum value at $x=3$

C. $f(x)$ decrease on $[-1, 5]$

D. $f(x)$ attains its maximum value at $x=5$

Solution: $f'(x)\ge 0$, on $(-\infty,-1]$ and $[5,\infty)$



$\therefore f(x)$ is increasing on $(-\infty,-1]$ and $[5,\infty)$

- $f'(x)\le 0$, on $[-1, 5]$

$\therefore f(x)$ is decreasing on $[-1, 5]$

- $f'(-1)=f'(5)=0$
- f' changes the sign from (+) to (−) at $x=-1$

  $\therefore f(x)$ has local maximum at $x=-1$.
- f' changes the sign from (−) to (+) at $x=5$

  $\therefore f$ has local minimum at $x=5$.

Answer: C

52. Let $f(x)=x^3-12x+1$ be defined on $[0, 3]$. What is the range of f:

A. $(-\infty,\infty)$ C. $[-15, 1]$

B. $(-\infty, 1]$ D. $[-15, 17]$

Solution: To find critical number set $f'(x)=0$

$\Rightarrow 3x^2-12=0\Rightarrow x^2=\frac{12}{3}=4\Rightarrow x=2\in[0,3]$

$\Rightarrow f(0)=0-0+1=1\leftarrow$ maximum of f

$\Rightarrow f(2)=8-24+1=-15\leftarrow$ minimum of f

$\Rightarrow f(3)=27-36+1=-8$

$\therefore$ The range of f is $[-15, 1]$

Answer: C

Telegram: https://t.me/Extreme Series

53. If $f(x) = \ln(2 + \sec x)$, then $f'(x)$ is equal to:

A. $\dfrac{\tan x}{2+\sec x}$ C. $\dfrac{\cos x}{1+2\cos x}$

B. $\dfrac{\tan x}{2+\cos x}$ D. $\dfrac{\tan x}{1+2\cos x}$

Solution: $f'(x) = \dfrac{d}{dx}f(x) = \dfrac{d}{dx}\ln u = \dfrac{1}{u} = \dfrac{du}{dx}$, let $u = 2 + \sec x \Rightarrow \dfrac{du}{dx} = \sec x.\tan x$ so that:

$$\frac{d}{dx}\ln(u) = \frac{1}{u}\frac{du}{dx} = \frac{\sec x \tan x}{2+\sec x}$$

$$f'(x) = \frac{(\sec x \times \tan x).\cos x}{2\cos x + 1} = \frac{\tan x}{2\cos x + 1}$$

Answer: D

54. Which of the following is the set of all critical numbers of $f(x) = \begin{cases} x-1, x<1 \\ x^2 - x, x \geq 1 \end{cases}$:

A. $\{1\}$ B. $\left\{\frac{1}{2}\right\}$ C. $\left\{\frac{1}{2}, 1\right\}$ D. $\{\}$

Solution: $f'(x) = \begin{cases} 1, x<1 \\ 2x-1, x \geq 1 \end{cases}$

Set $f'(x) = 0 \Rightarrow f'(x) \neq 0$, for all $x \in [1, \infty) \Rightarrow x = \dfrac{1}{2} \notin [1, \infty)$

$\therefore f$ has no critical number

Answer: D

55. Let $h(x) = x + 3^x$ and $F(x) = (f \circ h)(x)$ such that $f'(4) = -1$. Then $F'(1) =$ _____

A. $1 + 3\ln 3$ C. $4 + 12\ln 3$

B. $-1 - 3\ln 3$ D. $-4 - 12\ln 3$

Solution: $F'(x) = f'(h(x)).h'(x)$ and $h(1) = 1 + 3 = 4$

$\Rightarrow h'(x) = 1 + 3^x \ln 3 \Rightarrow h'(1) = 1 + 3\ln 3$

$\therefore F'(1) = f'(h(1)).h'(1) = f'(4).(1 + 3\ln 3) = -1 - 3\ln 3$,

Answer: B

56. The rate of certain chemical reaction is given by $r(x) = kx(a - x)$ for $0 \leq x$, $a > 0$ and $k > 0$. For what value of x does the rate of reaction increase?

A. $0 \leq x \leq \dfrac{a}{2}$ C. $-\infty \leq x \leq \dfrac{a}{2}$

B. $\dfrac{a}{2} \leq x \leq a$ D. $\dfrac{a}{2} \leq x \leq \dfrac{3a}{4}$

Solution: $r(x) = akx - kx^2 \Rightarrow r'(x) = ak - 2kx$

$\Rightarrow r'(x) \geq 0 \Rightarrow ak - 2kx \geq 0 \Rightarrow 2kx \leq ak$

$\Rightarrow x \leq \dfrac{a}{2} \Rightarrow 0 \leq x \leq \dfrac{a}{2}$

Answer: A

57. Which of the following is equal to $\int_0^{\pi/3} \dfrac{\tan x}{\cos x} dx$:

A. $\dfrac{1}{2}$ B. 1 C. -1 D. $-1 + \dfrac{2\sqrt{3}}{3}$

Solution: $\int_0^{\pi/3} \dfrac{\tan x}{\cos x} dx = \int_0^{\pi/3} \dfrac{\sin x}{\cos^2 x} dx$,

Let $u = \cos x$, $du = -\sin x\, dx$

$$\therefore \int_1^{\frac{1}{2}} \frac{-du}{u^2} = \frac{1}{u}\Big|_1^{\frac{1}{2}} = 2 - 1 = 1$$

58. The area of the region bounded between the graph of $f(x) = x^3$ and the x-axis form $x = -1$ to $x = 2$ is:

A. $\dfrac{15}{4}$ B. $\dfrac{17}{4}$ C. 4 D. 16

Solution: $\int_{-1}^{0} f(x)dx + \int_0^2 f(x)dx = -\int_{-1}^{0} x^3 dx + \int_0^2 x^3 dx$

$$\Rightarrow -\left(\frac{x^4}{4}\right)\Big|_{-1}^{0} + \left(\frac{x^4}{4}\right)\Big|_0^2 = 0 - \left(0 - \frac{1}{4}\right) + \left(\frac{2^4}{4} - 0\right) = \frac{1}{4} + 4 = \frac{17}{4}$$

Answer: B

Telegram: https://t.me/ExtremeSeries

59. $\int \frac{(e^x+1)^2}{e^x}dx$ equal to:

A. $e^x+e^{-x}+2x+c$ C. $\frac{1}{3}(e^x+1)^3+c$

B. $e^x-e^{-x}+2x+c$ D. $\frac{1}{3}(1-e^{-x})^3+c$

Solution: $\Leftrightarrow \int \frac{e^{2x}+e^x+1}{e^x}dx$

$\Rightarrow \int (e^x+e^{-x}+2)dx = e^x-e^{-x}+2x+c$

Answer: B

1. What are the respective values of A, B and C so that $\frac{x-1}{x^3+x}=\frac{A}{x}+\frac{Bx+C}{x^2+1}$?

A. $-1, 1, 1$ B. $-1, 1, 2$ C. $1, -2, 0$ D. $3, 1, 2$

Solution:

Multiplying both side by $x(x^2+1)$

$\Rightarrow x-1=A(x^2+1)+Bx^2+Cx,$

$Ox^2+x-1=Ax^2+Bx^2+Cx+A$ ← comparing the coefficient of variable with left side.

$\Rightarrow A+B=0$ and $Cx=x \Rightarrow C=1$, and $A=-1, B=1$

Answer: A

2. The solution set of $1-\frac{5}{x^2+4}=\frac{x-1}{x^2-x}-\frac{1}{x}$.

A. $\{1,-1\}$ B. $\{2\}$ C. $\{-1\}$ D. $\{1, 2, -\}$

Solution: Multiply by LCM both side by $(x^2+4)(x^2-x)x$, gives.

$\Rightarrow (x^2+4)(x^2-x)(x)-5(x^2-x)x$

$=(x^{2+}+4)(x-1)x-(x^2+4)(x^2-x)$

$\Rightarrow (x^2-x)x[x^2+4-5]=(x^2+4)[x^2-x-(x^2-x)]$

$\Rightarrow (x^2-x)(x^2-1)=0 \Leftrightarrow x(x-1)(x-1)(x+1)=0$

$\Rightarrow x=0, x=1, x=-1$

$\therefore$ The solution set $=\{-1\}$ since Domain $=\{x: x\in\mathbb{R}\mid 0, 1\}$

Answer: C

3. The simplified form of $\frac{t-\frac{1}{t}}{1+\frac{t}{1-t}}\div\frac{t^2-2t+1}{t}, t\neq 0,1$.

A. $t+1$ B. $t-1$ C. $1-t$ D. $-t-1$

Solution: The expression the same as

$$\Rightarrow \left(\frac{\frac{t^2-1}{t}}{\frac{1-t+t}{1-t}}\right)\cdot\frac{t}{(t-1)^2} \Leftrightarrow \frac{(t-1)(1-t)(t+1)}{t}\cdot\frac{t}{(t-1)(t-1)}$$

$\Rightarrow -(t+1)=-t-1,$

Answer: D

Telegram: https://t.me/ExtremeSeries

4. Which of the following is not true about the graph of $g(x)=\frac{x^2-1}{x^2+1}$?

A. The range of g is $(-\infty, 1)$.
B. g is an even function.
C. $y=1$ is H. A.
D. As $x \to \infty, g(x) \to 1$

Solution: To find the range, put $y = g(x)$ and solve for x.

$$\Rightarrow y=\frac{x^2-1}{x^2+1} \Leftrightarrow x^2y+y=x^2-1 \Rightarrow y+1=x^2-x^2y$$

$$\Leftrightarrow x^2(1-y)=y+1 \Rightarrow x^2=\frac{y+1}{1-y} \Rightarrow x=\sqrt{\frac{1+y}{1-y}}$$

A. The range, $=\left\{y:\frac{1+y}{1-y}\geq 0\right\}=\{y: -1 \leq y<1\}$

B. $g(-x)=g(x) \leftarrow$ it is even.

C. $\frac{x^2-1}{x^2+1}=\frac{x^2+1}{x^2+1}-\frac{2}{x^2+1}=1-\frac{2}{x^2+1}$, $\therefore y=1 \leftarrow$ H.A

D. As $x \to \infty, g(x) \to 1$, from c

Answer: A

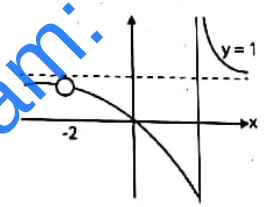
5. Which of the following functions could most likely be drawn in the figure below?

A. $f(x)=\frac{x}{x-2}$

B. $f(x)=\frac{x^2+2x}{x^2-4}$

C. $\frac{x+1}{x-2}$

D. $\frac{x^3+2x}{x^2-4}$



Solution: $\frac{x}{x-2}$ and $\frac{x+1}{x-2}$ has no hole at $x = -2$ and $\frac{x^3+2x}{x^2-4}$ has no horizontal asymptote thus, $f(x)=\frac{x^2+2x}{x^2-4}=\frac{x(x+2)}{(x-2)(x+2)}$ satisfy the graph.

Answer: B

6. The equation of the circle tangent to the x-axis at (6, 0) whose center is on the line $y=\frac{1}{2}x$ is:

A. $(x-6)^2+y^2=9$
B. $(x-6)^2+(y-3)^2=36$
C. $(x-6)^2+y^2=36$
D. $(x-6)^2+(y-3)^2=9$

Solution: Here,

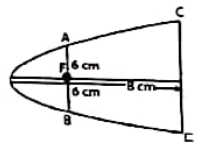
$\Rightarrow$center(h, k)$=\left(h, \frac{1}{2}h\right)=(2k, k)$, since lies on $y = 2x$

$\Rightarrow (x-h)^2+(y-k)^2=r^2 \leftarrow$ eqn of circle
$\Rightarrow (x-h)^2+(y-k)^2=k^2 \leftarrow$ tangent to x-axis
$\Rightarrow (x-2k)^2+(y-k)^2=k^2 \leftarrow$ put $h = 2k$
$\Rightarrow (6-2k)^2+(0-k)^2=k^2 \leftarrow$ (6, 0) on circle
$\Rightarrow (6-2k)^2+k^2=k^2 \Leftrightarrow (6-2k)^2=0$
$\Rightarrow k = 3$, thus (6, 3) is center
$\therefore (x-6)^2+(y-3)^2=9$

Answer: D

7. Across section of a parabolic reflector is shown below. The opening at the focus, AB is 12 cm. What is the diameter of the opening $\overline{CD}$, 8 cm from the vertex.

A. $8\sqrt{6}$ cm
B. 12 cm
C. 9 cm
D. $6\sqrt{8}$ cm



Solution: $\overline{AB} = 4P = 12 \Rightarrow P = 3$; $4Px = y^2 \leftarrow$ the eqn of parabola
$\Rightarrow 4(3)x=y^2 \Leftrightarrow y^2=12x \leftarrow$ eqn of parabola.
$y^2 = 12(8) = 96 \Rightarrow y=\pm 4\sqrt{6}$
$\Rightarrow \overline{CD}=2y=2\left(4\sqrt{6}\right)=8\sqrt{6}$

Answer: A

8. The equation $3x^2-6x=y^2$ represents:

A. an ellipse with center at (1, 0)
B. a hyperbola with center (1, 0)
C. an ellipse with center at (–1, 0)
D. a hyperbola with one of its foci at (1, 0)

Solution: $3x^2-6x=y^2 \Leftrightarrow 3x^2-y^2-6x=0$
$\Leftrightarrow 3x^2-6x-y^2=0 \Rightarrow 3(x^2-2x)-y^2=0$
$\Rightarrow 3(x^2-2x+1)-y^2=0+3 \leftarrow$ completing square

Telegram: https://t.me/ExtremeSeries

$\Rightarrow 3(x-1)^2 - y^2 = 3 \Leftrightarrow (x-1)^2 - \frac{y^2}{3} = 1$

$\Rightarrow$ hyperbola with center (1, 0) **Answer: B**

9. A mirror has an elliptic edge. If the shortest and longest distance between any two points on the edge of the mirror through the geometric center of the mirror are respectively 6 cm and 10 cm, which of the following is a possible equation of the edge of the mirror?

A. $\frac{x^2}{9} + \frac{(y-1)}{5} = 1$ C. $\frac{(x+1)^2}{36} + \frac{y^2}{100} = 1$

B. $\frac{(x-1)^2}{25} + \frac{y^2}{9} = 1$ D. $\frac{(x+1)^2}{36} + \frac{(y-1)^2}{25} = 1$

Solution: $2a = 10$ cm and $2b = 6$ cm

$\Rightarrow a = 5$ cm and $b = 3$cm $\Rightarrow$ so that $a^2 = 25$ and $b^2 = 9$

$\Rightarrow$ The possible equation will be $\frac{(x-1)^2}{25} + \frac{y^2}{9} = 1$

Answer: B

10. If $(\neg p \vee Q) \Rightarrow (R \wedge \neg R)$ is true then which of the following is necessarily true?

A. $R \Rightarrow Q$ C. $P \wedge \neg R$

B. $(Q \Rightarrow R) \wedge R$ D. $(Q \wedge R) \Rightarrow P$

Solution: since $R \wedge \neg R \equiv F$. Hence $\neg p \vee Q = F$,

$\therefore \neg P \equiv F,\ Q \equiv F, P \equiv T$

$\therefore (Q \wedge R) \Rightarrow P \equiv T,$ **Answer: D**

11. Suppose $A = \begin{pmatrix} 4 & 0 \\ 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 2 \\ 6 & -4 \end{pmatrix}$. If x is 2×2 matrix such that $x + \frac{1}{2}B^t = AB$, then x is equal to:

A. $\begin{pmatrix} 7 & 5 \\ 5 & -4 \end{pmatrix}$ B. $\begin{pmatrix} 5 & -3 \\ 5 & 0 \end{pmatrix}$ C. $\begin{pmatrix} 5 & -3 \\ 5 & -2 \end{pmatrix}$ D. $\begin{pmatrix} 7 & 5 \\ 7 & 0 \end{pmatrix}$

Solution: $x + \frac{1}{2}B' = AB \Rightarrow x = AB - \frac{1}{2}(B')$

$\Rightarrow x = \begin{pmatrix} 4 & 0 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 2 & 2 \\ 6 & -4 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 2 & 6 \\ 2 & -4 \end{pmatrix}$

$\Rightarrow x = \begin{pmatrix} 8 & 8 \\ 8 & -2 \end{pmatrix} - \begin{pmatrix} 1 & 3 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} 7 & 5 \\ 7 & 0 \end{pmatrix}$ **Answer: D**

12. $\begin{vmatrix} 1 & -a & b-a \\ 1 & b & a+b \\ 0 & c & c+c \end{vmatrix}$ is equal to:

A. 0 B. $-2ac$ C. $2bc$. D. $2c(b-a)$

Solution: Adding or multiplying of row (column) of matrix from another row (or column) the value of the determinant remains unchanged. Therefore the given matrix determinant is the same as:

$$\begin{vmatrix} 1 & -a & b \\ 1 & b & a \\ 0 & c & c \end{vmatrix} = 0\begin{vmatrix} -a & b \\ b & a \end{vmatrix} + c(-1)\begin{vmatrix} 1 & b \\ 1 & a \end{vmatrix} + c\begin{vmatrix} 1 & -a \\ 1 & b \end{vmatrix}$$

$\Rightarrow 0 - c(a-b) + c(b+a) = 2bc$ **Answer: C**

13. If θ is the angle between the vectors $\vec{A} = 2\vec{i} - \vec{j}$ and $\vec{B} = \vec{i} + \vec{j}$, then $\cos\theta$ is _____

A. $\frac{1}{\sqrt{10}}$ B. $\frac{1}{2}$ C. $\frac{\sqrt{3}}{2}$ D. $\frac{2}{\sqrt{5}}$

Solution: $\cos\theta = \frac{\vec{A}.\vec{B}}{|\vec{A}.\vec{B}|} = \frac{2-1}{\sqrt{5}\sqrt{2}} = \frac{1}{\sqrt{10}}$

Answer: A

14. If $\vec{A} = (3, -3)$ and $\vec{B} = (1, -3)$, what is the unit vector in the direction of $\vec{C} = 3\vec{A} - \vec{B}$.

A. $\left(\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\right)$ C. $\left(\frac{4}{5}, \frac{-3}{5}\right)$

B. $\left(\frac{1}{\sqrt{10}}, \frac{-3}{\sqrt{10}}\right)$ D. $\left(\frac{-6}{10}, \frac{8}{10}\right)$

Solution:

$$\frac{\vec{C}}{|\vec{C}|} = \frac{3(3,-3) - (1,-3)}{|3\vec{A} - \vec{B}|} = \frac{(8,-6)}{\sqrt{8^2 + 6^2}} = \left(\frac{4}{5}, \frac{-3}{5}\right)$$

Answer: C

Telegram: https://t.me/ExtremeSeries

15. If $\vec{u}$ is vector in the direction as vector $\vec{v} = (-3, 4)$ having length 3 times the length of $\vec{v}$, what is the length of the vector $\vec{u} - \frac{3}{2}\vec{v} - 2\vec{j}$, where $\vec{j}$ is unit vector in the direction of the positive y-axis.

A. $\frac{1}{2}\sqrt{185}$ B. $\frac{1}{2}\sqrt{145}$ C. $\frac{21}{2}$ D. $\frac{17}{2}$

Solution: Let $\vec{w} = 3(-3,4) - \frac{3}{2}(-3,4) - 2(0,1) = \left(\frac{-9}{2}, 4\right)$

$\therefore |\vec{w}| = \sqrt{\left(\frac{-9}{2}\right)^2 + (4)^2} = \sqrt{\frac{81}{4} + 16} = \frac{1}{2}\sqrt{145}$ **Answer: B**

16. Consider a circle given by $x^2 + y^2 = 2$ and the line ℓ given by parametric vector equation $(x, y) = (2, 0) + t(-1, 1)$. Which of the following is true?

A. ℓ is tangent to the circle at $\left(\frac{1}{2}, \frac{\sqrt{7}}{2}\right)$.

B. ℓ intersects with the circle at two distinct point

C. ℓ and the circle have no common point

D. The distance from the center of the circle to ℓ is $\sqrt{2}$.

Solution: $(x, y) = (2, 0) + t(-1, 1)$ ← parametric eqn of ℓ

$\Rightarrow x = 2 - t$ and $y = 0 + t$

$\Rightarrow x = 2 - y$ ← solving for t

$\Rightarrow x + y = 2$ ← eqn of ℓ

Eqn of circle and the line ℓ intersect at $x^2 + y^2 = 2$

$\Leftrightarrow x^2 + (2 - x)^2 = 2$ ← put $y = 2 - x \Leftrightarrow x^2 + x^2 - 4x + 4 = 2$

$\Leftrightarrow 2x^2 - 4x + 2 = 0 \Rightarrow x^2 - 2x + 1 = 0 \Rightarrow (x - 1)^2 = 0 \Rightarrow x = 1, y = 1$

$\therefore (1, 1)$ ← point of tangency of ℓ to circle

The distance from center (0, 0) of circle to ℓ to: $x + y - 2 = 0$ is

$\frac{|0 + 0 - 2|}{\sqrt{1^2 + 1^2}} = \frac{2}{\sqrt{2}} = \sqrt{2}$

Answer: D

17. The reflection of the parabola $y = x^2 + 1$ in the line $y = -x$.

A. $y = -x^2 - 1$ C. $x = y^2 + 1$

B. $y = x^2 - 1$ D. $x = -y^2 - 1$

Solution: Recall, reflection in the line $y = -x$

• M $((x, y)) = (-y, -x)$, so that $y \to -x$, $x \to -y \Rightarrow -x = (-y)^2 + 1$

$\Rightarrow -x = y^2 + 1 \Leftrightarrow x = -y^2 - 1$ **Answer: D**

18. If $\theta = \arcsin\left(\frac{1}{2}\right)$ then $\tan(2\theta)$ is equal to:

A. $\frac{1}{\sqrt{3}}$ B. 1 C. $\sqrt{3}$ D. $\frac{\sqrt{3}}{2}$

Solution: $\theta = \sin^{-1}\left(\frac{1}{2}\right) \Leftrightarrow \sin\theta = \frac{1}{2} \Rightarrow \cos^2\theta = 1 - \sin^2\theta =$

$1 - \frac{1}{4} = \frac{3}{4} \Rightarrow \cos\theta = \frac{\sqrt{3}}{2}$

$\therefore \tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{1}{\sqrt{3}} \Rightarrow \tan 2\theta = \frac{2\tan\theta}{1 - \tan^2\theta} = \frac{\frac{2}{\sqrt{3}}}{1 - \frac{1}{3}} = \sqrt{3}$

Answer: C

19. What is the amplitude and period, respectively of the graph of $f(x) = 2 - 3\sin\left(\frac{x + \pi}{4}\right)$?

A. $2, \frac{\pi}{4}$ B. $-3, 8\pi$ C. $3, \frac{\pi}{2}$ D. $3, 8\pi$

Solution: Amplitude, $|a| = |-3| = 3$

Period, $P = \frac{2\pi}{k} = \frac{2\pi}{\frac{1}{4}} = 8\pi$ **Answer: D**

20. The simplified form $(1 + \tan^2 x)\sin\left(x - \frac{3}{2}\pi\right)$.

A. $\sec x$ B. $\cos x$ C. $-\sec x$ D. $-\cos x$

Solution: $1 + \tan^2 x = \sec^2 x$ and

Telegram: https://t.me/Extremeseries

$$\sin\left(x-\frac{3\pi}{2}\right)=\sin x\cos\left(\frac{3\pi}{2}\right)-\sin\frac{3\pi}{2}\cos x=\cos x.$$

$$\therefore (\sec^2 x)\cos x=\frac{\cos x}{\cos^2 x}=\frac{1}{\cos x}=\sec x$$ **Answer: A**

21. The solution set of $\sqrt{3}\sin 2x=\cos 2x$ on $[0,\pi]$.

A. $\left\{\frac{\pi}{6},\frac{\pi}{12}\right\}$ B. $\left\{\frac{\pi}{6},\frac{\pi}{3}\right\}$ C. $\left\{\frac{\pi}{12},\frac{7\pi}{12}\right\}$ D. $\left\{\frac{\pi}{6},\frac{2\pi}{3}\right\}$

Solution: $\frac{\sqrt{3}\sin 2x}{\cos 2x}=\frac{\cos 2x}{\cos 2x}\Rightarrow\tan 2x=\frac{1}{\sqrt{3}}$

$\Rightarrow 2x=\frac{\pi}{6},\frac{\pi}{6}+\pi k\Rightarrow x=\frac{\pi}{12},\frac{\pi}{12}+\frac{\pi}{2}k$, $\Rightarrow$ when $k=1$, the solution set $=\left\{\frac{\pi}{12},\frac{7\pi}{12}\right\}$ **Answer: C**

22. Let $\{a_n\}_{n=1}^{\infty}$ be a sequence such that $a_{n+1}=-na_n$, $a_1=1$; $n=1,2,3,\ldots$ which of the following is not true about the sequence.

A. $a_3=2$ C. $a_n>0$ if n is odd

B. $a_4=-3$ D. $a_n<0$ if n is even

Solution: $a_{1+1}=a_2=-a_1=-1\,(1)=-1$

$a_{2+1}=a_3=-2a_2=-2\,(-1)=2$

$a_{3+1}=a_4=-3a_3=-3\,(2)=-6$ **Answer: B**

23. Let $S_n=4n^2$, $n=1,2,3,\ldots$ be the sum of first n-terms of a sequence $\{a_n\}_{n=1}^{\infty}$. Then a_3:

A. 8 B. 12 C. 16 D. 20

Solution: $a_n=S_n-S_{n-1}\Rightarrow a_3=S_3-S_2=4\,(3)^2-4(2)^2=36-16=20$ **Answer: D**

24. Let $\{a_n\}_{n\geq 1}$ be sequence $a_1=2$, and $a_n=3a_{n-1}$ for all $n\geq 2$. Then $\sum_{i=1}^{20}a_i$ is equal to:

A. $3^{20}-1$ B. $1-\frac{1}{3^{20}}$ C. $1-\frac{1}{2^{21}}$ D. $3^{-20}-1$

Solution: $S_{20}=\frac{2[3^{20}-1]}{3-1}=3^{20}-1$ **Answer: A**

25. $\lim_{n\to\infty}\frac{n+\cos 3n}{2-n}$ equal:

A. -1 B. 0 C. 1 D. 3

Solution: $\lim_{n\to\infty}\frac{\frac{n+\cos 3n}{n}}{\frac{2-n}{n}}=\lim_{n\to\infty}\frac{1+\frac{\cos 3n}{n}}{\frac{2}{n}-1}=\frac{1+0}{0-1}=-1$

Answer: A

26. The left hand limit, $\lim_{x\to 0^-}\frac{|x|-2xe^x}{x}$ is equal to:

A. 0 B. -2 C. -3 D. doesn't exist

Solution: $x\to 0^-$ means $x<0\Rightarrow |x|=-x$

$\Rightarrow\lim_{x\to 0^-}\frac{-x-2xe^x}{x}=\lim_{x\to 0^-}(-1-2e^x)=-1-2=-3$

Answer: C

27. $\lim_{x\to\pi}\frac{x\sin 2x}{x-\pi}$ equal to _____

A. 2π B. π C. 0 D. 1

Solution: Let $y=x-\pi\Rightarrow x=y+\pi, x\to\pi, y\to 0$

$$\therefore\lim_{y\to 0}(y+\pi)\frac{\sin(2y+2\pi)}{y}=\lim_{y\to 0}\left[\frac{(\sin 2y\cos 2\pi)}{y}+\frac{\cos 2y\sin 2\pi}{y}\right].$$

Answer: A

28. Let $f(x)=\frac{\sqrt{x}-2}{\frac{-4}{x}+1}$. Then $\lim_{x\to 4^+}f(x)$ is equal to

A. 0 B. $\frac{1}{4}$ C. 1 D. $-\infty$

Solution: $\lim_{x\to 4^+}\left(\frac{\sqrt{x}-2}{x-4}\right)x=\lim_{x\to 4^+}\frac{(\sqrt{x}-2)(\sqrt{x}+2)}{(x-4)(\sqrt{x}+2)}(x)$

$\Rightarrow\lim_{x\to 4+}\frac{(x-4)x}{(x-4)\sqrt{x}+2}=\lim_{x\to 4+}\frac{x}{\sqrt{x}+2}=\frac{4}{\sqrt{4}+2}=\frac{4}{2+2}=\frac{4}{4}=1$

Answer: C

Telegram: https://t.me/ExtremeSeries

29. $\lim_{x\to 2}\frac{6x-12}{x^2-x-2}$ is equal to:

A. 0 B. 2 C. 1 D. ∞

Solution: $\lim_{x\to 2}\frac{6x-12}{x^2-x-2}=\lim_{x\to 2}\frac{6(x-2)}{(x-2)(x+1)}=\lim_{x\to 2}\frac{6}{x+1}=\frac{6}{3}=2$

Answer: B

30. Let $f(x)=\begin{cases}3a-x \text{ if } x<1\\ b+1 \text{ if } x=1\\ ax \text{ if } x>1\end{cases}$ then the value of b:

A. $-\frac{1}{2}$ B. $\frac{1}{2}$ C. -1 D. 3

Solution: since it is continuous at x = 1.

$\Rightarrow \lim_{x\to 1^+} f(x)=\lim_{x\to 1^-} f(x)\Leftrightarrow \lim_{x\to 1^+} ax=\lim_{x\to 1^-} 3a-x$

$\Leftrightarrow a=3a-1\Rightarrow -2a=-1\Rightarrow a=\frac{1}{2}$ and

- $\lim_{x\to 1^+} f(x)=f(1)\Leftrightarrow \lim_{x\to 1^+} ax=b+1\Rightarrow a=b+1$

$\Leftrightarrow \frac{1}{2}=b+1\Rightarrow b=-\frac{1}{2}$

Answer: A

31. On which of the following interval is f continuous where

$f(x)=\begin{cases}\frac{x-x^2}{x-1}, x<0\\ 3-x,\ x\geq 0\end{cases}$:

A. $(-\infty, 0]$ B. $[-2, 1]$ C. $[-1, 2]$ D. $[0, \infty)$

Solution: **To show continuity in $(-\infty, 0]$**

i) $\lim_{x\to 0^-} f(x)=f(0)\Rightarrow \lim_{x\to 0^-}\frac{x-x^2}{x-1}=3-0\Leftrightarrow \lim_{x\to 0^-}\frac{x(1-x)}{x-1}\Rightarrow 0\neq 3,$

Therefore it is not continuous from the left to 0.

To show continuity on $(-\infty, 0)$ let $C\in(-\infty, 0)$

$\lim_{x\to C} f(x)=f(C)\Leftrightarrow \frac{C-C^2}{C-1}=f(C)$

$\therefore f$ is cont. on $(-\infty, 0)$ but not cont. $(-\infty, 0]$.

Answer: A

ii) f is continuous on $[0, \infty)$ because

$\lim_{x\to 0^+} f(x)=f(0)\Leftrightarrow \lim_{x\to 0^+} 3-x=f(0)\Rightarrow 3=3$ and let

$C\in[0, \infty)$, so that $\lim_{x\to C} f(x)=f(C)\Rightarrow 3-C=3-C$

$\therefore$ f is cont. on $[0, \infty)$

Note: f is continuous on $(-\infty, 0)$ and $[0, \infty)$.

Answer: D

32. If $f(x)=x^5+2x$, then $\lim_{h\to 0}\frac{f(1+h)-f(1)}{h}$ is equal to

A. 1 B. 7 C. 5 D. ∞

Solution: Recall $f'(a)=\lim_{h\to 0}\frac{f(a+h)-f(a)}{h}$ it is concept of derivative. So that $f'(1)=\lim_{h\to 0}\frac{f(1+h)-f(1)}{h}$

$\Rightarrow f'(x)=5x^4+2\Rightarrow f'(1)=5+2=7$

Answer: B

33. Let $f(x)=e^{3x-1}$ and $g(x)=\tan x$. If $F(x)=xf(x)+f(g(x))$. Then F '(0):

A. $3e^{-1}$ B. $4e^{-1}$ C. $\frac{1}{e}\left(1+\frac{1}{e}\right)$ D. $\frac{1}{e}\left(1+\frac{3}{e}\right)$

Solution: $f(x)=e^{3x-1}\Rightarrow f'(x)=e^{3x-1}(3)=3e^{3x-1}$, $g(x)=\tan x$

$\Rightarrow g'(x)=\sec^2 x$. Then: $F(x)=xf(x)+f(g(x))$

$\Rightarrow F'(x)=x'f(x)+xf'(x)+f'(g(x))g'(x)$

$\Rightarrow F'(x)=1.f(x)+xf'(x)+f'(g(0))g'(0)$

$\Rightarrow F'(0)=f(0)+0f'(0)+f'(g(0))g'(0)=e^{-1}+0+f'(0)(1)$

$=e^{-1}+e^{-1}(3)=4e^{-1}$

Answer: B

34. Let $f(x)=g\left(\sqrt{4x+1}\right)$. If $g'(3)=\frac{1}{2}$, then f ' (2) is equal to:

A. $\frac{2}{3}$ B. $\frac{1}{3}$ C. $\frac{1}{2}\sqrt{5}$ D. $\frac{1}{2}\sqrt{13}$

Solution: $f'(x)=g'\left(\sqrt{4x+1}\right).\left(\sqrt{4x+1}\right)'(4x+1)'$

$\Rightarrow f'(x)=g'\left(\sqrt{4x+1}\right)\frac{1}{2}(4x+1)^{-\frac{1}{2}}(4)$

$\Rightarrow f'(2)=g'(3)(2)\left(\frac{1}{9}\right)=\left(\frac{1}{2}\right)(2)\frac{1}{3}=\frac{1}{3}$

Answer: B

Telegram: https://t.me/ExtremeSeries

35. The second derivative of $f(x) = \ln(x^2 + 1)$.

A. $\dfrac{1-2x^2}{x^2+1}$ C. $\dfrac{-2(x^2-1)}{(x^2+1)^2}$

B. $\dfrac{-2(x^2-1)}{x^2+1}$ D. $\dfrac{2-4x^2}{(x^2+1)^2}$

Solution: $f(x) = \ln(x^2+1) \Rightarrow$

$$f'(x) = \frac{2x}{x^2+1} \Rightarrow f''(x) = \frac{2(x^2+1)-2x(2x)}{(x^2+1)^2} = \frac{-2(x^2-1)}{(x^2+1)^2}$$

Answer: C

36. Let $f(x) = x\,|x|$, for all $x \in (-\infty, \infty)$. Which of the following is true?

A. f is not differentiable at x = 0
B. $f'(x) = 2|x|$ at each $x \in (-\infty, \infty)$
C. f is differentiable only on $(0, \infty)$
D. $f'(x) = 2x$ at each $x \in (-\infty, \infty)$

Solution: $|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases} \Rightarrow f(x) = x|x| = \begin{cases} x^2, x \geq 0 \\ -x^2, x < 0 \end{cases}$

$$\Rightarrow f'(x) = \begin{cases} 2x, x \geq 0 \\ -2x, x < 0 \end{cases} \Rightarrow f'(x) = 2|x|$$

$\therefore f$ is differentiable on $(-\infty, \infty)$

Answer: B

37. Let $f(x) = x^4 - 6x^2 + 1$ on $[-2, 3]$. The local maximum value of f.

A. $f(0)$ B. $f(\sqrt{3})$ C. $f(3)$ D. $f(-\sqrt{3})$

Solution: $f'(x) = 4x^3 - 12x \Rightarrow 4x(x^2 - 3) = 0$, $x = 0$, $x = \sqrt{3}$, $-\sqrt{3} \leftarrow$ critical number

- $f(-2) = 16 - 6(4) + 1 = -6$
- $f(3) = 3^4 - 6(3^2) + 1 = 81 - 54 + 1 = 28 \leftarrow$ local maximum
- $f(0) = 1$
- $f(-\sqrt{3}) = 9 - 6(3) + 1 = -8 \leftarrow$ local minimum
- $f(\sqrt{3}) = -8$

$\therefore f(3) = 28$ is local max

Answer: C

38. Let $f(x) = x^4 - \dfrac{16}{3}x^3 + 6x^2$, for all $x \in (-\infty, \infty)$ what is the range of f.

A. $(-\infty, \infty)$ B. $[0, \infty)$ C. $[-9, \infty)$ D. $\left[\dfrac{5}{3}, -9\right]$

Solution: $f'(x) = 4x^3 - 16x^2 + 12x \Leftrightarrow 4x(x^2 - 4x + 3) = 0$

$\Leftrightarrow 4x(x-3)(x-1) = 0 \Rightarrow x = 0, x = 1, x = 3$

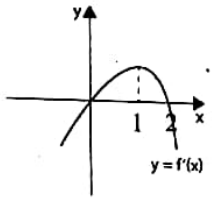
$f(0) = 0, f(1) = 1 - \dfrac{16}{3} + 6 = \dfrac{5}{3}$

$f(3) = 81 - 144 + 54 = -9 \leftarrow$ the minimum value

$\therefore$ The range of f $[-9, \infty)$

Answer: C

39. Suppose $f(x)$ is differentiable at all $x \in (-\infty, \infty)$ and the graph of $y = f'(x)$ is as shown in the figure below. Which of the following is true?



A. $f(x)$ is increasing on $(-\infty, 1)$
B. $f(x)$ attains its maximum value at $x = 2$
C. $f(x)$ attains its maximum value at $x = 1$
D. $f(x)$ is decreasing on $(0, 2)$

Solution: From the sign of derivative of $y = f'(x)$

	$-\infty$	0	2	∞
$f'(x)$	−	+	−	
conclusion about $f(x)$	decrease	increase	decrease	

Hence:

- $f(x)$ is increasing on $(0, 2)$
- $f(x)$ has local minimum at $x = 0$
- $f(x)$ has maximum value at $x = 2$
- $f(x)$ is decreasing $(-\infty, 0)$ and $(2, \infty)$

Answer: B

40. The radius of a right circular cylinder of height 10 cm increases at a rate of 3 cm/min. What is the rate of change of the volume of cylinder when the radius is 8 cm?

A. $80\pi cm^3/min$ C. $480\ \pi cm^3/min$

B. $320\pi cm^3/min$ D. $640\ \pi\ cm^3/min$

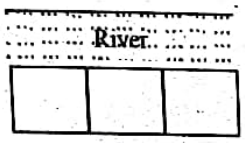
Solution: $\dfrac{dr}{dt} = 3$ cm/min, h = 10 cm, $v = \pi r^2 h$

Telegram: https://t.me/ExtremeSeries

$\Rightarrow v = \pi r^2 h \Rightarrow \frac{dv}{dt} = 20\pi r \frac{dr}{dt}$, when $r = 8$

$\Rightarrow \frac{dv}{dt} = 20\,\pi\,(8)(3)\ cm^3/min = 480\ \pi cm^3/min$ **Answer: C**

41. A farmer wants to make three identical rectangular enclosures (same width and same length) along a straight river, as in the diagram shown below. If he has a fencing material of 1,200 meters and the sides along the river need no fence, what should be the dimensions (length ℓ and width w) of each enclosure if the total area of the enclosures is to be maximized?



A. $\ell = 100$ m, w = 225 m

B. $\ell = 300$ m, w = 75 m

C. $\ell = 200$m, w 200 m

D. $\ell = 200$m, w = 150 m

Solution: Let x be each segment of length ℓ let y be the width

Perimeter: $3x + 4y = 1200 \Rightarrow 4y = 1200 - 3x = y = 300 - \frac{3}{4}x$

Area $= (3x)(y) \Rightarrow A(x) = (3x)\left(300 - \frac{3}{4}x\right) = 900x - \frac{9x^2}{4}$

$A'(x) = 900 - \frac{9}{2}x \Rightarrow A'(x) = 0$ if $x = 200$ m and

$y = w = 300 - \frac{3}{4}(200) = 150$ m **Answer: D**

42. Which of the following is an anti-derivative of $f(x) = \frac{3}{x\sqrt{x}}$?

A. $\frac{6}{\sqrt{x}} + C$ B. $-6\sqrt{x} + C$ C. $\frac{-6}{\sqrt{x}} + c$ D. $\frac{-3}{2\sqrt{x}} + 3$

Solution:

$f(x) = \frac{3}{x\sqrt{x}} \Rightarrow f(x) = 3x^{-3/2}$.

An anti derivative of $f(x)$ is $\frac{3x^{-1/2}}{\frac{-3}{2}+1} = (3)(-2)x^{-1/2} = \frac{-6}{\sqrt{x}} + C$

Answer: C

43. Which of the following is equal to $\int \frac{x^2-1}{x^3+x}dx$?

A. $\ln|x^2+1| + \ln|x| + C$

B. $\frac{1}{2}\ln(x^2+1) - \ln|x| + C$

C. $\ln(x^2+1) - \ln|x| + C$

D. $2\ln(x^2+1) - \ln|x| + C$

Solution: $\frac{x^2-1}{x^3+x} = \frac{x^2-1}{x(x^2+1)} = \frac{A}{x} + \frac{Bx+C}{x^2+1}$

$\Rightarrow x^2 - 1 = A(x^2+1) + Bx^2 + Cx \leftarrow$ Multiply both side by x

$x(x^2+1) = x^2 - 1 = Ax^2 + Bx^2 + A + Cx$

$\Rightarrow x^2 - 1 = (A+B)x^2 + Cx + A \leftarrow$ factoring x^2

$\Rightarrow A + B = 1$, and $A = -1$, $C = 0$ and $B = 2$

Now, $\int \frac{x^2-1}{x^3+x}dx = \int \frac{-dx}{x} + \int \frac{2xdx}{x^2+1}$. Note $\int \frac{-dx}{x} = -\ln x$

To find $\int \frac{2xdx}{x^2+1}$, $u = x^2 + 1 \Rightarrow du = 2x\,dx$

$\Rightarrow \int \frac{du}{u} = \ln|u| = \ln(x^2+1)$

$\therefore \int \frac{x^2-1}{x^3+x}dx = \ln|x| = \ln(x^2+1) + C \Rightarrow \ln(x^2+1) - \ln|x| + C$

Answer: C

44. $\int (2 + \sin^2 x)\cot x\,dx$ is equal to:

A. $2\csc x - \sec^2 x + C$ C. $2\ln|\sin x| - \cos^2 x + C$

B. $2\ln|\sin x| + \frac{1}{2}\sin^2 x + C$ D. $2\csc x + \sin x\cos x + C$

Solution: $\int (2 + \sin^2 x)\cot x\,dx = \int (2\cot x + \sin x\cos x)dx$

$\Rightarrow \int \frac{2\cos x}{\sin x} + \int \sin x\cos x\,dx$, let $u = \sin x \Rightarrow du = \cos x\,dx$

Telegram: https://t.me/ExtremeSeries

$$\Rightarrow \int \frac{2\,du}{u} + \int u\,du = 2\ln|u| + \frac{u^2}{2} + C = 2\ln|\sin x| + \frac{\sin^2 x}{2} + C$$

Answer: B

45. $\int_1^2 x\left(\frac{1}{x^2} + 2\ln x\right)dx$ is equal to:

A. $2\ln 2 - 1$ C. $2\ln 2 - \frac{5}{2}$

B. $5\ln 2 + \frac{5}{2}$ D. $5\ln 2 - \frac{3}{2}$

Solution: $\int_1^2\left(\frac{1}{x} + 2x\ln x\right)dx = \int_1^2 \frac{dx}{x} + \int_1^2 2x\ln x\,dx$

To find $\int 2x\ln x\,dx$, let $u = \ln x$, $dv = 2x\,dx \Rightarrow du = \frac{1}{x}, v = x^2$

$$\Rightarrow \int_1^2 2x\ln x\,dx = x^2\ln x - \int x\,dx = x^2\ln x - \frac{x^2}{2}\Big|_1^2 = 4\ln 2 - \frac{3}{2}$$

$$\therefore \int_1^2 \frac{dx}{x} + \int_1^2 2x\ln x\,dx = \ln x\Big|_1^2 + 4\ln 2 - \frac{3}{2} = 5\ln 2 - \frac{3}{2}$$

Answer: D

46. $\int_0^1 \frac{dx}{e^{-x}(1+e^x)}$ is equal to:

A. $\ln(e-1)$ B. $\ln(1+e) - \ln 2$ C. $\ln(2+e)$ D. $\ln 2$

Solution:

$\int_0^1 \frac{e^x dx}{(1+e^x)}$, since $\frac{1}{e^{-x}} = e^x$ let $u = 1 + e^x \Rightarrow du = e^x dx$

$$\Rightarrow \int \frac{du}{u} = \ln u = \ln(1+e^x)\Big|_0^1 = \ln(1+e) - \ln 2$$

Answer: B

47. The area of the region bounded between the graph of $y = x^2$ and $y = \sqrt{x}$ is:

A. 35 q unit B. $\frac{1}{3}$ sq. unit C. $\frac{2}{3}$ sq. unit D. 1 sq. unit

Solution:

The intersection $\begin{cases} y = x^2 \\ y = \sqrt{x} \end{cases}$ is (0, 0) and (1, 1).

$$\therefore A = \int_0^1 \left(\sqrt{x} - x^2\right)dx = \frac{2}{3}x^{3/2} - \frac{x^3}{3}\Big|_0^1 = \frac{2}{3} - \frac{1}{3} = \frac{1}{3} \text{ sq. unit}$$

48. Which one of the following points is outside the sphere $x^2 + y^2 + z^2 - 4z = 0$?

Answer: B

A. (0, −1, 1) B. (−2, 0, 2) C. (1, −1, 3) D. (−1, 2, 0)

Solution: $x^2 + y^2 + z^2 - 4z + 4 = 4 \Rightarrow (x-0)^2 + (y-0)^2 + (z-2)^2 = 4$

A. (0, −1, 1): $0^2 + 1^2 + 1^2 = 2 < 4$
∴ (0, −1, 1) lies inside the sphere.

B. (−2, 0, 2): $(-2)^2 + 0^2 + 0^2 = 4 = 4$
∴ (−2, 0, 2) lies on the sphere.

C. (1, −1, 3): $1^2 + (-1)^2 + (3-2) = 3 < 4$
∴ (1, −1, 3) lies inside sphere.

D. (−1, 2, 0): $(-1)^2 + (2)^2 + 0 = 5 > 4$
∴ (−1, 2, 0) lies outside sphere

49. Which of the following is true about the points of intersection of the sphere of radius 3 units with center at (−1, 3, 2) and z axis.

Answer: D

A. (0, 0, 3) is a point of intersection
B. There are two points of intersection.
C. There is no point of intersection.
D. There are many points of intersection.

Solution: $(x+1)^2 + (y-3)^2 + (z-2)^2 = 3^2 \leftarrow$ eqn of sphere.
Point on z axis is (0, 0, z). So that intersection of sphere with z axis:

$\Rightarrow (0+1)^2 + (0-3)^2 + (z-2)^2 = 9 \Rightarrow (z-2)^2 = 9 - 10 = -1$
$\Rightarrow (z-2)^2 = -1 \leftarrow$ no point of intersection

Answer: C

Telegram: https://t.me/ExtremeSeries

1. If $x<0$, then the simplest form of $f(x)=\dfrac{x-|x|}{x}$ is equal to:

A. $2x$ B. 2 C. -2 D. 0

Solution: $|x|=-x$ if $x<0 \Rightarrow f(x)=\dfrac{x-(-x)}{x}=\dfrac{2x}{x}=2$

Answer: B

2. If $f(x)=\dfrac{\sqrt{x+2}}{x+2}$, and $g(x)=\dfrac{1}{x}-2$, then $f(g(x))$ is equal to:

A. $\sqrt{x}-2$ B. $\sqrt{x}+2$ C. $\sqrt{x}$ D. $\dfrac{\sqrt{x}}{x}$

Solution:

$$f(g(x))=\frac{\sqrt{g(x)+2}}{g(x)+2}=\frac{\sqrt{\frac{1}{x}-2+2}}{\frac{1}{x}-2+2}=\frac{x}{\sqrt{x}}=\sqrt{x}$$

Answer: C

3. If $f(x)=\ln\left(\dfrac{x}{x-1}+2\right)$, for $x>1$, then the inverse of the function f

A. $g(x)=\dfrac{e^x-2}{e^x-3}$ C. $g(x)=\dfrac{e^x}{e^x+1}$

B. $g(x)=\dfrac{e^x-2}{e^x+1}$ D. $e^{\frac{x}{x-1}}-2$

Solution:

To find f^{-1}, set: $x\to y, y\to x$

$\Rightarrow y=\ln\left(\dfrac{x+2(x-1)}{x-1}\right)=\ln\left(\dfrac{3x-2}{x-1}\right)$, so $x\to y, y\to x$

$\Rightarrow x=\ln\left(\dfrac{3y-2}{y-1}\right)\Rightarrow e^x=\dfrac{3y-2}{y-1}\Rightarrow e^xy-e^x=3y-2$

$\Rightarrow e^xy-3y=e^x-2\Leftrightarrow y(e^x-3)=e^x-2\Rightarrow y=\dfrac{e^x-2}{e^x-3}$

Answer: A

4. The simplified form of $\dfrac{4-(a-2)^2}{a^4+2a^2+1}\div\dfrac{1-\dfrac{1}{a^2+1}}{a}$.

A. $\dfrac{4-a}{a}$ B. $\dfrac{-a}{a^2+1}$ C. $\dfrac{-1}{a^2(a^2+1)}$ D. $\dfrac{4-a}{a^2+1}$

Solution: The expression the same as:

$$\Leftrightarrow\frac{4-(a^2-4a+4)}{(a^2+1)^2}\div\frac{a^2+1-1}{(a^2+1)a}=\frac{-a^2+4a}{(a^2+1)^2}\cdot\frac{a(a^2+1)}{a^2}$$

$$\Rightarrow\frac{a(4-a)}{a^2}\cdot\frac{a(a^2+1)}{(a^2+1)^2}=\frac{4-a}{a^2+1}$$

Answer: D

5. If $f(x)=\dfrac{x-1}{(x-2)^2(x+1)}$ which of the following is true about f.

A. Its graph has an oblique asymptote.
B. The graph of f does not meet its asymptote.
C. As, $x\to -1^+, f(x)\to -\infty$
D. As $x\to 2^-, f(x)\to -\infty$

Solution: Here

A. Has no oblique asymptote since the degree of numerator less than denominator

B. $y=0\leftarrow$ H.A, so that the graph meet horizontal asymptote at $x=1$

C. $x\to -1^+, \dfrac{-1^+-1}{(-1-2)^2(-1^++1)}=\dfrac{-2}{(9)(0^+)}=-\infty\Rightarrow f(x)\to -\infty$

D. $x\to 2^-, f(x)=\dfrac{2-1}{(2^--2)^2(2+1)}=\dfrac{1}{(0^+)(3)}=+\infty$

$\therefore x\to 2^-, f(x)\to\infty$,

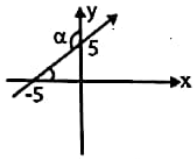
Answer: C

6. Line ℓ passes through (0, 5) and (–5, 0). What is the angle between the y-axis and ℓ in radian measure?

A. $\dfrac{\pi}{4}$ B. $\dfrac{\pi}{3}$ C. $\dfrac{\pi}{2}$ D. $\dfrac{3}{4}\pi$

Solution: Slope of $\ell=\dfrac{5-0}{0-(-5)}=\dfrac{5}{5}=1$

Telegram: https://t.me/ExtremeSeries



Angle of inclination, $\tan\theta = 1$

$\Rightarrow \theta = \frac{\pi}{4}$ ← the angle measured from positive x-axis to line ℓ.

$\Rightarrow \alpha = \pi - \frac{\pi}{4} = \frac{3}{4}\pi$ ← the angle between y-axis and ℓ measured C.C.W from y to ℓ. **Answer: D**

7. Consider a circle whose center is on the x-axis. If a line given by $y = x$ is tangent to the circle at point (2, 2), what is the equation of the circle?

A. $x^2 + y^2 = 8$ C. $(x-4)^2 + y^2 = 8$
B. $(x-2)^2 + y^2 = 4$ D. $(x-1)^2 + y^2 = 5$

Solution: Let the center (a, 0)

- Slope of the line $y = x$ is 1.
- Slope of the line passing through (2, 2) and (a, 0) is −1.

$$\Rightarrow \frac{2-0}{2-a} = -1 \Rightarrow 2 = -2 + a \Rightarrow a = 4$$

$\therefore$ The center of circle is (4, 0).

- The radius $r = \sqrt{(2-0)^2 + (2-4)^2} = \sqrt{8} \Rightarrow r^2 = 8$

$\Rightarrow$ Equation of circle: $(x-4)^2 + (y-0)^2 = 8$ **Answer: C**

8. What is the vertex and the equation of the direction, respectively of the parabola $x + y^2 + 2y + 1 = 0$?

A. $(0, -1), x = -\frac{1}{4}$ C. $(0, -1), x = \frac{1}{4}$
B. $(-1, 0), x = -\frac{1}{4}$ D. $(-1, 0), y = \frac{1}{4}$

Solution: Here $x + y^2 + 2y + 1 = 0 \Rightarrow y^2 + 2y + 1 = -x$

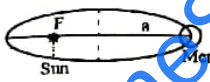
$\Rightarrow (y+1)^2 = -x \Rightarrow 4p = |-1| \Rightarrow p = \frac{1}{4}$ ← focal length

Vertex (0, −1) and directix, $x = \frac{1}{4}$ **Answer: C**

9. The orbit of Mercury around the sun forms an ellipse with eccentricity 0.206, length of the major axis 1.16×10^8 km and the sun at one focus. The best approximation of the maximum distance from Mercury to the sun.

A. 7.596×10^7 km C. 8.695×10^7 km
B. 5.695×10^7 km D. 6.995×10^7 km

Solution: Let 2a = distance between major axis.
2c = distance between focus
e = eccentricity



$\Rightarrow 2a = 1.16 \times 10^8$ km $\Rightarrow a = 5.8 \times 10^7$ km

$e = \frac{c}{a} \Rightarrow c = ea = (0.206)(5.8 \times 10^7 \text{ km}) = 1.195 \times 10^7$ km

The maximum distance, $a + c = 5.8 \times 10^7 + 1.95 \times 10^7 \approx 6.995 \times 10^7$ km

Note: the minimum distance $a - c = (5.8 - 1.195) \times 10^7 = 4.61 \times 10^7$ km **Answer: D**

10. Suppose proposition $p \Rightarrow \neg q$ is false (F), which of the following is true?

A. $\neg q \wedge (p \Rightarrow q)$ C. $(\neg q \vee p) \Leftrightarrow q$
B. $\neg p \vee (q \Rightarrow \neg p)$ D. $(p \wedge q) \Leftrightarrow \neg p$

Solution: $p \Rightarrow \neg q \equiv F$, if $p \equiv T$, $\neg q \equiv F$, and $q \equiv T$ $(\neg q \vee p) \Leftrightarrow q \equiv (F \vee T) \Leftrightarrow T \equiv T \Leftrightarrow T \equiv T$

Answer: C

11. Suppose $e(x) \equiv x$ is even, $p(x) \equiv x$ is prime, $d(x) \equiv x$ is divisible by 2. Which one of the following has the truth value F on the set of natural number?

A. $(\exists x)[e(x) \wedge \neg d(x)]$ C. $(\exists x)[e(x) \wedge \neg d(x)]$
B. $(\forall x)[e(x) \Rightarrow d(x)]$ D. $(\forall x)[e(x) \vee \neg d(x)]$

Solution: A. True, let $x = 2$
B. True D. True

Answer: C

12. Consider the following argument form production is high if rain continues. Rain does not continue. Therefore, either production is low or rain continues. Let p ≡ production is low, q ≡ rain continues. The following table is also given about p and q.

Row	p	q	¬p	¬q	¬p⇒q	q⇒¬p	p ∨q
1	T	T	F	F	T	F	T
2	T	F	F	T	T	T	T
3	F	T	T	F	T	T	T
4	F	F	T	T	F	T	F

Telegram: https://t.me/ExtremeSeries

Which of the following is necessary true:

A. argument form is valid due to row 2
B. argument form is valid due to row 2 and 3
C. argument form is invalid due to row 4
D. argument form is invalid due to rows 1 and 3

Solution: Argument form: $q \Rightarrow \neg p, \neg q \vdash p \vee q$. The premise $q \Rightarrow \neg p, \neg q$ are true in row 4 but the conclusion $p \vee q$ is false in row 4.

Answer: C

13. If x and y are non-negative integers which of the following is not true?

A. $(\forall x)(\exists x)(y > x^2 - 1)$ C. $(\exists y)(\forall x)(y \leq x^2 - 1)$
B. $(\exists x)(\forall x)\ (y > x^2 - 1)$ D. $(\exists y)(\exists x)(y \leq x^2 - 1)$

Solution:

A. Let $x = 2 \Rightarrow y > 3$, True
B. False, let fixed $x = 1 \Rightarrow y > 0$ is false, because when $y = 0$
C. True, let fixed number be $y_o \Rightarrow y_o \geq x^2 - 1 \Rightarrow y_o + 1$. Since x and y are non-negative so that let $y_o = 0 \Rightarrow x^2 \geq 1$ is true for all x.

Answer: B

14. A school has three class rooms for grade for grade 11 namely 11A, 11B and 11C. The number of students in these class rooms is 28, 20 and 22 respectively. All the students took an examination and the average score of the students of 11A, 11B and 11C is 60, 70 and 70 respectively. What is the average score in this examination for all grade 11 students?

A. 66 B. 66.67 C. 65 D. 65.67

Solution: The combined mean $\bar{x}$ is calculated by

$$\bar{x} = \frac{60 \times 28 + 20 \times 70 + 22 \times 70}{28 + 20 + 22} = \frac{4620}{70} = 66$$

Answer: A

15. You are given a data on the age of students, in a primary school.

Age	8	10	11	12	13
Number of students	5	5	8	10	12

Which of the following is NOT true about the data?

A. the median is 10.5 C. the mean is 10.5
B. the mode is 10 D. the range is 5

Solution: Median of raw data is given by

$$M_d = \begin{cases} \left(\frac{n+1}{2}\right)^{th} \text{ item if } n \text{ is odd} \\ \dfrac{\left[\left(\frac{n}{2}\right)^{th} \text{ item} + \left(\frac{n}{2}+1\right)^{th} \text{ item}\right]}{2} \text{ item if } n \text{ is even} \end{cases}$$

A. median $= \dfrac{\left[\left(\frac{40}{2}\right) + \left(\frac{40}{2}+1\right)\right]}{2}$ item $= \dfrac{10+11}{2} = 10.5$

B. mode $= 10 \leftarrow$ the most frequenct value

C. mean $= \dfrac{8\times5 + 10\times15 + 11\times8 + 12\times10 + 13\times2}{5+15+8+10+2} = 10.6$

D. range = max − min = 13 − 8 = 5

Answer: C

16. The following is a table of simple frequency distribution of a data with variable x.

x	1	3	4	5	7
Frequency	2	5	6	5	2

The standard deviation of the data is equal to:

A. $\sqrt{3}$ B. 3 C. $\sqrt{2.3}$ D. $\sqrt{3.6}$

Solution: Standard deviation = $\sqrt{\text{variance}}$ and mean $(\bar{x}) = 4$

$$S.D = \sqrt{\frac{\sum f_i (x_i - \bar{x})^2}{\sum f_i}}$$

$$= \sqrt{\frac{2(1-4)^2 + 5(3-4)^2 + 6(4-4)^2 + 5(5-4)^2 + 2(7-4)^2}{2+5+6+5+2}}$$

$$= \sqrt{\frac{18+5+0+5+18}{20}} = \sqrt{\frac{23}{10}} = \sqrt{2.3}$$

Answer: C

17. A three digit library identification card is to be printed from numbers 0, 1, 2, 3, 4, 5 in a such away that the first is non-zero and no number is to be repeated. How many such cards can be printed?

A. 100 B. 120 C. 150 D. 180

Solution: $5 \times 5 \times 4 = 100$

Answer: A

Telegram: https://t.me/Extremeseries

18. A student needs to select 3 books form 3 different mathematics, 3 different physic and 1 history books. What is the probability that one of them is mathematics and the other two are either physics or history books?

A. $\frac{3}{35}$ B. $\frac{9}{25}$ C. $\frac{15}{35}$ D. $\frac{18}{35}$

Solution: Use combination rule

$$\Rightarrow \frac{C(3,1).C(3,1).C(1,1)}{C(7,3)} + \frac{C(3,1)C(3,2).C(1,0)}{C(7,3)} = \frac{9}{35} + \frac{9}{35} = \frac{18}{35}$$

Answer: D

19. Items produced by a certain company are subjected to two kinds of defect D_1 and D_2. Out of the total production if 5% have defect D_1, 10% have defect D_2 and 2% have both defects then what is the probability for an item to have defect D_2 given that it has defect D_1?

A. 0.2 B. 0.05 C. 0.1 D. 0.4

Solution: P (D_1 and D_2) = 0.02, P (D_1) = 0.05 $\Rightarrow$ P (D_1 and D_2) = P (D_1). P (D_2/D_1) $\Rightarrow$ 0.02 = (0.05) P ($D_2.D_1$)

$$\Rightarrow P(D_2/D_1) = \frac{0.02}{0.05} = \frac{2}{5} = 0.4$$

Answer: D

20. Let $A = \begin{pmatrix} -2 & 0 & x \\ 2y & x+y & -4 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & -y \\ 0 & 3 \\ 1-x & 2 \end{pmatrix}$ such that $A + 2B^T = 0$. Then which of the following is the value of y?

A. 0 B. $\frac{-13}{2}$ C. -8 D. any real number

Solution: $B^T = \begin{pmatrix} 1 & 0 & 1-x \\ -y & 3 & 2 \end{pmatrix} \Rightarrow 2B^T = \begin{pmatrix} 2 & 0 & 2-2x \\ -2y & 6 & 4 \end{pmatrix}$

$$\Rightarrow A + 2B^T = \begin{pmatrix} -2 & 0 & x \\ 2y & x+y & -4 \end{pmatrix} + \begin{pmatrix} 2 & 0 & 2-2x \\ -2y & 6 & 4 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$\Rightarrow x + (2 - 2x) = 0$ and $x + y + 6 = 0 \Rightarrow -x = -2 \Rightarrow x = 2$, so that to find y put $x = 2$ in $x + y + 6 = 0 \Rightarrow 2 + y + 6 = 0 \Rightarrow y + 8 = 0$

$\Rightarrow y = -8$

Answer: C

21. Let A and B be 3 × 3 matrices such that $A = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 5 & 0 \\ 0 & 1 & \frac{1}{2} \end{pmatrix}$ and $|B| = \frac{1}{10}$ then $|2AB^T|$ equal to _

A. 1 B. 4 C. 100 D. 400

Solution: $|2AB^T| = 2^3 |A||B^T| = 8\left(2 \times 5 \times \frac{1}{2}\right)\left(\frac{1}{10}\right) = 4$

Explanation:

3 is the order to matrix $|A| = |A| = 2 \times 5 \times \frac{1}{2}$ ← determinant of triangular matrix.

22. What is the solution set of the following system of equations $\begin{cases} x + y + 2z = 1 \\ x + 2y + z = -2 \\ -2x - 2y - 4z = -2 \end{cases}$?

A. {(0, 1, 0)} C. {(−3km k+1,k)},k∈(−∞, ∞)

B. (−∞, ∞) D. {(3k, k − 1, k)}, k ∈(−∞, ∞)

Solution: $D = \begin{vmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ -2 & -2 & -4 \end{vmatrix} = -2\begin{vmatrix} 1 & 1 & 2 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{vmatrix} = 0$ since row 1 and 3 are the same.

$D_x = \begin{vmatrix} 1 & 1 & 2 \\ 2 & 2 & 1 \\ -2 & -2 & -4 \end{vmatrix} = -2\begin{vmatrix} 1 & 1 & 2 \\ 2 & 2 & 1 \\ 1 & 1 & 2 \end{vmatrix} = 0$, and $D_y = 0$, $D_z = 0$.

$\Rightarrow$ The system has infinite many solution.

$$-\begin{cases} x + y + 2z = 1 \\ x + 2y + z = 2 \end{cases} \quad \text{and} \quad +\begin{cases} x + 2y + z = 2 \\ -2x - 2y - 4z = -2 \end{cases}$$

$0 + -y + z = -1 \Rightarrow y = z + 1$ $\quad -x + 0 - 3z = 0 \Rightarrow x = -3z$

$\therefore$S.S = $\{(x, y, z)\} = \{(-3z, z + 1, z)\} = \{(-3k, k + 1, k)\}$ $k \in \mathbb{R}$

Answer: C

Telegram: https://t.me/ExtremeSeries

23. If $\begin{pmatrix} \alpha & 2 & \beta \\ 2 & 1 & 3 \\ -1 & 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -5 \\ 5 \\ 0 \end{pmatrix}$ and the determinant of the coefficient matrix $|A| = -5$ then the value of x is equal to:

A. 3 B. $\alpha + \beta$ C. -5α D. 5

Solution: Let $x = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ and $C = \begin{pmatrix} -5 \\ 5 \\ 0 \end{pmatrix}$, $|A| = -5 \Rightarrow Ax = C$

$\Rightarrow x = A^{-1}C$, But $A^{-1} = \frac{adj.A}{|A|} \Rightarrow x = \frac{adj.A}{|A|}(C)$ and to find adj. A first find the cofactor.

- $A_{11} = (-1)^{1+1}\begin{vmatrix} 1 & 3 \\ 0 & 1 \end{vmatrix} = 1$ $\quad A_{21} = 2$
- $A_{12} = (-1)^{1+2}\begin{vmatrix} 2 & 3 \\ -1 & 1 \end{vmatrix} = -5$ $\quad A_{22} = (-2)^{2+2}\begin{vmatrix} \alpha & \beta \\ -1 & 1 \end{vmatrix} = \alpha + \beta$
- $A_{13} = (-1)^{1+3}\begin{vmatrix} 2 & 1 \\ -1 & 0 \end{vmatrix} = 1$ $\quad A_{23} = (-1)^{2+3}\begin{vmatrix} \alpha & 2 \\ -1 & 0 \end{vmatrix} = 2$
- $A_{31} = (-1)^{3+1}\begin{vmatrix} 2 & \beta \\ 1 & 3 \end{vmatrix} = 6 - \beta$ $\quad A_{32} = (-1)^{3+2}\begin{vmatrix} \alpha & \beta \\ 2 & 3 \end{vmatrix} = 2\beta - 3\alpha$
- $A_{33} = (-1)^{3+3}\begin{vmatrix} \alpha & 2 \\ 2 & 1 \end{vmatrix} = \alpha - 4$,
- Adjoins of A is transpose of cofactor.

$\Rightarrow$ cofactor of $A = \begin{pmatrix} 1 & -5 & 1 \\ 2 & \alpha+\beta & 2 \\ 6-\beta & 2\beta-3\alpha & \alpha-4 \end{pmatrix}$,

adj. $A = \begin{pmatrix} 1 & 2 & 6-\beta \\ -5 & \alpha+\beta & 2\beta-3\alpha \\ 1 & 2 & \alpha-4 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{adj.A}{|A|}(C) = \frac{-1}{5}\begin{pmatrix} 1 & 2 & 6-\beta \\ -5 & \alpha+\beta & 2\beta-3\alpha \\ 1 & 2 & \alpha-4 \end{pmatrix}$

$\Rightarrow x = \begin{pmatrix} \frac{-1}{5} & \frac{2}{5} & \frac{6-\beta}{5} \end{pmatrix}\begin{pmatrix} -5 \\ 5 \\ 0 \end{pmatrix} = 1 + 2 + 0 = 3$

$\Rightarrow y = \begin{pmatrix} -1 & \frac{\alpha+\beta}{-5} & \frac{2\beta-3\alpha}{-5} \end{pmatrix}\begin{pmatrix} -5 \\ 5 \\ 0 \end{pmatrix} = 5 - \alpha - \beta + 0 = 5 - (\alpha + \beta$

$\Rightarrow z = \begin{pmatrix} \frac{-1}{5} & \frac{-2}{5} & \frac{-\alpha+4}{5} \end{pmatrix}\begin{pmatrix} -5 \\ 5 \\ 0 \end{pmatrix} = 1 - 2 + 0 = -1$

$\Rightarrow$S.S = $\{(x, y, z)\} = \{(3, 5 - (\alpha + \beta)$ or S.S = $\{(3, 5) - (\alpha + \beta), -1)\}$

Answer: A

24. In the set of complex numbers, the solution set of $x^2 - 2x + 5 = 0$ is:

A. ϕ C. $\{1 + 2i, 1 - 2i\}$

B. $\{2 + i, 2 - i\}$ D. $\{2 + 4i, 2 - 4i\}$

Solution: $x^2 - 2x + 5 = 0 \Leftrightarrow x^2 - 2x = -5 \Leftrightarrow x^2 - 2x + 1 = -5 + 1 \Rightarrow (x-1)^2 = -4 \Rightarrow x - 1 = \pm\sqrt{-4} = \pm 2i \Rightarrow$ and $x = 1 \pm 2i$

Answer: C

25. The simplified form of $\frac{4-3i}{3+4i} + \overline{1-2i}$ is equal _____

A. 1 + i B. 1 + 3i C. 2 – i D. 1 – 3i

Solution: $\frac{4-3i}{3+4i}\cdot\frac{3-4i}{3-4i} + 1 + 2i = \frac{(12-16i-9i-12)}{9+16} + 1 + 2i$

$\Rightarrow -i + 1 + 2i = 1 + i$,

Answer: A

26. Let z be complex number. Then the solution set of $z^3 - iz = 0$.

A. $\left\{\pm\frac{1}{\sqrt{2}}(1+i)\right\}$ C. $\left\{\pm\sqrt{2}(1+i)\right\}$

B. $\left\{0, \pm\frac{1}{\sqrt{2}}(1-i)\right\}$ D. $\left\{0, \pm\frac{1}{\sqrt{2}}(1-i)\right\}$

Telegram: https://t.me/ExtremeSeries

Solution: $z^3 - iz = 0 \Leftrightarrow z(z^2 - i) = 0 \Rightarrow z = 0$ or $z^2 - i = 0$ but $z = x + yi$ so that $z^2 - i = 0 \Leftrightarrow z^2 = i \Rightarrow (x + yi)^2 = i$

$\Rightarrow x^2 + 2xyi - y^2 = i \leftarrow$ comparing the real part and i

$\Rightarrow x = \pm y$ and $2xy = 1$ put in $2xy = 1$

$\Rightarrow 2x^2 = 1 \Rightarrow x^2 = \frac{1}{2} \Rightarrow x = \pm\frac{1}{\sqrt{2}}$ and $y = \pm\frac{1}{\sqrt{2}}$

$\therefore z = x + yi = \pm\frac{1}{\sqrt{2}}(1 + i)$ and $z = x + yi = 0$

$\therefore S.S = \left\{0, \pm\frac{1}{\sqrt{2}}(1 + i)\right\}$ **Answer: D**

27. Which of the following is a vector that lies on the line through (0, 0) and (2, 4)?

A. $\bar{u} = (2, 1)$ C. $\bar{u} = \left(\frac{1}{2}, 2\right)$

B. $\bar{u} = (-1, -2)$ D. $\bar{u} = (-2, -6)$

Solution:

Let A = (0, 0) and B = (2, 4) $\Rightarrow \overrightarrow{AB} = (2 - 0, 4 - 0) = (2, 4)$ vector equation of ℓ: $(x, y) = (0,0) + t(2, 4) = (2t, 4t)$, $t \in \mathbb{R}$

$\Rightarrow$ The set of all $(x, y) = (2t, 4t)$, if $t = -\frac{1}{2} \Rightarrow u = (-1, -2)$

Answer: B

28. Let i and j be the standard unit vector in the directions of positive x-axis and positive y-axis respectively and $\overrightarrow{AB}$ be a vector from the point A (−1, 0) to the point B (2, 2). If $\bar{v} = 3\overrightarrow{AB} - 3i + 2j$, then the unit vector in the direction of $\bar{v}$ is equal to:

A. $\left(\frac{3}{5}, \frac{4}{5}\right)$ B. $\left(\frac{-3}{5}, \frac{-4}{5}\right)$ C. $\left(\frac{-3}{5}, \frac{4}{5}\right)$ D. $\left(\frac{3}{5}, \frac{-4}{5}\right)$

Solution: $\overrightarrow{AB} = (2, 2) - (-1, 0) = (3, 2) = 3i + 2j$

$\bar{v} = 3(3i + 2j) - 3i + 2j = 6i + 8j = (6, 8)$

Unit vector in the direction of $\bar{v}$ is $\frac{\bar{v}}{|\bar{v}|} = \frac{(6,8)}{\sqrt{6^2 + 8^2}} = \left(\frac{3}{5}, \frac{4}{5}\right)$.

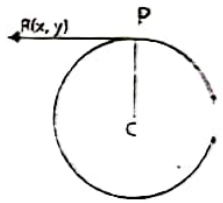
Answer: A

29. Which of the following is a vector equation of the line tangent to the circle $x^2 + y^2 + 2x - 7 = 0$?

A. $(x, y) = (0, 3) + \lambda(-1, 2)$ C. $(x, y) = (0, 3) + \lambda(1, -1)$

B. $(x, y) = (1, 2) + \lambda(2, -1)$ D. $(x, y) = (1, 2) + \lambda(-1, 2)$

Solution: $x^2 + y^2 + 2x - 7 = 0 \Rightarrow (x + 1)^2 + (y - 0)^2 = 8 \Rightarrow$ center C (−1, 0). Let R (x, y) and P (1, 2) point on tangent line,



$\overrightarrow{CP} = (1 - (-1), (2 - 0) = (2, 2)$

$\overrightarrow{CP} \cdot \overrightarrow{PR} = 0 \Rightarrow (2, 2) \cdot (x - 1, y - 2) = 0$

$\Rightarrow 2x - 2 + 2y - 4 = 0 \Rightarrow 2x + 2y = 6$

$\Rightarrow x + y = 3$. The direction vector $\bar{v} = (a, b)$ parallel to the line $x + y = 3$ is $x + y = 0$

$\Rightarrow y = -x$, $\bar{v} = (a, b) = (a, -a) = \lambda(1, -1)$

Take any arbitrary point on $x + y = 3$ say (0, 3).

$\therefore$ Vector equation of ℓ: $(x, y) = (0, 3) + \lambda(1, -1)$.

Answer: C

30. Let ℓ be the line whose equation $2x - y = 10$ whichy one of the following is the equation of the image of ℓ after reflection in the line $y = 2x - 5$ followed by a rotation through the angle of 90 about the origin?

A. $x + 2y = 0$ C. $x + 2y = 5$

B. $2x + y = 0$ D. $x - 2y = 5$

Solution: ℓ: $y = 2x - 10$ is parallel to the reflecting (line): $y = 2x - 5$. Hence, the image ℓ': $y = 2x + k$, now we need to find k. Let (a, b) be any point on ℓ, say $(0, -10)$.

Let the image of $(0, -10) = (a', b')$ which lies on ℓ': $y = 2x + k$.

$\Rightarrow M((0, -10)) = (a', b') \Rightarrow \frac{b' + 10}{a' - 0} = -\frac{1}{2} \leftarrow$ slope of perpendicular bisector of the two line $y = 2x - 10$ and $y = 2x + k$.

$\Rightarrow a' = -2b' - 20$. Also, the mid-point of (0, −10) and (a', b')

$\Rightarrow \left(\frac{0 + a'}{2}, \frac{b' - 10}{2}\right)$ which lies on reflecting axis $y = 2x - 5$

$\Rightarrow \left(\frac{a'}{2}, \frac{b' - 10}{2}\right) \Rightarrow \frac{b' - 10}{2} = 2\left(\frac{a'}{2}\right) - 5$

Telegram: https://t.me/ExtremeSeries

$\Rightarrow b' = 2a'$ but $a' = -2b' - 20 \Rightarrow a' = -2(2a') - 20$ but $a' = -2b' \Rightarrow a' = -2(2a') - 20 \Rightarrow 5a' = -20 \Rightarrow a' = -4$ and $b' = 2a' = 2$

$\therefore$ The image of M (0, −10) = (4, 8). Since (4, 8) lies on $y = 2x + k$, so that $\Rightarrow 8 = 2(4) + k \Rightarrow k = 0$.

$\therefore$ The image of $y = 2x - 10$ on reflecting line $y = 2x - 5$ is $y = 2x$. Now rotate $y = 2x$ on $\theta = 90°$ about (0, 0)

$\Rightarrow$ R $(x, y) = (-y, x) \Rightarrow x \to -y$, $y \to x$. Thus, the image after rotastion $\Rightarrow x = -2y \Leftrightarrow x + 2y = 0$ **Answer: A**

31. Let $\bar{a} = -i + 3k$ and $\bar{b} = -i + j$ be vector in the space. Which one of the following is the cosine of the angle between $\bar{a}$ and $\bar{a} - \bar{b}$.

A. $\frac{9}{10}$ B. $\frac{3}{5}$ C. $\frac{3}{\sqrt{10}}$ D. $\frac{-9}{10}$

Solution: $\bar{a} = -i + 0j + 3k$, and $\bar{b} = -i + j + 0k$. Let θ be angle between $\bar{a}$ and $\bar{a} - \bar{b} = \bar{a} - \bar{b} = 0i - j + 3k$, thus

$$\cos\theta = \frac{\bar{a}.(\bar{a}.\bar{b})}{|\bar{a}||\bar{a}-\bar{b}|} = \frac{(-i+0j+3k)}{\sqrt{1+9}} \cdot \frac{(0i-j+3k)}{\sqrt{1+9}} = \frac{9}{\sqrt{10}}$$

i.e. $\cos\theta = \frac{9}{\sqrt{10}}$

Answer: A

32. Let $\bar{v} = 3i - 4k$, when $\bar{i}$ and $\bar{k}$ are the standared unit vectors in the directions of positive x-axis and positive z-axis respectively, and $\overline{AB}$ be a vector from the point A (0, 1, 2) to a point B in space. If $\overline{AB}$ is parallel to $\bar{v}$ and $\|\overline{AB}\| = 10$, then point B is at:

A. (−6, −1, 10) C. (6, 1, −6)
B. (6, −1, −10) D. (−6, −1, 6)

Solution: Let B = (x, y, z). Since

$\overline{AB} \| \bar{V} \Rightarrow \overline{AB} = t\bar{v} \Rightarrow |\overline{AB}| = |t\bar{v}|$

$\Rightarrow t\bar{v} = t(3, 0, -4) \Rightarrow (3t, 0, -4t)$

$\Rightarrow |t\bar{v}| = 10 \Rightarrow \sqrt{9t^2 + 0 + 16t^2} = \sqrt{25t^2} = 5t$

$\Rightarrow 5t = 10 \Rightarrow t = 2$

$\Rightarrow \overline{AB} = \bar{B} - \bar{A} = (x, y, z) - (0, 1, 2) = (3, 0, -4)$

$\Rightarrow x = 6, y - 1 = 0 \Rightarrow y = 1$ and $z - 2 = -8 \Rightarrow z = -6$

$\Rightarrow$ B = (x, y, z) = (6, 1, −6) **Answer: C**

33. Let ℓ_1 and ℓ_2 be two lines in space intersecting at the origin, (0, 0, 0). If l_1 and l_2 pass through point A (1, 1, 0) and B (0, 1, 1) respectively, then the angle between ℓ_1 and ℓ_2 is equal to:

A. 30° B. 45° C. 60° D. 90°

Solution: Let origin be O = (0, 0, 0) $\Rightarrow \overline{OA} = (1, 1, 0)$ and $\overline{OB} = (0, 1, 1)$. Let θ be angle between $\overline{OA} = \ell_1$ and $\ell_2 = \overline{OB}$ is

$$\cos\theta = \frac{\overline{OA}.\overline{OB}}{|\overline{OA}||\overline{OB}|} = \frac{(1,1,0).(0,1,1)}{\sqrt{1+1+0}\sqrt{0+1+1}} = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$

Answer: C

34. The simplified form of $\csc\left(\frac{\pi}{2} - x\right)\cos x - \sin^2 x$.

A. $2\cos x$ B. $\cos^2 x$ C. $2\sin x$ D. $\sin 2x$

Solution: $\csc\left(\frac{\pi}{2} - x\right)\cos x - \sin^2 x = (\sec x)\cos x - \sin^2 x$

$\Rightarrow 1 - \sin^2 x = \cos^2 x$ **Answer: B**

35. $\cos(\tan^{-1}(0.5))$ is equal to:

A. $\frac{1}{\sqrt{5}}$ B. $\frac{1}{\sqrt{3}}$ C. $\frac{2}{\sqrt{5}}$ D. $\frac{2}{\sqrt{3}}$

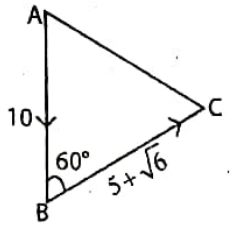
Solution: Let $\theta = \tan^{-1}(0.5) \Rightarrow \tan\theta = \frac{1}{2} \Rightarrow \cos\theta = \frac{3}{\sqrt{5}}$ and $\sin\theta = \frac{1}{\sqrt{5}}$ **Answer: C**

36. A patrol that on a sea sailed from its station 10 km to south and then changed is course and sailed $5 + \sqrt{6}$ km in the direction N60°E. Then, what is the distance the boat should travel in order to return to its station by the short route?

A. $4\sqrt{6}$ km B. 9 km C. $3\sqrt{6}$ km D. 7 km

Solution: $AC^2 = AB^2 + BC^2 - 2(AB)(BC)\cos 60°$

Telegram: https://t.me/ExtremeSeries



$$= 100 + \left(5+\sqrt{6}\right)^2 - 2(10)\left(5+\sqrt{6}\right)\left(\frac{1}{2}\right)$$

$$= 100 + 25 + 10\sqrt{6} + 6 - 50 - 10\sqrt{6} = 81$$

$$AC = \sqrt{81}\text{ km} = 9\text{ km}$$

Answer: B

37. Let $a_n = n + \cos n\pi$, $n = 1, 2, 3, \ldots$ be the n^{th} term of a sequence. Then which of the following is true?

A. $a_5 = 6$ C. $a_7 + a_8 = 15$

B. $a_{10} = 9$ D. $a_7 - a_6 = 1$

Solution:

A. $a_5 = 5 + \cos(5\pi) = 5 + (-1) = 4$

B. $a_{10} = 9 + \cos(9\pi) = 9 + (-1) = 8$

C. $a_7 + a_8 = [7 + \cos(7\pi)] + [8 + \cos 8\pi] = 7 + (-1) + (8+1) = 15$

D. $a_7 + a_6 = [7 + \cos 7\pi] - [6 + \cos 6\pi] = 7 + (-1) - [6+1] = 1$

Answer: C

38. If $\{A_n\}$ is an arithmetic sequence with first term $A_1 = 5$ and $A_5 = 21$, then $\sum_{n=1}^{20} A_n$.

A. 760 B. 780 C. 860 D. 870

Solution: Use, $d = \frac{A_k - A_m}{k-m} \Rightarrow d = \frac{A_5 - A_1}{5-1} = \frac{21-5}{4} = 4$

Use, $S_n = \frac{n}{2}[2A_1 + (n-1)d] \Rightarrow S_{20} = \frac{20}{2}[2(5) + (20-1)4] = 860$

Answer: C

39. What is the sum of the series $\sum_{n=0}^{\infty} 2^n 3^{2-n}$?

A. 6 B. 18 C. 27 D. ∞

Solution: $\sum_{n=0}^{\infty} 2^n 3^{2-n} = \sum_{n=0}^{\infty} 2^n . 3^{-n} . 3^2 = \sum_{n=0}^{\infty} 9\left(\frac{2}{3}\right)^n$

$\Rightarrow G_1 = 9, r = \frac{2}{3} \Rightarrow S = \frac{G_1}{1-r} = \frac{9}{1-\frac{2}{3}} = 9 \times 3 = 27$

Answer: C

40. A business association gets a net profit of Birr 3,000 at the end of each month. Just after the fifth year, its amount was Birr 230,000. What was the starting amount of the business?

A. Birr 47,000 C. Birr 50,000

B. Birr 53,000 D. Birr 56,000

Solution: Monthly profit = 3000 Birr

Yearly profit = $12 \times 300 = 36{,}000$ Birr

$\Rightarrow$ Total amount in 5 year = 230,000 (given)

Let A_0 = initial investment

Let A_1 = 1st year profit = 36,000

Let A_2 = 2nd year profit = $A_1 + 36{,}000$

A_3 = 3rd year profit = $A_2 + 36{,}000 \Rightarrow A_1 + 36{,}000 + 36{,}000$

Let A_5 = 5th year profit = $(36000)5 = 180{,}000$

$\Rightarrow$ Total profit in 5 year = 180,000 $\Rightarrow$ Total amount = initial + total profit (at 5 year) $230{,}000 = A_0 + 180{,}000$

$\Rightarrow A_0 = 230{,}000 - 180{,}000 = 50{,}000$

Answer: C

41. $\lim_{n\to\infty} \frac{3\sqrt{n} - 6n + 5}{4n+1}$ equal to:

A. $-\frac{3}{2}$ B. 0 C. $\frac{3}{4}$ D. $\frac{5}{4}$

Solution: Divide the numerator and denominator by n.

$$\lim_{n\to\infty} \frac{\frac{3\sqrt{n}}{n} - \frac{6n}{n} + \frac{5}{n}}{\frac{4n}{n} + \frac{1}{n}} = \lim_{n\to\infty} \frac{\frac{3}{n} - 6 + \frac{5}{n}}{4 + \frac{1}{n}} = \frac{0 - 6 + 0}{4 + 0} = -\frac{6}{4} = \frac{-3}{2}$$

Answer: A

42. $\lim_{x\to 1^-} \frac{x^4 - 1}{|x-1|}$ equal to:

A. 4 B. 0 C. −1 D. −4

Solution: $|x-1| = \begin{cases} x-1 \text{ if } x \geq 1 \\ -(x-1) \text{ if } x < 1 \end{cases}$ so that $\lim_{x\to 1^-} \frac{x^4-1}{|x-1|}$

$$= \lim_{x\to 5} \frac{(x-1)(x^2+1)(x+1)}{-(x-1)} = -\left(\lim_{x\to 1^-}(x^2+1)(x+1 = (-4)\right)$$

Answer: D

Telegram: https://t.me/Extremeseries

43. $\lim_{x\to 0}\frac{x\ \csc\ 3x}{x+1}$ is equal to:

A. $\frac{1}{3}$ B. 0 C. 1 D. 3

Solution: $\lim_{x\to 0}\frac{x\ \csc\ 3x}{x+1}=\lim_{x\to 0}\frac{x}{\sin\ 3x}.\lim_{x\to 0}\frac{1}{x\ +\ 1}=\frac{1}{3}$

Answer: A

44. $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{2x+4}$ is equal to:

A. e B. e^2 C. e^4 D. ∞

Solution: $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{2x+4}=\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{2x}.\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{4}=e^2.1=e^2$

Answer: B

45. Let $f(x)=\begin{cases}\frac{a}{x^2+3} & \text{if } x\le 1\\ \frac{x-1}{\sqrt{x}-1} & \text{if } x>1\end{cases}$ what is the value of a if f is continuous at $x=1$.

A. 0 B. 2 C. 4 D. 8

Solution: Since f is continuous at $x=1$.

$\Rightarrow \lim_{x\to 1^+} f(x)=\lim_{x\to 1^-} f(x)\Rightarrow \lim_{x\to 1^+}\frac{x-1}{\sqrt{x}-1}=\lim_{x\to 1^-}\frac{a}{x^2+3}$

$\lim_{x\to 1^+}\frac{(x-1)(\sqrt{x}+1)}{x-1}=\frac{a}{4}\Rightarrow 2=\frac{a}{4}\Rightarrow a=8$ **Answer: D**

46. If $f(x)=\pi^2$ then $f'(1)$ is equal to:

A. 2π B. e^2 C. 1 D. 0

Solution: $f(x)=\pi^2$ is constant function $\Rightarrow f'(x)=0\Rightarrow f'(1)=0$

Answer: D

47. Let $f(x)=\frac{6x}{x+a}$, for what value of a is $f'(a)=1$.

A. $\frac{1}{3}$ B. $\frac{2}{3}$ C. $\frac{3}{2}$ D. 3

Solution:

$f'(x)=\frac{6(x+a)-6x}{(x+a)^2}\Rightarrow f'(x)=\frac{6a}{(x+a)^2}\Rightarrow f'(a)=\frac{6a}{4a^2}=1$

$\Leftrightarrow 4a=6\Rightarrow a=\frac{6}{4}=\frac{3}{2}$ **Answer: C**

48. Which one of the following is necessarily true about a function $f(x)$?

A. If f is continuous at $x=a$, then f is differentiable at $x=a$.

B. If f is not differentiable at $x=a$ then $\lim_{x\to a^-} f(x)\neq\lim_{x\to a^+} f(x)$.

C. If f is differentiable at x = a then $\lim_{x\to a^-} f(x)=f(a)=\lim_{x\to a^+} f(x)$.

D. If the derivative f '(a) = 0, then f attains its maximum value at $x=a$.

Solution:

A. False, example, $f(x)=|x-1|$, cont. at $x=1$ but not differentiable at $x=1$.

B. False

C. True, if function f is differentiable at $x=a$ then f is continuous at $x=a$.

D. False, example $f(x)=x^3+5\Rightarrow f'(x)=3x^2$. f' has the same sing for all x.

Answer: C

49. If $g(x)=\frac{f(x)}{x+1}+(f(x))^2, f(1)=8$ and $f'(1)=2$ then $g'(1)$ is equal to:

A. 36 B. 31 C. 25 D. 16

Solution: We have $f(1)=8$ and $f'(1)=2$.

$g'(x)=\frac{f'(x)(x+1)-f(x)(x+1)'}{(x+1)^2}+2f(x)f'(x)$

$\Rightarrow g'(1)=\frac{f'(1)(1+1)-f(1)(1)}{(1+1)^2}+2f(1)f'(1)$

$=\frac{(2)(2)-8}{4}+2(8)(2)=-1+32=31$

Answer: B

50. If $f(x)=xe^{3x}-\cos(2x)$, then $f''(0)$ is equal to:

A. 0 B. 2 C. 6 D. 10

Telegram: https://t.me/ExtremeSeries

Solution:

$f'(x) = (x')\,e^{3x} + x\,(e^{3x})' + \sin(2x)\,.\,2 = e^{3x} + 3 \times e^{3x} + 2\sin(2x)$

$f''(x) = 3e^{3x} + 3e^{3x} + (3)(3x)e^{3x} + 4\cos(2x)$

$f''(0) = 3 + 3 + 0 + 4 = 10$ **Answer: D**

51. At what value (s) of x does $f(x) = \frac{1}{3}x^3 - \frac{1}{2}x^2 - 2x + 1$ have a local maximum?

A. $x = 2$ B. $x = -1$ C. $x = 5$ D. $x = 2, x = -1$

Solution: $f'(x) = x^2 - x - 2 = (x - 2)(x + 1)$ critical number,

$f'(x) = 0 \Rightarrow x = 2$ and $x = -1$.

$f''(x) = 2x - 1 \Rightarrow f''(2) = 4 - 1 = 3 > 0 \Rightarrow x = 2$ is local minimum value and $f''(-1) = -2 - 1 = -3 < 0 \Rightarrow$ at $x = -1$ is local maximum

Answer: B

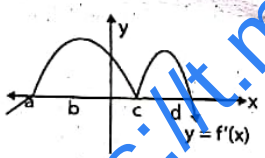
52. Look at the following graph of $f'(x)$. Which of the following is true about f?

A. f is increasing on $[c, d]$

B. f is decreasing on $[b, c]$

C. f has local minimum at d

D. f has a local extreme value at c



Solution: Remember that:

- If $f'(x) > 0$ for all x in (m, n), then f is increasing on $[m, n]$.
- If $f'(x) < 0$ for all x in (m, n) then f is decreasing on $[m, n]$.
- If $f'(x_o) = 0$ and f' changes the sign:
  - i) from (+) to (–) then f has local max at x_o
  - ii) from (–) to (+) then f has local min at x_o

Now, from the above graph we have:

Interval	$-\infty < x < a$	$a < x < b$	$b < x < c$	$c < x < d$	$d < x < \infty$
sign of $f'(x)$	–	+	+	+	–
Conclusion of $y = f(x)$	decreasing	increasing	decreasing	increasing	Decreasing

- f is increasing on $[c, d]$, $[a, b]$, f is decreasing on $[b, c]$
- f has local minimum at $x = a$
- f has local maximum at $x = d$
- no local extreme value at b and c because no sign change

Answer: A

53. On which of the following interval is the graph of $f(x) = \ln(x^2 + 1)$ concave upward?

A. $(-\infty, -1][1, \infty)$ B. $[0, \infty)$ C. $(-\infty, 0]$ D. $(-1, 1)$

Solution: If $f''(x) > 0$ for all x in I, then the graph of f concave upward in I. If $f''(x) < 0$ for all x in I, then graph of f is concave down ward in I.

Now that,

$f'(x) = \frac{2x}{x^2 + 1}$ and $f''(x) = \frac{2(x^2 + 1) - 4x^2}{(x^2 + 1)^2} = \frac{2 - 2x^2}{(x^2 + 1)^2}$

Because $f''(x) = 0$ when $x = \pm 1$, so that test f'' in $(-\infty, -1)$, $(-1, 1)$ and $(1, \infty)$.

	$-\infty$	-1	1	∞
$2 - 2x^2$	–	+	–	
$(x^2 + 1)^2$	+	+	+	
$f''(x)$	–	+	–	
conclusion of f	concave down	concave upward	concave down ward	

$\therefore f$ is concave down ward on $(-\infty, -1)$ and $(1, \infty)$.

$\therefore f$ is concave upward on $(-1, 1)$ **Answer: D**

54. Water is poured into a cylindrical tanker of radius 5 meters at a rate of 10 meter cube/min. What is the rate of change of the height of the level of water when it rises to 3 meters?

A. $\frac{3}{25\pi}$ m/min C. $\frac{1}{2\pi}$ m/min

B. $\frac{2}{5\pi}$ m/min D. $\frac{3}{5\pi}$ m/min^{-1}

Solution: $v = \pi r^2 h \Rightarrow v = \pi(5^2)h = 25\pi h$

$\Rightarrow \frac{dv}{dt} = 25\pi \frac{dh}{dt}$ ← derivate w.r.t time

$\Rightarrow 10m^3 / \min = 25\pi \frac{dh}{dt} \Rightarrow \frac{10}{25\pi} = \frac{2}{5\pi} m / \min$ **Answer: B**

55. A rectangular field of length ℓ and width (w) meters for $w < \ell$ has perimeter 400 meters. If a circular region of area w^2 is to be reserved for office purpose what should be the length of eth field so that the area of the remaining region is maximum.

A. 50m B. 100m C. 120m D. 150m

Telegram: https://t.me/ExtremeSeries

Solution:

i) A (rectangle) = lw ii) A (reserve) = w^2

iii) P (perimeter of rectangle) = $2w + 2\ell = 400$

$\Rightarrow w + \ell = 200 \Rightarrow \ell = 200 - w$

iv) The remaining area A = A (rectangle) – A (reserve)

$\Rightarrow A(\ell, w) = \ell w - w^2 \Rightarrow A(w) = (200 - w)w - w^2$

$= 200w - w^2 - w^2 \Rightarrow A(w) = 200w - 2w^2$

$\Rightarrow A'(w) = 200 - 4w \Rightarrow A'(w) = 0 \Rightarrow 200 - 4w = 0 \Rightarrow 4w = 200$

$\Rightarrow w = 50$ and the length (ℓ) of the field, $\ell = 200 - w$

$\Rightarrow \ell = 200 - 50 = 150$

Answer: D

56. The anti derivative of f (x) = tan x.

A. $\frac{1}{2}\tan^2 x$ B. $\sec^2 x$ C. $-\ln(\cos x)$ D. $\ln|\sec x|$

Solution: $\int \tan x dx = \int \frac{\sin x}{\cos x} dx$, Let $u = \cos x \Rightarrow du = -\sin x\, dx$

$\Rightarrow \int \frac{-du}{u} = -\ln|u| + c \Rightarrow -\ln|\cos x| + c$

$\Rightarrow \ln(\cos x)^{-1} = \ln\left(\frac{1}{\cos x}\right) = \ln|\sec x|$

Answer: D

57. $\int \frac{x}{x^2 + 2x + 1} dx$ equal to:

A. $\ln(x^2 + 2x + 1) + c$ C. $\ln(x^2 + 2x + 1) - \frac{1}{x+1} + c$

B. $\ln|x+1| + \frac{1}{x+1} + c$ D. $\ln(x^2 + 2x + 1) + \frac{1}{x+1} + c$

Solution: $\int \frac{x}{x^2 + 2x + 1} dx = \int \frac{x}{(x+1)^2} dx$, Let $u = x + 1$

$\Rightarrow x = u - 1 \Rightarrow du = dx$

$\Rightarrow \int \frac{u-1}{u^2} du = \int \frac{1}{u} du - \int \frac{du}{u^2} = \ln|u| + \frac{1}{u} + c$

$\Rightarrow \ln|x+1| + \frac{1}{x+1} + c$

Answer: B

58. $\int \frac{e^{2x} - 4x}{xe^{2x}} dx$ is equal to _____

A. $\ln|x| + 2e^{-2x} + c$ C. $\ln|x| + 4e^{-2x} + c$

B. $\ln|x| - 4e^{2x} + c$ D. $-\frac{1}{x^2} + 2e^{-2x} + c$

Solution: $\int \frac{e^{2x} - 4x}{xe^{2x}} dx = \int \left(\frac{e^{2x}}{xe^{2x}} - \frac{4x}{xe^{2x}}\right) dx$

$= \int \frac{dx}{x} - \int \frac{4}{e^{2x}} dx = \ln|x| + 2e^{-2x} + c$

Explanation: $\boxed{\int e^{mx} dx = \frac{e^{mx}}{m}}$

Answer: A

59. $\int_0^1 \frac{(x-1)^2}{x+1}$ equal to:

A. $-\frac{3}{2} + \ln 6$ C. $-\frac{5}{2} + 4\ln 2$

B. $-\frac{5}{2} + \ln 15$ D. $\frac{5}{2} + \ln 16$

Solution: $\int_0^1 \frac{(x-1)^2}{x+1} dx = \int_0^1 \frac{x^2 - 2x + 1}{x+1} dx = \int_0^1 \left(x - 3 + \frac{4}{x+1}\right) dx$

$\Rightarrow \frac{x^2}{2} - 3x + 4\ln|x+1| \Big|_0^1 = \frac{1}{2} - 3 + 4\ln 2 = -\frac{5}{2} + 4\ln 2$

Answer: C

60. If $f(x) = \begin{cases} \sin x, & x \le 0 \\ 3x\sqrt{x^2+1} & x > 0 \end{cases}$ then $\int_{-\frac{\pi}{2}}^{\sqrt{8}} f(x) dx$ equal:

A. 17 B. 25 C. 27 D. $3\sqrt{8}$

Solution: $\int_{-\frac{\pi}{2}}^{\sqrt{8}} f(x) dx = \int_{-\frac{\pi}{2}}^{0} f(x) dx + \int_0^{\sqrt{8}} f(x) dx$

$= \int_{-\frac{\pi}{2}}^{0} \sin x\, dx + \int_0^{\sqrt{8}} 3x\sqrt{x^2+1}\, dx$

Let $u = \sqrt{x^2+1} \Rightarrow u^2 = x^2 + 1 \Rightarrow 2udu = 2x\, dx \Rightarrow udu = x\, dx$ when $x = \sqrt{8}$ then $u = \sqrt{8+1} = 3$, $x = 0$, then $u = \sqrt{0+1} = 1$

Telegram: https://t.me/ExtremeSeries

$$\int_{-\frac{\pi}{2}}^{0}\sin x\,dx+\int_{1}^{3}3u^2du=-\cos x\Big|_{-\frac{\pi}{2}}^{0}+3\frac{u^3}{3}\Big|_{1}^{3}$$

$=-1-0+3^3-1^3=25$ **Answer: B**

61. The volume of the solid which is generated when the region bounded by $y=\sqrt{x+1}$ and x-axis from $x=0$ to $x=2$ is rotated about x-axis is equal to:

A. 4π B. 3π C. $\frac{4}{3}\pi$ D. $\frac{3}{4}\pi$

Solution: Volume $=\int_a^b\pi[f(x)]^2dx=\int_0^2\pi\left(\sqrt{x+1}\right)^2dx$

$$\Rightarrow\pi\int_0^2(x+1)dx=\pi\left[\frac{x^2}{2}+x\right]_0^2=\pi\left[\frac{4}{2}+2\right]=4\pi$$

Answer: A

62. If a sphere with center C (0, 1, 1) intersects the z-axis at P (0, 0, 3) then the radius of the sphere is equal to:

A. 5 B. 3 C. $\sqrt{3}$ D. $\sqrt{5}$

Solution: Equation of sphere:

$\Rightarrow(x-0)^2+(y-1)^2+(z-1)^2=r^2\Rightarrow(0-0)^2+(0-1)^2+(3-1)^2=r^2$

$\Rightarrow r^2=5\Rightarrow r=\sqrt{5}$ **Answer: D**

63. Suppose A and B are the end points of a diameter of the sphere with equation $x^2+y^2+(z+2)^2=1$. If A $(1, 0, -2)$ then B is equal to:

A. $(0, 1, -2)$ B. $(0, 0, -1)$ C. $(-1, 0, -2)$ D. $(0, -1, -2)$

Solution: Let center of sphere P $(0, 0, -2)$. Let B $= (x, y, z)$ and P $(0, 0, -2)$ is mid pt. of AB.

$$\Rightarrow\left(\frac{x+1}{2},\frac{y+0}{2},\frac{z-0}{2}\right)=(0,0,-2)\Rightarrow x+1=\Rightarrow x=-1,y=0,$$

$$\frac{z-2}{2}=-2\Rightarrow z-2=-4\Rightarrow z=-4+2=-2$$

$\therefore B(x, y, z)=(-1, 0, -2)$

Answer: C

1. If $f(x)=\frac{1}{e^x+1}$, then which of the following is equal to $f^{-1}(x)$

A. $\ln(1-x)-\ln(x)$ C. $\ln\left(\frac{1}{x+1}\right)$

B. e^{-x} D. $\frac{1}{x^{-x}+1}$

Solution:

Remember that, to find $f^{-1}(x)$, first interchange $x\to y$ and $y\to x$.

Then $y=\frac{1}{e^x+1}\Rightarrow x=\frac{1}{e^y+1}\Rightarrow xe^y+x=1$

$$\Rightarrow xe^y=1-x\Rightarrow e^y=\frac{1-x}{x}$$

$$\Rightarrow y=\ln\left(\frac{1-x}{x}\right)\Rightarrow f^{-1}(x)=\ln(1-x)-\ln x.$$ **Answer: A**

2. Which one of the following function has no vertical asymptote?

A. $f(x)=\ln(x+1)$ C. $f(x=\frac{x^2-9}{x-3}$

B. $f(x)=\frac{x^2+1}{x^3+8}$ D. $f(x)=\frac{x-1}{x^3-x}$

Solution:

A. $x+1=0\Rightarrow x=-1\leftarrow$ vertical asymptote of $\ln(x+1)$

B. $x^3+8=0\Rightarrow x=-2\leftarrow$ Vertical asymptote.

C. $\frac{x^2-9}{x-3}=\frac{(x-3)(x+3)}{x-3}=x+3,x\neq3$ No vertical asymptote

D. $\frac{x-1}{x(x-1)}=\frac{1}{x},x\neq0,1$ thus $x=0\leftarrow$ Vertical asymptote.

Answer: C

3. The value of $|x|+2x$ if $x<0$

A. $-3x$ B. $3x$ C. $-x$ D. x

Telegram: https://t.me/ExtremeSeries

Solution: $x<0 \Rightarrow |x| = -x$, thus

$|x|+2x = -x+2x = -x+2x = x$ **Answer: D**

4. Given $f(x) = \ln(x-1)$ and $g(x) = \sqrt{1-2x}$, then the domain of fog is equal to _____

A. $\{x \in \mathbb{R} : x > 1\}$ C. $\{x \in \mathbb{R} : x < 0\}$

B. $\left\{x \in \mathbb{R} : x \le \frac{1}{2}\right\}$ D. $\left\{x \in \mathbb{R} : x > \frac{1}{2}\right\}$.

Solution: Remember that, to find the domain of (fog)(x). A number x is in the domain of (fog)(x) is and only if both g(x) and $f(g(x))$ are defined. Hence, $g(x) = \sqrt{1-2x}$ is defined on

$\{x : 1-2x \ge 0\} = \left\{x : x \le \frac{1}{2}\right\}$

$f(g(x)) = \ln(g(x)-1) = \ln\left(\sqrt{1-2x}-1\right)$.

is defined $\sqrt{1-2x}-1 > 0 = \sqrt{1-2x} > 1 \Rightarrow 1-2x > 1 \Rightarrow x < 0$

Therefore both $g(x)$ and $f(g(x))$ is defined on $x < 0$,

$\Rightarrow$ Domain f(g(x)) $\{x : x < 0\}$. **Answer: C**

- **Exercise left for you**: what is the domain (gof)?

5. The solution set of $\frac{2}{x} - \frac{x-2}{x^2-2x} = 1 - \frac{2x-2}{3x-2}$?

A. $\{1,-2\}$ B. $\{1,2\}$ C. $\{-1\}$ D. $\{1\}$

Solution: The expression is the same as

$\left(\frac{2}{x}\right)\left(\frac{x-2}{x-2}\right) - \frac{x-2}{x^2-2x} = \frac{3x-2-2x+2}{3x-2}; x \ne 0, 2, \frac{2}{3}$

$\Leftrightarrow \frac{x-2}{x(x-2)} = \frac{x}{3x-2}$

$\Rightarrow \frac{1}{x} = \frac{x}{3x-2} \Leftrightarrow 3x-2 = x^2 \Rightarrow x^2-3x+2 = 0$

$\Rightarrow (x-2)(x-1) = 0 \Rightarrow x = 2$, x = 1, but $x \ne 2$

Therefore: S.S = {1}

Answer: D

6. If a line with angle of inclination of $\frac{3\pi}{4}$ passes through (0,1), which one of the following is the equation of the line

A. $y = -x+1$ B. $y = x+1$ C. $y = -x-1$ D. $y = x-1$

Solution: We know that Slope $= \tan\left(\frac{3\pi}{4}\right) = -1$

$\Rightarrow$ equation of line l: $y-1 = -1(x-0) \Rightarrow y = y = -x+1$

7. If $x^2 - 6x + y^2 + k = 0$ is equation of circle with radius 2, then what is the value of k?

A. 13 B. 5 C. 4 D. -4

Solution: Use completing square method

$\Rightarrow x^2 - 6x + 9 + y^2 = -k + 9$ ← Adding 9 to both sides

$\Rightarrow (x-3)^2 + (y-0)^2 = r^2 = -k+9$, but $r = 2 \Rightarrow r^2 = -k+9$

$\Leftrightarrow 4 = -k+9 \Rightarrow k = 5$. **Answer: B**

8. A parabola with focus at (3,-1) has directrix, $y = 3$, then the equation of parabola

A. $(x-3)^2 = -4(y+1)$ C. $(x-3)^2 = 4(y+1)$

B. $(x-3)^2 = -8(y-1)$ D. $(x-3)^2 = 8(y-1)$

Solution:

Since the focus below the directrix, $y=3$ then the parabola open down ward. Equation of parabola is given by $(x-h)^2 = -4p(y-k)$

- Distance between focus and directrix is $|2p| \Rightarrow 2p = 3-(-1) = 4 \Rightarrow p = 2$
- Vertex $(h, k) = (3,-1+2) = (3,1) \Rightarrow (x-3)^2 = -4(2)(y-1)$

$\Rightarrow (x-3)^2 = -8(y-1)$

Answer: B

9. **Exercise left for you**: Find the directrix, focus, of question number 8 for choice A, C and D.

10. A satellite moves along a hyperbolic curve whose horizontal transverse axis is 24km and an asymptote, $y = \frac{5}{12}x + 2$ then, what is the eccentricity of the hyperparabola?

Telegram: https://t.me/ExtremeSeries

A. $\frac{5}{13}$ B. $\frac{12}{13}$ C. $\frac{13}{12}$ D. $\frac{15}{3}$

Solution:

The distance between vertices of hyperbola (transverse axis) is $2a=24 \Rightarrow a=12$.

The equation with center (h, k) is given by: $\frac{(x-h)^2}{a^2}-\frac{(y-k)^2}{b^2}=1$

- Asymptote given by $y-k=\pm\frac{b}{a}(x-h)$ with a=12. But we have: $y=\frac{5}{12}x+2$, so that $b=5, a=12$
- Eccentricity (e)

$$y(e)=\frac{c}{a}=\frac{\sqrt{a^2+b^2}}{a}=\frac{\sqrt{144+25}}{12}=\frac{13}{12}$$

$$y(e)=\frac{13}{12}$$

Answer: C

11. ***Exercise left for you (Tip).*** From question number 10, what is the distance between foci?

12. Let p, q and r be proposition such that $p \Rightarrow (r \vee \neg q)$ is false. Then, which one of the following proposition is true?

A. $p \Rightarrow r$ B. $\neg p \Leftrightarrow q$

C. $\neg r \Rightarrow q$ D. $q \Leftrightarrow r$

Solution: $p \Rightarrow (r \vee \neg q)$ is false if and only if $r \vee \neg q \equiv F$ and $P \equiv T$, so that $(r \vee \neg q)$ is false if $r \equiv F$ and $\neg q \equiv F$, imply, $q \equiv T$, thus,

A. $p \Rightarrow r \equiv r \equiv T \Rightarrow F \equiv F$

B. $\neg p \Leftrightarrow q \equiv F \Leftrightarrow T \equiv F$

C. $\neg r \Rightarrow q \equiv T \Rightarrow T \equiv T$

D. $q \Leftrightarrow r \equiv T \Leftrightarrow F \equiv F$

Answer: B

13. Consider the following argument:
If he does not love her, she will not marry him. He loves her. Therefore she will marry him."
If '$P \equiv$ he love her' and '$q \equiv$ she will marry him' Which one of the following is the correct representation of the argument and its validity?

A. $p \Rightarrow \neg q, p \square q$; Valid argument

B. $p \Rightarrow q, p \square q$; Valid argument

C. $\neg p \Rightarrow \neg q, p \square q$; invalid argument

D. $p \Rightarrow q, p \square q$; Invalid argument

Solution: $\neg p \Rightarrow \neg q, p \vdash q$

1) $p \equiv True \leftarrow$ premise

2) $\neg p \Rightarrow \neg q \equiv T \leftarrow$ premise

3) $\neg p \equiv F$ and $\neg q \equiv F$ then $q \equiv T$ and if $\neg p \equiv T$, then $q \equiv F$ thus, $\neg p \Rightarrow \neg p, p \vdash q$ is invalid,

Answer: C

14. For real number x and y, which one of the following statements is true

A. $(\forall x)(\exists y)(x^2+y+1=0)$ C. $(\exists y)(\forall x)(x^2+y+1=0)$

B. $(\exists x)(\forall y)(x^2+y+1=0)$ D. $(\forall y)(\exists x)(x^2+y+1=0)$

Solution:

A. For every x, we can find y, that satisfies $x^2+y+1=0$ is true, $\Rightarrow y=-x^2-1$

B. There is fixed x which stands for all y that satisfies $x^2+y+1=0$ is false.

C. False

D. For every y, we can find x that satisfies $x^2+y+1=0$ is false.

Let $y=0 \Rightarrow -x^2-1=0 \Rightarrow -x^2=1 \Rightarrow x^2=-1$ is false.

Answer: A

Telegram: https://t.me/Extremeseries

15. Among students who took a quiz, 15 students scored 6, 20 student scored 7, 10 students scored 8, 5 student scored 10. What is the average score of the student?

A. 7.8 B. 7.5 C. 7.2 D. 7.0

Solution:

$$\text{Average score} = \frac{15\times6+20\times7+10\times8+5\times10}{15+20+10+5}$$

$$= \frac{90+140+80+50}{50} = \frac{360}{50} = 7.2,$$ **Answer: C**

16. The market that students scored in an examination is grouped in class intervals shown in the following table.

lass interval (mark)	55 - 64	65 - 74	75 - 84	85 – 94	95 – 100
mber of student	8	12	20	6	4

What is the median of the mark?

A. 25.0 B. 75.5 C. 77.0 D. 79.5

Solution:

Remember that the median of data is given by

$$M_d = B_L + \left(\frac{\frac{n}{2} - Cf_b}{f_c}\right)i$$

- Median class is $\left(\frac{n}{2}\right)^{th} = \left(\frac{50}{2}\right)^{th} = (25)^{th}$ item, it lies on 3^{rd} class (75 - 84).
- Lower boundary of median class denoted by $B_L = 74.5$
- Cumulative frequency just before the median class denoted by $Cf_b = 8+12=20$
- Frequency of median class, $f_c = 20$
- Class interval (width), $i = 65\text{-}55 = 10$
- Total frequency, $n=50$

Therefore $Md = 74.5 + \left(\frac{\frac{50}{2} - 20}{20}\right)10 = 77.0$

Answer: C

- **Exercise left for you (Tip).** What is the mean, the mode, of the mark?

17. How many four digit even numbers can be formed from 1,2,3,4 and 5 if the number start with 3?

A. 40 B. 50 C. 100 D. 120

Solution:

Let the digit be ABCD but A = 3, is 1 way
B = we can fill either of student = 5 way
C = we can fill either of 5 digit = 5 way
D = we can fill either 2 or 4 = 2 way.
Thus, $1\times5\times5\times2 = 50$ way. **Answer: B**

- **Exercise left for you (Tip).** How many 5 digit even numbers can be formed from 1,2,3,4 and5 if the number starts with 3 and 1?

18. A committee consisting of 3 students is to be selected from 10 candidates among which 4 are girls. What is the probability that at least one girl is selected.

A. $\frac{5}{6}$ B. $\frac{2}{3}$ C. $\frac{1}{3}$ D. $\frac{1}{6}$

Solution: Remember that P(At least one girl = 1 – P (All boys)

$$\text{P (All boys)} = \frac{c(6,3)c(4,0)}{c(10,3)} = \frac{20}{120} = \frac{1}{6}$$

$$\Rightarrow \text{P(At least one girl)} = 1 - \frac{1}{6} = \frac{5}{6}$$ **Answer: A**

19. A box contains 5 white, 6 red and 4 black balls of identical size. If 3 balls are randomly taken out of the box one after the other, what is the probability that the first ball is white and both the second and third balls are red?

A. $\frac{2}{15}$ B. $\frac{3}{15}$ C. $\frac{4}{75}$ D. $\frac{5}{91}$

Solution:

Let us assume that drawing is without replacement.
Let W = White, let Red = R_2, R_3 (2^{nd}, 3^{rd} draw) $\Rightarrow$ P(W R_2 R_3)

Telegram: https://t.me/Extremeseries

$$= P(W).P(R_2/W).P(R_3/WR_2) = \frac{5}{15}.\frac{6}{14}.\frac{5}{13} = \frac{5}{91}$$

Answer: D

- Find the probability of getting the ball:
  A. All 3 are White
  B. The ball drawn in the order (Red, White, Black)

20. Suppose $A = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$. If X is a 2×2 matrix such that $AX - A^t = 2A$. then what is the value of X

A. $\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$ B. $\begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix}$ C. $\begin{pmatrix} 3 & 6 \\ 6 & 9 \end{pmatrix}$ D. $\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$

Solution: Let $X=\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $A^t=\begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$ then, $AX - A^t = 2A$

$$\Rightarrow \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} - \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ 4 & 6 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} a+2c & b+2d \\ 2a+3c & 2b+3d \end{pmatrix} = \begin{pmatrix} 3 & 6 \\ 6 & 9 \end{pmatrix}$$

$$\Rightarrow \begin{cases} a+2c=3 \\ 2a+3c=6 \end{cases} \text{ and } \begin{cases} b+2d=6 \\ 2b+3d=9 \end{cases}$$

$$\Rightarrow \underset{-c=0\Rightarrow a=3}{+\begin{cases} -2a-4c=-6 \\ 2a+3c=6 \end{cases}} \qquad \underset{-d=-3\Rightarrow d=3\Rightarrow b=0}{+\begin{cases} -2b-4d=-12 \\ 2b+3d=9 \end{cases}}$$

Thus, $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$

Answer: D

21. Suppose that A and B are 3×3 matrices I is the identity matrix of order 3 such that AB=2I. If det B = |B|=6. What is the det (A^t)?

A. $\frac{1}{3}$ B. $\frac{4}{3}$ C. 12 D. 48

Solution: Recall that $|AB|=|2I| \Rightarrow |A||B| = 8(1)$ since $|2I| = 2^3(1) = 8$, and 3 is order

$\Rightarrow |A|(6) = 8 \Rightarrow |A| = \frac{8}{6} = \frac{4}{3} \leftarrow |A^t| = |A|$

Answer: B

22. For what value of b does the parabola $P(x) = ax^2+x+b$, pass through the points (-1, 5) and (2, -1)

A. 9 B. 3. C. -3 D. -15

Solution:

$$\left.\begin{aligned} P(-1) = a-1+1+b = 5 \Rightarrow a+b=6 \\ P(2) = 4a+2+b = -1 \Rightarrow 4a+b=-3 \end{aligned}\right\}$$

$-3a = 9 \Rightarrow a = -3$ Thus, -3+b=6 $\Rightarrow$ b=9

Answer: A

23. Consider the system $\begin{cases} ax+y+z=1 \\ x+2y+4z=0 \\ 5x-y+z=0 \end{cases}$

If the determinant of the coefficient matrix is 2 then what is the solution of the system of equation:

A. $\left(3\alpha, \frac{19\alpha}{2}, \frac{-11\alpha}{2}\right)$ C. $\frac{3}{\alpha}, \frac{-19}{2}, \frac{11}{2}$

B. $\left(3, \frac{19}{2}, \frac{-11}{2}\right)$ D. $\left(\frac{3}{2}, \frac{19}{2}, \frac{-9}{2}\right)$

Solution: D=2, remember, $|A^t| = |A|$

$$x = \frac{D_x}{D} = \frac{\begin{vmatrix} 1 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & -1 & 1 \end{vmatrix}}{2} = \frac{\begin{vmatrix} 1 & 0 & 0 \\ 1 & 2 & -1 \\ 1 & 4 & 1 \end{vmatrix}}{2} = \frac{\begin{vmatrix} 2 & -1 \\ 4 & 1 \end{vmatrix}}{2} = \frac{2+4}{2} = 3$$

$$Y = \frac{D_y}{D} = \frac{\begin{vmatrix} \alpha & 1 & 1 \\ 1 & 0 & 4 \\ 5 & 0 & 1 \end{vmatrix}}{2} = -\frac{\begin{vmatrix} 1 & \alpha & 1 \\ 0 & 1 & 4 \\ 0 & 5 & 1 \end{vmatrix}}{2} \leftarrow \text{Interchanging row } (R_1 \leftrightarrow R_2)$$

Telegram: https://t.me/Extremeseries

$$y=\frac{D_y}{D}=-\frac{\begin{vmatrix}1&0&0\\ \alpha&1&5\\ 1&4&1\end{vmatrix}}{2}=\frac{\begin{vmatrix}1&5\\4&1\end{vmatrix}}{2}=-\frac{(1-20)}{2}=\frac{19}{2}$$

$$P_z=\begin{vmatrix}\alpha&1&1\\1&2&0\\5&-1&1\end{vmatrix}=1(-1-10)=-11$$

$$z=\frac{-11}{2}$$

$$\therefore S.S=\left\{\left(3,\frac{19}{2},\frac{-11}{2}\right)\right\}$$

Answer: B

24. Which one of the following is the simplest form of $|3+4i|-\frac{25i}{3+4i}$?

A. $5-5i$ B. $5+5i$ C. $1+3i$ D. $1-3i$

Solution: We know that

$$|3+4i|=\sqrt{3^2+4^2}=5\Leftrightarrow 5-\frac{25i}{3+4i}\cdot\frac{3-4i}{3-4i}=5-\frac{75i-100}{25}$$

$$=5-4-3i=1-3i$$

Answer: D

25. In the set of complex numbers, which one of the following is the solution set of $z^3-iz^2+2z=0$?

A. {0} B. {0, -i} C) {0, -i, 2i} (D) {0, i, -2i}

Solution:

$z^3-iz^2+2z=0\Rightarrow z(z^2-iz+2)=0$

$\Rightarrow z\ (z^2-iz+2)=0\Rightarrow z=0$ or $z^2-iz+2=0$

$\Rightarrow z=i\pm\sqrt{(i)^2-4(2)}$

$\Rightarrow z=\frac{i+3i}{2}\Rightarrow z=4i$ or $z=2i^2$

$\therefore$ S.S = {0, -i, 2i}

Answer: D

26. If $z=\cos\left(\frac{\pi}{10}\right)+i\sin\left(\frac{\pi}{10}\right)$, then what is the value of z^5?

A. $\frac{\pi}{2}+\frac{\pi}{2}i$ B. $\frac{1}{2}+\frac{1}{2}i$ C. i D. 1 + i

Solution:

$$z^5=\left[\cos\left(\frac{\pi}{10}\right)+i\sin\left(\frac{5\pi}{10}\right)\right]=\cos\frac{\pi}{2}+i\sin\left(\frac{\pi}{2}\right)=0+i=i$$

Answer: C

27. If $\overline{U}=(-3,\ x)$ and $\overline{V}=(x,\ y-2)$ are vectors what is the value of y so that $\overline{U}+\overline{V}=3\overline{U}-\frac{1}{2}\overline{V}$

A. $\frac{2}{3}$ B. $\frac{-10}{3}$ C. -4 D. $\frac{-22}{3}$

Solution: $\overline{U}+\overline{V}=(-3+x,x+y-2)=(-9,3x)-\left(\frac{1}{2}x,\frac{y-2}{2}\right)$

$$=\left(-3+x,x+y-2\right)=\left(-\frac{1}{2}x-9,3x-\frac{y+2}{2}\right)$$

$$\Rightarrow -3+x=-\frac{1}{2}x-9 \text{ and } x+y-2=3x-\frac{y+2}{2}$$

$\Rightarrow -12=3x\Rightarrow x=-4$ and $x+y-2=3x-\frac{y}{2}+1$, since

$x=-4\Rightarrow -4+y-2=-12-\frac{y}{2}+1$

$$\Rightarrow -6+y=-11-\frac{y}{2}\Rightarrow -6+11=-\frac{y}{2}-y$$

$$\Rightarrow -\frac{3y}{2}=5\Rightarrow y=\frac{-10}{3}$$

Answer: B

28. Suppose $\overline{A}=2j-k$ and $\overline{B}=5\overline{i}+15\overline{k}$, where $\overline{i},\overline{j}$ and $\overline{k}$ are the standard unit vectors in the directions of positive x, y and z axis respectively. Which one of the following is the unit vector in the direction of $\overline{A}+\frac{1}{5}\overline{B}$?

Telegram: https://t.me/ExtremeSeries

A. $\frac{3}{5}\vec{i}+\frac{4}{5}\vec{k}$ C. $\frac{4}{5}\vec{j}-\frac{3}{5}\vec{k}$

B. $\frac{1}{3}\vec{i}+\frac{2}{3}\vec{j}+\frac{2}{3}\vec{k}$ D. $\frac{2}{3}\vec{i}-\frac{1}{3}\vec{j}+\frac{2}{3}\vec{k}$

Solution: Unit vector in the direction of $\vec{A}+\frac{1}{5}\vec{B}$ is given by

$$\frac{\vec{A}+\frac{1}{5}\vec{B}}{\left|\vec{A}+\frac{1}{5}\vec{B}\right|}=\frac{2\vec{j}-\vec{k}+\frac{1}{5}\left(5\vec{i}+15\vec{k}\right)}{\left|2\vec{j}+\vec{i}+2\vec{k}\right|}$$

$$=\frac{i+2j+2k}{\sqrt{1+4+4}}=\frac{\vec{i}}{3}+\frac{2}{3}j+\frac{2}{3}\vec{k}$$ **Answer: B**

29. A line given by a vector equation $r(t)= (0,3) + t(1,1)$ is tangent to circle at point (0,3). If the radius of the circle is $\sqrt{2}$, which one of the following is the centre of circle?

A. (1,4) B. (1,-4) C. (-1,2) D. (1,2)

Solution: The position vector of (0,3) and centre (x, y) is $(x,y-3)$, since it is perpendicular to the direction vector t(1,1)

$\Rightarrow (x,y-3).(1,1)=0 \Rightarrow x+y-3=0 \Rightarrow x+y=3$, hence

A. $(1+4)=5$ B. $1+(-4)=-3$ D. $1+2=3$

Answer: D

30. Let $\vec{a}=2\vec{i}+(x-1)\vec{j}+\vec{k}$ and $\vec{c}=\vec{i}-\vec{j}+y\vec{k}$ be vectors.

If $\vec{a}.\vec{c}=0$ and $\|\vec{a}\|=3$ then the possible value of y?

A. -4 B. -1 C. 3 D. 4

Solution:

$\vec{a}.\vec{c}=0 \Rightarrow (2i+(x-1)j+k)(i-j+yk)=0 \Rightarrow 2-1(x-1)+y=0 \Rightarrow 2-x+1+y=0 \Rightarrow y-x=-3$,

However

$\|a\|=3 \Rightarrow \|a\|=\sqrt{4+(x-1)^2+1}=3 \Rightarrow 4+(x-1)^2+1=9 \Rightarrow (x-1)^2=9-5=4 \Rightarrow x-1=\pm\sqrt{4}=\pm 2 \Rightarrow x=3$ or $x=-1$ thus from $y-x=-3$ let $x=3 \Rightarrow y-3=-3 \Rightarrow y=0$ and Let $x=-1$ then $y+1=-3 \Rightarrow y=-3-1=-4$

Answer: A

31. If $\vec{A}$ is perpendicular to $\vec{B}$, what is the cosine of the angle between $\vec{A}$ and $\vec{A}-\vec{B}$

A. $\frac{\|\vec{A}-\vec{B}\|}{\|\vec{A}\|}$ C. $\frac{\|\vec{A}-\vec{B}\|}{\|\vec{B}\|}$

B. $\frac{\|\vec{A}\|}{\|\vec{A}-\vec{B}\|}$ D. $\frac{\|\vec{B}\|}{\|\vec{A}-\vec{B}\|}$

Solution:

$\vec{A}.(\vec{A}-\vec{B})=|\vec{A}||\vec{A}-\vec{B}|\cos\theta \Rightarrow \vec{A}.\vec{A}-\vec{A}.\vec{B}=|\vec{A}||\vec{A}-\vec{B}|\cos\theta$

$$\Rightarrow \cos\theta=\frac{\vec{A}.\vec{A}}{|\vec{A}||\vec{A}-\vec{B}|}=\frac{|\vec{A}|^2}{|\vec{A}||\vec{A}-\vec{B}|}=\frac{|A|}{|\vec{A}-\vec{B}|}$$

Note, If $\vec{A}\perp\vec{B}$ then $\vec{A}.\vec{B}=0$ **Answer: B**

32. Which one of the following is necessarily true?

A. If $|\vec{A}|=|\vec{B}|$, then $\vec{A}=\vec{B}$

B. If $|\vec{KA}|=K|\vec{A}|$, for any real number K

C. If $\vec{A}$ is parallel to $\vec{B}$, then $\vec{A}.\vec{B}=0$

D. If $\vec{U}$ is a unit vector in the direction of $\vec{A}$, then $\vec{A}.\vec{U}=|\vec{A}|$

Solution:

A. False, example, let $\vec{A}=3i+4j$ and $\vec{B}=0i+5j$

B. False, example: $K=-3<0$

C. False, $\vec{A}.\vec{B}=\pm|A||B|$

D. True **Answer: D**

33. Suppose P(1,2,1) and Q(1,0,2) are points in space and $\vec{A}=\vec{PQ}$. If $\vec{B}$ is parallel to $\vec{PQ}$ and $\vec{A}.\vec{B}=-10$, then which

A. $\vec{A}$ and $\vec{B}$ has the same direction. C. $|\vec{B}|=\frac{1}{10}|\vec{A}|$

B. $|\vec{B}|=10|\vec{A}|$ D. $|\vec{B}|=2|\vec{A}|$

Solution:

Telegram: https://t.me/ExtremeSeries

$\vec{A} = \overrightarrow{PQ} = (1, 0, 2) - (1, 2, 1) = (0, -2, 1)$

$\vec{A} = \overrightarrow{PQ} \doteq \sqrt{0^2 + (-2)^2 + 1^2} = \sqrt{5}$

- If $\vec{B} \parallel \overrightarrow{PQ}$, then $\vec{B} = k\ \overrightarrow{PQ}$, $k \in \mathbb{R}$

  $\Rightarrow \vec{B} = k\ \vec{A} \Leftrightarrow |\vec{B}| = |k|\,|\vec{A}| = \sqrt{5}\,|k|$

- If $\vec{A}.\vec{B} = |\vec{A}|\ |\vec{B}| \cos\theta = -10$

  $\Leftrightarrow (\sqrt{5})(\sqrt{5})|k|\cos\theta = -10$

  $\Rightarrow 5|k|\cos\theta = -10 \Leftrightarrow |k| = 2$ and $\cos\theta = -1$

- Because $\vec{A}.\vec{B} = -10 < 0$ and $\vec{A} \parallel \vec{B}$, so that $\theta = \pi$

  $\Rightarrow \cos\theta = -1$, therefore $\vec{B} = 2\,|A|$. **Answer: D**

34. What is the image of the ellipse whose equation is $2(x+2)^2 + (y-1)^2 = 2$ under a translation that takes (2,1) to (4,0) followed by a rotation 90^0.

A. $x^2 + 2y^2 = 2$ C. $2(x-4)^2 + y^2 = 2$

B. $2x^2 + y^2 = 2$ D. $(x-4)^2 + 2y^2 = 2$

Solution:

$2(x+2)^2 + (y-1)^2 = 2 \Rightarrow (x+2)^2 + \frac{(y-1)^2}{2} = 1$ (y – ellipse)

- Translation of vector: $\vec{V} = (4-2, 0-1) = (2, -1)^2$
- Centre of ellipse (-2,1), (0,0), $\xrightarrow{\vec{V}=(2,-1)}$

The new ellipse becomes: $x^2 + \frac{y^2}{2} = 1 \Leftrightarrow 2x^2 + y^2 = 2$

- When the ellipse $2x^2 + y^2 = 1$ is rotated by $\theta = 90^0$

  $\Rightarrow$ y-ellipse $\xrightarrow{\theta = 90^0}$ x-ellipse

  $2x^2 + y^2 = 2 \xrightarrow{\theta = 90^0} x^2 + 2y^2 = 2$ **Answer: A**

35. If $f(x) = 2 - \frac{1}{2}\sin\left(\frac{\pi}{2} - x\right)$, then which one of the following is the amplitude and period of f, respectively?

A. $\frac{1}{2}$ and 4 B. $-\frac{1}{2}$ and 4 C. 2and Π D. $\frac{1}{2}$ and Π

Solution: The amplitude $\left|-\frac{1}{2}\right| = \frac{1}{2}$,

- Period is $\frac{2\pi}{K} = \frac{2\pi}{\frac{\pi}{2}} = 2\pi.\frac{2}{\pi} = 4$ **Answer: A**

36. **Exercise left for you (Tip).** Find the range, the maximum and minimum value.

37. Which one of the following is equal to $\sec\left(\frac{\pi}{2} - x\right)\sin^3 x + \cos 2x$?

A. 2cosx B. 2sinx C. $\cos^2 x$ D. $\sin^2 x$

Solution: Recall that $\sec\left(\frac{\pi}{2} - x\right) = \csc x = \frac{1}{\sin x}$

$\Rightarrow \frac{\sin^3 x}{\sin x} + \cos^2 x - \sin^2 x = \cos^2 x$

Answer: C

38. The solution set of $\cos^2 x + \frac{1}{2}\sin 2x = 1$ in the interval $[0, 2\pi]$

A. $\left\{0, \frac{\pi}{4}, \pi, \frac{5\pi}{4}\right\}$ C. $\{0, \pi, \}$

B. $\left\{0, \frac{\pi}{4}, \frac{3\pi}{4}, \pi\right\}$ D. $\left\{0, \frac{\pi}{4}, \pi\right\}$

Solution: We know that

$\sin 2x = \sin(x + x) = \sin x \cos x$ and $\cos^2 x = 1 - \sin^2 x$. Now

$\Leftrightarrow 1 - \sin^2 x + \frac{1}{2}(2)\sin x \cos x = 1 \Rightarrow \sin x \cos x - \sin^2 x = 0$

$\Rightarrow \sin x(\cos x - \sin x) = 0 \Leftrightarrow \sin x = 0$ or

$\cos x - \sin x = 0 \Rightarrow \sin x = 0 \Rightarrow x = 0, \pi$ or

$\cos x = \sin x \Rightarrow x = \frac{\pi}{4}, \frac{5\pi}{4}$

$s.s = \left\{0, \frac{\pi}{4}, \pi, \frac{5\pi}{4}\right\}$

Answer: A

Telegram: https://t.me/ExtremeSeries

39. What is $\cot(\arcsin x)$ if $0 < x < 1$

A. $\frac{x}{\sqrt{1-x^2}}$ C. $\sqrt{1-x^2}$

B. $\frac{\sqrt{1-x^2}}{x}$ D. $\frac{1}{\sqrt{1-x^2}}$

Solution: $\cot(\sin^{-1}x)$ let $\sin^{-1}x=\theta \Rightarrow \sin\theta = x$ we know that:

$$\cos^2\theta+\sin^2\theta=1 \Rightarrow \frac{\cos^2}{\sin^2\theta}+1=\frac{1}{\sin^2\theta}$$

$$\cot^2\theta+1=\frac{1}{x^2} \Rightarrow \cot^2\theta=\frac{1}{x^2}-1=\frac{1-x^2}{x^2}$$

$$\Rightarrow \cot\theta=\sqrt{\frac{1-x^2}{x^2}}=\frac{\sqrt{1-x^2}}{x}$$

Answer: B

40. If $\{a_n\}$ is a sequence such that $a_1=2$ and $a_n+1=a_n+4$ for all $n\geq 1$ then $\sum_{n=1}^{35}a_n$ is equal to:

A. 2460 B. 2458 C. 2450 D. 2442

Solution:

$\Rightarrow a_2=a_{1+1}=a_1+4=2+4=6 \Rightarrow a_3=a_{2+1}=a_2+4$

$=6+4=10 \Rightarrow d=10-6=4$

Hence, $\sum_{n=1}^{35}a_n=a_1+a_2+a_3+....+a_{35} \Rightarrow S_n=\frac{n}{2}[2A_1+(n-1)d]$

hence $s_{35}=\frac{35}{2}[2(2)+(35-1)4]=2450$

Answer: C

41. The population of certain country is currently 80 million with growth rate of 25 per year.

Given $(0.02)^9=5.12\times10^{-16}$, $(0.02)^{10}=1.024\times10^{-17}$

$(1.02)^9=1.19$ and $(1.02)^{10}=1.22$ which one of the following is the best approximation of the population after 10 year (in million)?

A. 81.3 million C. 95.2 million

B. 86.8 million D. 97.6 million

Solution:

$G_1=80$ million, rate $=2\%=0.02$, after, t-year, the proportion

$G_t=G_1(1+\text{rate})^t$

$\Rightarrow G_{10}=80(1+0.02)^{10}=80(1.02)^{10}$ million $=97.6$ million

Answer: D

42. What is the 50^{th} term of the sequence 3,10,17,24,31,...?

A. 310 B. 346 C. 510 D. 531

Solution: $A_n=A_1+(n-1)d \Rightarrow A_{50}=3+(50-1)7=346$

Answer: B

43. What is the sum of the series $\sum_{n=1}^{\infty}\frac{2^{2n+1}}{5^{n-1}}$

A. 40 B. 20 C. 10 D. 0

Solution:

$$\sum_{n=1}^{\infty}\frac{2^{2n}.2}{5^n.5^{-1}}=10\sum_{n=1}^{\infty}\left(\frac{4}{5}\right)^n, G_1=\frac{4}{5}, r=\frac{4}{5}<1$$

$$\Rightarrow 10\sum_{n=1}^{\infty}\left(\frac{4}{5}\right)=10\left(\frac{G_1}{1-r}\right)=10\left(\frac{\frac{4}{5}}{1-\frac{4}{5}}\right)=40$$

Answer: A

44. Which one of the following is equal to $\lim_{n\to\infty}\left(\frac{1-n-3n^2}{6n^2+1}\right)$?

A. $\frac{1}{6}$ B. $-\frac{1}{2}$ C. $-\frac{1}{6}$ D. $-\infty$

Solution: Dividing the numerator and denominator by n^2,

$$\Rightarrow \lim_{n\to\infty}\left(\frac{\frac{1}{n^2}-\frac{1}{n}-3}{6+\frac{1}{n^2}}\right)=\frac{0-0-3}{6+0}=-\frac{1}{2}$$

Answer: B

45. $\lim_{x\to1}\frac{\sqrt{x}-1}{x^2-1}$ is equal to

A. ∞ B. $\frac{1}{2}$ C. $\frac{1}{4}$ D. 0

Telegram: https://t.me/ExtremeSeries

Solution: $\lim_{x\to 1}\frac{\sqrt{x}-1}{(x-1)(x+1)}=\lim_{x\to 1}\frac{\sqrt{x}-1}{(x-1)(x+1)}\cdot\frac{\sqrt{x}+1}{\sqrt{x}+1}$

$=\lim_{x\to 1}\frac{x-1}{(x-1)(x+1)(\sqrt{x}+1)}=\frac{1}{(2)(2)}=\frac{1}{4}$

Answer: C

46. $\lim_{x\to 1}\frac{1-x}{1-\frac{1}{x^2}}=$_____

A. 1 B. 0 C. $-\frac{1}{2}$ D. Does not exist

Solution: $\lim_{x\to 1}\frac{(1-x)x^2}{x^2-1}=\lim_{x\to 1}\frac{(1-x)(x^2)}{(x-1)(x+1)}=-\frac{1}{2}$

Answer: C

47. $\lim_{x\to 1}\left(1+\frac{1}{x}\right)^{-\frac{x}{2}}$ is equal to

A. $\frac{1}{\sqrt{e}}$ B. $\sqrt{e}$ C. e^{-2} D. ∞

Solution: $\lim_{x\to 1}\left(\left(1+\frac{1}{x}\right)^{x}\right)^{-\frac{1}{2}}=e^{-\frac{1}{2}}=\frac{1}{\sqrt{e}}$

Answer: A

48. Let $f(x)=\begin{cases}3^x+k \text{ if } x\le 0\\ \frac{3\sin(2x)}{x} \text{ if } x>0\end{cases}$

if f is continuous at $x=0$, then what is the value of k?

A. 6 B. 5 C. 2 D. 0

Solution: Since it is continuous at $x=0$ implies $\lim_{x\to 0} f(x)$ exist

$\Rightarrow \lim_{x\to 0^+} f(x)=\lim_{x\to 0^-} f(x)$

$\Leftrightarrow \lim_{x\to 0^+}\frac{3\sin(2x)}{x}=\lim_{x\to 0^-}3^x+k\Rightarrow 3(2)=1+k\Rightarrow k=6-1=5$

Answer: B

49. If $f(x)=x^2+2\ln x$, then what is $\lim_{h\to 0}\frac{f(2+h)-f(2)}{h}$

A. 5 B. 4 C. 2 D. 0

Solution: Recall that $f'(a)=\lim_{h\to 0}\frac{f(a+h)-f(a)}{h}$

$\Rightarrow f'(x)=2x+\frac{2}{x}\Rightarrow f'(2)=4+\frac{2}{2}=5$

Answer: A

50. The equation of the line tangent to the graph of $f(x)=\frac{1}{x+1}+\cos x$ at (0, f(0))?

A. $x+y=1$ B. $x-y=-2$ C. $x+y=2$ D. $x+4y=2$

Solution: $f'(x)=-\frac{1}{(x+1)^2}-\sin x\Rightarrow f'(0)=-1-0=-1$

- Equation of tangent line is given by $y-f(0)=f'(0)(x-0)\Rightarrow y-2=-1(x)\Leftrightarrow y+x=2$

Answer: C

51. If $g(x)=xf(x)-\sqrt{f(x)}$ and $f(2)=f'(2)=4$ then which of the following is equal to, $g'(2)$

A. 11 B. 8 C. 2 D. 0

Solution:

$g(x)=xf(x)-(f(x))^{\frac{1}{2}}$ use the product and quotient rule.

$g'(x)=x'f(x)+xf'(x)-\frac{f'(x)}{2\sqrt{f(x)}}$

$\Rightarrow g'(2)=1.f(2)+2f'(2)-\frac{f'(2)}{2\sqrt{f(2)}}=4+2(4)-\frac{4}{4}=12-1=11$

Answer: A

52. Which one of the following is true about the derivative of $f(x)=x|x|$

A. f is not differentiable at $x=0$

B. $f'(x)=2|x|$, for every $x\in(-\infty,\infty)$

C. $f'(x)=2x$, for every $x\in(-\infty,\infty)$

D. $f'(x)=|x|+x$, for every $x\in(-\infty,\infty)$

Telegram: https://t.me/ExtremeSeries

Solution: Let $x \geq 0$, then, $|x| = x$. Then

$\Rightarrow f(x) = x|x| \Rightarrow f(x) = x^2 \Rightarrow f'(x) = 2x,\ x > 0,$

Let $x < 0$, then, $|x| = -x$.

Then $f(x) = x(-x) = -x^2 \Rightarrow f'(x) = -2x, x < 0,$

$$f'(x) = \begin{cases} 2x, x > 0 \\ -2x, x < o \end{cases} \Rightarrow f'(x) = 2|x| \text{ for all } x \in (-\infty, \infty)$$

Answer: B

53. If $f(x) = e^{2x} + x - 3\cos x$, then $f''(x)$?

A. $e^{2x} + 1 - 3\sin x$ C. $4e^{2x} - 3\cos x$

B. $e^{2x} + 1 + 3\sin x$ D. $4e^{2x} + 3\cos x$

Solution: $f'(x) = 2e^{2x} + 1 + 3\sin x \leftarrow (e^{2x})' = 2e^{2x}$

$f''(x) = 4e^{2x} + 3\cos x$

Answer: D

54. On which one of the following intervals does $f(x) = x^4 + 4x$ increasing on:

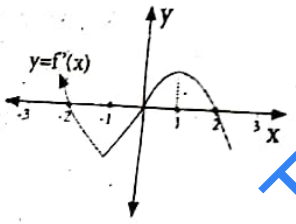
A. $(-\infty, -1]$ B. $(-\infty, 0]$ C. $[-1, \infty)$ D. $(-\infty, \infty)$

Solution: $f(x)$ is increasing on $[a,b]$ if $f'(x) \geq 0$ for all $x \in (a,b) \Rightarrow f'(x) = 4x^3 + 4 \geq 0 \Rightarrow 4(x^3 + 1) \geq 0$

For all $x \geq -1$, therefore f is increasing on $[-1, \infty)$.

Answer: C

55. Suppose f is differentiable on $(-\infty, \infty)$ and the graph of its derivative is as shown below



- Which one f the following is true about f

A. f is decreasing on $(-\infty, \infty] \cup [1, \infty)$

B. f has a local minimum at $x = -2$

C. f is concave down $[0, \infty)$

D. f is concave up on $(-1,1)$

Solution: $f'(x) \leq 0$ on $[-2,0] \cup [2, \infty) \Rightarrow f$ is decreasing

- $f'(x) \geq 0$. on $(-\infty, -2] \cup [0,2]$ so that f is increasing.
- f has local maximum at $x = -2$ because f is increasing on $x \leq -2$ then decreasing on $x \geq -2$.
- The graph of $y = f'(x)$ is increasing on $(-1,1)$. so that $f''(x) > 0$ on $(-1,1)$ thus f is concave up on $(-1,1)$.

Answer: D

56. A box seen below is to have a square base an open top and volume of 32 cubic units. If x is the length of each side of its base and y is its height, how many units should x and y be in order to make the box with the smallest amount of material.

A. $x = 4, y = 2$ C. $x = \sqrt{8}, y = 4$

B. $x = 2, y = 8$ D. $x = \sqrt{2}, y = 16$

Solution: Volume of box $= (B)(h) = x^2 y = 32 \Rightarrow y = \frac{32}{x^2}$.

surface area of box $A = 4xy + x^2$

$\Rightarrow A(x) = (4x)\left(\frac{32}{x^2}\right) + x^2 \Rightarrow A(x) = \frac{128}{x} + x^2$

$\Rightarrow A'(x) = -\frac{128}{x^2} + 2x$, set, $A'(x) = 0$

$\Rightarrow \frac{2x^3 - 128}{x^2} = 0 \Rightarrow 2x^3 - 128 = 0 \Rightarrow x^3 = 64 \Rightarrow x = 4$, then

$y = \frac{32}{16} = 2$

Answer: A

57. The volume V of a melting ice cube after t seconds is $V = 2000 - 40t + 0.2t^2 \left(\text{in cm}^3\right)$. How fast is the volume changing when t = 40 second?

A. $24 cm^3/\text{sec}$ B. $15 cm^3/\text{sec}$ C. $-15 cm^3/\text{sec}$ D. $-24 cm^3/\text{sec}$

Telegram: https://t.me/ExtremeSeries

Solution: $\frac{dV}{dt} = -40 + 0.4t$ when t = 40se

$\Rightarrow -40 + 0.4(40) = -40 + 16 = -24\, cm^3/sec$ **Answer: D**

58. At which value(s) of x does $f(x) = \frac{1}{4}x^4 - 2x^2$ have a local maximum.

A. $x = 4$ C. $x = -2$ and $x = 2$

B. $x = 0$ D. $x = 0$ and $x = 2$

Solution:

$f'(x) = x^3 - 4x \Rightarrow x(x^2 - 4) = 0 \Rightarrow x = 0, x = -2, x = 2$

- $f(0) = 0 - 0 = 0 \leftarrow$ Local maximum.
- $f(-2) = \frac{16}{4} - 8 = -4 \leftarrow$ Local minimum.
- $f(2) = -4$

Answer: B

59. What is an anti-derivative of $f(x) = \frac{2}{4x^2 + 4x + 1}$

A. $\frac{-1}{2x+1}$ C. $-\frac{2}{4x^2 + 4x + 1}$

B. $-\frac{2}{2x+1}$ D. $\ln(4x^2 + 4x + 1)$

Solution: Use $\int k(ax+b)^n dx = \frac{k}{a(n+1)}(ax+b)^{n+1} + C$

$\Rightarrow \int \frac{2}{4x^2 + 4x + 1} dx = \int 2(2x+1)^{-2} dx$

$= \frac{2}{2(-2+1)}(2x+1)^{-2+1} + C = -\frac{1}{2x+1}$ **Answer: A**

60. $\int (1+x)3^x dx$ is equal to

A. $(1+x)3^x - 3^x + C$

B. $(1+x)3^x + (\log 3^e)3^x + C$

C. $(1+x)3^x \log 3^e - (\log 3^e)^2 3^x + C$

D. $(1+x)3^x \log 3^e - 3x(\log e^3) + C$

Solution: $\int (1+x)3^x dx$, using table method

Derivative		Anti-derivative
1+x	+	3^x
1	-	$3^x \log 3^e$
0		$3^x (\log 3^e)^2$

$\Rightarrow \int (1+x)3^x dx = (1+x)3^x \log 3^e - 3^x (\log 3^e)^2 + C$ **Answer: C**

61. Which one of the following is equal to $\int \frac{\ln(xe^x)}{x} dx$

A. $\ln|x| + \frac{1}{2}e^x + C$ C. $\ln|x| + e^{2x} + C$

B. $\frac{1}{2}(\ln x)^2 + x + C$ D. $-\frac{1}{x^2} + (\ln x)^2 + x + C$

Solution: Recall that $\ln(xe^x) = \ln x + \ln e^x = \ln x + x$

$\Rightarrow \int \frac{\ln(xe^x)}{x} dx = \int \left(\frac{\ln x}{x} + \frac{x}{x}\right) dx = \int \left(\frac{\ln x}{x} + 1\right) dx$

$= \frac{(\ln x)^2}{2} + x + C$ **Answer: B**

62. $\int \frac{1}{x^2 + x} dx$ is equal to _____

A. $\ln|x^2 + x| + C$ C. $\ln|x| - \ln|x+1| + C$

B. $2\ln|x+1| + \ln|x| + C$ D. $\ln|x| + \ln|x+1| + C$

Solution: Using partial decomposition:

$\frac{1}{x^2 + x} = \frac{1}{x(x+1)} = \frac{A}{x} + \frac{B}{x+1} \Leftrightarrow A(x+1) + Bx = 1$

- Let $x = 0 \Rightarrow A(0+1) + 0 = 1 \Rightarrow A = 1$
- Let $x = -1 \Rightarrow A(-1+1) + B(-1) = 1 \Rightarrow B = -1$

$\Rightarrow \int \frac{1}{x^2 + x} dx = \int \left(\frac{1}{x} - \frac{1}{x+1}\right) dx = \ln|x| - \ln|x+1| + C$

Answer: C

Telegram: https://t.me/ExtremeSeries

63. The value of $\int_0^{\pi/2} 2x\cos x\,dx$ is equal to

A. $\Pi-2$ B. $\frac{\Pi}{2}+1$ C. $\Pi+2$ D. $\frac{\Pi}{2}-1$

Solution: Use integration by part $\int udv = uv - \int vdu$

Let $u = 2x$, $du = 2dx$, $dv = \cos x\,dx$, $v = \sin x$

$$\Rightarrow \int_0^{2\Pi} 2x\cos x dx = 2x\sin x - \int 2\sin x dx = 2x\sin x + 2\cos x$$

$$\int_0^{\pi/2} = (\Pi\sin\frac{\Pi}{2} + 0) - (0 + 2\cos 0) = \Pi - 2$$ **Answer:** A

64. What is the area of the region between the graphs of $y = x^2$ and $y = -x+2$ for $0 \le x \le 2$

A. 3 B. 2 C. $\frac{3}{2}$ D. $\frac{2}{3}$

Solution: Area, $A = \int_0^2 (-x+2)dx = -\frac{x^2}{2} + 2x\Big|_0^2 = -2+4 = 2$

Answer: B

65. Which one of the following points is closer to the sphere $x^2 + y^2 + Z^2 - 2x + 6Z + 9 = 0$

A. (1,0,0) B. (0,0,0) C. (0,-1,0) D. (0,0,-1)

Solution

$x^2 - 2x + y^2 + Z^2 + 6Z = -9 \Rightarrow x^2 - 2x + 1 + y^2 + Z^2 + 6Z + 9$

$= -9 + 9 + 1 \Rightarrow (x-1)^2 + (y-0)^2 + (Z+3)^2 = 1$,

distance between point and center (1,0,-3)

A. Distance between (1,0,0) and (1,0,-3) is $d = \sqrt{0+0+9} = \sqrt{9}$

B. Distance between (0,0,0) and (1,0,-3) is $d = \sqrt{1+0+9} = \sqrt{10}$

C. Distance between (0,0,-1) and (1,0,-3) is $d = \sqrt{1+0+4} = \sqrt{5}$, It is closest.

Answer: D

1. Which of the following expression is a polynomial expression?

A. $x^2 - 3x + \sin x$ C. $\frac{2+\pi}{1+\pi^2}$

B. $\frac{4x^3 + 12x^2 - x}{\pi x^2}$ D. $2 - 3x^{2/3} + 7x^{5/2} + 3x^{-1}$

Hint:

$\frac{2+\pi}{1+\pi^2}$ is constant – polynomial function. **Answer: C**

2. If $f(x) = \sqrt[3]{1+e^{-x}}$, then $f^{-1}(x)$ equals _____

A. $\ln\left(\frac{1}{x^3-1}\right)$ B. $\frac{1}{\ln(x^3-1)}$ C. $\ln(1-x^3)$ D. $(1+e^{-x})^3$

Hint: Let $y = f(x)$, then $y = \sqrt[3]{1+e^{-x}}$

To find the inverse interchange $y \to x$ and $x \to y$

$\Rightarrow x = \sqrt[3]{1+e^{-y}} \Leftrightarrow x^3 = 1 + e^{-y}$

$\Rightarrow e^{-y} = x^3 - 1 \Leftrightarrow -y = \log_e^{(x^3-1)} = \ln(x^3-1)$

$\Rightarrow y = -\ln(x^3-1) = \ln(x^3-1)^{-1} = \ln\left(\frac{1}{x^3-1}\right)$

$\therefore f^{-1}(x) = \ln\left(\frac{1}{x^3-1}\right)$

Answer: B

3. If $f(x) = \frac{x+1}{x-1}$ and $f(a) = 5$, then $f(2a)$ is equal to:

A. 2 B. 4 C. 6 D. 8

Hint: Here at:

$f(x) = \frac{x+1}{x-1}$ then $f(a) = \frac{a+1}{a-1} = 5$

$\Rightarrow a + 1 = 5a - 5 \Leftrightarrow -4a = -6;\ a = \frac{6}{4} = \frac{3}{2}$

$f(2a) = f\left(2\left(\frac{3}{2}\right)\right) = f(3)$

$\Rightarrow f(3) = \frac{3+1}{3-1} = \frac{4}{2} = 2$

Answer: A

Telegram: https://t.me/ExtremeSeries

4. Let $f(x)=\dfrac{x^2-1}{x-x^2}$ which of the following is true about $f(x)$.

A. $x=0$ and $x=1$ are its vertical asymptotes.
B. $y=1$ is its horizontal asymptote.
C. $y=x-1$ is its oblique asymptote.
D. It is almost the same as the horizontal line $y=-1$ as $x\to\pm\infty$.

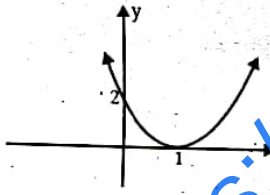
Hint:

$$y=\frac{x^2-1}{x-x^2}=\frac{(x-1)(x+1)}{x(1-x)}=-\left(\frac{x+1}{x}\right)=-1-\frac{1}{x}$$

i) $x=0$ is the vertical asymptote.
ii) $y=-1$ is the horizontal asymptote.
iii) as $x\to\pm\infty, y\to-1$

Answer: D

5. Which of the following equations below is represented by the following parabola?



A. $y=x^2+2$
B. $y=(2x-1)^2$
C. $y=2(x-1)^2$
D. $y=(2x+1)^2$

Hint:

$y=a(x-b)^2+c$

- The vertex of the parabola is (1, 0).
- The y – intercept is (0, 2).
- The value of c is 0.

$\therefore y=2(x-1)^2+0$

Answer: C

6. If the equation $(x-2)^2-(y-2)^2=1$ represents a hyperbola which one of the following represents equation of an asymptote to the hyperbola?

A. $y=4-x$ B. $x+y=1$ C. $x=2-y$ D. $x+2y=3$

Hint:

$$(x-2)^2-(y-2)^2=1\Leftrightarrow\frac{(x-2)^2}{1}-\frac{(y-2)^2}{1}=1$$

Asymptote: $y-2=\pm\dfrac{1}{1}(x-2)$

$\Leftrightarrow y-2=x-2$ and $y-2=-x+2$

$y-x=0$ and $y+x=4\Rightarrow y=4-x$

Answer: A

7. What is the distance from the origin to the line that passes through (1, 0) and (0, 1)?

A. $\frac{1}{2}\sqrt{2}$ B. 1 C. $\frac{1}{2}$ D. $\sqrt{2}$

Hint:

The equation of the line through (1, 0) and (0, 1) is $y-1=\left(\dfrac{0-1}{1-0}\right)(x-0)\Leftrightarrow x+y+1=0.$

The distance from a point (x_o, y_o) to the line $Ax + By + C = 0$ is obtained by the formula:

$$d=\left|\frac{Ax_o+By_o+C}{\sqrt{A^2+B^2}}\right|=\left|\frac{1(0)+1(0)+1}{\sqrt{1^2+1^2}}\right|=\frac{1}{\sqrt{2}}=\frac{\sqrt{2}}{2}$$

Answer: A

8. If two lines $y = x$ and $y = x - 4$ are tangent to the circle at (2, 2) and (4, 0), respectively, then what is the equation of the circle?

A. $(x-2)^2+y^2=4$ C. $(x-3)^2+(y-1)^2=2$
B. $(x-4)^2+(y-2)^2=4$ D. $(x-1)^2+(y+1)^2=10$

Hint: Let (h, k) be center of circle.

☞ The distance between the center (h, k) and point on the circle (2, 2) and (4, 0) would be:
$(h-2)^2+(k-2)^2=(h-4)^2+(k-0)^2$-
$\Rightarrow h^2-4h+4+k^2+4=h^2-8h+16+k^2$ ← canceling
$\Leftrightarrow 4h-4k=8\Rightarrow h-k=2\ \ldots(*)$

☞ The slope of the line passing through (h, k) and (2, 2) is perpendicular to the line $y = x$. i.e., $\dfrac{k-2}{h-2}=-1$

$\Rightarrow k-2=-h+2\Leftrightarrow -h+2\Leftrightarrow k+h=4\ \ldots(**)$

Solving (*) and (**) will be:

$$\begin{cases}h-k=2\\h+k=4\end{cases}\Rightarrow 2h=6, h=3, k=1$$

Radius: $r^2=(h-2)^2+(k-2)^2=(3-2)^2+(1-2)^2=2$

$\therefore$The equation of circle $(x-3)^2+(y-1)^2=2$

9. What is the equation of the directix for the parabola whose equation is $y^2 + 8x + 6y + 25 = 0$?

A. $y=3$ B. $x=2$ C. $x=0$ D. $x=4$

Telegram: https://t.me/ExtremeSeries

Hint:
$y^2 + 6y + 8x + 25 = 0 \Leftrightarrow y^2 + 6y + 9 = -8x - 25 + 9$
$(y+3)^2 = -8(x+2)$← opens to the left.
∴Directx: $x + 2 = 2 \Rightarrow x = 0$.

10. A semi elliptical arc over a tunnel for a road through a mountain has a major axis of length 80 meters and a height of 30 meters at the center. What is the equation of the semi – elliptical arc over the tunnel, if the center is considered as the origin?

A. $\frac{x^2}{6400} + \frac{y^2}{900} = 1$ C. $\frac{x^2}{1600} + \frac{y^2}{900} = 1$

B. $\frac{x^2}{900} + \frac{y^2}{6400} = 1$ D. $\frac{x^2}{8100} + \frac{y^2}{6400} = 1$

Hint: Let $2a$ = length of major axis = 80 m
$2b$ = length of semi minor axis = 60 m
$\Rightarrow 2a = 80\ m \Rightarrow a = 40$m and $b = 30$m

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow \frac{x^2}{1600} + \frac{y^2}{900} = 1$ **Answer: C**

11. Let the equation $x^2 + 2x + y^2 = 8$ represents a circle. Then which one of the following lines cut the circle at exactly two points?

A. $4x + 3y + 19 = 0$ C. $2y = 5x + 43$
B. $3x + 4y + 14 = 0$ D. $2x = y - 50$

Hint: $x^2 + 2x + y^2 = 8 \Rightarrow x^2 + 2x + 1 + y^2 = 8 + 1$
$\Rightarrow (x+1)^2 + (y+0)^2 = 9$, center = $(-1, 0)$, $r = 3$
The distance between the center (-1, 0) and a given line is:

A. $d = \frac{|Ax_o + By_o + C|}{\sqrt{A^2 + B^2}} = \frac{|4(-1) + 3(0) + 19|}{\sqrt{4^2 + 3^2}} = \frac{15}{5} = 3$

Since: $d = r = 3$, therefore the circle and the line $4x + 3y + 19 = 0$ intersect exactly at one point.

B. $d = \frac{|3(-1) + 4(0) + 14|}{\sqrt{3^2 + 4^2}} = \frac{14}{5} < r$

Therefore the line $3x + 4y + 14 = 0$ and the circle intersect exactly at two point.

C. $d = \frac{|5(-1) + (-2)(0) + 43|}{\sqrt{5^2 + (-2)^2}} = \frac{38}{\sqrt{29}} > r$

Therefore the line do not intersect the circle.

D. $d = \frac{|2(-1) - 1(0) + 50|}{\sqrt{(2)^2 + (-1)^2}} = \frac{48}{\sqrt{5}} > r$ →do not intersect

Answer: B

12. If $(P \vee q) \Rightarrow (\neg r \wedge r)$ is true, then which one of the following is necessarily true?

A. $(P \vee r) \Rightarrow q$ B. $\neg q \wedge r$ C. $\neg P \Rightarrow r$ D. $\neg P \vee r$

Hint: $\neg r \wedge r$ is always false. Thus $P \vee q \Rightarrow (\neg r \wedge q)$ is true.
$P \vee q \equiv F$, therefore, $P \equiv F$ and $q \equiv F$.
A. $(F \vee F) \Rightarrow F \equiv T$ **Answer: A**

13. Which one of the following is a valid logical argument?

A. $P \Rightarrow q,\ q \vdash P$ C. $\neg P \wedge q,\ q \Rightarrow r \vdash r$
B. $P \Leftrightarrow q,\ P \Rightarrow q \vdash q$ D. $\neg P,\ P \vee q,\ r \Rightarrow q \vdash r$

Hint:
1) $\neg P \wedge q,\ q \equiv T$, ⟵ premise
$\neg P \equiv T$ and $q \equiv T$
2) $q \Rightarrow r \equiv T$ ⟵ premise
$T \Rightarrow r \equiv T$, so that $r \equiv T$
$\neg P \wedge q,\ q \Rightarrow r \vdash r$ is valid argument. **Answer: C**

14. Which one of the following is equivalent to $\neg[(\forall x)(P(x) \Rightarrow Q(x))]$?

A. $(\forall x)(\neg P(x) \Rightarrow \neg Q(x))$ C. $(\exists x)(\neg P(x) \wedge Q(x))$
B. $(\exists x)(\neg P(x) \Rightarrow \neg Q(x))$ D. $(\exists x)((P(x) \wedge \neg Q(x))$

Hint: Recall that: $P \Rightarrow Q \equiv \neg P \vee Q$
$\neg(P \Rightarrow Q) \equiv (\neg P \vee Q) \equiv P \wedge \neg Q$
$\neg[(\forall x)(P(x) \Rightarrow Q(x))] \equiv (\exists x)[P(x) \wedge \neg Q(x)]$ **Answer: D**

15. If the list of a measurement is 10, α, 5, α, 5, 10, 20, 15, 20, 5 with mean $\bar{x}$, then what is the value of α in terms of $\bar{x}$?

A. $10\bar{x} - 90$ B. $9\bar{x} - 90$ C. $5\bar{x} - 90$ D. $5\bar{x} - 45$

Hint:
$\bar{x} = \frac{\alpha + \alpha + 5 + 5 + 5 + 10 + 10 + 15 + 20 + 20}{10} = \frac{2\alpha + 90}{10}$.
$10\bar{x} = 2\alpha + 90 \Rightarrow \alpha = 5\bar{x} - 45$ **Answer: D**

16. The following is the frequency distribution of a grouped data.

Class interval	3 – 7	8 – 12	13 – 17	18 – 22
Frequency (f)	2	2	10	6

Telegram: https://t.me/ExtremeSeries

What is the mean and standard deviation of the distribution respectively?

A. $15, 2\sqrt{5}$ B. $15, 2\sqrt{7.5}$ C. $12.5, 5\sqrt{2}$ D. $12.5, \sqrt{15}$

Hint:

Class	f	Class mark (x_C) (Midpoint)	Fx_C	$(x_C - \bar{x})^2$	$f(x_C - \bar{x})^2$
3 – 7	2	5	10	$(-10)^2 = 100$	200
8 – 12	2	10	20	$(-5)^2 = 25$	50
13 – 17	10	15	150	$0^2 = 0$	0
18 – 22	6	20	120	$5^2 = 25$	150

- Mean $(\bar{x}) = \dfrac{\sum fx_C}{\sum f} = \dfrac{300}{20} = 15$
- Variance $\dfrac{\sum f(x_C - \bar{x})^2}{\sum f} = \dfrac{200 + 50 + 0 + 150}{20} = 20$
- Standard deviation = $\sqrt{\text{variance}} = \sqrt{20} = 2\sqrt{5}$.

Answer: A

17. If Q_1, D_i and P_i are respectively the i^{th} quartile, decile and percentile of data arranged in an increasing order, then which one of the following is necessarily true?

A. $Q_2 = \dfrac{Q_1 + Q_3}{2}$ B C. $P_{25} > Q_1$

B. $D_3 > P_{25}$ D. Q_2= mean of the data

Explanation:

Bear in mind that:

The set of a given data:

i) Divided into two equal part is **Median.**
ii) Divided into four equal part is **Quartile.**
iii) Divided into ten equal part is **Decile.**
iv) Divided into hundred equal part is **percentile.**

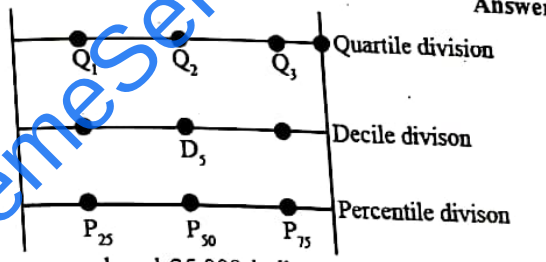
Recall that:

i) $Q_2 = D_5 = P_{50}$ = Median of the data
ii) $Q_1 = P_{25}$
iii) $Q_3 = P_{75}$

- $D_3 = \dfrac{3}{10}(100) < 30\%$ of the data from the lower.

- $P_{25} = \dfrac{25}{100}(100\%) < 25\%$ of the data from the lower.

Answer: B



18. A company produced 25,000 bulbs and randomly started 2% of the product. Among the tested bulbs, if 40 have defect to type D_1, 60 have defect of type D_2 and 25 have both types of defects. What is the probability that a bulb produced by the company has none of the defects?

A. 0.95 B. 0.80 C. 0.85 D. 0.20

Solution:

2% of 25,000 is 500 P (non – defects) = 1 – P (all defect)

$= 1 - (P(D_1) + P(D_2) - P(D_1 n D_2))$

$= 1 - \left[\dfrac{40}{500} + \dfrac{60}{500} - \dfrac{25}{500}\right]$

$= 1 - 0.15 = 0.85$

Answer: C

19. If S is a set with 10 elements and ACS, what is the probability that A has 3 or more elements?

A. $\dfrac{7}{10}$ B. $\dfrac{8}{11}$ C. $\dfrac{121}{128}$ D. $\dfrac{7}{128}$

Solution:

P (3 or more elements) = 1 – P(1 – element) – P (2 – element)

$= 1 - \dfrac{1}{10} - \dfrac{2}{10} = 1 - \dfrac{3}{10} = \dfrac{7}{10}$

Answer: A

Telegram: https://t.me/ExtremeSeries

20. If $A=\begin{pmatrix}2&0&-1\\1&2&0\\0&0&-1\end{pmatrix}$ and $(2A+B)^T=A^TA$ then which one fo the following is equal to B.

A. $\begin{pmatrix}1&0&-2\\2&0&0\\0&0&4\end{pmatrix}$ B. $\begin{pmatrix}0&0&0\\0&0&0\\0&0&0\end{pmatrix}$ C. $\begin{pmatrix}8&0&-4\\4&8&0\\0&0&-4\end{pmatrix}$ D. $\begin{pmatrix}1&2&0\\0&0&0\\-2&0&4\end{pmatrix}$

Solution:

Recall that:

i) $(A-B)^t=A^t+B^t$

ii) $(AB)^t=B^tA^t$

$(2A+B)^T=A^TA \Leftrightarrow 2A^T+B^T=A^TA$

To find B make transpose both side

$(2A^T+B^T)^T=(A^TA)^T \Leftrightarrow 2A+B=A^TA$

$$2\begin{pmatrix}2&0&-1\\1&2&0\\0&0&-1\end{pmatrix}=(A^TA)^T\Leftrightarrow 2A+B=A^TA$$

$$2\begin{pmatrix}2&0&-1\\2&4&0\\-2&4&0\end{pmatrix}+B=\begin{pmatrix}2&1&0\\0&2&0\\-1&0&1\end{pmatrix}\begin{pmatrix}2&0&-1\\1&2&0\\0&0&-1\end{pmatrix}=\begin{pmatrix}5&2&-2\\2&4&0\\-2&0&2\end{pmatrix}$$

$$B=\begin{pmatrix}5&2&-2\\2&4&0\\-2&0&2\end{pmatrix}-\begin{pmatrix}4&0&-2\\2&4&0\\0&0&-2\end{pmatrix}=\begin{pmatrix}1&2&0\\0&0&0\\-2&0&4\end{pmatrix}$$

Answer: D

21. If $M=\begin{pmatrix}0&1&2\\3&-1&0\\5&2&4\end{pmatrix}$ and $A^TM=2I$, where A is 3×3 matrix and I is the identity matrix of order 3, then what is det (A) ?

A. 0.2 B. $\frac{4}{17}$ C. 0.8 D. $\frac{1}{17}$

Solution: **Recall that:** $|A^T|=|A|$

$A^TM=2I \Rightarrow |A^TM|=|2I|$

$\Leftrightarrow |A^T||M|=2^3|I|=8(1)$. Also $|M|=0\begin{vmatrix}-1&0\\2&4\end{vmatrix}-1\begin{vmatrix}3&0\\5&4\end{vmatrix}+2\begin{vmatrix}3&-1\\5&2\end{vmatrix}$

$=0-12+2(6+5)=10$

$\Leftrightarrow |A|(10)=8$, therefore, $|A|=\frac{8}{10}=0.8$

Answer: C

22. What should be the values of k so that the system of equation $\begin{cases}x-y+z=1\\-x+5y-4z=1\\2x+2y-z=k\end{cases}$ has a solution?

A. 0 B. 1 C. -4 D. 4

Solution:

The augmented matrix for system is $\left[\begin{array}{ccc|c}1&-1&1&1\\-1&5&-4&1\\2&2&-1&k\end{array}\right]$ reduce this matrix to echelon form. Add the first row to the second row an -2 times the first row to the third row. That is $R_2\to R_1+R_2$ and $R_3\to 2R_1+R_3$.

$$\left[\begin{array}{ccc|c}1&-1&1&1\\0&4&-3&2\\0&4&-3&-2+k\end{array}\right]\ R_3\to -R_2+R_3\ \left[\begin{array}{ccc|c}1&-1&1&1\\0&4&-3&2\\0&0&0&-4+k\end{array}\right]$$

Therefore, $-4+k=0\Rightarrow k=4$

Answer: D

23. Suppose $A_x=b$, where A is 3×3 matrix, $b=(b_1,b_2,b_3)^T$ and $x=(x,y,z)^T$. Which one of the following is necessarily true about this system of linear equation?

A. The system has a solution only when det (A) $\neq 0$.

B. The cramer's ruel is suitable to solve the system if two rows of A are identical.

C. If det (A) $\neq 0$ and the second column of A is a multiple of b then $x=0$.

D. If $b=0$, then $x=(0,0,T)^T$ is the only solution of the system.

Telegram: https://t.me/ExtremeSeries

Solution:

Let $\begin{bmatrix} a_1 & a_{12} & a_{12} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_2 \end{bmatrix}$

A. False, because if $D + 0, D_x = D_z = 0$ has infinite solution.

B. False

C. True, $D \neq 0$, and $x = \frac{D_x}{D} = 0$ because, first and second column of A are proportional, so that $D_x = 0$.

Answer: D

24. If $z = x + yi$ is a complex number, then $|z|^2 + \frac{1}{2}(z - \bar{z})^2 = 1$ is equivalent to which one of the following equations?

A. $x^2 - y^2 = 1$ C. $x^2 - y^2 = 2$

B. $x^2 - 3iy^2 = 1$ D. $2x^2 - y^3 = 2$

Solution: $|z| = \sqrt{x^2 + y^2}$

$|z|^2 + \frac{1}{2}(z - \bar{z}) = 1$ implies that

$\left(\sqrt{x^2 + y^2}\right) + \frac{1}{2}(x + yi - (x - yi))^2 = 1$

$\Rightarrow x^2 + y^2 + \frac{1}{2}(2yi)^2 = 1$

$\Rightarrow x^2 - y^2 = 1$

Explanation:

$(2yi)^2 = -4y^2$

Answer: A

25. If $W = \frac{16i}{1+i} + (1 - 3i)^2$ and $Z = |W| + \bar{W}$ the simplest form of Z is _____.

A. $\sqrt{2} + 2i$ B. $2 + 2i$ C. $4 - 2i$ D. $2 - 2i$

Solution:

W is equivalent to

$W = \frac{16i(1-i)}{(1+i)(1-i)} + 1 - 6i - 9 = \frac{16i + 16}{1+1} - 8 - 6i$

$\Rightarrow W = 8i + 8 - 8 - 6i = 2i, z = |W| + \bar{W} = 2 - 2i$ **Answer: D**

26. What is the equation of a line that pass through the point $(-1, 2)$ and parallel to the vector $v\ (1, -1)$?

A. $2x - y = 1$ C. $x - 2y = 3$

B. $x + y - 1 = 0$ D. $y - 2x + 1 = 0$

Solution:

$(x, y) = (x_o, y_o) + t(a, b)$ ← vector equation line

$\Rightarrow (x, y) = (-1, 2) + t(1, -1)$ ← vector equation

$\Rightarrow x = -1 + t$ and $y = 2 - t$

Solving for t, $t = x + 1$ and $t = 2 - y$

$\Rightarrow x + 1 = 2 - y \Leftrightarrow x + y - 1 = 0$

Answer: B

27. Suppose $A = 3i - 4j$ an dB is a vector in the xy – plane such that the angle between A and B is $\pi/3$. If $\vec{u}$ is the unit vector in the direction of B, then $\vec{A}.(\vec{A} - 2u)$ is equal to:

A. 20 B. 5 C. 15 D. 30

Solution: Here recall that: $|u| = 1$.

$\vec{A}.(\vec{A} - 2u) = \vec{A}.\vec{A} - 2\vec{A}\vec{u} = |A|^2 - 2|A||u|\cos\left(\frac{\pi}{3}\right)$

$\Rightarrow \left(\sqrt{3^2 + 4^2}\right)^2 - 2\sqrt{3^2 + 4^2}(1)\left(\frac{1}{2}\right) = 25 - (5) = 20$

Answer: A

28. If ℓ is the line passing through (0, 2) and parallel to $v = i + 3j$, which one of the following is true about (ℓ) and the circle $(x - 2)^2 + (y - 1)^2 = 5$?

A. ℓ is tangent of the circle at (0, 2).

B. ℓ is tangent to the circle to the at some point P, where $P \neq (0, 2)$.

C. ℓ intersects the circle at two distinct points.

D. The distance between l and the center of the circle is greater than $\sqrt{5}$.

Solution:

$(x, y) = (a, b) + \overrightarrow{tv}$

$\Rightarrow (x, y) = (0, 2) + t(1, 3)$

$\Leftrightarrow x = 0 + 1$ and $y = 2 + 3t$, solving for t

Telegram: https://t.me/ExtremeSeries

$y = 2 + 3x \Leftrightarrow 3x - y + 2 = 0$

center of circle = (2, 1) and radius = $\sqrt{5}$.

The perpendicular distance from the center (2, 1) to the line (ℓ):

$3x - y + 2 = 0,\ d = \dfrac{|Ax + By + C|}{\sqrt{A^2 + B^2}} = \dfrac{|3(2) - 1(1) + 2|}{\sqrt{3 + (1)}} = \dfrac{7}{\sqrt{10}}$

We can see that: $d < r \Rightarrow \dfrac{7}{\sqrt{10}} < \sqrt{5}$

We conclude that ℓ intersect the circle at two point.

Answer: C

29. If A = (−2, 3), B = (3, 1) and C s any other point on the plane, then which one of the following is coordinate form of $\overrightarrow{AC} - \overrightarrow{BC}$.

A. (-5, 2) B. (5, -2) C. (1, 4) D. (-1, -4)

Solution: Recall that BC = –CB

$\overrightarrow{AC} - \overrightarrow{BC} = \overrightarrow{AC} + \overrightarrow{CB} = \overrightarrow{AB}$

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = (3, 1) - (-2, 3) = (5, -2)$

Answer: B

30. Suppose $A = 2i - j + 2k$ and B is a vector in space such that $|\vec{B}| = \vec{A}.\vec{B}$. If $\vec{u}$ is the unit vector in the direction of $\vec{B}$, then $|\vec{A} + \vec{u}|^2$ is equal to:

A. 16 B. 12 C. 10 D. 14

Solution: We have, $|\vec{A}| = \sqrt{4 + 1 + 4} = 3$

$|\vec{B}| = |\vec{A}.\vec{B}| = |\vec{A}||\vec{B}| \cos\theta \Rightarrow \cos\theta = \dfrac{|B|}{|\vec{A}||\vec{B}|} = \dfrac{1}{|\vec{A}|}$

$|\vec{A} + \vec{u}| = \left(\sqrt{|A| + 2|\vec{A}||\vec{u}|\cos|\vec{u}|}\right)$

$= (3)^2 + 2(3)(1)\left(\dfrac{1}{3}\right) + 1^2 = 12$

Explanation:

$\cos\theta = \dfrac{1}{|A|} = \dfrac{1}{3}$

31. If a translation T takes the circle $x^2 + y^2 - 2x + 6y + 3 = 0$ into the circle whose equation is $(x + 2)^2 + (y - 4)^2 = 7$, then what is the image of the origin under T?

A. (−3, 7) B. (1, 2) C. 1, -3 D. (-2, 4)

Solution: Applying completing the square

$x^2 - 2x + 1 + y^2 + 6y + 9 = -3 + 1 + 9$

$\Leftrightarrow (x - 1)^2 + (y + 3)^2 = 7$

Translating the center of circle: (1, -3)→(-2, 4),

Translating vector: v = (-2 − 1, 4 − (-3)) = (-3, 7)

$(0, 0) \xrightarrow{(-3,7)} (-3, +0, 7 + 0) = (-3, 7)$ **Answer: A**

32. What is the image of the line given by (x, y) = (1, 0) + t (3, 6), under the translation that takes (1, 0) to (0, 1) followed by the reflection about the line $y = 2x$?

A. $y = 2x + 3$ B. $y = 2x - 3$ C. $y = 2x + 6$ D. $y = 2x - 5$

Solution: $(x, y) = (1, 0) + t(3, 6)$

$\Rightarrow x = -1 + 3t$ and $y = 0 + 6i$, solving for t

$\dfrac{x+1}{3} = \dfrac{y-0}{6} \Leftrightarrow y = 2x + 2$

$(1, 0) \rightarrow (0, 1)$, translating vector v = (-1, 1)

Taking any point on line $y = 2x + 2$, let (1, 4)

$(1, 4) \xrightarrow{-1,1} (0, 5)$

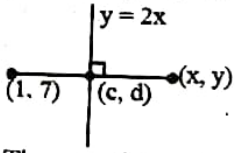
Translation is parallel movement so that the image of $y = 2x + 2$ is $y = 2x + b$.

To find b, put (0, 5) on $= 2x + b$

$\Rightarrow 5 = 2(0) + b \Leftrightarrow b = 5$

$\therefore y = 2x + 5$

To find the image of $y = 2x + 5$ under reflecting line $y = 2x$ is let $y = 2x + b$. Let take (1, 7) on $y = 2x + 5$.



The equation of line through point (1, 7) and (x, y) is

$\dfrac{y-7}{x-1} = -\dfrac{1}{2} \Leftrightarrow y = -\dfrac{1}{2}x + \dfrac{15}{2}$

To find (c, d): $\begin{cases} y = 2x \\ y = -\dfrac{1}{2}x + \dfrac{15}{2} \end{cases} = \begin{cases} d = 2C \\ d = -\dfrac{1}{2}C + \dfrac{15}{2} \end{cases}$

$2C = -\dfrac{1}{2}C + \dfrac{15}{2} \Leftrightarrow 2C + \dfrac{1}{2}C = \dfrac{15}{2}$

C = 3 and d = 6

(C, d) = (3, 6) are midpoint between (1, 7) and (x, 7) and (x, y).

Telegram: https://t.me/Extreme Series

$\frac{x+1}{2}=3$ and $\frac{y+7}{2}=6$

$\Rightarrow x=5$ and $y=5$

To find b, in $y=2x+b$ is

$\Rightarrow 5=2(5)+b \Rightarrow b=-5$

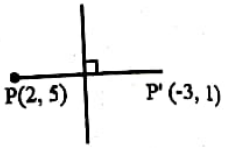
$\therefore$The image: $y=2x-5$

Answer; D

33. If a point (2, 5) is reflected under a line to the point (-3, 1) what si the line of reflection?

A. $2x+3y=7$ C. $8y+10x=19$

B. $x+3y=7$ D. $2x+3y+5=0$



Solution: Let P = (2, 5) and P' = (-3, 1)

- Slope of $PP'=\frac{1-5}{-3-2}=\frac{-4}{-5}=\frac{4}{5}$
- Slope of lien perpendicular to PP' is $\frac{5}{4}$
- Midpoint of PP' is $\left(\frac{-3+2}{2},\frac{1+5}{2}\right)=\left(-\frac{1}{2},3\right)$
- Equation of line passing through is $\frac{y-3}{x+\frac{1}{2}}=-\frac{5}{4}$

$\Rightarrow y-3=-\frac{5}{4}\left(x+\frac{1}{2}\right)\Leftrightarrow 4y-12=-5x-\frac{5}{2}$

$\Rightarrow 8y+10x=19$

Answer: C

34. If $\theta=2\arctan\left(\frac{1}{2}\right)$, then which one of the following is equal to $\sec\theta$?

A. $\frac{25}{3}$ B. $\frac{4}{5}$ C. $\frac{5}{3}$ D. $\frac{5}{4}$

Solution: $\theta=2\tan^{-1}\left(\frac{1}{2}\right)\Leftrightarrow\tan\left(\frac{\theta}{2}\right)=\frac{1}{2}$

Recall that: $\sin^2\left(\frac{\theta}{2}\right)+\cos^2\left(\frac{\theta}{2}\right)=1$

Dividing both side by $\cos^2\left(\frac{\theta}{2}\right)$ will get

$\Rightarrow\tan^2\left(\frac{\theta}{2}\right)+1=\sec^2\left(\frac{\theta}{2}\right)$

$\Rightarrow\frac{1}{4}+1=\sec\left(\frac{\theta}{2}\right)\Leftrightarrow\sec\left(\frac{\theta}{2}\right)=\frac{\sqrt{5}}{2}$

$\Rightarrow\cos\left(\frac{\theta}{2}\right)=\frac{2}{\sqrt{5}}$ since we have $\cos\left(\frac{\theta}{2}+\frac{\theta}{2}\right)=\cos\theta$

$\Rightarrow\cos\theta=\cos\left(\frac{\theta}{2}\right)\cos\left(\frac{\theta}{2}\right)-\sin\left(\frac{\theta}{2}\right)\sin\left(\frac{\theta}{2}\right)=\cos^2\frac{\theta}{2}-\sin^2\frac{\theta}{2}$

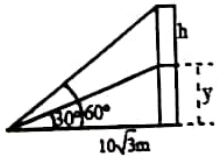
$=\cos^2\left(\frac{\theta}{2}\right)-\left(1-\cos^2\left(\frac{\theta}{2}\right)\right)=2\cos^2\left(\frac{\theta}{2}\right)-1$

$\Rightarrow\cos\theta=2\left(\frac{4}{5}\right)-1=\frac{3}{5}\Rightarrow\sec\theta=\frac{5}{3}$

Answer: C

35. An observer on level ground is at a distance $10\sqrt{3}$m from a building. The angles of elevation to the bottom of the windows on the second and third floors are 30° and 60°, respectively. What is the distance h between the bottoms of the windows?

A. 15m B. 20m C. $15\sqrt{3}$m D. 32m



Solution: $\tan 30°=\frac{y}{10\sqrt{3}}$

$\Rightarrow y=\left(10\sqrt{3}\right)\tan 30°=10m$

- $\tan 60°=\frac{h+y}{10\sqrt{3}}$

$\Rightarrow h+y=10\sqrt{3}\tan 60°=30$m

$\therefore h=30\text{m}-y=30-10\text{m}=20\text{m}$

Answer: B

Telegram: https://t.me/ExtremeSeries

36. Suppose that an air plane is descending at a speed of 50 miles per hour at an angle of 30° below the horizontal line. What is the x and y components respectively, of the velocity of the plane?

A. $(50\sqrt{3}, 25)$ C. $(25, -25\sqrt{3})$

B. $(-25, \sqrt{50}, 3)$ D. $(-25, \sqrt{3}, -25)$

Solution:

- $\cos 30° = \frac{x}{50} \Rightarrow x = 50\cos 30° = -25\sqrt{3}$
- $\sin 30° = \frac{y}{50} \Rightarrow y = -50\cos 30° = -25$

$\therefore (x, y) = (-25\sqrt{3}, -25)$

37. Suppose P and Q are points in space such that the mid – point of PQ is on the negative z – axis and the distance between P and Q is 6. If P = (2, -1, 0), then what is the coordinate of Q?

A. (-2, 1, 4) B. (2, -1, 6) C. (2, -1, -6) D. (-2, 1, -4)

Solution:

Let Q = (x, y, z) and P = (2, -1, 0)

Midpoint of PQ $= \left(\frac{x+2}{2}, \frac{y-1}{2}, \frac{z}{2}\right) = (0, 0, -k)$

$\Rightarrow x = -2, y = 1, z = -2k \Rightarrow Q = (x, y, z) = (-2, 1, -2k)$

To find k, use distance between P and Q

$(-2-2)^2 + (-1-1)^2 + (-2k)^2 = 36$

$4k^2 = 36 - 20 = 16 \Rightarrow k = 2$, therefore, Q = (-2, 1, -4)

38. The sequence $\left\{\frac{(n-1)(2n+1)}{1-n^2}\right\}_{n=1}^{\infty}$ converges to:

A. $-\infty$ B. -2 C. 0 D. 1

Hint: Here: Multiplying the numerator will be:

$$\left\{\frac{(n-1)(2n+1)}{1-n^2}\right\}_{n=1}^{\infty} = \left\{\frac{2-\frac{1}{n}-\frac{1}{n^2}}{\frac{1}{n^2}-1}\right\}_{n=1}^{\infty} \quad \ldots \text{(dividing by } n^2\text{)}$$

As, $n \to \infty$; $\frac{2-0-0}{0-2} = -2$

Answer: B

39. Which one of the following represents a geometric sequence?

A. $3, 1, \frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \ldots$ C. $1, 3, 6, 10, 15, \ldots$

B. $\frac{1}{2}, -\frac{1}{3}, \frac{1}{4}, -\frac{1}{5}, \frac{1}{6}, \ldots$ D. $-3, 6, -9, 12, -15, \ldots$

Hint: Geometric sequence characterized by common ratio between successive term.

Use: $r = \frac{G_2}{G_1} = \frac{G_3}{G_2} = \frac{G_4}{G_3} = \ldots = \frac{G_n}{G_{n-1}}$

Thus $r = \frac{\frac{1}{3}}{1} = \frac{\frac{1}{9}}{\frac{1}{3}} = \frac{\frac{1}{27}}{\frac{1}{9}} = \ldots = \frac{1}{3}$

Answer: A

40. What is the value of the sum: $\sum_{n=1}^{\infty}\left(\frac{2^n + 5^n}{10^n}\right)$?

A. 0.325 B. 1 C. $\frac{5}{4}$ D. $\frac{37}{9}$

Hint:

The expression is the same as:

$$\sum_{n=1}^{\infty}\left(\frac{2^n}{10^n}\right) + \sum_{n=1}^{\infty}\left(\frac{5^n}{10^n}\right) = \sum_{n=1}^{\infty}\left(\frac{2}{10}\right)^n + \sum_{n=1}^{\infty}\left(\frac{5}{10}\right)^n$$

(i) $\sum_{n=1}^{\infty}\left(\frac{2}{10}\right)^n = \frac{1}{5} + \frac{1}{25} + \frac{1}{125} + \ldots, G_1 = \frac{1}{5}, r = \frac{1}{5}$

$$S_{\infty} = \frac{G_1}{1-r} = \frac{\frac{1}{5}}{1-\frac{1}{5}} = \frac{1}{4}$$

(ii) $\sum_{n=1}^{\infty}\left(\frac{5}{10}\right)^n = \frac{G_1}{1-r} = \frac{\frac{1}{10}}{1-\frac{1}{10}} = \frac{\frac{1}{2}}{\frac{1}{2}} = 1$

Therefore from (i) and (ii): $\frac{1}{4} + 1 = \frac{5}{4}$

Answer: C

Telegram: https://t.me/ExtremeSeries

41. The sum of the series $\sum_{n=1}^{\infty}(-1)^n 3^{-2n}$ is equal to _____

A. $-\frac{1}{8}$ B. -0.13 C. -0.1 D. $\frac{1}{8}$

Hint: Here $\sum_{n=1}^{\infty}(-1)^n 3^{-2n} = -\frac{1}{3^2}+\frac{1}{3^4}-\frac{1}{3^6}+\ldots \Rightarrow G_1 = -\frac{1}{9}, r = -\frac{1}{9}$

$$S_\infty = \frac{G_1}{1-r} = \frac{-\frac{1}{9}}{1-\left(-\frac{1}{9}\right)} = \frac{-1}{10} = -0.1$$

Answer: C

42. Let $a_n = \left(\frac{n+3}{n+1}\right)^n$, then $\lim_{n\to\infty} a_n$ is equal to:

A. 1 B. $\frac{1}{2}e$ C. e^2 D. ∞

Hint: Use: $\lim_{n\to\infty}\left(1+\frac{a}{kn}\right)^n = e^{\frac{a}{k}}$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty}\left(\frac{n+3}{n+1}\right)^n = \lim_{n\to\infty}\left(\frac{\frac{n}{n}+\frac{3}{n}}{\frac{n}{n}+\frac{1}{n}}\right)^n$$

$$\Rightarrow \lim_{n\to\infty} a_n = \frac{\lim_{n\to\infty}\left(1+\frac{3}{n}\right)^n}{\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n} = \frac{e^3}{e^1} = e^2$$

Answer: C

43. Given that $\lim_{x\to 3} f(x) = 5$ and $\lim_{x\to 3} g(x) = 11$. What is the value of

$$\lim_{x\to 3}\left[\frac{(f(x)-g(x))(g(x)-2f(x))}{(g(x)-(f(x))^2}\right]?$$

A. $\frac{-66}{96}$ B. $\frac{-1}{6}$ C. 0 D. does not exist

Hint: The expression is the same as:

$$\lim_{x\to 3}\left[\frac{(f(x)-g(x))(g(x)-2f(x))}{(g(x)-(f(x)(g(x)f(x))}\right] = \lim_{x\to 3}\frac{g(x)-2f(x)}{g(x)-f(x)}(-1)$$

$$\Rightarrow (-1)\left[\frac{\lim_{x\to 3} g(x) - 2\lim_{x\to 3} f(x)}{\lim_{x\to 3} g(x) + \lim_{x\to 3} f(x)}\right] = (-1)\left(\frac{11-2(5)}{11+5}\right) = \frac{-1}{6}$$

Answer: B

44. $\lim_{x\to\infty}\left(\frac{x}{x+2}\right)^{-3x}$ is equal to _____

A. e^6 B. e^{-3} C. $e^{-3/2}$ D. e^{-6}

Hint: The expression is the same as:

$$\lim_{x\to\infty}\left(\frac{1}{1+\frac{2}{x}}\right)^{-3x} = \frac{1}{\lim_{x\to\infty}\left(1+\frac{2}{x}\right)^{-3x}} = \frac{1}{(e^2)^{-3}} = e^6$$

45. If $f(x) = x^2\sqrt{2x+12}$, what is the slope of the tangent line to the graph of f at $x = 2$.

A. -4 B. 2 C. 18 D. 17

Hint: Recall that $\sqrt{x} = x^{1/2}$

$$f(x) = x^2\sqrt{2x+12} = \sqrt{x^4(2x+12)} = \sqrt{2x^5+12x^4}$$

$$f'(x) = \frac{1}{2}(2x^5+12x^4)^{-1/2}10x^4 + 48x^3$$

$$\Rightarrow f'(2) = \frac{1}{2}[2(2)^5+12(2^4)]^{-1/2}[10(2)^4+48(2^3)]$$

$$\Rightarrow f'(2) = \frac{1}{2}[64+192]^{-1/2}(544) = \frac{1}{2}\cdot\frac{544}{\sqrt{256}} = 17$$

46. Suppose, $g(2) = 1$, $g'(2) = 10$ and $f(x) = \frac{x^2}{1+xg(x)}$, then the value of $f'(2)$ is equal to _____

A. -8 B. $-\frac{8}{9}$ C. $\frac{4}{3}$ D. $\frac{8}{9}$

Hint: Use quotient rule

$$f'(x) = \frac{(x^2)'[1+xg(x)] - x^2(1+xg(x))'}{[1+xg(x)]^2}$$

$$\Rightarrow f'(x) = \frac{2x'[1+xg(x)] - x^2[1'+x'g(x)+xg'(x))]}{[1+xg(x)]^2}$$

Telegram: https://t.me/ExtremeSeries

$$f'(x)=\frac{2x[1+xg(x)]-x^2[0+1g(x)+xg'(x)]}{[1+xg(x)]^2}$$

$$f'(2)=\frac{2(2)[1+2g(2)]-(2)^2[0+g(2)+2g'(2)]}{[1+2g(2)]^2}$$

$$f'(2)=\frac{4[1+2(1)]-(4)[0+1+2(10)]}{[1+2(1)]^2}=\frac{12-84}{9}=-8$$

47. Let $f(x)=\begin{cases}\dfrac{a\sin x}{x-|x|}, \text{if } x<0\\ e^{-x}+\cos x, \text{if } x\geq 0\end{cases}$

If f is continuous at $x=0$, then what is the value of a?

A. 4 B. 2 C. $\frac{1}{2}$ D. -4

Hint: Since f is cont. at x = 0

$$\lim_{x\to 0^-} f(x)=f(0)\Leftrightarrow \lim_{x\to 0^-}\frac{a\sin x}{x-|x|}=e^{-0}+\cos(0)$$

$$\lim_{x\to 0^-}\frac{a\sin x}{2x}=1+1=2\Rightarrow\frac{a}{2}=2\Rightarrow a=4$$

48. The derivative of $f(x)=\dfrac{1+\sin x}{\cos x}$ is

A. $\sec x+\tan x$ C. $\dfrac{1}{1+\tan x}$

B. $\dfrac{1+\sin x}{\cos^2 x}$ D. $\dfrac{\cos x}{\sin^2 x}$

Hint: $f'(x)=\dfrac{(1+\sin x)'(\cos x)-(1+\sin x)(\cos x)'}{(\cos x)^2}$

$$\Rightarrow f'(x)=\frac{(\cos x)(\cos x)-(1+\sin x)(-\sin x)}{\cos^2 x}$$

$$\Rightarrow f'(x)=\frac{\cos^2 x+\sin x+\sin^2 x+\sin^2 x)}{\cos^2 x}=\frac{1+\sin x}{\cos^2 x}$$

49. If F $(x) = f(2x + 2)$. $g(1 - x^2)$, with $f(2) = -3$, $f'(2) = 4$, $g(1) = -5$, and $g'(1) = 1$, then what is the value of F '(0)?

A. -40 B. -20 C. 0 D. 19

Hint:

F $'(x) = (f(2x + 2))'$. $g(1 - x^2) + f(2x + 2)(g(1 - x^2))'$

F $'(x) = f'(2x + 2)(2)g(1 - x^2) + f(2x + 2)\, g'(x)$

F $'(0) = [f'(2)](2)\, g(1 - 0) + f(2)\, g'(1)[0]$

F $'(0) = 4(2)(-5) + 0 = -40$

Answer: A

50. If $f(x) = e^{2x}\sin x$, then $f''(x)$ is _____.

A. $3e^{2x}\sin x - 4e^{2x}\cos x$ C. $e^{2x}(3\sin x + 4\cos x)$

B. $4e^{2x}\sin x + 2e^{2x}\cos x$ D. $e^{2x}(4\sin x - \cos x)$

Hint: $f'(x) = (e^{2x})'\sin x + (e^{2x})(\sin x)$

$\Rightarrow f'(x) = 2e^{2x}\sin x + e^{2x}\cos x = e^{2x}(2\sin x + \cos x)$

$\Rightarrow f''(x) = (e^{2x})'(2\sin x + \cos x) + e^{2x}(2\sin x + \cos x)'$

$\Rightarrow f''(x) = 2e^{2x}(2\sin x + \cos x) + e^{2x}(2\cos x - \sin x)$

$= e^{2x}[4\sin x + 2\cos x + 2\cos x - \sin x] = e^{2x}[3\sin x + 4\cos x]$

51. Let $y = \sin(3x^2)$, then $\dfrac{d^2y}{dx^2}$ equal to _____

A. $-6\sin(3x^2)$

B. $\cos(6x) - 6\sin(3x)^2$

C. $6\cos(3x^2) - 36x^2\sin(3x^2)$

D. $x^2\cos(3x^2) + 6\sin(3x^2)$

Hint:

$$\frac{dy}{dx}=\frac{d}{dx}[\sin(3x^2)]=\cos(3x^2)\frac{d}{dx}(3x^2)$$

$$\Rightarrow \frac{dy}{dx}=6x\cos(3x^2) \text{ and } \frac{d^2y}{dx^2}=\frac{d}{dx}[6x\cos(3x^2)]$$

$$=\frac{d}{dx}(6x)\cos(3x)^2+\frac{d}{dx}[\cos(3x^2)]$$

$$=6\cos(3x^2)-6x\sin(3x^2)\frac{d}{dx}(3x^2)$$

$$\therefore \frac{d^2y}{dx^2}=6\cos(3x^2)-36x^2\sin(3x^2)$$

Answer: C

52. Which one of the following is the set of all critical numbers of $f(x)=\frac{1}{3}x^3-|4x-1|$?

A. $\left\{\frac{1}{4},2\right\}$ B. $\left\{-2,\frac{1}{4},2\right\}$ C. $\{-2, 2\}$ D. $\left\{\frac{1}{4}\right\}$

Telegram: https://t.me/ExtremeSeries

Hint: $f(x)=\frac{1}{3}x^3-|4x-1|=\begin{cases}\frac{1}{3}x^3-(4x-1) & \text{if } x\geq\frac{1}{4}\\ \frac{1}{3}x^3+4x-1 & \text{if } x<\frac{1}{4}\end{cases}$

$$f'(x)=\begin{cases}x^2-4 & \text{if } x>\frac{1}{4}\\ \nexists & \text{if } x=\frac{1}{4}\\ x^2+4 & \text{if } x<\frac{1}{4}\end{cases}$$

$f'(2)=0$ and $f'(-2)=0$ And $f'_+\left(\frac{1}{4}\right)\neq f'_-\left(\frac{1}{4}\right)$, since $f'\left(\frac{1}{4}\right)$ does not exist. Therefore, the critical numbers are -2, $\frac{1}{4}$ and 2.

Answer: B

53. The total cost (in Birr) of producing x iron sheets per day is $C(x) = 1{,}000 + 100x - 0.5x^2$, $0 \leq x \leq 100$. What is the marginal (rate of change of) cost at production level of 80 iron sheet?

A. 8.5 B. 20 C. 1800 D. 5800

Hint: $C(x) = 1{,}000 + 100x - 0.5x^2$

The rate of change of cost at x = 80 is equal to $C'(x) = 100 - x$.

At $x = 80$, $C'(80) = 100 - 80 = 20$. **Answer: B**

54. If a box with square base and open top is made from 1200 cm^2 material, what is the largest volume of the box in cm^3?

A. 4000 B. 8000 C. 15000 D. 3000

Hint: Let x = the side of the square

h = height of the box

V = (base area)(height) = x^2h

Surface area = (Area of base) + (Area of 4 face)

$\Rightarrow s = x^2 + 4xh = 1200\ cm^2 \Rightarrow x^2 + 4xh = 1200\ cm^2$

$h=\frac{1200-x^2}{4x}$, however

$$v=x^2h=x^2\left(\frac{1200-x^2}{4x}\right)=300x-\frac{x^3}{4}$$

$$v(x)=300x-\frac{x^3}{4}\Rightarrow v'(x)=300-\frac{3x^3}{4}$$

$v'(x) = 0$, when $\frac{3x^3}{4}=300\Rightarrow x^2=400$

$\Rightarrow x = 20$ cm, $h=\frac{1200+400}{80}=10$ cm

$\therefore V_{max}$ = (Base area)(height) = (400)(10) cm^3 = 4000 cm^3

55. A water tank is a circular cylinder with base radius 2m and height 3m. If the tank is empty and water is pumped into it at rate of $2m^3/min$, how long does it take for the tank to be full?

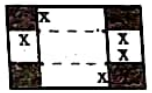
A. 1.5min B. $\frac{3}{2}\pi$ min C. 6πmin D. 12 min

Hint: We have: rate =2 m^3/min; $V_{cylinder} = \pi r^2 h = \pi(4m^2)(3m) = 12\pi m^3$; Time taken = $\frac{\text{Vol}}{\text{rate}}=\frac{12\pi m^3}{2m^3/min}=6\pi$ min

56. Suppose that equal squares cut from each of the four corners of a square card board whose sides are 72 cm long. The resulting flaps are then folded up to form a box without a top. How long should be each of the four squares that has to be cut off to maximize the volume of the box?

A. 6 cm B. 12 cm C. 15 cm D. 24 cm

Hint:



$V(x) = (72-2x)^2(x) = (5184 - 288x + 4x^2)(x)$

$V(x) = 5184x - 288x^2 + 4x^3 \Rightarrow v'(x)$

$= 12x^2 - 576x + 5184$

$V''(x) = 24x - 576$ and $V''(x) = 0$, when $x = 24$ **Answer: D**

57. If $F(x)$ is an ant derivative of $f(x) = 1-\frac{2}{x^2}$ and F (1) = 0 then F (2) is equal to:

A. 0 B. $\frac{1}{2}$ C. $-\frac{1}{2}$ D. 3

Hint: $F(x)=\int\left(1-\frac{2}{x^2}\right)dx=\int(1-2x^{-2})dx$

$$\Rightarrow F(x)=x-\frac{2x^{-2+1}}{-2+1}+C=x+\frac{2}{x}+C$$

Telegram: https://t.me/ExtremeSeries

$F(1) = 1 + \frac{2}{1} + C = 0 \Rightarrow C = 0 - 3 = -3$

$\Rightarrow F(x) = x + \frac{2}{x} - 3$, therefore $F(2) = 2 + \frac{2}{2} - 3 = 0$ **Answer: A**

58. $\int \frac{x + \ln(x+1)}{(x+1)^2} dx$ is equal to:

A. $\ln(x+1) + \frac{x}{x+1} + C$ C. $(x+1)^2 - \frac{1}{x+1} + C$

B. $(x+1)^2 + \frac{1}{x+1} + C$ D. $\frac{x\ln(x+1)}{x+1} + C$

Hint: The expression is the same as:

$\int \frac{x}{(x+1)^2} dx + \int \frac{\ln(x+1)}{(x+1)^2} dx$, let $u = x + 1 \Rightarrow du = dx, x = u - 1$

$\Rightarrow \int \frac{u-1}{u^2} du + \int \frac{\ln u}{u^2} du = \int \frac{1}{u} du - \int \frac{du}{u^2} + \int u^{-2} \ln u du$

$= \ln u + \frac{1}{u} + \int u^{-2} \ln u du \ldots (*)$

Applying integration by part for $\int u^{-2} \ln u du$.

$f(u) = \ln u$ and $g'(u) = u^{-2} du$ $f'(u) = \frac{1}{u}$ and $g(u) = \frac{1}{u}$ Therefore

$\int u^{-2} \ln u du = \frac{-\ln u}{u} + \int \frac{du}{u^2} = \frac{-\ln u}{u} - \frac{1}{u} + C$

Substituting this in (*) would be:

$\ln u + \frac{1}{u} - \frac{\ln u}{u} - \frac{1}{u} + c = \ln u - \frac{\ln u}{u} + C$

$\Rightarrow \frac{u \ln u - \ln u}{u} + C = \frac{(\ln u)(u-1)}{u} + C$

$\Rightarrow \frac{\ln(x+1)(x+1-1)}{x+1} + C = \frac{x\ln(x+1)}{x+1} + C$ **Answer: D**

59. $\int_0^{\pi/2} \frac{x - \sin x}{\sec x} dx$ is equal to:

A. $\frac{\pi - 3}{2}$ B. $\frac{\pi - 1}{2}$ C. $\frac{3 - \pi}{2}$ D. $\frac{\pi + 3}{2}$

Hint: The expression is the same as: $\int_0^{\pi/2} x \cos x \, dx - \int_0^{\pi/2} \sin x \cos x \, dx$

Part 1: Evaluating $\int_0^{\pi/2} x \cos x \, dx$ **by parts.**

Let $u = x$, $dv = \cos x \, dx$; $du = 1$; $v = \sin x$ $\| \boxed{\int u dv = uv - \int v du}$

$x \sin x - \int \sin x dx = x \sin x + \cos x \Big|_0^{\pi/2}$

$\Rightarrow \frac{\pi}{2} \sin \frac{\pi}{2} + \cos \frac{\pi}{2} - (0 + \cos 0) = \frac{\pi}{2} + 0 - 1 = \frac{\pi}{2} - 1$

Part (2): Evaluating $\int_0^{\pi/2} \sin x \cos x \, dx$ **by subtraction.**

Let $u = \sin x \Rightarrow du = \cos x \, dx$. $\int u du = \frac{u}{2} + c = \frac{\sin x}{2} \Big|_0^{\pi/2} = \frac{1}{2}$

From part (1) and part (2): $\frac{\pi}{2} - 1 - \frac{1}{2} = \frac{\pi - 3}{2}$. **Answer: A**

60. What is the derivate of $f(x) = \int_0^{x^2 + \pi} \frac{dt}{\sin t + 1}$?

A. $\frac{\cos x}{\sin(x^2 + \pi) + 1}$ C. $\frac{2x \cos x}{\sin(x^2 + \pi) + 1}$

B. $\frac{2x}{\sin(x^2 + \pi) + 1}$ D. $\int_0^{2x} \frac{dt}{\sin t + 1}$

Hint: Note: $\frac{d}{dx} \int_{h(x)}^{g(x)} f(t) dt = f(g(x)) g'(x) - f(h(x)) h'(x)$

Let $f(t) = \frac{1}{\sin t + 1}$, $g(x) = x^2 + \pi$, $h(x) = 0$ and $g'(x) = 2x$.

$f(g(x)).g'(x) - f(h(x))h'(x)$

$\frac{1}{\sin g(x) + 1} . 2x - \frac{1}{\sin 0 + 1} . 0 = \frac{2x}{\sin(x^2 + \pi) + 1}$ **Answer: B**

61. What is the area of the region between the graphs of $y = -x^2 + 2$ and $y = |x|$ where $-1 \le x \le 2$?

A. $\frac{11}{6}$ B. $\frac{25}{6}$ C. $\frac{7}{3}$ D. $\frac{11}{3}$

Telegram: https://t.me/ExtremeSeries

Hint: The intersection of $y = -x^2 + 2$ and $y = |x|$ is $x = -1$ and $x = 1$. The area of the shaded region would be:

$$A = \int_{-1}^{1}[-x^2+2-(-x)]dx + \int_{0}^{1}(-x^2+2-x)dx + \int_{1}^{2}(x+x^2-2)dx$$

$$= \frac{-x^3}{3} + 2x + \frac{x^2}{2}\Big|_{-1}^{0} + \frac{-x^3}{3} + 2x - \frac{-x^2}{2}\Big|_{0}^{1} + \frac{-x^2}{2} + \frac{-x^3}{2} - 2x\Big|_{1}^{2}$$

$$= (0+0+0) - \left(\frac{1}{3} - 2 + \frac{1}{2}\right) + \left(\frac{-1}{3} + 2 - \frac{1}{2} - 0\right) + \left(\frac{4}{2} + \frac{8}{3} - 4 - \frac{1}{2} - \frac{1}{3} + 2\right)$$

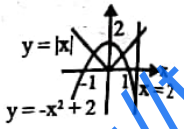
$$= \frac{5}{3} - \frac{3}{2} + 4 = \frac{25}{6}$$

Answer: B

62. What is the area of the region between the graph of $f(x) = -x^2 + 4x - 3$ and the x – axis from $x = 0$ to $x = 3$?

A. $\frac{-2}{3}$ B. $\frac{2}{3}$ C. $\frac{4}{3}$ D. $\frac{8}{3}$

Hint: The area of the bounded region is



$$A = \int_{0}^{3} f(x)dx = \int_{0}^{1} f(x)dx + \int_{1}^{3} f(x)dx$$

Let $A = |A_1| + |A_2|$

i) $A_1 = \int_{0}^{1}[0-(-x^2+4x-3)dx = \int_{0}^{1} x^2 - 4x + 3)dx$

$$= \frac{x^3}{3} - \frac{4x^2}{2} + 3x\Big|_{0}^{1} = \frac{1}{3} - 2 + 3 - 0 = \frac{4}{3}$$

ii) $A_2 = \int_{1}^{3}(-x^2+4x-3)dx = \frac{-x^3}{3} + \frac{4x^2}{2} - 3x\Big|_{1}^{3} = -9 + 18 - 9 - \left(-\frac{1}{3} + 2 - 3\right) = \frac{4}{3}$ Therefore, $A = A_1 + A_2 = \frac{4}{3} + \frac{4}{3} = \frac{8}{3}$

Answer: D

1. Which of the following functions is a non-to-one correspondence?

A. $f: \mathbb{R} \to \mathbb{R}, f(x) = \tan x$, where $\mathbb{R}$ is the domain of f

B. $g: \mathbb{R} \to \mathbb{R}, g(x) = 2^x$

C. $h: [0,\infty) \to [0,\infty), h(x) = x^2$

D. $r: [0,\infty) \to [0,\infty), r(x) = x + 5$

Explanation:

(i) $h: [0,\infty) \to [0,\infty)$ ← Range of $h = [0, \infty)$

(ii) Range of $h(x) = x^2$ is $[0, \infty)$

Range of h in first set (i) is equal to range of $h(x)$ in 2nd set. (ii) → One-to-one correspondence exists between two sets h: $[0, \infty)$ and $h(x) = x^2$

Answer: C

Exercise: For what value of K:

a) $g: \mathbb{R} \to K, g(x) = 2^x$ is one-to-one correspondence

b) $r: [0,\infty) \to K, r(x) = x + 5$ is one-to-one correspondence?

2. Different codes, each of which consisting of five characters, are to be generated in such a way that the first two characters are any of the English capital letters (A to Z) and the remaining three are any of the digits (0, 1,…,9). How many distinct codes can be generated so?

A. 468,000 B. 260 C. 676,000 D. 25!× !10

Explanation: English letters A to Z are 26 and the digit are 10, then:

→ The first two characters are $26 \times 26 = 676$

→ The next three characters are $10 \times 10 \times 10 = 1000$

→ Using multiplication rule we can generate $676 \times 1000 = 676{,}00$

Answer: C

3. A certain meeting hall has 20 rows of seats. There are 20 seats in the first row, 22 seats in the second row, 24 seats in the third row, and so on. How many seats are there on the last (20^{th}) row of the hall?

A. 46 B. 58 C. 760 D. 5240

Explanation: $A_1 = 20, A_2 = 22, A_3 = 24, A_{20} = ?$

$A_n = A_1 + (n-1)d \Leftrightarrow A_{20} = 20 + (20-1)(2)$

$\Rightarrow A_{20} = 20 + 38 = 58$

Answer: B

4. Two perpendicular lines ℓ_1 and ℓ_2 are intersecting at (-1, 2). If the angle of inclination of ℓ_1 is 45°, then what is the equation of ℓ_2?

A. $y = -x + 3$ C. $y = -x + 1$

B. $y = x + 3$ D. $y = x + 1$

Telegram: https://t.me/ExtremeSeries

Explanation: slope of $\ell_1 = \tan 45° = 1$ and

slope of $\ell_2 = \tan 135° = -1$

$\therefore$ The equation of $\ell_2 : \dfrac{y-2}{x-(-1)} = -1$

$\Rightarrow y - 2 = -(x+1)$ therefore, $\ell_2 : y = -x + 1$ **Answer: C**

5. What is the area of the region between the graph of $y = \sin x$ and x-axis where $0 \le x \le 2\pi$?

A. 4 B. 4π C. 2 D. 2π

Explanation:

The intersection of x-axis ($y = 0$) and $y = \sin x$ on $[0, 2\pi]$

→ $\sin x = 0 \Rightarrow x = 0, \pi, 2\pi$

The area bounded by x-axis and $y = \sin x$ is

$$A = \int_0^{\pi} (\sin x - 0)\,dx + \int_{\pi}^{2\pi} (0 - \sin x)\,dx =$$

$$\Rightarrow A = -\cos x\Big|_0^{\pi} + \cos x\Big|_{\pi}^{2\pi} = -(-1-1) + (-1(-1)) = 4$$

Answer: A

6. The following is set of data representing the average mark of 13 students: 91, 89, 93, 91, 87, 94, 92, 85, 91, 90, 96, 93, 89. Then which one of the following statements is true about the data?

A. The median is 90.5.
B. The upper quartile is 92.
C. The range of the marks is 11.
D. The mean is 91.5.

Explanation:

→ To find the median arrange the data from least to greatest. Median of a data is the "middle" value of data

85, 87, 89, 89, 90, 91 **91** 91, 92, 93, 93, 94, 96

$\therefore$ Median = 91

→ The median of the upper part of a data is called upper quartile

$(Q_3) = 91, 92, \boxed{93, 93,}\ 94, 96$

$$\therefore Q_3 = \frac{93+93}{2} = \frac{186}{2} = 93 \leftarrow \text{upper quartile.}$$

→ Range = upper extreme – lower extreme = 96 – 85 = 11

→ $\text{Mean} = \dfrac{\text{sum of all data}}{\text{number}} = \dfrac{1181}{13} = 90.85$

Answer: C

Exercise: What is the value of lower quartile

7. If $f(x) = \ell n\left(\sqrt{x^2+1}\right)$, which of the following is equal to $f'(x)$?

A. $\dfrac{x}{\sqrt{x^2+1}}$ C. $\dfrac{2x}{\sqrt{x^2+1}}$

B. $\dfrac{x}{x^2+1}$ D. $\dfrac{2x}{x^2+1}$

Explanation: Let $f(x) = \ln\sqrt{g(x)}$

Recall that $\left(\sqrt{g(x)}\right)' = \dfrac{g'(x)}{2\sqrt{g(x)}}$ and $(\ln x)' = \dfrac{1}{x}$

$$f'(x) = \frac{1}{\sqrt{x^2+1}} \cdot \left(\frac{2x}{2\sqrt{x^2+1}}\right) = \frac{x}{x^2+1}$$

→ since $\left(\sqrt{x^2+1}\right)' = \dfrac{2x}{2\sqrt{x^2+1}}$

Answer: B

8. If $z = \dfrac{3+i}{i-2}$ is a given complex number, then what is the conjugate, $\overline{z}$, of z?

A. $\overline{z} = \dfrac{-3+i}{2+i}$ C. $\overline{z} = -6-2i$

B. $\overline{z} = \dfrac{i-3}{2-i}$ D. $\overline{z} = -1-i$

Explanation: $\overline{z} = \dfrac{3-i}{-2-i} = \dfrac{(3-i)}{-(2+i)} = \dfrac{-3+i}{2+i}$

9. What is the principal argument of $(5+5i)^{11}$?

A. $\dfrac{\pi}{2}$ B. $\dfrac{2\pi}{3}$ C. $\dfrac{\pi}{4}$ D. $\dfrac{3\pi}{4}$

Explanation: Let Z = 5 + 5i,

Telegram: https://t.me/ExtremeSeries

$$Arg(z)=\tan^{-1}\left(\frac{5}{5}\right)=\tan^{-1}(1)=\frac{\pi}{4}$$

$$\therefore Arg(z)^{11}=11\left(Arg(z)\right)=11\left(\frac{\pi}{4}\right)=\frac{11\pi}{4}$$

$$\therefore \text{Principal } Arg(z)=\frac{11\pi}{4}-2\pi=\frac{3\pi}{4}$$

Answer: D

10. An object is moving along the parabola $y=\sqrt{2x}$ in xy-plane. At what point on its path does the object becomes closest to the point (2, 0)?

A. $(3,\sqrt{6})$ C. $(1,\sqrt{2})$

B. (1, 1) D. (2, 2)

Explanation: Let d be the closest distance between the point (2, 0) and (x, y) on the parabola $y=\sqrt{2x}$.

$$\therefore d=\sqrt{(x-2)^2+(y-0)^2}\Rightarrow d(x)=\sqrt{(x-2)^2+\left(\sqrt{2x}\right)^2}$$

$$d(x)=\sqrt{x^2-4x+4+2x}=\sqrt{x^2-2x+4}$$

Let $g(x)=x^2-2x+4\Rightarrow g'(x)=2x-2$

$$g'(x)=0\Rightarrow 2x-2=0\Rightarrow x=1$$

$\therefore$ The closest point will be when $x=1$; $(1,\sqrt{2})$ **Answer: C**

11. For arbitrary propositions p and q, which one of the following is a valid equivalence?

A. $\neg(p\Rightarrow q)\equiv(p\Rightarrow q)$

B. $[\neg(p\Rightarrow q)\wedge p]\equiv(p\wedge\neg q)$

C. $[p\vee\neg q]\equiv(p\Rightarrow q)$

D. $[(p\vee q)q\Rightarrow p]\equiv(p\Rightarrow\neg q)$

Explanation: A. $\neg(p\Rightarrow q)\equiv(q\Rightarrow p)$

$\neg[\neg p\vee q]\equiv\neg q\vee p$

$p\wedge\neg q\neq p\vee\neg p$

B. $\neg[(p\Rightarrow q)\wedge p]\equiv p\wedge\neg q$

$\neg(\neg p\vee q)\wedge p$

$(p\wedge\neg q)\wedge p$

$(p\wedge q)\wedge\neg q\equiv p\wedge\neg q$

Answer: B

12. Which one of the following is to $\lim_{x\to 9}\frac{x-9}{3-\sqrt{x}}$?

A. 6 B. −3 C. −6 D. ∞

Explanation: $\lim_{x\to 9}\frac{x-9}{3-\sqrt{x}}=\lim_{x\to 9}\frac{x-9}{3-\sqrt{x}}\cdot\frac{3+\sqrt{x}}{3+\sqrt{x}}$

$$\lim_{x\to 9}\frac{(x-9)\left(\sqrt{x}+3\right)}{9-x}=-\lim_{x\to 9}\left(\sqrt{x}+3\right)=-6$$

Answer: C

13. The ozone level (in ppb – parts per billion) on a sunny day in a metropolitan area is given by the formula; $p(t)=80+12t-t^2$, where t is time in hours and $t=0$ corresponds to 9 A.M. What is the rate of increase of the ozone level after 3 hours (i.e. at 12 A.M.)?

A. 6 ppb C. 107 ppb

B. 12 ppb D. 113 ppb

Explanation: $p(t)=80+12t-t^2$

$p'(t)=12-2t$ and $p'(3)=12-2(3)=6$

Answer: A

14. Suppose that p represents the statement "*He missed the tournament*", q represents the statement "*He got the gold medal.*" And r represents the statement "*He took a trip abroad.*". Then which one of the following symbolic expression represents the statement: "If he takes a trip abroad and he does not miss the tournament, then he will get the gold medal."?

A. $(r\Rightarrow q)\vee\neg p$ C. $(r\Rightarrow q)\wedge\neg p$

B. $r\wedge(p\Rightarrow q)$ D. $\neg(r\vee p)\vee q$

Answer: C

15. Let $f(x)=\begin{cases}a\dfrac{\sin 2x}{x}, & \text{if } x<0\\ e^{2x}-2, & \text{if } x\geq 0\end{cases}$

If f is continuous at $x=0$, then what is the value of a?

Telegram: https://t.me/ExtremeSeries

A. $\frac{1}{2}$ B. 2 C. $-\frac{1}{2}$ D. -2

Solution: If f is cont. at $x = 0$, then

$$\lim_{x\to 0} f(x) = f(0) \Leftrightarrow \lim_{x\to 0^-} f(x) = f(0)$$

$$\therefore \lim_{x\to 0} a\frac{\sin 2x}{x} = e^{2(0)} - 2 \Rightarrow 2a = 1 - 2 = -1$$

$$a = -\frac{1}{2}$$

Answer: C

16. Which one of the following is an arithmetic sequence?
   A. 3, 5, 7, 9, 11, …
   B. 3, 6, 12, 24, 48,…
   C. −3, 6, −9, 12, −15,…
   D. 1, 3, 6, 10, 15, 21,…

Answer: A

17. If $f(x) = 2x(x^2+1)^4$, then which of the following is an anti derivative of $f(x)$?

A. $\frac{2x}{5}(x^2+1)^5 + c$ C. $\frac{x}{5}(x^2+1)^5 + 1$

B. $\frac{2}{5}(x^2+1)^5 + c$ D. $\frac{1}{5}(x^2+1)^5 - 1$

Solution: $\int 2x(x^2+1)^4 dx$; Let $U = x^2+1 \Rightarrow dU$

$$\therefore \int U^4 dU = \frac{U^5}{5} + c = \frac{(x^2+1)^5}{5} + c$$

Answer: D

18. Which one of the following sequence is a convergent sequence?

A. $1, \frac{1}{2}, 1, \frac{1}{3}, 1, \ldots$ C. $\left\{10^{109} - \frac{1}{100}n\right\}_{n=1}^{\infty}$

B. $\{(-1)^n\}_{n=1}^{\infty}$ D. $\left\{\sin\left(\frac{1}{n}\right)\right\}_{n=1}^{\infty}$

Solution: A sequence $\{a_n\}$ is convergent if $\lim_{n\to\infty} f(a_n)$ exist

$$\therefore \lim_{n\to\infty} \sin\left(\frac{1}{n}\right) = 0$$

Answer: D

19. What is the absolute maximum value of $f(x) = 2x^2 - x^4 - 4$ on $[0, 2]$?

A. -3 B. 3 C. -4 D. 12

Solution: $f'(x) = 4x - 4x^3 = 4x(1 - x^2)$

$f'(x) = 0, \Rightarrow x = 0, x = 1$, but $x = -1 \notin [0,2]$

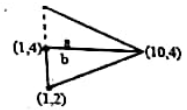
Critical numbers are 0 and 1

$f(0) = -4$, $f(1) = -3$ and $f(2) = -12$

$\therefore f(1) = -3 \leftarrow$ Absolute maximum.

Answer: A

20. The equation of an ellipse with center at (1, 4), vertices at (10, 4) and (1, 2) is:



A. $4(x-1)^2 + 81(y-4)^2 = 324$

B. $(x-1)^2 + 9(y-4)^2 = 4$

C. $9(x-1)^2 + 4(y-4)^2 = 1$

D. $2(x-1)^2 + 9(y-4)^2 = 4$

Solution: Center (1, 4)

a = 10 − 1 = 9 and b = 4 − 2 = 2, (h, k) = (1, 4)

$$\therefore \frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1 \Leftrightarrow \frac{(x-1)^2}{81} + \frac{(y-4)^2}{4} = 1$$

$$324\left(\frac{(x-1)^2}{81} + \frac{(y-4)^2}{4}\right) = 324 \Leftrightarrow 4(x-1)^2 + 81(y-4)^2 = 324$$

Answer: A

21. If $p(x) = 3x^2$ and $q(x) = x^2 + x$, then what is solution set of $\frac{p(x)}{3q(x)} - \frac{1}{x} = \frac{1}{q(x)}$?

A. $\{-1, 2\}$ B. $\{2\}$ C. $\{-1, -2\}$ D. $\{-3\}$

Solution: Multiplying both side by $q(x)$, get:

$$\Rightarrow \frac{p(x)}{3} - \frac{q(x)}{x} = 1 \Leftrightarrow \frac{3x^2}{3} - \left(\frac{x^2+x}{x}\right) = 1$$

$$\Rightarrow x^2 - (x+1) = 1 \Leftrightarrow x^2 - x - 1 - 1 = 0$$

$$\Rightarrow x^2 - x - 2 = 0 \Leftrightarrow (x-2)(x+1) = 0$$

Telegram: https://t.me/ExtremeSeries

$\Rightarrow x=2$ and $x=-1$, but $q(-1)=0$

$\therefore$ The solution set is $\{2\}$ only.

Answer: B

22. What are the values of u and v that satisfy the equation: $\frac{u+3i}{4-2i}=\frac{2+vi}{20}$?

A. $u=2, v=3$ C. $u=2, v=16$

B. $u=-6, v=10$ D. $u=-4, v=6$

Solution: Multiplying both side by $q(x)$

$$\frac{u+3i}{4-2i}=\frac{2+vi}{20}\Leftrightarrow\frac{(u+3i)(4+2i)}{(4-2i)(4+2i)}=\frac{2+vi}{20}$$

$$\Rightarrow\frac{4u+2ui+12i-6}{16+4}=\frac{2+vi}{20}$$

$$\Rightarrow(4u-6)+(2u+12)i=2+vi$$

$\therefore u=\frac{8}{4}=2$ and $v=2(2)+12=16$ **Answer: C**

23. What is the sum of all multiples of 3 between 20 and 200?

A. 7,227 C. 6,150

B. 6,570 D. 5,166

Solution:

Multiple of 3 between 20 and 200 are 21, 24, 27, 30, ..., 198

$$A_n=A_1+(n-1)d\Leftrightarrow 198=21+(n-1)(3)\Rightarrow n=\frac{180}{3}=60$$

$$S=\frac{60}{2}(21+198)=6,570$$ **Answer: B**

24. The inverse of the function defined by $g(x)=\frac{2x}{x+3}$ is equal to

A. $g^{-1}(x)=\frac{2x}{x-3}$ C. $g^{-1}(x)=-\frac{x+3}{2x}$

B. $g^{-1}(x)=-\frac{3x}{x-2}$ D. $g^{-1}(x)=-\frac{x+2}{3x}$

Solution: $g(x)=\frac{2x}{x+3}\Leftrightarrow y=\frac{2x}{x+3}$ interchange $\boxed{\begin{matrix}x\to y\\ y\to x\end{matrix}}$

$\Rightarrow x=\frac{2y}{y+3}\Leftrightarrow xy+3x=2y$, solving for y

$\Rightarrow 3x=2y-xy$ implies $y(2-x)=3x\Rightarrow y=\frac{3x}{2-x}=\frac{-3x}{x-2}$

$\therefore g^{-1}(x)=-\frac{3x}{x-2}$ **Answer: B**

25. If $A=\begin{pmatrix}0 & x & 0\\ 1 & -1 & 1\\ 0 & y & -1\end{pmatrix}$ and $A^{-1}=\begin{pmatrix}1 & 1 & 1\\ 3 & 0 & 0\\ 2 & 0 & -1\end{pmatrix}$, then what are the values of x and y?

A. $x=3, y=-2$ C. $x=-3, y=2$

B. $x=\frac{2}{3}, y=\frac{1}{3}$ D. $x=\frac{1}{3}, y=\frac{2}{3}$

Solution: We know that $A.A^{-1}=I$

$$AA^{-1}=\begin{pmatrix}0 & x & 0\\ 1 & -1 & 1\\ 0 & y & -1\end{pmatrix}\begin{pmatrix}1 & 1 & 1\\ 3 & 0 & 0\\ 2 & 0 & -1\end{pmatrix}=\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}$$

$$AA^{-1}=\begin{pmatrix}3x & 0 & 0\\ 0 & 1 & 0\\ 3y-2 & 0 & 1\end{pmatrix}=\begin{pmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}\Rightarrow 3x=1 \text{ and } 3y-2=0$$

$x=\frac{1}{3}$ and $y=\frac{2}{3}$ **Answer: D**

26. If $f(2)=-3$, $f'(2)=4$, $g(1)=-5$, $g'(1)=1$ and $F(x)=f(2x+2).g(1-x^2)$, then what is the value of $F'(0)$?

A. 19 B. 0 C. –20 D. –40

Solution: Here at

$$F'(x)=f'(2x+2)(2)g(1-x^2)+f(2x+2)g'(1-x^2)\cdot(-2x)$$

$$\therefore F'(0)=f'(0+2)(2)g(1-0)+f(2)\cdot g'(1)(0)$$

$$=f'(2)(2)g(1)+f(2)g'(1)(0)$$

$$=(4)(2)(-5)+(-3)(1)(0)=-40+0=-40$$

Telegram: https://t.me/ExtremeSeries

27. A ladder 6 m long rests against a vertical wall. If the bottom of the ladder slides away form the wall at a rate (speed) of $\frac{1}{2}$ m/sec, how fast is the angle between the top of the ladder and the wall changing when the angle is $\frac{\pi}{4}$ rad?

A. $\frac{\sqrt{2}}{12}$ rad / seo
C. $\frac{\sqrt{2}}{6}$ rad / sec
B. $\frac{\sqrt{2}}{2}$ rad / sec
D. $\frac{\sqrt{2}}{3}$ rad / sec

Solution:

$\sin\theta = \frac{x}{6}$ and $\frac{dx}{dt} = \frac{1}{2}$ m / sec

$$\frac{d}{dt}(\sin\theta) = \frac{d}{dt}\left(\frac{x}{6}\right) \Leftrightarrow \cos\theta.\frac{d\theta}{dt} = \frac{1}{6}\frac{dx}{dt}$$

$$\frac{d\theta}{dt} = \frac{1}{6}\frac{dx}{dt}\cdot\frac{1}{\cos\theta} = \frac{1}{6}\cdot\frac{1}{2}\cdot\frac{1}{\cos\theta\frac{\pi}{4}} = \frac{1}{6\sqrt{2}}$$

$$\therefore \frac{d\theta}{dt} = \frac{1}{6\sqrt{2}} = \frac{1}{6\sqrt{2}}\cdot\frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{12}\text{rad/sec}$$

Answer: A

28. The value (s) of x where the graph of the function $y = \frac{x^2-1}{x^3}$ cross its horizontal asymptote is (are);

A. $x = -2$
B. $x = -1$ and $x = 1$
C. $x = 1$
D. $x = -\sqrt{2}$ and $x = 1 + \sqrt{2}$

Solution: The horizontal asymptote is $y = 0$, so set $\frac{x^2-1}{x^3} = 0$

$\Rightarrow x^2 - 1 = 0$ at $x = 1$ and $x = -1$

$\therefore$ The graph crosses its horizontal asymptote at $x = 1$ and $x = -1$

Answer: B

29. If $A = \begin{pmatrix} 3 & -2 & 8 \\ 0 & 6 & 7 \\ 0 & 4 & 5 \end{pmatrix}$, then $\det(A^T A)$ is equal to

A. 12
B. 36
C. 30
D. 15

Solution:

$\det(A^T A) = \det(A^T)\det(A)$

det(A) by expanding first column:

$$\det(A) = 3\begin{vmatrix} 6 & 7 \\ 4 & 5 \end{vmatrix} - 0\begin{vmatrix} -2 & 8 \\ 4 & 5 \end{vmatrix} + 0\begin{vmatrix} -2 & 8 \\ 6 & 7 \end{vmatrix}$$

$$= 3(30 - 28) - 0 + 0 = 6$$

However, $\det(A) = \det(A^T)$

$\therefore \det(A^T A) = \det(A^T)\det(A) = 6 \times 6 = 36$

Answer: B

30. Suppose that a function f has the property that $f(x+y) = f(x)f(y)$ for all values of x and y and that $f(0) = 2$, $f'(0) = 1$. Then which one of the following represents the formula for the derivative $f'(x)$?

A. $f'(x) = 2f'(x) + 1$
B. $f'(x) = f(x) + 2f'(x)$
C. $f'(x) = f(x) + 2$
D. $f'(x) = 2f(x) - 1$

Solution: $f(x+y) = f(x).f(y)$ Let $y = 0$

$f(x+0) = f(x)f(0)$

$\Rightarrow f'(x) = f(x)f'(0) + f'(x)f(0)$

$\Rightarrow f'(x) = f(x)(1) + f'(x)(2)$

$\therefore f'(x) = f(x) + 2f'(x)$

Answer: B

31. A ball is thrown vertically from ground up to a height of 16m. Each time it drops h meters, it rebounds 0.80h m. Nothing that the ball travels every height of h twice. What is the total vertical distance travelled by the ball before it comes to rest?

A. 40m
B. 80m
C. 160m
D. 320m

Solution: $h = 16\text{m}, r = 0.8$

Telegram: https://t.me/ExtremeSeries

If ball is thrown vertically upward:

→ Upward series = $16 + (0.8) + 16 + (0.8)^2 16 + \ldots$

→ Downward series = $16 + (0.8) + 16 + (0.8)^2 16 + \ldots$

∴ Total vertical distance = sum of the two series

$$= 2\left(\frac{G_1}{1-r}\right) = \frac{2(16)}{1-0.8} = \frac{32}{0.2} = 160\text{m}.$$

Answer: C

Exercise: From Q #31, if a ball is dropped from the same height 16m and it rebounds 0.8m. what is the total vertical distance before comes to rest.

32. Suppose that the first 3 letters (A, B, and C) and number digits are to be used to form car plates in small town. How many different plates can be formed in total that contain 1, 2 or 3 letters and then followed by 3 digits?

A. 3,000 C. 39,000

B. 27,000 D. 100,000

Solution: The car plate has the form:

i	ii	iii	iv

→ in the (i) place the digit can be $0-9 = 10$ way

→ in the (ii) place the digit can be $0-9 = 10$ way

→ in the (iii) place the digit can be $0-9 = 10$ way

→ in the (iv) place either A, B or C = 3 way

∴ Using multiplication rule, the car plates can be $3 \times 10 \times 10 \times 10 = 3{,}000$

Answer: A

33. The derivative of the function $F(x) = \int_{-x}^{x} \frac{1}{1+t}\,dt$?

A. $\frac{2}{1-x^2}$ C. $\frac{1}{1+x}$

B. $\ln|1+x|$ D. $\ln\left|\frac{1+x}{1-x}\right|$

Explanation: Fundamental theorem of calculus II.

i) $F(x) = \int_a^{g(x)} f(t)$ then $F'(x) = f\left(g(x)\right)\cdot g'(x)$

$$\Rightarrow F(x) = \int_{-x}^{x} \frac{1}{1+t}dt = \int_{-x}^{0} \frac{1}{1+t}dt + \int_{0}^{x} \frac{1}{1+t}dt.$$

$$\Rightarrow F(x) = -\int_0^{-x} \frac{1}{1+t}dt + \int_0^x \frac{1}{1+t}dt$$

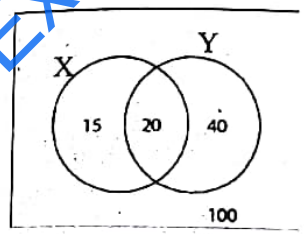
$$F'(x) = -\left(\frac{1}{1-x}(-1)\right) + \frac{1}{1+x}\cdot 1$$

$$= \frac{1}{1-x} + \frac{1}{1+x} = \frac{1+x+1-x}{1-x^2} = \frac{2}{1-x^2}$$

Answer: A

34. A city has two daily newspapers. X and Y. The following information was obtained from a survey of 100 residents of the city: 35 people subscribe to X, 60 people subscribe to Y and 20 subscribe to both newspapers. Then how many of the people in the survey do not subscribe to either of the newspapers?

A. 5 B. 25 C. 40 D. 55



Explanation:
People in the survey do not subscribe to either of the newspapers.
$= 100 \quad (15 + 20 + 40) = 25$

35. Which one of the following is necessarily true?

A. If $f'(x) = 0$ for all x in an interval I, then $f(x) = 0$ for all x in 1.

B. If $f(x) = x^2 \sin x + 5$, then there is $c \in (0, \pi)$ such that $f'(c) = 0$.

C. If $f(x) = e^x + x^2$ is increasing on $(-\infty, \infty)$.

D. If $f'(c) = 0$, then f attains its maximum or minimum value at $x = c$.

Explanation: From Roll's theorem $f(0) = 5$ and $f(\pi) = 0 + 5 = 5$

Since $f(0) = f(\pi)$, then there is a number $c \in (0, \pi)$ such that $f'(c) = 0$

Exercise:

a) If $f'(x) = 0$ for all x in interval, I, then what can you conclude about $f(x)$?

Telegram: https://t.me/ExtremeSeries

b) On what interval $f(x)=e^x+x^2$ is increasing and decreasing.

Answer:
a) f is constant for all x.
b) increasing on $[0,\infty)$ and decreasing on $(-\infty,0]$

36. What is the focus of the parabola $y^2+4y+8x=4$?

A. $(1,-2)$ C. $(3,-2)$
B. $(-1,-2)$ D. $(-3,-2)$

Solution: $y^2+4y=-8x+4$

$y^2+4y+4=-8x+4+4=-8x+8$

$(y+2)^2=-8(x-1)\rightarrow(h,k)=(1,-2)$

$\therefore 4p=8\Rightarrow p=2$, and

Focus is $(h,-p,k)=(1-2,-2)=(-1,-2)$

Answer: B

37. Which one of the following is equal to $\int\frac{\ell n x+x^2e^x}{x}dx$?

A. $\frac{1}{2}\ell n^2x+e^x(x^2-1)+c$

B. $\frac{1}{2}\ell n^2x+e^x(x-1)+c$

C. $\frac{1}{x^2}\ell nx+e^x(x-1)+c$

D. $\frac{-1}{x^2}\ell nx+e^x(x^2-1)+c$

Solution: $\int\left(\frac{\ell n x+x^2e^x}{x}\right)dx=\int\frac{\ell n x}{x}dx+\int(xe^xdx)$

First : $\int\frac{\ell n x}{x}dx$ let $U=\ell nx\Rightarrow du=\frac{1}{x}dx$

$\Rightarrow\int UdU=\frac{U^2}{2}+c=\frac{(\ell nx)^2}{2}+c$

second : $\int xe^xdx$, Let $U=x$, $dV=e^xdx$

$dU=1dx$ $V=e^x$

$\therefore\int UdV=Uv-\int VdU=xe^x-\int e^xdx=xe^x-e^x$

Therefore, $\int\left(\frac{\ell n x^2+x^2e^x}{x}\right)dx=\frac{(\ell nx)^2}{2}+xe^x-e^x+c$

38. For what values of a and b is the function $f(x)=\begin{cases}1-3x^2, & \text{for } x\le 1\\ ax+b, & \text{for } x>1\end{cases}$ differentiable at $x=1$?

Answer: B

A. $a=6, b=0$
B. $a=-3, b=1$
C. $a=0, b=-2$
D. $a=-6, b=4$

Explanation:
Here, differentiable at $x=1$
$f'_+(1)=f'_-(a)$ and $f(x)=ax+b$
$\Rightarrow f'(x)=a$, for $x>1$
Also, $f(x)=1-3x^2\Rightarrow f'(x)=-6x$, for $x<1$
$\Rightarrow f'_+(1)=a=-6=f'_-(1)$
Differentiability implies continuity
$\therefore\lim_{x\to1}f(x)=f(1)$
$\Rightarrow\lim_{x\to1}(ax+b)=1-3(1)^2=-2$
$\Rightarrow a+b=-2$, but $a=-6$
$-6+b=-2\Rightarrow b=-2+6=4$

Answer: D

39. Consider the following system of equations: $\begin{cases}ax+by=2\\ x+3y+2z=0\\ 2x+y+z=0\end{cases}$

If the determinant of the coefficient matrix is 2, then what is the solution set of the system?

A. $\{(1,3,-5)\}$ C. $\{(-2,-6,10)\}$
B. $\left\{\left(\frac{1}{a},\frac{1}{b},0\right)\right\}$ D. $\varnothing$

Telegram: https://t.me/ExtremeSeries

Solution: Here at

$$D=\begin{vmatrix}a & b & 0\\ 1 & 3 & 2\\ 2 & 1 & 1\end{vmatrix}=2 \text{ coefficient of matrix}$$

$$\Rightarrow x=\frac{D_x}{D}=\frac{\begin{vmatrix}2 & b & 0\\ 0 & 3 & 2\\ 0 & 1 & 1\end{vmatrix}}{2}=\frac{2\begin{vmatrix}3 & 2\\ 1 & 1\end{vmatrix}-b\begin{vmatrix}0 & 2\\ 0 & 1\end{vmatrix}+0}{2}=\frac{2}{2}=1$$

$$\Rightarrow y=\frac{D_y}{D}=\frac{\begin{vmatrix}a & 2 & 0\\ 1 & 0 & 2\\ 2 & 0 & 1\end{vmatrix}}{2}=\frac{a\begin{vmatrix}0 & 2\\ 0 & 1\end{vmatrix}-2\begin{vmatrix}1 & 2\\ 2 & 1\end{vmatrix}+0}{2}=-\frac{6}{2}=3$$

$$z=\frac{D_z}{D}=\frac{2(1-6)}{2}=-\frac{10}{2}=-5$$

$$\therefore S.S=\{(1,3,-5)\}$$

Answer: A

40. If f and g are continuous on $\mathbb{R}$ and $a, b \in \mathbb{R}$, which one of the following is necessarily true?

A. If $\int_a^b f(x)dx=\int_a^b g(x)dx$, then $f(x)=g(x)$ for all $x\in[a,b]$.

B. If $f'(x)=g'(x)$ for all $x\in[a,b]$, then $\int_a^b f(x)dx=\int_a^b g(x)dx$.

C. $f(x)\geq 3$ for all $x\in[-2,2]$, then $\int_{-2}^{2} f(x)dx\geq 12$.

D. $\int_a^b f(x)dx=\int_b^a f(x)dx$.

Answer: A

41. Which of the following functions could most likely be drawn as in the Figure 1 below?

A. $\frac{x+3}{x+2}$

B. $\frac{x^2-2x}{x^2-4}$

C. $\frac{-x^2-x+6}{x^2-4}$

D. $\frac{x^2+x-6}{x^2-4}$

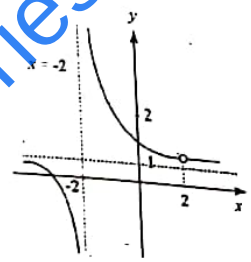


Figure 1

Solution: $\frac{x^2+x-6}{x^2-4}=\frac{(x+3)(x-2)}{(x-2)(x+2)}=\frac{x+3}{x+2}$

Domain $=\mathbb{R}\setminus\{-2,2\}$

- Vertical asymptote is only $x=-2$
- The graph has hole at $x=2$
- Horizontal asymptote is $y=1$,

Answer: D

42. A measurement is grouped into five class intervals with the following frequency distribution.

Class Interval	5–15	15–25	25–35	35–45	45–55
Frequency	22	40	68	50	204

What are the first quartile Q_1 and the 75th percentile p_{75} of the measurement?

A. $Q_1=20, P_{75}=40$ C. $Q_1=20, P_{75}=39$

B. $Q_1=22, P_{75}=40$ D. $Q_1=22, P_{75}=39$

Solution:

Class Interval	5–15	15–25	25–35	35–45	45–55
Frequency	22	40	68	50	20
Cumulative frequency	22	62	130	180	200

First quartile class $=\left(\frac{1\times 200}{4}\right)$ item $=50^{th}$ item

$\rightarrow$ 50^{th} item lies on 2^{nd} class $=15-25$

Telegram: https://t.me/ExtremeSeries

$\Rightarrow B_L = 15, \frac{n}{4} = 50, cf_b = 22, f_c = 40$, and $i = 25 - 15 = 10$

$$\therefore Q_K = B_L + \frac{\left(\frac{k \times n}{4} - cf_b\right) i}{f_c} \text{ where, } K = 1,\ n = 200$$

$$\therefore Q_1 = 15 + \left(\frac{50 - 22}{40}\right) \times 10 = 22$$

Exercise: What is the value of Q_3?

Percentile, $Pt = B_L + \left[\frac{tn}{100} - cf_b\right] i$, where $t = 75, n = 200$

$$P_{75} \text{ class} = \left(\frac{t \times n}{100}\right)^{\text{item}} = \left(\frac{75 \times 200}{100}\right)^{\text{item}} = \left(150^{\text{th}}\right)\text{item}$$

$\rightarrow$ 150^{th} item lies on 4^{th} class $= 35 - 45$

$$B_L = 35; \frac{tn}{100} = 150,\ cf_b = 130, f_c = 50,\ i = 10$$

$$\therefore P_{75} = 35 + \left[\frac{150 - 130}{50}\right] \times 10 = 39$$

Exercise: calculate

a) P_{25}, P_{50} and P_{90}.

Answer: D

43. The number of shoes s that a factory can produce per day is a function of the number of hours t it operates:

$$s(t) = 40t \text{ for } 0 \le t \le 12.$$

The daily cost c, in Birr, to manufacture s shoes is given by the function

$$c(s) = 0.1s^2 + 90s + 800.$$

If the factory operates for 10 hours, what is the cost it incurs in producing as much shoes it can within this time?

A. Birr 400 C. Birr 52,800

B. Birr 1,600 D. Birr 124,600

Solution: $S(10) = 40(10) = 400$

$$C(400) = (0.1)(400)^2 + 90(400) + 800$$
$$= 16{,}000 + 36{,}000 + 800$$
$$= 52{,}800,$$

Answer: C

44. Three persons P_1, P_2 and P_3 are firing at a target independently, and have a probability 0.7, 0.5 and 0.4, respectively of hitting the target. What is the probability that at least one of them hits the target?

A. 0.95 B. 0.85 C. 0.91 D. 0.99

Solution: Here.

$P(P_1) = 0.7 \Rightarrow P(\text{not } P_1) = 1 - 0.7 = 0.3$

$P(P_2) = 0.5 \Rightarrow P(\text{not } P_2) = 1 - 0.5 = 0.5$

$P(P_3) = 0.4 \Rightarrow P(\text{not } P_3) = 1 - 0.4 = 0.6$, then

P(at least one of them hit)

$= 1 - P$ (none)

$= 1 - P$ (not P_1) $\times P$ (not P_2) $\times P$ (not P_3)

$= 1 - P(0.3 \times 0.5 \times 0.6)$

$= 1 - 0.09 = 0.91$

Answer: C

45. A company manufactures x computer sets per month. The monthly marginal profit (in Birr) is given by:

$$P'(x) = 165 - 0.1x, \text{ for } 0 \le x \le 400.$$

The company is currently manufacturing 10 sets of computers per month, but is planning to increase production. What is the total change in the monthly profit if the monthly production increased to 60 sets?

A. Birr 500 C. Birr 8,075

B. Birr 1,865 D. Birr 18,635.

Explanation: Marginal-profit is the derivative of the profit function.

$$P(x) = \int P'(x)dx = \int (165 - 0.1x)dx = 165x - 0.1\frac{x^2}{2} + c$$

$\therefore$ Change in profit $= P(60) - P(10)$

$$= \left[165(60) - 0.1\left(\frac{60^2}{2}\right)\right] - \left[165(60) - \frac{0.1}{2}(10^2)\right]$$

$$= 9720 - 1645 = 8075$$

Answer: C

46. Which one of the following is equal to $\lim_{x \to \infty}\left(\frac{3x}{3x+2}\right)^{-3x}$?

A. e^2 B. e^{-3} C. e^{-2} D. e^3

Telegram: https://t.me/ExtremeSeries

Solution: $\lim_{x\to\infty}\left(\frac{3x}{3x+2}\right)^{-3x} = \lim_{x\to\infty}\left(\frac{1}{1+\frac{2}{3x}}\right)^{-3x}$

$$= \frac{1}{\lim_{x\to\infty}\left(\frac{1}{1+\frac{2}{3x}}\right)^{-3x}} = \frac{1}{\left(e^{\frac{2}{3}}\right)^{-3}} = \frac{1}{e^{-2}} = e^2$$

Answer: A

47. The total cost (in birr) of producing x radio sets per day is given by the expression $\frac{1}{4}x^2+35x+25$ and the price per set at which they may be sold is given by $50-\frac{1}{2}x$. What should be the daily output to obtain a maximum total profit?

A. 50 sets per day C. 10 sets per day
B. 23 sets per day D. 7 sets per day

Explanation: Profit (P) = sale – total cost

$$= (x)(P) - \text{cost}$$

$$\Rightarrow P = x\left(50-\frac{1}{2}x\right) - \left(\frac{1}{4}x^2+35x+25\right)$$

$$= 50x - \frac{1}{2}x^2 - \frac{1}{4}x^2 - 35x - 25$$

$$P(x) = 15x - \frac{3}{4}x^2 - 25$$

For maximizing profit:

$$\frac{dP}{dx} = 0 \Rightarrow 15 - \frac{3x}{2} = 0$$

$$\therefore \frac{3}{2}x = 15 \Rightarrow x = \frac{2\times 15}{3} = 10$$

Answer: C

48. Let $f(x) = \begin{cases} 3-e^{2x} & \text{if } x < 0.5 \\ \frac{2^x-5}{x+1}, & \text{if } x \geq 0.5 \end{cases}$. If c is a zero of f, that is, $f(c) = 0$, then which one of the following intervals must contain c?

A. $(-\infty, 0]$ B. $[0, 1]$ C. $[1, 2]$ D. $[2, 3]$

Solution:

For $x < 0.5$ let $x = 0$

$f(0) = 3 - e^0 = 3 - 1 = 2 \leftarrow$ Positive

For $x \geq 0.5$, let $x = 1$

$f(1) = \frac{2^1-5}{2+1} = \frac{2-5}{3} = \frac{-3}{3} = -1 \leftarrow$ negative

$\therefore$ We can find $c \in (0,1)$ such that $f(c) = 0$

Answer: B

49. What is the solution set of the system $\begin{cases} x+y-z=1 \\ x+2y-3z=1 \\ 2x+3y-4z=2 \end{cases}$?

A. $\{(0,2,1)\}$ C. $\{(2k+1,-k,k)k \in \mathbb{R}\}$
B. $\{(1-k,2k,k)k \in \mathbb{R}\}$ D. $\varnothing$

Solution: Consider the system:

$$\begin{cases} x+y-z=1 & \ldots\ldots \text{A} \\ x+2y-3z=1 & \ldots\ldots \text{B} \\ 2x+3y-4z=2 & \ldots\ldots \text{C} \end{cases}$$

We choose to add equation (A) and (B) $+\begin{cases} x+y-z=1 \\ x+2y-3z=1 \end{cases}$

$$2x+3y-4z=2 \ldots (*)$$

Since equation (C) and (*) are the same, therefore, the system must be **dependent**, provided that the system has infinitely many solutions:

→ From equation A solve for x:

$$x+y-z=1 \Rightarrow x=1-y+z \ldots\ldots (**)$$

Now replace x be $1-y+z$ in equation B.

$$\Rightarrow (1-y+z)+2y-3z=1$$

Telegram: https://t.me/ExtremeSeries

$y-2z=0$, hence $\boxed{y=2z.}$

Replace $y=2z$ in (**)

$\Rightarrow x=1-y+z=1-(2z)+z=1-z$

$\therefore x=1-z$

Thus, for any real number $z=k\in\mathbb{R}$

$(x,y,z)=(1-z,2z,z)=\{(1-k,2k,k)/k\in\mathbb{R}\}$

Answer: B

50. Which one of the following is NOT a tautology?

A. $[p\vee(q\Rightarrow r)]\Rightarrow[\neg p](q\Rightarrow r)$

B. $p\vee(q\Rightarrow\neg p)$

C. $p\Rightarrow(p\Rightarrow q)\vee q$

D. $[p\Rightarrow(q\wedge\neg r)]\Leftrightarrow[\neg p\Rightarrow(q\neg\vee r)]$

Explanation: At autology is true value for any possible combination.

Recall that $a\Rightarrow b\equiv\neg a\vee b$

A. $[p\vee(q\Rightarrow r)]\Rightarrow[\neg p](q\Rightarrow r)$

$p\vee(\neg q\vee r)\Rightarrow[p\vee(\neg q\vee r)\leftarrow$ tautology

B. $p\vee(q\Rightarrow\neg p)\equiv p\vee(\neg q\vee\neg p)$

$=(p\vee\neg p)\vee\neg q$

$=T\vee\neg q$

T $\leftarrow$ tautology

C. $p\Rightarrow(p\Rightarrow q)\vee q$

$p\Rightarrow(\neg p\vee q)\vee q$

$p\Rightarrow(\neg p\vee(q\vee q)$

$p\Rightarrow(\neg p\vee q)$

$\neg p\vee(\neg p\vee q)$

$\neg p\vee q\leftarrow$ not tautology

Answer: C

51. Which one of the following is true about a conic section represented by the equation $\frac{x^2}{k}+\frac{y^2}{k-9}=1$?

A. It is a circle whose center is at origin for some $k\in\mathbb{R}$.

B. It is an ellipse whose major axis is vertical when $k>9$.

C. It is a hyperbola whose foci are at (-3, 0) and (3k, 0) when $0<k<9$.

D. It is a hyperbola whose foci are at (-3k, 0) and (3k, 0) when $0<k<9$.

Solution: $\frac{x^2}{k}+\frac{y^2}{k-9}=1$

A. It cannot be a circle, because $k\neq k-9$

B. When $k>9$, the expression will be horizontal ellipse because $k>k-9$

C. For $0<k<9$, the equation becomes

$\frac{x^2}{k}+\frac{y^2}{k-9}=1\leftarrow$ hyperbola

Now, $a^2=k$ and $b^2=9-k$

$c^2=a^2+b^2=k+9-k=9\Rightarrow c=\pm3$

$\therefore$ The foci are (-3, 0) and (3, 0)

Answer: C

52. Two ships, one with angle of depression 60° due east and the other with 30° due west are observed from a plane 1,000 meters above a sea. If the two ships are on the same line, what is the distance between the two ships?

[*you may use the values:*

$\sin 30°=\cos 60°=\frac{1}{2}$ *and* $\sin 60°=\cos 30°=32.$]

A. $\frac{4{,}000}{\sqrt{3}}$m

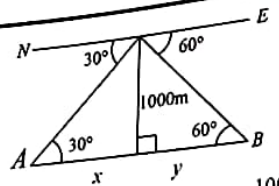
B. 2,000 m

C. $\sqrt{3}500$m

D. $\frac{1}{\sqrt{3}}600$m

Solution: Here at,

Telegram: https://t.me/ExtremeSeries



$$\tan 30° = \frac{1000}{x} \text{ and } \tan 60° = \frac{1000}{y}$$

$$\Rightarrow x = \frac{1000}{\tan 30°} = 1000\sqrt{3}\text{m and } y = \frac{1000}{\tan 60°} = \frac{1000\text{m}}{\sqrt{3}}$$

- The distance between A and B $= x + y$

$$= 1000\sqrt{3} + \frac{1000}{\sqrt{3}} = 1000\left(\sqrt{3} + \frac{1}{\sqrt{3}}\right) = \frac{4000}{\sqrt{3}}\text{m}.$$

Answer: A

53. What is the image of the ellipse $(x-1)^2 + 4y^2 = 1$ under the translation that takes (1, 1) to (0, 2) followed by the reflection through the x-axis?

A. $x^2 + 4(y-1)^2 = 1$ C. $x^2 + 4(y+1)^2 = 1$

B. $4x^2 + (y-1)^2 = 1$ D. $4x^2 + (y+1)^2 = 1$

Solution: Translating vector (v) $= (0-1, 2-1) = (-1, 1)$
When you translating the center of an ellipse (1, 0)

$$(1,0) \xrightarrow{v(-1,1)} (0,1)$$

When you reflect (0, 1) on x-axis is (0, −1)
Since translation is parallel movement, hence the given ellipse is horizontal.

$\therefore$ the image become $x^2 + 4(y+1)^2 = 1$

54. What is the amplitude and period, respectively, of the graph of $f(x) = -6\sin x.\cos x$?

Answer: C

A. $3, \pi$ C. $3, \frac{\pi}{2}$

B. $6, \pi$ D. $6, 2\pi$

Solution: Recall that: $\sin 2x = 2\sin x \cos x$

$$\therefore -3\sin(2x) = -6\sin x \cos x$$

$$\Rightarrow \text{Amplitude} = |-3| = 3 \text{ and Period} = \frac{2\pi}{2} = \pi$$

Answer: A

55. What is the work done (in Joules) when a force of 50 Newton is used to pull a crate 20 meters along a level path if the force is at an angle of 60°?
[Remember that: Work = (Force) × (distance travelled).]

A. 360 B. 500 C. 760 D. $1500\sqrt{2}$

Solution: $W = F \cdot S = |F||S|\cos\theta$

$$\therefore W = (50N)(20)\cos 60° = 500J.$$

Answer: B

56. Suppose ℓ is the line through the center of the sphere $x^2 + y^2 + (z-2)^2 = 9$ and the intersects sphere at (1, 2, 4). What is the cosine of the angle between ℓ and positive z-axis?

A. $\frac{2}{3}$ B. $\frac{1}{3}$ C. $\frac{3}{5}$ D. $\frac{4}{5}$

Solution: Center of sphere c (0, 0, 2) point on the sphere P(1, 2, 4) are on ℓ.

→ Position vector $\overline{CP} = (1-0, 2-0, 4-2) = (1,2,2)$

→ Point on z-axis R = (0, 0, z).

Angle between ℓ and z-axis.

$$\cos\theta = \frac{\overline{CP} \cdot \overline{R}}{|\overline{CP}||\overline{R}|} = \frac{(1,2,2)\cdot(0,0,z)}{\sqrt{1^2+2^2+2^2}\cdot\sqrt{0^2+0^2+z^2}}$$

$$\therefore \cos\theta = \frac{0+0+2z}{\sqrt{1+4+4}\cdot\sqrt{z^2}} = \frac{2z}{\sqrt{9}\cdot\sqrt{z^2}} = \frac{2}{3}.$$

Answer: A

57. Suppose that the equation $x^2 + y^2 + z^2 + 2x + Bz = 6(y+1)$, represents a sphere. Where is the point (1, -1, 4) located relative to the sphere?

A. Inside the sphere
B. On the sphere
C. At the center of the sphere
D. Outside the sphere

Telegram: https://t.me/ExtremeSeries

Solution: Here at,

$x^2+y^2+z^2+2x+8z-6y=6$

$\Rightarrow (x^2+2x+1)+(y^2-6y+9)+z^2+8z+16$

$=6+1+9+16$

$\Rightarrow (x+1)^2+(y-3)^2(z+4)^2=32$; Replace the

Point $(1,-1,4):(1+1)^2+(-1-3)^2+(4+4)^2=84>32$

$\therefore$ the point (1, -1, 4) lies outside the sphere

Answer: D

58. If $P=(3,\alpha-1,\alpha+2)$ and $Q=(2\alpha+1,3,3\alpha)$ are points in space, what should be the value (s) of α so that the distance between the two points is 6?

A. $\alpha=-2$ or $\alpha=5$ C. $\alpha=-1$ or $\alpha=3$

B. $\alpha=0$ or $\alpha=5$ D. $\alpha=-3$ or $\alpha=2$

Solution: Here distance $PQ=d=6$

$\Rightarrow d=\sqrt{(2\alpha+1-3)^2+(3-(\alpha-1))^2+(3\alpha-\alpha-2)^2}=6$

$\Rightarrow \sqrt{(2\alpha-2)^2+(4-\alpha)^2+(2\alpha-2)^2}=6$

$\Rightarrow \sqrt{(4\alpha^2-8\alpha+4)+(16-8\alpha-\alpha^2)+(4\alpha^2-8\alpha+4)}=6$

Squaring both side:

$9\alpha^2-24\alpha+24=36 \Leftrightarrow 3\alpha^2-8\alpha+8=12$

$3\alpha^2-8\alpha-4=0$

$\therefore \alpha=\frac{8\pm\sqrt{64+48}}{6}=\frac{8\pm4\sqrt{7}}{6}=\frac{4\pm2\sqrt{7}}{3}$

59. If (−1, 2, 2) and (1, 0, -2) are endpoints of a diameter of a sphere, then which one of the following is true about the sphere?

A. (0, 1, 0) is a point on the sphere

B. The equation of the sphere is $x^2+(y-1)^2+z^2=6$

C. The equation of the sphere is $x^2+(y-2)^2+z^2=6$

D. The radius of the sphere is 6.

Solution: Center $=\left(\frac{1+(-1)}{2},\frac{2+0}{2},\frac{-2+2}{2}\right)=(0,1,0)$

radius $r=\sqrt{(0-1)^2+(1-0)^2(0+2)^2}=6$

Equation: $x^2+(y-1)^2+z^2=6$ **Answer: B**

60. What is the possible value of x that solves the equation: $\sin^{-1}x+\cos^{-1}\left(\frac{3}{5}\right)=\pi$?

A. $\frac{\pi}{3}$ B. $\frac{3}{5}$ C. $\frac{5\pi}{2}$ D. $\frac{4}{5}$

Solution: $\sin^{-1}x+\cos^{-1}\left(\frac{3}{5}\right)=\pi$

$\Rightarrow \sin^{-1}x=\pi-\cos^{-1}\left(\frac{3}{5}\right)$

$\Rightarrow x=\sin\left(\pi-\cos^{-1}\left(\frac{3}{5}\right)\right)$

$\Rightarrow x=\sin\pi\cos\left(\cos^{-1}\left(\frac{3}{5}\right)\right)-\cos\pi\sin\left(\cos^{-1}\left(\frac{3}{5}\right)\right)$

$\Rightarrow x=0-(-1)\sin\left(\cos^{-1}\left(\frac{3}{5}\right)\right)$, Let $\cos^{-1}\left(\frac{3}{5}\right)=\theta$

$\Rightarrow \cos\theta=\frac{3}{5}$, therefore $\sin\theta=\frac{4}{5}$

$\therefore x=\frac{4}{5}$ **Answer: D**

61. If $\vec{A}=4\vec{i}-3\vec{j}$ and $\vec{u}$ is a unit vector such that $\left|\vec{A}+\vec{u}\right|^2=27$, then the cosine of the angle between $\vec{A}$ and $\vec{u}$ is equal to ___.

A. 0.1 B. 0.2 C. 0.3 D. 0.4

Solution: $\cos\theta=\frac{\vec{A}\cdot\vec{U}}{\left|\vec{A}\right|\left|\vec{U}\right|}$, Recall that

$\left|\vec{A}+\vec{U}\right|^2=\left(\vec{A}+\vec{U}\right)\cdot\left(\vec{A}+\vec{U}\right)=27=|A|^2+2A\cdot U+|U|^2=27$

$\Rightarrow \sqrt{4^2+3^2}^2+2A\cdot U+1=27$

Telegram: https://t.me/ExtremeSeries

$$\therefore \vec{A}\cdot\vec{U} = \frac{27-26}{2} = \frac{1}{2}$$

$$\cos\theta = \frac{\vec{A}\cdot\vec{U}}{\|\vec{A}\|\|\vec{U}\|} = \frac{\frac{1}{2}}{\sqrt{25}\cdot 1} = \frac{1}{2\times 5} = \frac{1}{10} = 0.1$$

Answer: A

62. If each the compound propositions $P \vee Q, P \Rightarrow R$ and is $\neg R$ is True, then which one of the following is True?

A. P B. Q C. $Q \Rightarrow P$ D. $P \wedge \neg R$

Solution: $\neg R \equiv T, R = F$

i) $P \Rightarrow R \equiv T$

$P \Rightarrow F \equiv T, P \equiv F$

ii) $P \vee Q \equiv T$ and $P \equiv F$

$F \vee Q \equiv T$, therefore, $Q \equiv T$

Answer: B

63. If $A = (1,-2), B = (-3,2)$ and $\vec{V}$ is a position vector such $2\vec{V} + \overrightarrow{AB} = \vec{0}$, then $\vec{V}$ is equal to _____.

A. (2, 0) B. (−1, 0) C. (−2, 2) D. (2, −2)

Solution: $2\vec{V} + \overrightarrow{AB} = \vec{0}$

$2\vec{V} = -\overrightarrow{AB} \Rightarrow 2\vec{V} = \overrightarrow{BA} = (1-(-3), -2-2) = (4,-4)$

$\therefore V = (2,-2)$

Answer: D

64. If angle θ is an acute angle of a right triangle, what is the length of the side adjacent to θ, given that the hypotenuse has 6 units length $\sec\theta = 10/3$?

A. 1.8 units C. 18 units

B. 2 units D. 20 units

Solution: $\sec\theta = \frac{10}{3} \Rightarrow \cos\theta = \frac{3}{10}$

$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenus}} = \frac{x}{6} = \frac{3}{10}$$

$$\therefore x = 6 = \left(\frac{3}{10}\right) = \frac{18}{10} = 1.8 \text{ units.}$$

Answer: A

1. If the truth value of a proposition p is false, then which one of the following compound proposition has a truth value true?

A. $\neg p \wedge p$ C. $\neg(\neg p \vee p)$

B. $\neg p \Rightarrow p$ D. $p \Rightarrow \neg p$

Hint: $\neg p \Rightarrow p \equiv \neg(\neg p) \vee p \equiv p \vee p \equiv F \vee F \equiv F$

- $\neg(\neg p \vee p) \equiv p \wedge \neg p \equiv F \wedge T \equiv F$
- $p \Rightarrow \neg p \equiv \neg p \vee p \equiv T \vee F \equiv T$

Answer: D

2. What is the contra positive of "If $x \in \mathbb{N}$, then x is integer and $x > 0$."?

A. If x is not integer or $x < 0$, then $x \notin \mathbb{N}$.

B. If x is integer or $x > 0$, then $x \in \mathbb{N}$.

C. If x is not integer or $x \leq 0$, then $x \notin \mathbb{N}$.

D. If $x \notin \mathbb{N}$, then x is not integer and $x \leq 0$.

Hint: The contra positive of "If $x \in \mathbb{N}$, then x is integer and $x > 0$."? Let $p \equiv x \in \mathbb{N}$, $q \equiv$ is an integer, $r \equiv x > 0$.

- The equivalent statement: $p \Rightarrow (q \wedge r)$
  The contra positive of $p \Rightarrow (q \wedge r)$ is $\neg(q \wedge r) \Rightarrow \neg p$
  $\equiv (\neg q \vee \neg r) \Rightarrow \neg p$, the same as
- $\neg p \equiv x \notin \mathbb{N}$, $\neg r \equiv x \leq 0$.

If x is not integer or $x \leq 0$ then $x \notin \mathbb{N}$. **Answer: C**

3. Which one of the following compound propositions is a tautology?

A. $(q \vee \neg q) \Rightarrow p$ C. $p \vee (q \wedge \neg q)$

B. $p \Rightarrow (q \vee \neg q)$ D. $p \Rightarrow (q \wedge \neg q)$

Hint:

A. $(q \vee \neg q) \Rightarrow p \equiv T \Rightarrow p \equiv T$

B. $p \Rightarrow (q \vee \neg q) \equiv p \Rightarrow T \equiv T$....(tautology)

C. $p \vee (q \wedge \neg q) \equiv p \vee F \equiv F$

D. $p \Rightarrow (q \wedge \neg q) \equiv p \Rightarrow F$

Answer: B

Telegram: https://t.me/ExtremeSeries

4. The following is a simple frequency distribution of a data with variable X.

X	3	5	6	7
Frequency	2	5	2	1

What are the mean $(\bar{x})$ and variance δ^2 of the data?

A. $\bar{x}=5,\ \delta^2=0.7$
B. $\bar{x}=6,\ \delta^2=1.4$
C. $\bar{x}=6,\ \delta^2=0.7$
D. $\bar{x}=5,\ \delta^2=1.4$

Hint:

- Mean $(\bar{x})=\dfrac{3\times2+5\times5+6\times2+7\times1}{2+5+2+1}=\dfrac{50}{10}=5$
- Variance $=\dfrac{2(3-5)^2+5(5-5)^2+2(6-5)^2+1(7-5)^2}{10}$

$$=\frac{14}{10}=1.4$$

Answer: D

5. Which one of the following is a one-to-one correspondence function from A = [0, 1] to B = [1, 2]?

A. $f(x)=x$
B. $f(x)\ \frac{1}{3}x^3+1$
C. $f(x)=2x+1$
D. $f(x)=x^2+1$

Hint: Suppose $f: A\to B$
$\Rightarrow f:[0,1]\to[1,2]$ defined by $f(x)=x^2+1$ is one to one correspondence because, $f(0)=0^2+1=1$ and $f(1)=1^2+1=2$

Answer: D

6. What is the solution set of $\dfrac{1-\frac{1}{x}}{1-\frac{1}{x^2}}=3x^2-\dfrac{x}{1+\frac{1}{x}}$?

A. $\left\{-1,\frac{1}{3}\right\}$ B. $\left\{\frac{1}{3}\right\}$ C. $\left\{3,\frac{-1}{3}\right\}$ D. $\left\{\frac{-1}{3}\right\}$

Solution: Simplifying each side:

$$\frac{1-\frac{1}{x}}{1-\frac{1}{x^2}}=\frac{1-\frac{1}{x}}{\left(1-\frac{1}{x}\right)\left(1+\frac{1}{x}\right)}=\frac{1}{1+\frac{1}{x}}=\frac{x}{x+1}\ldots(*)$$

and

$$3x^2-\frac{x}{1+\frac{1}{x}}=3x^2-\frac{x^2}{x+1}=x^2\left(3-\frac{1}{x+1}\right)\ldots(**)$$

From (*) and (**) we have

$$\frac{x}{x+1}=x^2\left(3-\frac{1}{x+1}\right)\Rightarrow\frac{1}{x+1}=x\left(3-\frac{1}{x+1}\right)$$

$$\Rightarrow\frac{1}{x+1}=3x-\frac{x}{x+1}\Leftrightarrow 3x=\frac{1}{x+1}+\frac{x}{x+1}$$

$$\Rightarrow 3x=\frac{x+1}{x+1}=1\Rightarrow x=\frac{1}{3}$$

Answer: B

7. Suppose $f(x)=\dfrac{Q(x)}{x\left(x^2-1\right)}$ where $Q(x)$ is a quadratic function. Which one of the following is necessarily true about the graph of f?

A. $x=0$, $x=1$ and $x=-1$ are the vertical asymptotes of the graph of f.
B. The graph of f does not interest with its horizontal asymptote.
C. The vertical asymptote of the graph of f is only $x=-1$ if $Q(x)=x^2-x$.
D. The vertical asymptote of the graph of f is only $x=1$ if $Q(x)=2x^2$.

Answer: C

8. Which one of the following is equation of a circle whose center is on y-axis and radius is 3?

A. $x^2+y^2+6y=0$
B. $(x-2)^2+y^2=9$
C. $x^2+(y-2)^2=3$
D. $x^2-2x+y^2=8$

Hint: $x^2+y^2+6y=0\Leftrightarrow x^2+y^2+6y+9=0+9$
$x^2+(y+3)^2=9,\Leftrightarrow x^2+(y+3)^2=3^2$

Answer: A

Telegram: https://t.me/ExtremeSeries

9. The planet mercury's orbit around the sun is an ellipse with eccentricity 0.206, length of the major axis 1.16×10^8 km and the sun at one focus. what is the maximum distance from mercury to the sun?

A. 6.99×10^8 km C. 9.66×10^7 km
B. 6.99×10^7 km D. 9.66×10^8 km

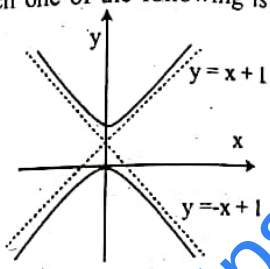
Hint: $2a = 1.16\times10^8$ km $\Rightarrow a = 0.58\times10^8$ km $\Rightarrow a = 5.8\times10^7$ km

$e=\frac{c}{a}\Rightarrow c = ea = (0.206)(0.58\times10^8\text{ km}) = \mathbf{1.195\times10^7\ km}$

Max-distance = a + c = $(5.8 + 1.95)\times10^7$ km = $\mathbf{6.99\times10^7\ km}$

Answer: B

10. The graph of a hyperbola and the lines of its asymptotes are as shown in the following figure. which one of the following is an equation of the hyperbola?



A. $y^2-2y-x^2=0$
B. $y^2-3y-x^2=0$
C. $x^2-(y-1)^2=1$
D. $(x-1)^2-y^2=1$

Hint: To find the center, solve

$\begin{cases} y = x+1 \\ y = -x+1 \end{cases} \Rightarrow x+1 = -x+1 \Rightarrow 2x = 0 \Rightarrow x = 0$ and $y = 0+1 = 1$

Thus, center (h, k)=(0, 1). Also $2a = 2 \Rightarrow a = 1$

Because transverse axis is parallel to y-axis, hence standard equation.

$$\frac{(y-k)^2}{a^2}-\frac{(k-h)^2}{b^2}=1\Leftrightarrow\frac{(y-1)^2}{1}-\frac{(x-0)^2}{1}=1$$

$$\Rightarrow y^2-2y+1-x^2=1\Leftrightarrow y^2-2y-x^2=0$$

$$\Rightarrow y^2-2y+1-x^2=1\Leftrightarrow y^2-2y-x^2=1-1=0$$

Answer: A

11. If $\{A_n\}_{n=1}^{\infty}$ is an arithmetic sequence such that its 1st term $A_1=-5$ and its 5th term $A_5=15$, then its 11th term A_{11} is equal to:

A. 40 B. 50 C. 45 D. 55

Hint: $d=\frac{A_n-A_m}{n-m}=\frac{A_5-A_1}{5-1}$

$$\Rightarrow d=\frac{15-(-5)}{5-1}=\frac{20}{4}=5.$$

$$A_n = A_1+(n-1)d \Leftrightarrow A_{11} = -5+(11-1)(5) = 45$$

Answer: C

12. What is the sum of all multiples of 4 that are between 30 and 301?

A. 12,882 B. 11,288 C. 6,288 D. 6,882

Hint: 32, 36, 40…300. Here, $A_1 = 32$, d = 4

$$\Rightarrow A_n = A_1+(n-1)d \Leftrightarrow 300 = 32+(n-1)4$$

$$\Rightarrow n=\frac{300-32+4}{4}=\frac{272}{4}=68.$$

Hence, $S_{68}=\frac{68}{2}[32+300]=\mathbf{11{,}288}$ **Answer: B**

13. The left hand side limit, $\lim\limits_{x\to0^-}\frac{xe^x-|x|}{x}$

A. 0 B. 2 C. 1 D. Does not exist

Hint: $\lim\limits_{x\to0^-}\frac{xe^x-|x|}{x}=\lim\limits_{x\to0^-}\left(\frac{xe^x}{x}-\frac{|x|}{x}\right)$

$$=\lim_{x\to0^-}e^x+(1)=e^0+1=1+1=2$$

Answer: B

14. Which one of the following is equal to $\lim\limits_{x\to0}\frac{2x+\tan x}{x\sec x}$?

A. 2 B. 0 C. 1 D. 3

Hint: $\lim\limits_{x\to0}=\frac{2x+\tan x}{x\sec x}=\lim\limits_{x\to0}\frac{\frac{2x+\tan x}{x}}{\frac{x\sec x}{x}}$

$$=\lim_{x\to0}\left(\frac{2+\frac{\tan x}{x}}{\sec x}\right)=\frac{2+1}{1}=3$$

Answer: D

Telegram: https://t.me/ExtremeSeries

15. A box contains 10 items of which 3 are defective. If 2 items are randomly taken out of the box, what is the probability that both items are not defective?

A. $\frac{7}{10}$ B. $\frac{4}{7}$ C. $\frac{7}{15}$ D. $\frac{49}{100}$

Hint: If 3 are defective then 7 are non-defective.

$$P(\text{both non-defective}) = \frac{C(7,2)}{C(10,2)} = \frac{7\times 6}{10\times 9} = \frac{7}{15}$$

Answer: C

16. Items produced by a certain company are subjected to two kinds of defects D_1 and D_2. Out of the total product, 5% have the defect D_1, 10% have the defect D_2, 2% have both defects. What is the probability that a randomly both defects. What is the probability that a randomly selected item has neither defect D_1 nor defect D_2?

A. 0.13 B. 0.5 C. 0.98 D. 0.87

Hint: $P(D_1) + P(D_2) - P(D_1 \cap D_2) + x = 100\%$

$\Rightarrow 5\% + 10\% - 2\% + x = 100\%$

$\Rightarrow x = 100\% - 13\% = 87\% = 0.87$

Answer: D

17. If $A = \begin{pmatrix} 2 & 7 \\ 1 & 3 \end{pmatrix}$ and $B^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$, then $(AB)^{-1}$ is equal to

A. $\begin{pmatrix} 4 & -3 \\ 4 & -5 \end{pmatrix}$ C. $\begin{pmatrix} -3 & 11 \\ 1 & -3 \end{pmatrix}$

B. $\begin{pmatrix} -2 & 5 \\ 2 & -4 \end{pmatrix}$ D. $\begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}$

Hint: If $x = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $D = ad - bc \neq 0$ then x is invertible and

$$A^{-1} = \begin{pmatrix} \frac{d}{D} & -\frac{b}{D} \\ -\frac{c}{D} & \frac{a}{D} \end{pmatrix}$$

If $A = \begin{pmatrix} 2 & 7 \\ 1 & 3 \end{pmatrix}$ and $D = 6 - 7 = -1$, then

$$A^{-1} = \begin{pmatrix} \frac{3}{-1} & \frac{-7}{-1} \\ \frac{-1}{-1} & \frac{2}{-1} \end{pmatrix} = \begin{pmatrix} -3 & 7 \\ 1 & -2 \end{pmatrix}$$

We have $(AB)^{-1} = B^{-1}A^{-1}$

$$\Rightarrow \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} -3 & 7 \\ 1 & -2 \end{pmatrix} = \begin{pmatrix} -3+1 & 7+(-2) \\ 0+2 & 0+(-4) \end{pmatrix} = \begin{pmatrix} -2 & 5 \\ 2 & -4 \end{pmatrix}$$

Answer: B

18. Let $A = \begin{pmatrix} 0 & \alpha & \beta \\ 2 & 2 & 1 \\ 3 & 1 & 2 \end{pmatrix}, b = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix}$ and $X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$. If det(A) = 3, then what is the solution of the system $AX = b$?

A. $\{(6, -2, -8)^T\}$ C. $\{(3, 1, 4)^T\}$

B. $\left\{\left(0, \frac{1}{a}, \frac{5}{b}\right)^T\right\}$ D. $\varnothing$

Hint: $AX = b \Rightarrow (A^{-1}A)X = A^{-1}B$

$X = A^{-1}b$

- Co-factor of a_{11} is $A_{11} = (-1)^{1+1}\begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = 4 - 1 = 3$
- Co-factor of a_{12} is $A_{12} = -\begin{vmatrix} 2 & 1 \\ 3 & 2 \end{vmatrix} = -(4 - 3) = -1$
- Co-factor of a_{13} is $A_{13} = 2 - 6 = -4$
- Co-factor of a_{21} is $A_{21} = -(2\alpha - \beta) = \beta - 2\alpha$
- Co-factor of a_{22} is $A_{22} = -3\beta$
- Co-factor of a_{23} is $A_{23} = -3\alpha$

Cofactor of $a_{31} = \alpha - 2\beta$,
Cofactor of a_{32} is $A_{32} = -2\beta$
Cofactor of a_{33} is $A_{33} = 2\alpha$.

Telegram: https://t.me/EXtremeSeries

Matrix cofactor of $A=\begin{bmatrix} 3 & -1 & -4 \\ \beta-2\alpha & -3\beta & -3\alpha \\ \alpha-2\beta & -3\alpha & 2\alpha \end{bmatrix}$

$\therefore$ adj of $A=\begin{bmatrix} 3 & \beta-2\alpha & \alpha-2\beta \\ -1 & -3\beta & -2\beta \\ -4 & -3\alpha & 2\alpha \end{bmatrix}$

$$\therefore X=A^{-1}b=\left(\frac{1}{|A|}adj.A\right)(b)=\frac{1}{3}\begin{pmatrix} 3 & \beta-2\alpha & \alpha-2\beta \\ -1 & -3\beta & -2\beta \\ -4 & -3\alpha & 2\alpha \end{pmatrix}\begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix}$$

$X=(6 \quad -2 \quad -8)^T$

Answer: A

19. In the set of complex numbers, what is the solution set of $x^2+4x+5=0$?

A. $\varnothing$
B. $\{1-2i, 1+2i\}$
C. $\{2-i, 2+i\}$
D. $\{-2-i, -2+i\}$

Solution: $x^2+4x+5=0$

$\Rightarrow x^2+4x+4+5-4=0 \Leftrightarrow (x+2)^2+1=0$

$\Rightarrow (x+2)^2=-1 \Rightarrow x+2=\sqrt{-1}=\pm i$

$\Rightarrow x=-2-i$ or $x=-2+I$

Answer: D

20. If $z=(1+i)^{10}$, then which of the following is equal to z?

A. $1+32i$ B. $32i$ C. $10i$ D. $1+10i$

Hint: Modulus of $1+i$ is $r=\sqrt{1^2+1^2}=\sqrt{2}$

$$\text{Arg}\,(1+i)=\theta=\tan^{-1}\left(\frac{1}{1}\right)=\tan^{-1}(1)=\frac{\pi}{4}$$

$$(1+i)^{10}=\left(\sqrt{2}\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)\right)^{10}$$

$$\left(\sqrt{2}^{10}\right)\left(\cos\frac{10\pi}{4}+i\sin\frac{10\pi}{4}\right)=32(0+i)=32i$$

Answer: B

21. A closed cylindrical can is to be made to hold 1000cm^3 of oil. What are the dimensions (radius r and height h) that will minimize the total surface area of the can?

A. $r=\frac{\sqrt[3]{50}}{\pi}, h=2\frac{\sqrt[3]{50}}{\pi}$
B. $r=\frac{\sqrt[3]{500}}{\pi}, h=2\frac{\sqrt[3]{500}}{\pi}$
C. $r=\sqrt[3]{\frac{50}{\pi}}, h=2\sqrt[3]{\frac{50}{\pi}}$
D. $r=\sqrt[3]{\frac{500}{\pi}}, h=2\sqrt[3]{\frac{500}{\pi}}$

Hint: Volume $=\pi r^2h=1000 \Rightarrow h=\frac{1000}{\pi r^2}$

Surface area: $A=2\pi r^2+2\pi rh$

$$A(r)=2\pi r^2+2\pi r\left(\frac{1000}{\pi r^2}\right)=2\pi r^2+\frac{2000}{r}$$

$$\Rightarrow A'(r)=4\pi r-\frac{2000}{r^2},\ set\ A'(r)=0$$

$$\Rightarrow r^3=\frac{500}{\pi} \Rightarrow r=\sqrt[3]{\frac{500}{\pi}}$$

Answer: D

Extreme Tip 1: The volume of a closed a cylindrical tin can is 16π cubic inches. If a minimum amount of tin is to be used to construct the can, what must be the height of the can in inches?

A. $2\sqrt[3]{2}$ B. $2\sqrt[3]{4}$ C. 4 D. 8

Solution: Volume $=\pi r^2h=16\pi \Rightarrow h=\frac{16}{r^2}$

Surface area: $A=2\pi r^2+2\pi rh$

$$A(r)=2\pi r^2+2\pi r\left(\frac{16}{r^2}\right)=2\pi r^2+\frac{32\pi}{r}$$

$$A'(r)=4\pi r-\frac{32\pi}{r^2} \longleftarrow \text{set } A'(r)=0$$

$$r^3=\frac{32\pi}{4\pi}=8 \Rightarrow r=\sqrt[3]{8}=2$$

$$h=\frac{16}{r^2}=\frac{16}{4}=4$$

Answer: C

Telegram: https://t.me/ExtremeSeries

22. If $f(x) = 3x^2\sqrt{x^3-1}$, then which of the following is an anti derivative of $f(x)$?

A. $\frac{3x^2}{2}(x^3-1)^{3/2}+c$

B. $\frac{2}{3}(x^3-1)^{3/2}$

C. $\frac{3}{2}(x^3-1)^{3/2}$

D. $\frac{3}{2}(x^3-1)^{2/3}+c$

Hint: Anti derivative of $f(x)$ is $\int f(x)\,dx$

$=\int 3x^2\sqrt{x^3-1}\,dx$, let $U = x^3 - 1$, thus $du = 3x2dx$

$$\Rightarrow \int \sqrt{U}\,dU = \frac{U^{1/2+1}}{\frac{1}{2}+1} = \frac{U^{3/2}}{\frac{3}{2}} + c = \frac{2}{3}U^{3/2} + c$$

$$\Rightarrow \int 3x^2\sqrt{x^3-1}\,dx = \frac{2}{3}(x^3-1)^{3/2} + c$$

Answer: B

Extreme Tip 2: The value of $\int \frac{x^2}{e^{x^3}}dx$ _____

Solution: Let $U = x^3 \Rightarrow dU = 3x^2dx$

$$\Rightarrow x^2dx = \frac{dU}{3}$$

$$\therefore \int \frac{x^2}{e^{x^3}}dx = \int \frac{dU}{3e^U} = \left(\frac{1}{3}\right)\left(\frac{e^{-U}}{-1}\right) = \frac{-1}{3}e^{-x^3} = \frac{-1}{3e^{x^3}} + c$$

23. Which of the following is equal to $\int \frac{(\ln x)^2 + x^2\cos x}{x}dx$?

A. $\frac{1}{x^2}\ln x + x\sin x - \cos x + c$

B. $\frac{1}{3}(\ln x)^3 + x\sin x - \cos x + c$

C. $\frac{1}{3}(\ln x)^3 + x\sin x + \cos x + c$

D. $\frac{1}{x^2}\ln x + x\sin x + \cos x + c$

Hint: $\int \frac{(\ln x)^2 + x^2\cos x}{x}dx$

$$= \int \frac{(\ln x)^2}{x}dx + \int \frac{x^2\cos x}{x}dx$$

First $\int \frac{(\ln x)^2\,dx}{x}$, Let $U = \ln x \Rightarrow dU = \frac{dx}{x}$

$$\int U^2 dU = \frac{U^3}{3} + c = \frac{(\ln x)^3}{3} + c$$

The second part $\int \frac{x^2\cos x}{x}dx = \int x\cos x\,dx$

Using integration by part

$\int U dv = Uv - \int v dU$

Let $U = x \Rightarrow dU = dx$

$dv = \cos x dx \Rightarrow \int dv = \int \cos x dx \Rightarrow v = \sin x$

$$\therefore \int x\cos x dx = x\sin x - \int \sin x dx$$

$$= x\sin x + \cos x + c$$

$$\therefore \int \frac{(\ln x)^2 + x^2\cos x}{x}dx = \frac{(\ln x)^3}{3} + x\sin x + \cos x + c$$

Answer: C

24. The value of the solid generated when the region bounded between the graph of $y = \begin{cases} x^2, & 0 \le x \le 2 \\ 4, & 2 \le x \le 3 \end{cases}$ and x-axis is rotated about the x-axis is:

A. $\frac{32\pi}{5}$ B. $\frac{112\pi}{5}$ C. $\frac{112\pi}{3}$ D. $\frac{64\pi}{5}$

Hint: $v_1 = \int_0^2 \pi\left(x^2\right)^2 dx = \pi\left[\frac{x^5}{5}\right]_0^2 = \frac{32\pi}{5}$

$$v_2 = \int_2^3 \pi(4)^2 dx = \pi[16x]_2^3 = 48\pi - 32\pi = 16\pi$$

$$\therefore v = v_1 + v_2 = \frac{32\pi}{5} + 16\pi = \frac{112\pi}{5}$$

Answer: B

Telegram: https://t.me/ExtremeSeries

Extreme Tip 3: The region is enclosed by the graph of $y = x^2$, the line $x = 2$, and the x-axis is revolved about the y-axis. The volume of solid generated is:

A. 8π B. $\frac{32\pi}{5}$ C. $\frac{16\pi}{3}$ D. 4π

Solution: Since it is revolved about y-axis.

$\Rightarrow y = x^2 \Rightarrow x = \sqrt{y}$ when $x = 2, y = 4$

$$\therefore v = \int_0^4 \pi f(y)^2\,dy = \int_0^4 \pi\left(\sqrt{y}\right)^2 dy = \pi\left[\frac{y^2}{2}\right]_0^4$$

$$= \pi\left[\frac{16}{2} - 0\right]_0^4 = 8\pi$$

Answer: A

25. Which one of the following is true about the horizontal asymptote(s) of the graph of $y = \frac{|x|+2}{x}$?

A. $y = 2$ is the only horizontal asymptote of the graph.
B. $y = 1$ and $y = -1$ are horizontal asymptotes of the graph.
C. $y = 2$ and $y = -2$ are the horizontal asymptotes of the graph.
D. $y = 1$ is the only horizontal asymptote of the graph.

Hint: $y = \frac{|x|-2}{x} = \begin{cases} 1 + \frac{2}{x} & \text{when } x > 0 \\ -1 + \frac{2}{x} & \text{when } x < 0 \end{cases}$

- As $x \to \infty$, $y = 1$ and
- As $x \to -\infty$, $y = -1$

Answer: A

26. If $f(x) = 2x^5 - 3x$, then $\lim_{x \to 1} \frac{f(x) - f(1)}{x - 1}$ is equal to _____.

A. 1 B. −1 C. 7 D. ∞

Hint: $\lim_{x \to 1} \frac{f(x) - f(1)}{x - 1} = f'(1)$

and $f'(x) = 10x^4 - 3$, so $f'(1) = 10(1)^4 - 3 = 7$

Answer: C

27. If $f(x) = e^{3x}\cos x - \frac{x+\pi}{x^2+2}$, then $f'(0)$ is equal to _____.

A. $3 - \frac{\pi}{2}$ B. $\frac{3}{2}$ C. $\frac{7}{4}$ D. $\frac{5}{2}$

Answer: D

28. If $f(x) = \ln\left(\sqrt{x^2-5}\right)$, which one of the following is equal to $f'(x)$?

A. $\frac{x}{x^2-5}$ B. $\frac{-x}{\sqrt{x^2-5}}$ C. $\frac{2x}{\sqrt{x^2-5}}$ D. $\frac{-x}{x^2-5}$

Hint:

Use $= \left(\ln\left(g(x)\right)\right)' = \left(n \ln g(x)\right)' = \frac{n\,g'(x)}{g(x)}$

$$\therefore \left(\ln\sqrt{x^2-5}\right)' = \left(\frac{1}{2}\ln\left(x^2-5\right)\right)' = \frac{1}{2}\left(\frac{2x}{x^2-5}\right) = \frac{x}{x^2-5}$$

Answer: A

29. Let f be twice differentiable function on $\Re$. Which one of the following is necessarily true?

A. If $f'(c) = 0$, at some $c \in \Re$, then f has a relative extreme value at $x = c$.
B. If $f'(x)$ is increasing, then the graph of $y = f(x)$ is concave upward.
C. If $f'(x) = 0$ for all $x \in \Re$, then $f(x) = 0$ for all $x \in \Re$.
D. If $f(x)$ is increasing, then $f''(x) \geq 0$ for all $x \in \Re$.

Hint:

If the graph of $f'(x)$ is increasing, so that $f''(x) > 0$, then the graph of $y = f(x)$ is concave upward.

Answer: B

For further explanation refer extreme maths page 716. Q#61

Telegram: https://t.me/ExtremeSeries

Extreme Tip 3: The region is enclosed by the graph of $y = x^2$, the line $x = 2$, and the x-axis is revolved about the y-axis. The volume of solid generated is:

A. 8π B. $\frac{32\pi}{5}$ C. $\frac{16\pi}{3}$ D. 4π

Solution: Since it is revolved about y-axis.

$$\Rightarrow y = x^2 \Rightarrow x = \sqrt{y} \text{ when } x = 2, y = 4$$

$$\therefore v = \int_0^4 \pi f(y)^2\, dy = \int_0^4 \pi\left(\sqrt{y}\right)^2 dy = \pi\left[\frac{y^2}{2}\right]_0^4$$

$$= \pi\left[\frac{16}{2} - 0\right]_0^4 = 8\pi$$

Answer: A

25. Which one of the following is true about the horizontal asymptote(s) of the graph of $y = \frac{|x|+2}{x}$?

A. $y = 2$ is the only horizontal asymptote of the graph.
B. $y = 1$ and $y = -1$ are horizontal asymptotes of the graph.
C. $y = 2$ and $y = -2$ are the horizontal asymptotes of the graph.
D. $y = 1$ is the only horizontal asymptote of the graph.

Hint: $y = \frac{|x|+2}{x} = \begin{cases} 1+\frac{2}{x} & \text{when } x > 0 \\ -1+\frac{2}{x} & \text{when } x < 0 \end{cases}$

- As $x \to \infty$, $y = 1$ and
- As $x \to -\infty$, $y = -1$

Answer: A

26. If $f(x) = 2x^5 - 3x$, then $\lim_{x \to 1} \frac{f(x) - f(1)}{x - 1}$ is equal to _____.

A. 1 B. −1 C. 7 D. ∞

Hint: $\lim_{x \to 1} \frac{f(x) - f(1)}{x - 1} = f'(1)$

and $f'(x) = 10x^4 - 3$, so $f'(1) = 10(1)^4 - 3 = 7$

Answer: C

27. If $f(x) = e^{3x}\cos x - \frac{x+\pi}{x^2+2}$, then $f'(0)$ is equal to _____.

A. $3 - \frac{\pi}{2}$ B. $\frac{3}{2}$ C. $\frac{7}{4}$ D. $\frac{5}{2}$

Answer: D

28. If $f(x) = \ln\left(\sqrt{x^2-5}\right)$, which one of the following is equal to $f'(x)$?

A. $\frac{x}{x^2-5}$ B. $\frac{-x}{\sqrt{x^2-5}}$ C. $\frac{2x}{\sqrt{x^2-5}}$ D. $\frac{-x}{x^2-5}$

Hint:

Use $= \left(\ln\left(g(x)\right)^n\right)' = \left(n \ln g(x)\right)' = \frac{n g'(x)}{g(x)}$

$\therefore \left(\ln\sqrt{x^2-5}\right)' = \left(\frac{1}{2}\ln\left(x^2-5\right)\right)' = \frac{1}{2}\left(\frac{2x}{x^2-5}\right) = \frac{x}{x^2-5}$

Answer: A

29. Let f be twice differentiable function on $\Re$. Which one of the following is necessarily true?

A. If $f'(c) = 0$, at some $c \in \Re$, then f has a relative extreme value at $x = c$.
B. If $f'(x)$ is increasing, then the graph of $y = f(x)$ is concave upward.
C. If $f'(x) = 0$ for all $x \in \Re$, then $f(x) = 0$ for all $x \in \Re$.
D. If $f(x)$ is increasing, then $f^n(x) \geq 0$ for all $x \in \Re$.

Hint:

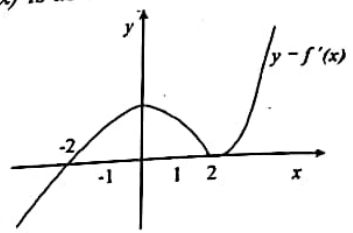
If the graph of $f'(x)$ is increasing, so that $f^n(x) > 0$, then the graph of $y = f(x)$ is concave upward.

Answer: B

For further explanation refer extreme maths page 716. Q#61

Telegram: https://t.me/ExtremeSeries

30. Suppose $f(x)$ is differentiable on $(-\infty, \infty)$ and the graph of its derivative, $y = f'(x)$ is as shown in the following figure.



Which one of the following is true about $f(x)$?

A. $f(x)$ is increasing on $(-\infty, 0) \cup (2, \infty)$.
B. $f(x)$ has a local extreme value at $x = 2$.
C. $f(x)$ has a local minimum value at $x = -2$.
D. $f(x)$ has a local maximum value at $x = 0$.

Hint: Using sign chart.

The sign of $y = f'(x)$	$-\infty$	$---$	-2	$+++$	2	$+++$	∞
$y = f(x)$		decrease		increase		increase	

- $f(x)$ is decreasing on $(-\infty, -2]$
- $f(x)$ is increasing on $[-2, 2]$, $[2, \infty]$
- The sign of $f'(x)$ change from negative to positive at $x = -2$
  $\Rightarrow f(x)$ attains a local min at $x = -2$.

Answer: C

☞ *Refer extreme maths page 685.*

31. For any n × n square matrix A, which one of the following is true?

A. $\det(A) = -\det(A^T)$, where A^T is the transpose of A.
B. If k is a scalar, then $\det(kA) = k^n \det(A)$
C. If B is a matrix obtained from A by interchanging of two rows of A, then $\det(B) = \det(A)$
D. If A is invertible, then $\det(A) = \det(A^{-1})$

Answer: B

32. The solution of the system of linear equation of $\begin{cases} x - 3y - 2z = 6 \\ 2x - 4y - 3z = 8 \\ -3x + 6y + 8z = -5 \end{cases}$ is:

A. $x = -1, y = -3, z = -2$
B. $x = -1, y = -3, z = 2$
C. $x = 1, y = -3, z = 2$
D. $x = 1, y = 3, z = -2$

Hint: By direct substitution $x = -1, y = -3, z = 2$

Answer: C

33. If $Z = \frac{2-4i}{1+i}$ then the modulus of the conjugate of Z, $|Z|$ is:

A. $\sqrt{10}$ B. $3\sqrt{2}$ C. $2\sqrt{3}$ D. $2\sqrt{2}$

Hint: $\bar{z} = \frac{z+4i}{1-i}$ then $|\bar{z}| = |z| = \sqrt{10}$

$$|Z| = \left|\frac{2-4i}{1+i}\right| = \frac{|2-4i|}{|1+i|} = \frac{\sqrt{2^2+4^2}}{\sqrt{1^2+1^2}} = \sqrt{10}$$

Answer: A

34. The n^{th} term of the sequence: 1, −4, 9, −16,...is:

A. $a_n = (-2)^n$
B. $a_n = (-1)^n n^2$
C. $a_n = (-1)^{2n} n^2$
D. $a_n = (-1)^{n-1} n^2$

Answer: D

35. If $f: A \to B$ and $g: B \to C$ are functions, then which one of the following is true about the composition function?

A. Domain of $(gof) \subseteq$ Domain of f
B. Range of $(gof) \not\subseteq$ Range of g
C. Domain of $(gof) \not\subseteq$ Domain of f
D. Range of $(gof) \subseteq$ Range of g

Answer: A

Telegram: https://t.me/ExtremeSeries

36. If the point (3, -2) is on the graph of $y = f(x)$, which point is on the graph of $y = f^{-1}(x)$?

A. $\left(\frac{1}{3}, -2\right)$ B. $(3, -1)$ C. $(-2, 3)$ D. $\left(3, -\frac{1}{2}\right)$

Answer: C

Extreme Tip 4: Let $f(x) = 1 + \frac{1}{2}\ln(x-3)$ then $f^{-1}(x)$ is equal to

Solution: We set $y = f(x) \Rightarrow y = 1 + \frac{1}{2}\ln(x-3)$

Interchange $x \to y$ and $y \to x$

$\Rightarrow x = 1 + \frac{1}{2}\ln(y-3)$, solve for y

$\Rightarrow \frac{1}{2}\ln(y-3) = x - 1 \Leftrightarrow \ln(y-3) = 2x - 2$

$y - 3 = e^{2(x-1)} \Leftrightarrow y = e^{2(x-1)} + 3$

$\therefore f^{-1}(x) = e^{2(x-1)} + 3$

37. The equation of the line that passes through (2, -1) and perpendicular to $3x + 4y = 6$ is:

A. $-4x + 3y = 5$ C. $4x + 3y = 11$

B. $4x - 3y = 5$ D. $-4x + 3y = -11$

Hint: Slope of $3x + 4y = 6$ is:

$m_1 = -\frac{3}{4}$, so that $m_2 = \frac{4}{3}$.

Thus, the equation through $(2, -1)$ is

$y + 1 = \frac{4}{3}(x-2) \Leftrightarrow -4x + 3y = -11$ Answer: D

38. There are three children in a room ages three, four, and five. If a four-year-old child enters the room then which one of the following is true?

A. mean age will stay the same but the standard deviation will increase.

B. mean age will stay the same but the standard deviation will decrease.

C. mean age and standard deviation will increase

D. mean age and standard deviation will stay the same.

Hint: mean age $= \frac{3+4+5}{3} = 4$

- Sd. dev. $= \frac{(3-4)^2+(4-4)^2+(5-4)^2}{3} = \frac{2}{3}$
- mean age and standard deviation after four year child enters.

$\rightarrow$ mean $= \frac{3+4+5+4}{4} = \frac{16}{4} = 4$

$\rightarrow$ S.D $= \sqrt{\frac{(3-4)^2+2(4-4)^2+(5-4)^2}{4}} = \sqrt{\frac{2}{4}} = \sqrt{\frac{1}{2}}$

Answer: B

39. In how many more ways can 4 people be arranged in a row than if they were arranged in a circle?

A. 1 B. 6 C. 18 D. 12

Hint: $4! - 3! = 4 \times 3 \times 2 \times 1 - (3 \times 2 \times 1) = 24 - 6 = 18$

Answer: C

40. Two machines A and B work independently. The probability that both machines A and B work is 0.4. If the conditional probability that machine B works given that machine A works is 0.5, then the conditional probability that machine A works given that machine B works is _____ .

A. 0.8 B. 0.3 C. 0.5 D. 0.7

Hint: $P(B/A) = \frac{P(A \cap B)}{P(A)} \Rightarrow 0.5 = \frac{0.4}{P(A)}$

$\therefore P(A) = \frac{0.4}{0.5} = 0.8$

Since, they are independent: $P(A/B) = P(A) = 0.8$ **Answer: A**

41. Which one of the following is ture about the fucntion f defined by $f(x) = x^2 + e^{2x}$?

A. f is decreasing for $x \geq 0$

B. f is increasing for $x \geq 0$

C. f has relative minimum at $x = 0$.

D. f has relative maximum at $x = 0$.

Telegram: https://t.me/ExtremeSeries

Hint: $f(x)=x^2+e^{2x} \Rightarrow f'(x)=2x+2e^{2x}$
$f'(x)>0$ for all $x \geq 0$.
$\therefore f$ is increasing on $x \geq 0$.

Answer: B

42. The value of $\int_0^1 (x+1)e^{x^2+2x}\,dx$ is:

A. $\frac{e^3}{2}$ B. $\frac{e^4-e}{2}$ C. $\frac{e^3-1}{2}$ D. e^3-1

Hint: Let $U=x^2+2x \Rightarrow dU=(2x+2)\,dx$
$\Rightarrow dU=2(x+1)\,dx$
$\therefore (x+1)dx=\frac{dU}{2}$

$$\int_0^1 (x+1)e^{x^2+2x}\,dx=\int e^U\frac{dU}{2}=\frac{e^U}{2}+c$$

$$\Rightarrow \int_0^1 (x+1)e^{x^2+2x}\,dx=\left.\frac{e^{x^2+2x}}{2}\right|_0^1=\frac{e^3}{2}-\frac{1}{2}=\frac{e^3-1}{2}$$

Answer: C

43. $\int_0^3 (x+1)^{\frac{1}{2}}\,dx =$ _____

A. $\frac{21}{2}$ B. $\frac{14}{3}$ C. 7 D. $\frac{16}{3}$

Hint: $\int_0^3 (x+1)^{\frac{1}{2}}\,dx=\frac{2}{3}(x+1)^{3/2}\Big|_0^3$
$=\frac{16}{3}-\frac{2}{3}=\frac{14}{3}$

Answer: B

44. Given $f(x)=\begin{cases} x+1, \text{for } x<0 \\ \cos \pi x, \text{for } x \geq 0 \end{cases}$, then $\int_{-1}^1 f(x)\,dx =$ _____.

A. $\frac{1}{2}+\frac{1}{\pi}$ B. $\frac{1}{2}-\frac{1}{\pi}$ C. $\frac{1}{2}$ D. $-\frac{1}{2}$

Hint: $\int_{-1}^1 f(x)\,dx=\int_{-1}^0 f(x)\,dx+\int_0^1 f(x)\,dx$
$=\int_{-1}^0 f(x)\,dx+\int_0^1 \cos \pi x\,dx$
$=\frac{x^2}{2}+x\Big|_{-1}^0+\frac{\sin \pi x}{\pi}\Big|_0^1=0-\left(\frac{1}{2}-1\right)+0=\frac{1}{2}$

Answer: C

45. The sum of $\sum_{n=0}^{\infty} 5\left(\frac{2}{3}\right)^n$ is _____.

A. 0 B. 15 C. $\frac{10}{3}$ D. 5

Hint: $G_1=5, r=\frac{2}{3}<1$. Convergent
$\therefore S=\frac{G_1}{1-r}=\frac{5}{1-\frac{2}{3}}=\frac{5}{\frac{1}{3}}=5\times 3=15$

Answer: B

46. In which interval the sequence $\left\{\frac{(-1)^n}{3n}\right\}_{n=1}^{\infty}$ is bounded?

$\left[\frac{-1}{9},\frac{1}{12}\right]$ B. $\left[\frac{-1}{3},\frac{1}{6}\right]$ C. $\left[\frac{-1}{6},\frac{1}{3}\right]$ D. $\left[\frac{-1}{12},\frac{1}{9}\right]$

Hint: $\left\{\frac{(-1)^n}{3n}\right\}_{n=1}^{\infty}=\left[\frac{-1}{3},\frac{1}{6},\frac{-1}{9},\frac{1}{12},\frac{-1}{15}\right]$

$\therefore \frac{-1}{3} \leq \left\{\frac{(-1)^n}{3n}\right\}_{n=1}^{\infty} \leq \frac{1}{6}$

Answer: B

47. Which one of the following is true about the function

$$f(x)=\begin{cases} \frac{x^2}{x}, x \neq 0 \\ 0, x=0 \end{cases}?$$

A. f is continuous except at $x=0$
B. f has an infinite discontinuity at $x=0$

Telegram: https://t.me/ExtremeSeries

C. f is continuous everywhere.
D. f has $x=0$ as a vertical asymptote.

Hint: $\lim_{x\to 0} f(x)=f(0)\Rightarrow \lim_{x\to 0^+} f(x)=0=f(0)$

$\lim_{x\to a} f(x)=f(a)\Rightarrow a=a$

Answer: C

48. $\lim_{x\to\infty} x\sin\frac{1}{x}$ is equal to:

A. 0 B. 1 C. ∞ D. -1

Hint: Let $U=\frac{1}{x}\Rightarrow x=\frac{1}{U}$, as $x\to\infty$, $U\to 0$.

$\therefore \lim_{x\to\infty} x\sin\frac{1}{x}=\lim_{U\to 0}\frac{\sin U}{U}=1$

Answer: B

49. $\frac{d}{dx}\left(\ln e^{2x}\right)$ is equal to:

A. $\frac{1}{e^{2x}}$ B. $\frac{2}{e^{2x}}$ C. $2x$ D. 2

Hint: $\frac{d}{dx}\left(\ln e^{2x}\right)=\frac{d}{dx}(2x\ln e)=\frac{d}{dx}(2x)=2$

Answer: D

50. If $f(x)=2+|x-3|$ for all x, then the value of the derivative $f'(x)$ at $x=3$ is _____.

A. -1 B. does not exist C. 1 D. 2

Answer: B

51. The graph of $y=5x^4-x^5$ has a point of inflection at:

A. (3, 162) only C. (0, 0) only
B. (4, 256) only D. (0, 0) and (3, 162)

Hint: $f'(x)=20x^3-5x^4$

$f''(x)=60x^2-20x^3$

$\Rightarrow f''(x)=0$ when $x=0$ or $x=3$

$\therefore (3,f(3))=(3,162)$ is inflection point.

Answer: A

52. Let the angle between $\vec{V}=-2\vec{i}-\vec{j}+2\vec{k}$ and $\overrightarrow{PQ}$ be 60°, where P and Q are points in space. If $\vec{V}\cdot\overrightarrow{PQ}=2$, then what is that distance between P and Q?

A. $\frac{3}{4}$ B. $\frac{4}{5}$ C. $\frac{4}{3}$ D. $\frac{5}{4}$

Hint: $\vec{V}\cdot\overrightarrow{PQ}=|V||PQ|\cos\theta=2$

$\Rightarrow\left|\overrightarrow{PQ}\right|=\frac{2}{|V|\cos\theta}=\frac{2}{\left(\sqrt{4+1+4}\right)\left(\frac{1}{2}\right)}=\frac{4}{3}$

Answer: C

53. Which one of the following is a valid assertion that can be proved by the principle of mathematical induction?

A. $2^n>10n$ for every integer n such that $n\geq 6$.
B. $r^2>0$ for every real number r such that $r\geq 1$.
C. $n^2>10n>2n^2$ for every natural number $n\geq 1$.
D. $2^n>8n$ for every integer n such that $n\geq 3$.

Answer: A

54. The image of a figure with vertices A(1, 2), B(3, 6), C(−1, 2) and D(−2, −2) after reflection across the x-axis is:

A. A′(1, −2), B′(−3, −6), C′(1, −2), and D′(2, 2)
B. A′(−1, 2), B′(−3, 6), C′(1, −2), and D′(2, −2)
C. A′(1, −2), B′(3, −6), C′(−1, −2), and D′(−2, 2)
D. A′(1, −2), B′(3, 6), C′(−1, 2), and D′(−2, −2)

Hint: $(x,y)\xrightarrow{x-axis}(x,-y)$

Answer: C

55. If $\vec{V}=\overrightarrow{AB}+3\overrightarrow{BA}$, where A and B are distinct points in the coordinate plane, then which one of the following is equal to $3\vec{V}$

A. $6\overrightarrow{AB}$ B. $-6\overrightarrow{AB}$ C. $12\overrightarrow{AB}$ D. $-12\overrightarrow{AB}$

Hint: We know that $3\overrightarrow{BA}=-3\overrightarrow{AB}$

$\therefore \vec{V}=\overrightarrow{AB}+3\overrightarrow{BA}=\overrightarrow{AB}-3\overrightarrow{AB}=-2\overrightarrow{AB}$

$3\vec{V}=3\left(-2\overrightarrow{AB}\right)=-6\overrightarrow{AB}$

Answer: B

Telegram: https://t.me/ExtremeSeries

56. If $\vec{u}$ is a unit vector in the direction of $\vec{A}$ and $|\vec{A}|=4$, then $\vec{A}\cdot\vec{u}$ is equal to _____.

A. $\frac{1}{4}$ B. 4 C. $\frac{1}{2}$ D. 2

Hint: $\vec{A}\cdot\vec{u} = |\vec{A}||\vec{u}|\cos 0° = (4)(1)(1) = 4$

Answer: B

57. If $\vec{A}$ and $\vec{B}$ are parallel vectors with opposite directions and $|\vec{B}|=|2\vec{A}|$, then $\vec{B}-\vec{A}$ is equal to.

A. $\vec{A}$ B. $-\vec{A}$ C. $-3\vec{A}$ D. $3\vec{A}$

Hint: $|\vec{B}|=|2\vec{A}|$ since parallel and opposite then $\vec{B}=-2\vec{A}$.

$\therefore \vec{B}-\vec{A}=-2\vec{A}-\vec{A}=-3\vec{A}$

Answer: C

58. If $\cot(\theta)=2$, then which of the following is equal to $\csc(\theta)$?

A. $\sqrt{5}$ B. $\frac{2}{\sqrt{5}}$ C. $\frac{1}{\sqrt{5}}$ D. $\frac{1}{2}$

Hint: $\csc^2\theta = \cot^2\theta + 1 = 4 + 1 = 5$

$\therefore \csc\theta = \sqrt{5}$

Answer: A

59. What is the amplitude and period, respectively, of the graph of $f(x)=4\sin\left(\frac{x}{3}\right)\cos\left(\frac{x}{3}\right)$?

A. $4, \frac{\pi}{3}$ B. $2, 3\pi$ C. $2, \frac{2\pi}{3}$ D. $4, 3\pi$

Hint: $f(x)=4\sin\left(\frac{x}{3}\right)\cos\left(\frac{x}{3}\right) = 2\sin\left(\frac{2x}{3}\right)$

- Amplitude 2, period, $P=\frac{2\pi}{\frac{2}{3}}=3\pi$

Answer: B

60. A boat on a sea sailed from its station toward North with constant speed of 80km/h. Another boat from the same station sailed 60°NE (North East) with constant speed of 100km/h. If the two boats started sailing at the same time, what is the straight distance between them after they have sailed for just 30 minutes?

A. $10\sqrt{42}km$ B. 90km C. $10\sqrt{41}km$ D. $10\sqrt{21}km$

Hint: We use $c^2 = a^2 + b^2 - 2ab\cos\theta$.

$\Rightarrow c^2 = 80^2 + 100^2 - 2(80)(100)\cos 60°$.

$= 6400 + 10{,}000 - 8000 = 8400$.

$c = \sqrt{8400} = 20\sqrt{21}km/hr$

distance between them $20\sqrt{21}\text{km} \times \frac{1}{2}\text{hr}$

$=10\sqrt{21}\text{km}$

Answer: D

61. What is the value k, for which the two vectors $\vec{u}=\begin{pmatrix}1\\k\\-3\end{pmatrix}$ and $\vec{v}=\begin{pmatrix}2k\\-5\\4\end{pmatrix}$ are perpendicular?

A. 4 B. –4 C. 3 D. –3

Hint: $\vec{u}\cdot\vec{u}=\begin{pmatrix}1\\k\\-3\end{pmatrix}\cdot\begin{pmatrix}2k\\-5\\4\end{pmatrix}=0$

$\Rightarrow 2k-5k-12=0 \Leftrightarrow -3k=12$

$\therefore k=-4$

Answer: B

62. If one of the end point of the line segment is (3, 2, -4) and the mid-point is (4, 1, -2), then the coordinate of the other end point is:

A. (5, 0, 0)
B. (2, 0, 5)
C. (5, 1, 2)
D. (3, 1, 0)

Telegram: https://t.me/ExtremeSeries

Hint: $\frac{3+x}{2}=4,\ \frac{2+y}{2}=1$ and $\frac{-4+z}{2}=-2$

$x=5, y=0$ and $z=0$

63. The valid conclusion form the premises:

$P \vee Q, Q \Rightarrow R, P \Rightarrow M, \neg M$ is _____

A. $P \wedge (R \vee R)$ C. $Q \wedge (P \wedge R)$

B. $P \wedge (R \wedge R)$ D. $R \wedge (P \vee Q)$

Hint: $\neg M \equiv$ True so, M = false

$P \equiv F, Q \equiv T, R \equiv T$

Answer:

64. What is the value of $\arcsin\left(\frac{-\sqrt{2}}{3}\right)$

A. $\frac{\pi}{4}$ B. $\frac{\pi}{2}$ C. $-\frac{\pi}{4}$ D. $-\frac{\pi}{2}$

Hint: $\sin^{-1}\left(-\frac{\sqrt{2}}{2}\right)=-\sin^{-1}\left(\frac{\sqrt{2}}{2}\right)=\frac{-\pi}{4}$

Answer: D

Answer: C

Extreme Tip 9: Let $f(x)=\begin{cases}\frac{\sin kx}{x}, & \text{if } x<0\\ e^{2x}-2 & \text{if } x\geq 0\end{cases}$

If f is continuous at $x=0$, then what is the value of k.

A. 2 B. −1 C. 2 D. −2

Solution: Since f is constant at $x=0$. Then

$\lim_{x\to 0^-} f(x)=\lim_{x\to 0^+} f(x)$

$\lim_{x\to 0^-}\frac{\sin kx}{x}=\lim_{x\to 0^+} e^{2x}-2$

$k=e^0-2=1-2=-1$

Answer:

Extreme Tip 5: An object is moving along the parabola $y=2\sqrt{x}$ in xy-plane. At what point on its path does the object becomes closest to the point (5, 0)?

A. (4, 4) B. (1, 2) C. $(3,\sqrt{6})$ D. $(3, 2\sqrt{3})$

Solution: Let d be the distance between the point (5, 0) and a point (x, y) on the graph $y=2\sqrt{x}$ is given by

$d=\sqrt{(x-5)^2+(y-0)^2}$

$\Rightarrow d(x)=\sqrt{(x-5)^2+\left(2\sqrt{x}\right)^2}=\sqrt{x^2-10x+25+4x}$

$\Rightarrow d(x)=\sqrt{x^2-6x+25}$, let $g(x)=x^2-6x+25$.

$g'(x)=2x-6 \Rightarrow g'(x)=0$ when $x=3$.

When $x=3,\ y=2\sqrt{3}$

$\therefore (3, 2\sqrt{3})$ is the closest point.

Answer: D

Extreme Tip 6: What is the area of the region between the graph of $y=-x^2+1$ and x-axis where $0\leq x\leq 2$?

A. $\frac{2}{3}$ B. $\frac{7}{2}$ C. 3 D. 2

Solution:

$\int_0^2 f\left(-x^2+1\right)dx=\left[\frac{-x^3}{3}+x\right]_0^2=\left|\frac{-8}{3}+\frac{6}{3}\right|=\frac{2}{3}$

Answer: A

Extreme Tip 7:

What is the equation of the tangent line to the graph of $f(x)=\frac{4}{x^2}-2x+4$ at $(2, f(2))$?

A. $3x+y=6$ C. $6x+y=13$

B. $3x+y=7$ D. $-6x+y=13$

Telegram: https://t.me/ExtremeSeries

Solution: $f(x) = 4x^{-2} - 2x + 4$

$f'(x) = -8x^{-3} - 2$ and $f'(2) = \frac{-8}{8} - 2 = -3$

$\therefore y - f(2) = f'(2)(x-2)$ is equation of tangent line

$y - 1 = -3(x-2) \Rightarrow 3x + y = 7$

Answer: B

Extreme Tip 5:

If $f(x) = \ln(x+1)$ and $g(x) = x^3 + 7$, then what is the domain of $(fog)(x)$?

A. $(-2, \infty)$ B. $(-1, \infty)$ C. $[-2, \infty)$ D. $\varnothing$

Solution: Domain of $(fog)(x)$ is $x \in Dg$ and $g(x) \in D$

$= \mathbb{R}$ and $x^3 + 7 \in (-1, \infty)$

$= \mathbb{R} \cap (-2, \infty) = (-2, \infty)$

Answer: A

1. Let p and q stand for the statements "Nejat is intelligent" and "Almaz is hardworking", respectively. Which of the following represent the statement "Almaz is hard working if Nejat is intelligent?"

A. $\neg p \wedge q$ B. $\neg p \vee q$ C. $p \wedge q$ D. $\neg q \vee q$

Hint: If Nejat is intelligent, then Almaz is hardworking.

$p \Rightarrow q = \neg P \vee q$

Answer: B

2. Suppose the eccentricity of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is a reciprocal to that of the eccentricity of the ellipse $x^2 + 4y^2 = 4$. If the hyperbola passes through a focus of the ellipse, then what is the equation of the hyperbola?

A. $x^2 - 2y^2 = 2$ C. $\frac{x^2}{3} - \frac{y^2}{2} = 1$

B. $x^2 - 3y^2 = 3$ D. $\frac{x^2}{2} - \frac{y^2}{3} = 1$

Hint: $x^2 + 4y^2 = 4 \Rightarrow \frac{x^2}{4} + \frac{y^2}{1} = 1$, so, $m^2 = 4, n^2 = 1$

- Eccentricity of ellipse: $e = \frac{c}{m} = \frac{\sqrt{m^2 - n^2}}{m}$

$= \frac{\sqrt{4-1}}{2} = \frac{\sqrt{3}}{2}$

- Foci of ellipse: F(C, 0) and (-C, 0) $= \left(-\sqrt{3}, 0\right)$ and $\left(\sqrt{3}, 0\right)$
- Since the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is intersect the ellipse at $\left(\sqrt{3}, 0\right)$ and $\left(-\sqrt{3}, 0\right)$.

$\frac{\left(\sqrt{3}\right)^2}{a^2} - \frac{0^2}{b^2} = 1$, then $a^2 = 3 \ldots (x)$

Given that eccentricity of hyperbola

$= \frac{1}{\text{eccentricity of ellipse}} = \frac{1}{\frac{\sqrt{3}}{2}} = \frac{2}{\sqrt{3}}$

Telegram: https://t.me/ExtremeSeries

$\Rightarrow$Eccentricity of hyperbola $= \frac{\sqrt{a^2+b^2}}{a} = \frac{2}{\sqrt{3}}$, but $a^2 = 3$

$\Rightarrow \frac{\sqrt{3+b^2}}{\sqrt{3}} = \frac{2}{\sqrt{3}} \Rightarrow 3+b^2 = 4$, then $b^2 = 1$

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ will be $\frac{x^2}{3} - \frac{y^2}{1} = 1 \Leftrightarrow x^2 - 3y^2 = 3$

3. What is the radius of the largest possible circle that can be inscribed the ellipse given by $5(x-1)^2 + 3y^2 = 15$? **Answer: B**

A. $\sqrt{3}$ B. $\sqrt{5}$ C. 3 D. 5

Hint: $5(x-1)^2 + 3y^2 = 15 \Rightarrow \frac{(x-1)^2}{3} + \frac{y^2}{5} = 1$

Circle is inscribed if the radius of circle is equal to the length of semi minor axis.

Now, $a^2 = 5 \Rightarrow a = \sqrt{5}$ and $b^2 = 3 \Rightarrow b = \sqrt{3}$

4. What are the values for the center (C) and radius (r) of the circle $x^2 + y^2 - 4x + 6y = 5$? **Answer: A**

A. $C = (-2, 3), r = 3\sqrt{2}$ C. $C = (2, -3), r = 2\sqrt{3}$

B. $C = (2, -3), r = 3\sqrt{2}$ D. $C = (-2, 3), r = 2\sqrt{3}$

Hint: $x^2 + y^2 - 4x + 6y = 5 \Rightarrow x^2 - 4x + y^2 + 6y = 5$

$\Rightarrow x^2 - 4x + 4 + y^2 + 6y + 9 = 5 + 4 + 9$

$\Rightarrow (x-2)^2 + (y+3)^2 = 18$

C (2, -3) and $r^2 = 18$, $r = 3\sqrt{2}$ **Answer: B**

5. What is the equation of line that passes through point (a, a) in the xy – plane if it is a parallel to a line that passes through points (a, b) and (b, a), where $a \neq b$? **Answer: C**

A. $y = x$ C. $y = -x + 2a$

B. $y = -x$ D. $y = 2x - a$

Hint: Slope: $m = \frac{a-b}{b-a} = -1$

Equation of parallel line is $y - a = -1\ (x - a)$

$\Rightarrow y = -x + 2a$

6. Which one of the following is true about the graph of $f(x) = \frac{x^3 - x}{x^3(x-1)}$? **Answer: C**

A. The vertical asymptotes of the graph are $x = 0$ and $x = 1$.

B. A horizontal asymptote of the graph is $y = 1$.

C. The graph intersects its horizontal asymptote at point (-1, 0).

D. The graph intersects the vertical line $x = 1$ at point (1, 2).

Hint: $f(x) = \frac{x^3 - x}{x^3(x-1)} = \frac{x(x^2-1)}{x^3(x-1)} = \frac{x(x-1)(x+1)}{x^3(x-1)} = \frac{x+1}{x^2}$

- Horizontal asymptote is $y = 0$
- $\frac{x+1}{x^2} = 0 \Rightarrow x = -1$

What is the solution set of $\frac{1}{1+\frac{1}{x}} - \frac{1}{1-\frac{1}{x}} = \frac{x+\frac{1}{x}}{x-\frac{1}{x}}$? **Answer: A**

A. {} B. {-1} C. {1} D. {-1, 1}

Solution: the expression is equivalent to $\frac{x}{x+1} - \frac{x}{x-1} = \frac{x^2+1}{x^2-1}$

$\Rightarrow \frac{x^2 - x - x^2 - x}{x^2 - 1} = \frac{x^2+1}{x^2-1} \Rightarrow -2x = x^2 + 1$

$\Rightarrow x^2 + 2x + 1 = 0 \Rightarrow (x+1)(x+1) = 0$

$x = -1 \leftarrow$ not in the domain

$\therefore$ S. S = { } = ϕ

8. Which one of the following is the inverse of $f(x) = 8x^3 + 2$? **Answer: B**

A. $f^{-1}(x) = \frac{1}{8x^3 + 2}$ C. $f^{-1}(x) = 8x^{-3} - 2$

B. $f^{-1}(x) = \frac{1}{2}\sqrt[3]{x-2}$ D. $f^{-1}(x) = \frac{1}{8}\sqrt[3]{x-2}$

Solution: $f(x) = 8x^3 + 2$, set $y = f(x)$

$\Rightarrow y = 8x^3 + 2$, interchange $x \rightarrow y$ and $y \rightarrow x \Rightarrow x = 8y^3 + 2$, solve for y

$\Rightarrow 8y^3 = x - 2$, then $y = \frac{\sqrt[3]{x-2}}{2}$

Telegram: https://t.me/ExtremeSeries

9. If $f(x)=\sqrt{x^3}$ and $(fog)(x)=\sqrt[4]{x}$, then what is the value of $g(8)$?
A. $\sqrt[3]{2}$ B. 2 C. $\sqrt{2}$ D. $2\sqrt{2}$

Hint: $(fog)(x)=f(g(x))=\sqrt[4]{x}\Rightarrow\sqrt{(g(x))^3}=\sqrt[4]{x}\Leftrightarrow g(x)=$
$\Rightarrow g(x)=x^{2/12}=x^{1/6}$, therefore, $g(8)=8^{1/6}=(2^3)^{1/6}=2^{1/2}$

Answer: C

10. Which of the following functions is a one to one correspondence?
A. $f:[0,\infty]\to\mathbb{R}$ defined by $f(x)=|x|$.
B. $f:\mathbb{R}\to[0,\infty)$ defined by $f(x)=x^2$.
C. $f:\mathbb{R}\to[0,\infty)$ defined by $f(x)=3^x$.
D. $f:(0,\infty)\to\mathbb{R}$ defined by $f(x)=\log_2^x$.

Hint:
i) If $f:A\to B$ is on to if and only if the range of $f=B$.
$\therefore B=\mathbb{R}$ and range of $f(x)=\log_2^x$ is $\mathbb{R}$
$\therefore$ Range of $f=B=\mathbb{R}$
$\therefore$ f is on to
ii) For all $x>0$, $f(x)=\log_2^x$ is one – to – one. Since f is both one – to – one and onto, then f is one – to – one correspondence.

Answer: D

11. If $A=(a_{ij})_{3\times3}$ is a square matrix with $A^{-1}=\begin{pmatrix}1&3&2\\1&4&3\\0&4&5\end{pmatrix}$, then what is the cofactor of a_{23}?
A. $-\frac{3}{4}$ B. $-\frac{2}{7}$ C. $\frac{2}{7}$ D. $-\frac{3}{7}$

Hint: Inverse of A^{-1} is A. i.e $(A^{-1})^{-1}=A$.

$$\text{Det } A^{-1}=|A^{-1}|=1\begin{vmatrix}1&3\\4&5\end{vmatrix}-1\begin{vmatrix}3&2\\4&5\end{vmatrix}+0=-7-7=-14$$

The cofactor of each element:

$$C_{11}=\begin{vmatrix}1&3\\4&5\end{vmatrix}=-7,\quad C_{12}=-\begin{vmatrix}1&3\\0&5\end{vmatrix}=-5,\quad C_{13}=4$$

$$C_{21}=-\begin{vmatrix}3&2\\4&5\end{vmatrix}=-7,\quad C_{22}=\begin{vmatrix}1&2\\0&5\end{vmatrix}=5,\quad C_{23}=-4$$

$$C_{31}=\begin{vmatrix}3&2\\1&3\end{vmatrix}=7,\quad C_{32}=-\begin{vmatrix}1&2\\1&3\end{vmatrix}=-1,\quad C_{33}=-2$$

$$\begin{bmatrix}-7&-5&4\\-7&5&-4\\7&-1&-2\end{bmatrix}^t=\begin{bmatrix}-7&-7&7\\-5&5&-1\\4&-4&-2\end{bmatrix}$$

$$\therefore A=\frac{\text{Adj.}A^{-1}}{|A^{-1}|}=-\frac{1}{14}\begin{bmatrix}-7&-7&7\\-5&5&-1\\4&-4&-2\end{bmatrix}=\begin{bmatrix}\frac{1}{2}&\frac{1}{2}&-\frac{1}{2}\\\frac{5}{14}&\frac{-5}{14}&\frac{1}{14}\\\frac{-2}{7}&\frac{2}{7}&\frac{1}{7}\end{bmatrix}$$

$$\therefore \text{Cof of } a_{23}=(-1)\begin{vmatrix}\frac{1}{2}&\frac{1}{2}\\\frac{-2}{7}&\frac{2}{7}\end{vmatrix}=-\left(\frac{1}{7}+\frac{1}{7}\right)=\frac{-2}{7}$$

Answer: B

12. A team of 10 researchers consists 4 biologists and 6 chemists. If 3 persons are chosen randomly from the team, what is the probability that at least one is a biologist?
A. $\frac{2}{3}$ B. $\frac{2}{5}$ C. $\frac{5}{6}$ D. $\frac{7}{10}$

Hint: P (at least one is biologist) = 1 – P(all chemist)

$$=1-\frac{C(6,3).C(4,0)}{C(10,3)}=1-\frac{1}{6}=\frac{5}{6}$$

Answer: C

13. The probability that an electronic device produced by a company does not function properly is equal to 0.1. If devices are bought, then what is the probability that at least 1 device function properly?
A. 0.81 B. 0.09 C. 0.18 D. 0.99

Hint: P(x successes in n trials) = $C(n,x).P^x.(1-P)^x$
P(at least 1 device function) = P(1) + P(2)
$= C(2,1)(0.9)^1(1-0.9)^1 + C(2,2)(0.9)^2(1-0.9)^{2-2}$

$$=2\left(\frac{9}{10}\right)\left(\frac{1}{10}\right)+(1)\left(\frac{9}{10}\right)^2(1)=\frac{18}{100}+\frac{81}{100}=0.99$$

Answer: D

Telegram: https://t.me/ExtremeSeries

14. Two machines A and B produce respectively 60% and 40% of the total number of items of a factory. The percentages of defective output of these machines are 2% and 5%, respectively. If an item is selected at random, then what is the probability that the item is defective?

A. 0.032 B. 0.07 C. 0.32 D. 0.426

Hint:

$P(A) = 0.6$, $P(B) = 0.4$, $P(D_1|A) = 0.02$ and $P(D_2/B) = 0.05$

$P(D) = P(A).P(D_1/A) + P(B).P(D_2|B)$

$= (0.6)(0.02) + (0.4)(0.05) = 0.012 + 0.02 = 0.032$

Answer: A

15. In how many ways can a committee of 3 members be formed from 7 candidates?

A. 7 B. 21 C. 28 D. 35

Hint: $C(7,3) = \frac{7!}{4!3!} = \frac{7\times6\times5}{3\times2} = 35$

Answer: B

16. The following is a frequency distribution table of a grouped data with variable X.

x	3 – 7	8 – 12	13 – 17	18 – 22
Frequency	4	6	8	2

What is the mean $(\bar{x})$ and the variance (s^2) of the data respectively?

A. $\bar{x} = 12, s^2 = 21$ C. $\bar{x} = 13, s^2 = 9$

B. $\bar{x} = 12, s^2 = 25$ D. $\bar{x} = 13, s^2 = 16$

Hint: Class mark: $X_C = 5, 10, 15$ and 20

$$\text{mean}(\bar{x}) = \frac{\Sigma \text{ of class mark}}{\Sigma \text{ of freq}} = \frac{\Sigma fx_C}{\Sigma f}$$

$$\therefore \bar{x} = \frac{4\times5+6\times10+8\times15+2\times20}{4+6+8+2} = \frac{240}{20} = 12$$

ii) Variance $S^2 = \frac{\Sigma f(X_C - \bar{X})^2}{\Sigma f}$

$$= \frac{4(5-12)^2+6(10-12)^2+8(15-12)^2+2(20-12)^2}{20} = 21$$

Answer: A

17. The expenditure of 100 families is given below.

Expenditure	0 – 9	10 – 19	20 – 29	30 – 39	40 – 49
No. of families	14	23	f_1	21	f_2

If the mode of the data is 23.5, what are the values of f_1 and f_2?

A. $f_1 = 27, f_2 = 15$ C. $f_1 = 25, f_2 = 17$

B. $f_1 = 15, f_2 = 27$ D. $f_1 = 17, f_2 = 25$

Hint: $m_o = B_L + i\left(\frac{d_1}{d_1+d_2}\right)$, where i = 10

$B_L = 19.5$, $d_1 = f_1 - 23$, and $d_2 = f_1 - 21$

$$\Rightarrow 23.5 = 19.5 + 10\left(\frac{f_1-23}{f_1-23+f_1-21}\right)$$

$$10\left(\frac{f_1-23}{2f_1-44}\right) = 23.5 - 19.5 = 4$$

$$\Rightarrow \frac{f_1-23}{2f_1-44} = 0.4 \Leftrightarrow f_1 - 23 = 0.8f_1 - 17.6$$

$\Rightarrow f_1 - 0.8f_1 = -17.6 + 23 = 5.4 \Leftrightarrow f_1 = 27$

To find f_2: $14 + 23 + 27 + 21 + f_2 = 100$

$\therefore f_2 = 100 - 85 = 15$

Answer: A

18. The first group of 10 children has a mean weight of 15.6kg, the second group of another 10 children has a mean weight of 16kg, and the third group of children has a mean weight of 20 kg. If the mean weight of all the children is 17 kg, what is the total number of children in all of the three groups?

A. 28 B. 29 C. 30 D. 32

Hint: Combined mean $(Z) = \frac{10\times15.6+10\times16+20x}{10+10+x} = 17$

$$\Leftrightarrow \frac{156+160+20x}{20+x} = 17 \Leftrightarrow 316+20x = 340+17x$$

$\Leftrightarrow 3x = 24$, then $x = 8$

$\therefore$ Total number of children = 10 + 10 + 8 = 28

Answer: A

19. Which of the following is a valid argument?

A. $\neg p \Rightarrow \neg q, q \vdash \neg p$ C. $\neg p \vee q, r \Rightarrow p, r \vdash \neg q$

B. $p \Rightarrow \neg q, p, r \Rightarrow q \vdash \neg r$ D. $\neg p, p \vee q, r \Rightarrow q \vdash \neg r$

Telegram: https://t.me/Extreme Series

Hint: P ≡ T, P ⇒ ¬q ≡T, then ¬q ≡ T, q ≡ F
r ⇒ q ≡ T, since q is false, then r is false
∴¬r is True

20. Consider the following open propositions: $P(x) = x$ is a prime number, $C(x) = x$ is a composite number, and $E(x) = x$ is an even number. Which one of the following has a truth value of True in the set of positive integers? **Answer: B**

A. $(\forall x)[P(x) \Rightarrow \neg E(x)]$
B. $\neg(\forall x)[C(x) \Rightarrow \neg P(x)]$
C. $(\exists x)[\neg P(x) \wedge \neg C(x)]$
D. $\neg(\exists x)[E(x) \wedge \neg C(x)]$

Hint: Let $x = 4$, $E(4) = T$, $C(4) = T$
$\neg[E(4) \wedge \neg(4)] = \neg(T \wedge F) = T$
$\therefore \neg(\exists x)[E(x) \wedge \neg C(x)]$ is true

21. Which one of the following is a convergent sequence? **Answer: D**

A. $\left\{\left(\frac{5}{3}\right)^n\right\}$ B. $\left\{\frac{2n}{n+1}\right\}$ C. $\left\{\frac{n^2}{n+1}\right\}$ D. $\left\{\frac{(-1)^n}{3}\right\}$

Hint: Bounded monotonic sequence are convergent.

22. Suppose a radioactive material loses one-third of its mass per year. If its current mass is 81g, then how much will its mass be just after 7 years? **Answer: B**

A. $27g$ B. $\frac{1}{27}g$ C. $\frac{128}{27}g$ D. $\frac{128}{81}g$

Hint: $81, 27, 9, 3, 1, \frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \ldots$

23. Which of the following is the sum of the series $5 - \frac{10}{3} + \frac{20}{9} - \frac{40}{27} + - \ldots$? **Answer: B**

A. -5 B. -3 C. 3 D. 5

Hint: $G_1 = 5, r = \frac{-2}{3}$

$$S_n = \frac{G_1}{1-r} = \frac{5}{1-\left(\frac{-2}{3}\right)} = \frac{5}{1+\frac{2}{3}} = \frac{5}{\frac{5}{3}} = 5 \times \frac{3}{5} = 3$$

Answer: C

24. What is the sum of $\sum_{n=1}^{30}(-1)^n\left(\frac{1}{n}+\frac{1}{n+1}\right)$?

A. $-\frac{29}{30}$ B. $\frac{29}{30}$ C. $\frac{30}{31}$ D. $-\frac{30}{31}$

Hint: $S_n = -1 - \frac{1}{n+1}$ when n is odd

$S_n = -1 + \frac{1}{n+1}$ when is even

$S_{30} = -1 + \frac{1}{31} = \frac{-30}{31}$ **Answer: D**

25. Which of the following relations holds for the sequence: -10, -3, 4, 11, ...?

A. $a_n = a_{n-1} - 8$
B. $a_n = a_{n-1} + 7$
C. $a_n = a_{n-1} - 7$
D. $a_n = a_{n-1} + 8$

Hint: $a_2 - a_1 = a_3 - a_2 = a_4 - a_3 = \ldots = 7$
$\Rightarrow a_n - a_{n-1} = 7 \Leftrightarrow a_n = a_{n-1} + 1$ **Answer: B**

26. If $z = \sqrt{2}\cos\left(\frac{\pi}{12}\right) + i\sqrt{2}\sin\left(\frac{\pi}{12}\right)$, then what is the value of z^3?

A. $2 + 2i$
B. $\sqrt{2} + \sqrt{2}i$
C. $2\sqrt{2} + 2\sqrt{2}i$
D. $3\sqrt{2} + 3\sqrt{2}i$

Hint: $Z^3 = \left(\sqrt{2}\right)^3\left[\cos\left(\frac{3\pi}{12}\right) + i\sin\left(\frac{3\pi}{12}\right)\right]$

$$= 2\sqrt{2}\left[\frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2}\right] = 2 + 2i$$

Answer: A

27. Which one of the following is the conjugate of $|3+4i| - \frac{25i}{3+4i}$?

A. $5 + 3i$ B. $1 + 3i$ C. $3 - 5i$ D. $1 - 3i$

Hint: $|3+4i| = \sqrt{3^2+4^2} = \sqrt{9+16} = 5$

$$5 - \frac{25i}{3+4i} = \frac{(3+4i)(5)}{3+4i} - \frac{25i}{3+4i} = \frac{15+20i-25i}{3+4i}$$

$$= \frac{15-5i}{3+4i} \Leftrightarrow \frac{5(3-i)}{3+4i} = \frac{5(3-i)}{3+4i}\cdot\frac{3-4i}{3-4i}$$

Telegram: https://t.me/ExtremeSeries

$$= \frac{5(9-3i-12i-4)}{9+16} = \frac{5(5-15i)}{25} = 1-3i$$

Conjugate will be $1 + 3i$

Answer: B

28. If $z_1 = \frac{2-i}{1+i}$ and $z_2 = \frac{1+i}{1-i}$, then what is the value of $z_1 + 2z_2$?

A. $1+i$ B. $\frac{i+1}{2}$ C. $\frac{1-i}{2}$ D. $1-i$

Hint: $z_1 + 2z_2 = \frac{2-i}{1+i} + 2\left(\frac{1+i}{1-i}\right) = \frac{2-i}{1+i} + \frac{2+2i}{1-i}$

$$= \frac{(2-i)(1-i)+(1+i)(2+2i)}{1+1} = \frac{1+i}{2}$$

Answer: B

29. When $\begin{vmatrix} a & b & c \\ a & -a & a \\ a & a & -a \end{vmatrix} = a^3$, and $a \neq 0$, what is the solution of

$$\begin{pmatrix} a & b & c \\ a & -a & a \\ a & a & -a \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}?$$

A. $x = 0, y = 2a, z = 2a$

B. $x = \frac{1}{a}, y = 2a, z = 2a$

C. $x = \frac{1}{a}, y = \frac{2}{a}, z = \frac{2}{a}$

D. $x = 0, y = \frac{2}{a}, z = \frac{2}{a}$

Hint: $D_x = \begin{vmatrix} 1 & b & c \\ 0 & -a & a \\ 0 & a & -a \end{vmatrix} = 0, x = \frac{D_x}{D} = 0$

$$D_y = \begin{vmatrix} a & 1 & c \\ a & 0 & a \\ a & 0 & -a \end{vmatrix} = -(a^2 - a^2) = 2a^2, y = \frac{D_y}{D} = \frac{2}{a}$$

Answer: D

$$D_z = \frac{2}{a}$$

30. What are the values of λ and μ so that the system $\begin{cases} x+y+z=6 \\ x+2y+3z=10 \\ x+2y+\lambda z=\mu \end{cases}$ has infinitely many solutions?

A. $\lambda \neq 3$ and $\mu \in \mathbb{R}$ C. $\lambda = 3$ and $\mu = 10$

B. $\lambda = 3$ and $\mu \neq 10$ D. $\lambda \in \mathbb{R}$ and $\mu = 10$

Hint: In matrix notation, the given system of equations can be written as $AX = B$ where $A = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{vmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} 6 \\ 10 \\ \mu \end{bmatrix}$.

Augmented matrix [A: B]

$$= \left[\begin{array}{ccc:c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array}\right] \text{ To change } a_{21} = 0 \text{ and } a_{31} = 0$$

$R_2 - R_1, R_3 - R_1$

$$= \left[\begin{array}{ccc:c} 1 & 1 & 1 & 2 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda-3 & \mu-6 \end{array}\right] \text{ To change } a_{32} = 0$$

$R_1 - R_2, R_3 - R_2$

$$= -\left[\begin{array}{ccc:c} 1 & 0 & -1 & 2 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & x-3 & \mu-10 \end{array}\right]$$

Case I: If $\lambda = 3$, $\mu \neq 10$, the system has no solution

Case II: If $\lambda \neq 3$, μ may have any value then the system has unique solution.

Case III: If $\lambda = 3$, $\mu = 10$, then the system has an infinite solution.

Answer: C

31. Let f be a differentiable function with f(1) = -1 and f '(1) = 1. If $g(x) = [f(2x + 1) + 2]^2$, then what is the value of g'(0)?

A. 4 B. 2 C. -2 D. -4

Hint: $g'(x) = 2[f(2x + 1) + 2].f'(2x + 1).(2)$

$g'(0) = 2[f(1) + 2].f'(1).2 = 2(-1 + 2).2 = 4$

Answer: A

Telegram: https://t.me/ExtremeSeries

32. If $f(x) = \ell n\,(x^2 + 2)$, then what is the value of $f''(1)$?

A. $\frac{3}{2}$ B. $\frac{5}{9}$ C. $\frac{2}{3}$ D. $\frac{2}{9}$

Hint: If $f(x) = \ell n\,(x^2 + 2)$ then

$$f'(x) = \frac{1}{x^2+2}.2x = \frac{2x}{x^2+2}$$

$$f''(x) = \frac{2(x^2+2) - 2x(2x)}{(x^2+2)^2} = \frac{-2x^2+4}{(x^2+2)^2}$$

$$\therefore f''(1) = \frac{-2(1)^2+4}{(1+2)^2} = \frac{-2+4}{9} = \frac{2}{9}$$

Answer: D

33. If $x^2 + xy = 10$, then what is the value of $\frac{dy}{dx}$ when $x = 2$?

A. $-\frac{7}{2}$ B. $\frac{2}{7}$ C. $\frac{3}{2}$ D. $\frac{7}{2}$

Hint: $x^2 + xy = 10 \Rightarrow y = \frac{10 - x^2}{x}$

$$\frac{dy}{dx} = \frac{(-2x)(x) - (10 - x^2)}{x^2} = \frac{-x^2 - 10}{x^2}$$

$$\left.\frac{dy}{dx}\right|_{x=2} = \frac{-4-10}{4} = \frac{-14}{4} = \frac{-7}{2}$$

Answer: A

34. What is the equation of the tangent line to the graph of $f(x) = 3x^2 + 4x - 5$ at $(1, 2)$?

A. $10x - y - 8 = 0$ C. $-10x - y - 8 = 10$

B. $-10x + y - 8 = 0$ D. $10x + y - 8 = 0$

Hint: $f'(x) = 6x + 4$ and $f'(1) = 6 + 4 = 10$

Equation of tangent line: $y - 2 = f'(1)(x - 1)$

$\Rightarrow y - 2 = 10(x - 1) \Leftrightarrow 10x - y - 8 = 0$

Answer: A

35. If $f(x) = \pi^2 + 1$, then what is the value of $f'(x)$?

A. $2\pi + 1$ B. 2π C. 2 D. 0

Hint: Derivative of constant function is zero.

Answer: D

36. Suppose f is continuous on [2, 6] and the only solutions of the equation $f(x) = 7$ are $x = 2$ and $x = 5$. If $f(3) = 9$, then one of the following **CANNOT** be the value of $f(4)$:

A. 5 B. 7.5 C. 8 D. 9

Answer: A

37. What is the value of k so that $f(x) = \begin{cases} \frac{\tan 2x}{x}, & \text{if } x > 0 \\ k - e^{2x}, & \text{if } x \le 0 \end{cases}$ is continuous at $x = 0$?

A. 0 B. 2 C. 3 D. 5

Hint: $\lim_{x \to 0^+} f(x) = \lim_{x \to 0^-} f(x)$

$$\Leftrightarrow \lim_{x \to 0^+} \frac{\tan 2x}{x} = \lim_{x \to 0^-} k - e^{2x}$$

$\Rightarrow 2 = k - 1 \Leftrightarrow k = 3$

Answer: C

38. If f is continuous at $x = 0$ and $g(x) = \sqrt{x}\left(2f(x) + \frac{3}{\sqrt{x}}\right)$ for all $x \neq 0$, then what is the value of $\lim_{x \to 0^+} g(x)$?

A. 0 B. 2 C. 3 D. 5

Hint: $\lim_{x \to 0^+} \sqrt{x}\left(2f(x) + \frac{3}{\sqrt{x}}\right) = 0$

Answer: A

39. What is the value of $\lim_{x \to 0} \frac{\sin x \cos 2x}{x^2 + 3x}$?

A. $\frac{1}{3}$ B. $\frac{2}{3}$ C. 1 D. 2

Hint: $\lim_{x \to 0} \frac{\sin x \cos 2x}{x^2 + 3x} = \lim_{x \to 0} \frac{\sin x}{x} \cdot \frac{\cos 2x}{(x+3)}$

$$\lim_{x \to 0} \frac{\sin x}{x} . \lim_{x \to 0} \frac{\cos 2x}{x+3} = 1 . \frac{1}{3} = \frac{1}{3}$$

Answer: A

40. If $a \neq 0$, then what is the value of $\lim_{x \to a} \frac{x^2 - a^2}{x^4 - a^4}$?

A. $\frac{1}{a}$ B. $\frac{1}{2a^2}$ C. $\frac{1}{6a^2}$ D. 0

Hint: $\lim_{x \to a} \frac{x^2 - a^2}{x^4 - a^4} = \lim_{x \to a} \frac{(x-a)(x+a)}{(x-a)(x+a)(x^2+a^2)}$

$$\lim_{x \to a} \frac{1}{x^2 + a^2} = \frac{1}{a^2 + a^2} = \frac{1}{2a^2}$$

Answer: B

Telegram: https://t.me/ExtremeSeries

41. If $f(0) = -1, f(1) = 2$ and $f'(x)$ us continuous on $[0, 1]$, then which of the following is equal to $\int_0^1 f'(x)\sqrt{2+f(x)}dx$?

A. $\frac{16}{3}$ B. $\frac{14}{3}$ C. $\frac{8}{3}$ D. $\frac{4}{3}$

Hint: Let $u = 2 + f(x) \Rightarrow du = f'(x)dx$

When $x = 1 \Rightarrow u = 2 + f(1) = 2 + 2 = 4$

When $x = 0 \Rightarrow u = 2 + f(0) = 2 + (-1) = 1$

$$\int_1^4 \sqrt{u}du = \frac{2}{3}u^{3/2}\Big|_1^4 = \frac{16}{3} - \frac{2}{3} = \frac{14}{3}$$

Answer: A

42. What is the value of $\int_0^{\ln\sqrt{3}} \frac{e^x}{e^{-x}+e^x}dx$?

A. $\frac{1}{2}\ln 2$ B. $\ln 4$ C. $\ln 2$ D. 1

Hint: $\frac{e^x}{e^{-x}+e^x} = \frac{e^x}{e^x}\left[\frac{e^x}{e^{-x}+e^x}\right] = \frac{e^{2x}}{1+e^{2x}}$

Let $u = 1 + e^{2x}$, $du = 2e^{2x}dx$

When $x = 0$, $u = 1 + e^0 = 1 + 1 = 2$

When $x = \ln\sqrt{3}$, $u = 1 + e^{2\ln\sqrt{3}} = 1 + (e^{\ln\sqrt{3}})^2 = 1 + 3 = 4$

Using integration by substitution.

$$\int_0^{\ln\sqrt{3}} \frac{e^{2x}}{1+e^{2x}}dx = \int_2^4 \frac{du}{2u} = \frac{1}{2}\ln u\Big|_2^4 = \frac{\ln 4 - \ln 2}{2} = \frac{\ln\left(\frac{4}{2}\right)}{2} = \frac{\ln 2}{2}$$

Answer: A

43. What is the value of the area of the region enclosed by the graph of $f(x) = e^x$ and $g(x) = x$ between the lines $x = -1$ and $x = 1$?

A. $\frac{e^2-1}{e}$ C. $e^2 - \frac{1}{e} + 2$

B. $e^2 - \frac{1}{e}$ D. $e - \frac{1}{e} + 2$

Hint: $A = \int_{-1}^1 (e^x - x)dx = e^x - \frac{x^2}{2}\Big|_{-1}^1$

$$= \left(e - \frac{1}{2}\right) - \left(e^{-1} - \frac{1}{2}\right) = e - \frac{1}{e} = \frac{e^2-1}{e}$$

Answer: A

44. Which of the following is equal to the volume of the solid generated when the region bounded by the graph of $y = 2\sqrt{x+1}$ and the x – axis when $0 \le x \le 2$, rotates the x – axis.

A. 8π B. $8\pi^2$ C. 16π D. $16\pi^2$

Hint: $V = \pi\int_a^b (f(x))^2 dx = \pi\int_a^b y^2 dx$

$$V = \pi\int_0^2 \left(2\sqrt{x+1}\right)^2 dx = \pi\int_0^2 4(x+1)dx$$

$$= 4\pi\left(\frac{x^2}{2} + x\right)\Big|_0^2 = 4\pi(4) = 16\pi$$

Answer: C

45. A particle moves along the x – axis with velocity given by $v(t) = 3t^2 + 6t$ for time $t \ge 0$. If the particle is at position $x = 2$ at time $t = 0$, what is the position of the particle at $t = 1$?

A. 6 B. 9 C. 11 D. 12

Hint: $V = \frac{dx}{dt} \Rightarrow dx = Vdt \Leftrightarrow \int dx = \int Vdt$

$$x(t) = \int (3t^2 + 6t)dt = \frac{3t^3}{3} + \frac{6t^2}{2} + C$$

$x(0) = 0 + 0 + C = 2 \Rightarrow C = 2 \Rightarrow x(t) = t^3 + 3t^2 + 2$

$x(1) = 1 + 3 + 2 = 6$

Answer: A

46. Which of the following is equal to $\int x(e^x + \sin(x^2))dx$?

A. $e^x(x+1) + \frac{1}{2}\sin(x^2) + C$

B. $e^x(x-1) - \frac{1}{2}\sin(x^2) + C$

C. $e^x(x-1) - \frac{1}{2}\cos(x^2) + C$

D. $e^x(x-1) + \frac{1}{2}\cos(x^2) + C$

Hint: The expression is the same as $\int xe^x dx + \int x\sin(x)^2 dx$

i) $\int xe^x dx$: Let $u = x$, $dv = e^x dx$

$du = dx$ $V = e^x$

$\int udv = uv - \int vdu = xe^x - e^x + c$

Telegram: https://t.me/ExtremeSeries

ii) $\int x\sin(x)^2\,dx$; Let $u = x^2$ then $du = 2x\,dx$

$$\int \frac{\sin u}{2}\,du = \frac{-\cos u}{2} + C = \frac{-\cos x^2}{2} + C$$

$$\therefore \int xe^x dx + \int x\sin(x)^2\,dx = xe^x - e^x - \frac{\cos x^2}{2} + C$$

47. Which one of the following is the set of critical numbers of $f(x) = \frac{4}{3}x^3 + |x|$? **Answer: C**

A. $\left\{\frac{1}{2}\right\}$ B. $\left\{0, \frac{1}{2}\right\}$ C. $\left\{0, -\frac{1}{2}\right\}$ D. $\left\{-\frac{1}{2}, 0, \frac{1}{2}\right\}$

Hint:

Recall that $|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$

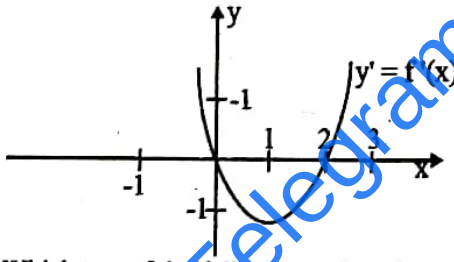
$f(x) = \frac{4}{3}x^3 + x$ for $x \geq 0$ and

$f(x) = \frac{4}{3}x^3 - x$ for $x \geq 0$

For $x < 0$: $f'(x) = 4x^2 - 1$ and $f'(x) = 0$

When $x = \frac{1}{2}$ and $-\frac{1}{2}$ **Answer: D**

48. Suppose $f: (-\infty, \infty) \to \mathbb{R}$ is differentiable and the graph of its derivative, $y' = f'(x)$, is as shown in the figure below.



Which one of the following is true about f?

A. f is increasing on $(1, \infty)$
B. f is concave upward on $(0, \infty)$
C. f has no relative maximum value
D. f has a relative minimum value at $x = 2$

Hint: Since f' changes from negative to positive at $x = 2$ then f has relative minimum at $x = 2$.

- Since $f' > 0$ on the interval $(-\infty, -1)$ and $(2, \infty)$, then f is increasing on $(-\infty, -1) \cup (2, \infty)$.
- $f' < 0$ on $(0, 2)$, then f is decreasing on $(0, 2)$.

Answer: D

49. A tin can of volume 54π cm^3 is to be made in the form of a right circular cylinder that has both flat top and flat bottom. What is the base radius of the tin if it is to be made of the least amount of metal?

A. 2cm B. 3cm C. 4cm D. 6cm

Hint: $V = \pi r^2 h$ and $A = 2\pi rh + 2\pi r^2$

$$\text{Vol} = \pi r^2 h = 54\pi \text{cm}^3 \Rightarrow h = \frac{54}{r^2}$$

$$A = 2\pi r\left(\frac{54}{r^2}\right) + 2\pi r^2 = \frac{108\pi}{r} + 2\pi r^2$$

$$A'(r) = \frac{-108\pi}{r^2} + 4\pi r = \frac{4\pi r^3 - 108\pi}{r^2}$$

$A'(r) = 0$ when $r = 3$ **Answer: B**

50. Air is being pumped into a spherical balloon so that its volume increases at a rate of 50cm^3/s. How fast is the radius of the balloon increasing when the diameter is 5cm?

A. $\frac{1}{50\pi}$cm/s B. $\frac{1}{25\pi}$cm/s C. $\frac{5}{\pi}$cm/s D. $\frac{2}{\pi}$cm/s

Hint: $V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$

$$\Leftrightarrow 50\text{cm}^3/\text{s} = 4\pi\left(\frac{25}{4}\right)\frac{dr}{dt}$$

$$\Rightarrow \frac{dr}{dt} = \frac{50}{25\pi} = \frac{2}{\pi}$$

Answer: D

51. If $2 \leq f'(x) \leq 4$ for all values of x, then the value of $f(8) - f(2)$ is between which of the following numbers?

A. 14 and 24 C. 12 and 18
B. 12 and 24 D. 8 and 10

Telegram: https://t.me/Extremaseries

Hint: $f'(x) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$

$\Rightarrow 2 \le f'(x) \le 4 = 2 \le \frac{f(8) - f(2)}{8 - 2} \le 4$

$\Leftrightarrow 2 \le \frac{f(8) - f(2)}{6} \le 4 \Leftrightarrow 12 < f(8) - f(2) \le 24$

Answer: B

52. Which of the following is true?

A. The amplitude of $f(x) = \sin 3x$ is 3.

B. The period of $f(x) = 2 \sin 4x$ is π.

C. The period of $f(x) = 3 \cos\left(\frac{1}{2}x - \frac{\pi}{3}\right)$ is 4π.

D. The amplitude of $f(x) = -5 \cos(3x + 2) - 2$ is 7.

Hint: $y = a \sin(bx + c) + d$, and $y = a \cos(bx + c)$

- Period is $\frac{2\pi}{b}$ and amplitude is $|a|$
- For $f(x) = 3 \cos\left(\frac{1}{2}x - \frac{\pi}{3}\right)$, the period $= \frac{2\pi}{\frac{1}{2}} = 4\pi$

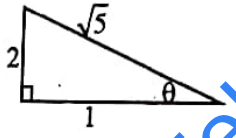
Answer: C

53. If $\theta = \arctan(2)$, then what is the value of $\sin(2\theta)$?

A. $\frac{2}{5}$ B. $\frac{4}{5}$ C. $4\sqrt{5}$ D. $\frac{2}{\sqrt{5}}$

Hint:

$\theta = \tan^{-1} 2 \Rightarrow \tan\theta = 2$



$\sin 2\theta = \sin(\theta + \theta)$

$= \sin\theta\cos\theta + \cos\theta\sin\theta = 2\sin\theta\cos\theta$

$\therefore \sin 2\theta = 2\left(\frac{2}{\sqrt{5}}\right)\left(\frac{1}{\sqrt{5}}\right) = \frac{4}{\sqrt{25}} = \frac{4}{5}$

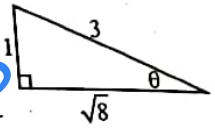
Answer: B

54. If $\cot\theta = \sqrt{8}$, and θ is a first quadrant angle, then what is the value of $\csc\theta$?

A. $\frac{1}{3}$ B. 3 C. $\frac{\sqrt{8}}{3}$ D. $\frac{1}{\sqrt{8}}$

Hint: $\cot\theta = \sqrt{8}$

- $1^2 + \sqrt{8}^2 = r^2 \Rightarrow r = \sqrt{9} = 3$
- $\sin\theta = \frac{1}{3}$, and $\csc\theta = 3$



Answer: B

55. What is the translation vector $u = (h, k)$ so that the equation $x^2 + 2y^2 + 6x - 8y + 15 = 0$ is transformed to an equation of the form $x^2 + 2y^2 + d = 0$, where d is a constant?

A. $u = (-3, 2)$ B. $u = (3, -2)$ C. $u = (-2, 3)$ D. $u = (2, -3)$

Hint: $x^2 + 2y^2 + 6x - 8y + 15 = 0$

$\Rightarrow x^2 + 6x + 9 + 2(y^2 - 4y + 4) = -15 + 9 + 8$

$\Rightarrow (x + 3)^2 + 2(y - 2)^2 = 2$

Answer: A

56. A line given by the vector equation $(x, y) = (-t, 6 + 2t)$, $t \in \mathbb{R}$, is tangent to a circle at a point (1, 4). What is the radius of the circle if its center is on the y-axis?

A. $\sqrt{5}$ B. $\frac{1}{2}\sqrt{5}$ C. $2\sqrt{5}$ D. $\sqrt{10}$

Hint: $V = (-t, 2t) = (-1, 2)t$

The position vector through center C (0, y) and tangent point T(1, 4) is $\overrightarrow{CT} = (1, 4 - y)$.

Then V. $\overrightarrow{CT} = 0 \Rightarrow (-1, 2).(1, 4 - y) = 0$

$\Rightarrow -1 + 8 - 2y = 0 \Leftrightarrow 2y = 7,\ y = \frac{7}{2}$

Hence, radius between point (1, 4) and $\left(0, \frac{7}{2}\right)$ is

$r = \sqrt{(1-0)^2 + \left(4 - \frac{7}{2}\right)^2} = \sqrt{1 + \frac{1}{4}} = \sqrt{\frac{5}{4}} = \frac{\sqrt{5}}{2}$

Answer: B

57. For two nonzero vectors $\vec{a}$ and $\vec{b}$ if $\|\vec{a} + \vec{b}\| = \|\vec{a}\|$, then which of the following is true?

A. $2\vec{a}.\vec{b} = \vec{b}.\vec{b}$

B. $2\vec{a} + \vec{b}$ and $\vec{b}$ are perpendicular

C. $2\vec{a} + \vec{b}$ and $2\vec{a} + \vec{b}$ are parallel

D. $\vec{a}.\vec{b} - \vec{b}.\vec{b} = 0$

Hint: $\|a + b\|^2 = (a + b).(a + b) = a.a + 2a.b + b.b$

Telegram: https://t.me/Extremeseries

$\|a\|^2 = a.a.$
$\|a+b\|^2 = \|a\|^2 \Leftrightarrow a.a + 2a.b + b.\ b = a.\ a$
$\Rightarrow 2a.b + b.\ b = 0 \Leftrightarrow b.\ (2a+b) = 0$

58. Let $\overrightarrow{PQ}$ be a vector with initial point P = (1, 5) and terminal point Q = (4, 0). If $\vec{v} = x\vec{i} + 2\vec{j}$ is parallel to $\overrightarrow{PQ}$, then what is the value of x?

A. $-\frac{6}{5}$ B. $-\frac{2}{5}$ C. -3 D. 3

Answer: B

Hint: $\overrightarrow{PQ} = (4-1, 0-5) = (3, -5)$ and $\vec{V} = (x, 2)$
$\overrightarrow{PQ}$ and $\vec{V}$ are paralle if $\overrightarrow{PQ} = \alpha\vec{V}$

Let $x = -\frac{6}{5}$, then $\alpha = \frac{-5}{2}$

$\Rightarrow -\frac{5}{2}\left(\frac{-6}{5}, 2\right) = (3, 5)$

Answer: A

59. Which one of the following is a valid assertion that can be proved by the principle of mathematical induction?

A. The sum of any two positive rational numbers is positive.
B. $r^2 \geq 1$, for every real number $r \geq 1$.
C. $n^2 \geq 4n$, for every integer $n \geq 4$.
D. $2^n \leq 4^n$, for every integer $n \leq 100$.

Answer: C

0. Consider the assertion: "The sum of positive irrational numbers is positive irrational number". Which one of the following is correct about the assertion?

A. Taking irrational numbers such as $\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{7}$, and so on, if we add any two of them, the sum is irrational. Therefore, the assertion is true.

B. The sum of $1+\sqrt{2}$ and $1-\sqrt{2}$ is 2, which is rational. This is a counter example that disproves the assertion.

C. The sum of $\sqrt{7}$ and $\sqrt{2}$ is a counter example that shows the assertion is false.

D. The assertion can be disproved by taking the sum of $1+\sqrt{2}$ and $2-\sqrt{2}$ as counter example.

Answer: D

61. Suppose "If $x \in A$, then $y \in B$" is a True statement. Then, which one of the following is necessarily true?

A. $y \in B$
B. If $x \notin A$, then $y \notin B$
C. If $y \in B$, then $x \in A$.
D. If $y \notin B$, then $x \notin A$

Answer: B

62. Let $\vec{A}$ and $\vec{B}$ be vectors in space such that $\vec{A}.\vec{B} = -2$ and $\vec{B} = 6\vec{i} - 7\vec{j} + \sqrt{15}\vec{k}$. If θ is the angle between $\vec{A}$ and $\vec{B}$, then what is the value of $|\vec{A}|$?

A. $\frac{1}{5}\cos\theta$ B. $\frac{1}{5\cos\theta}$ C. $\frac{-1}{5}\cos\theta$ D. $\frac{-1}{5\cos\theta}$

Hint: $\vec{A}.\vec{B} = |A||B|\cos\theta = -2$

$|B| = \sqrt{6^2 + 7^2 + \sqrt{15}^2} = \sqrt{36+49+15} = \sqrt{100} = 10$

$\Rightarrow 10|A|\cos\theta = -2$, then $|A| = \frac{-2}{10\cos\theta} = \frac{-1}{5\cos\theta}$.

Answer: D

63. If $P\left(2, \sqrt{5}, 1\right)$ and Q (3, 0, 9) are points on a sphere whose center is on z – axis, then which one of the following points is outside of the sphere?

A. (−4, 3, 5) B. (2, −2, 1) C. (3, 1, 1) D. (0, 0, 0)

Hint: Let (0, 0, z) center of sphere. Then:

$(0-2)^2 + \left(0-\sqrt{5}\right)^2 + (z-1)^2 = (0-3)^2 + 0 + (z-9)_2$

$\Rightarrow 4 + 5 + z^2 - 2z + 1 = 9 + 0 + z^2 - 18z + 81$

$\Rightarrow 16z = 90 - 10 = 80,\ z = 5$

Equation of sphere with center (0, 0, 5) is $(x-0)^2 + (y-0)^2 + (z-5)^2 = (0-3)^2 + (0-3)^2 + (0-0)^2 + (5-9)^2$

$x^2 + y^2 + (z-5)^2 = 25$

C (3, 1, 1): $3^2 + 1^2(1-5)^2 = 9 + 1 + 16 = 26 > 25$

Answer: C

64. If A(x, 0, 2), B(3, 0, 2) and $(C2, \sqrt{3}, 2)$ are vertices of equilateral triangle in space, then what is the value of x?

A. 5 B. 3 C. 2 D. 1

Hint: Side $\overline{AB} = \overline{AC} = \overline{BC}$

Side $\overline{AB} = \sqrt{(x-3)^2 + 0^2 + 0} = \sqrt{(x-3)^2}$

Telegram: https://t.me/ExtremeSeries

Side $\overline{AC} = \sqrt{(x-2)^2 + \left(0-\sqrt{3}\right)^2 + 0} = \sqrt{(x-2)^2 + 3}$

$\overline{AB} = \overline{AC} \Leftrightarrow (x-3)^2 = (x-2)^2 + 3$

$\Rightarrow x^2 - 6x + 9 = x^2 - 4x + 4 + 3$

$9 - 7 = -4x + 6x = 2x$

$\Rightarrow 2x = 2, x = 1$

Answer: D

65. A patrol boat on a sea sailed from its statin 7 km to the North; and changed its course and sailed $4\sqrt{2}$ km in the direction of 45° South – East. What is the shortest (straight) distance the boat should travel in order to return to its station?

A. 5km B. 7km C. $5\sqrt{2}$km D. $5+\sqrt{2}$km

Hint: $C^2 = a^2 + c^2 - 2(a)(c)\cos\theta$

$$C^2 = 7^2 + \left(4\sqrt{2}\right)^2 - 2(7)\left(4\sqrt{2}\right)\left(\frac{\sqrt{2}}{2}\right)$$

$C^2 = 49 + 32 - 56 = 25$

$\therefore C = 5$

Answer: A

1. If a function f is differentiable at a, then what is the value of $\lim_{x\to a} f(x)$?

A. $f(a)$ B. $f'(a)$ C. $f'(a) - f(a)$ D. 0

Solution: If f is differentiable at a, then f is continuous at a, the value of $\lim_{x\to a} f(x) = f(a)$

Answer: A

2. Which one of the following is a convergent sequence?

A. $\left\{\frac{1+2^n}{3^n}\right\}$ C. $\left\{\frac{1-3^n}{2^n}\right\}$

B. $\left\{\frac{1}{n} + \sin(n)\right\}$ D. $\left\{\frac{(-1)^n}{2}\right\}$

Solution:

A. $\lim_{n\to\infty}\left(\frac{1+2^n}{3^n}\right) = 0$ C. $\lim_{n\to\infty}\left(\frac{1-3^n}{2^n}\right) = -\infty$

B. $\lim_{n\to\infty}\left(\frac{1}{n} + \sin(n)\right) = \not\exists$ D. $\lim_{n\to\infty}\frac{(-1)^n}{2} = \pm\frac{1}{2}$

$\therefore \left\{\frac{1+2^n}{3^n}\right\}$ is convergent sequence.

Answer: A

3. What is the value of $\sum_{n=2}^{20}\left(\frac{1}{n-1} - \frac{1}{n}\right)$?

A. $\frac{17}{20}$ B. $\frac{23}{20}$ C. $\frac{21}{20}$ D. $\frac{19}{20}$

Solution: $\sum_{n=2}^{20}\left(\frac{1}{n-1} - \frac{1}{n}\right)$

$$= \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \ldots + \left(\frac{1}{n-1} - \frac{1}{n}\right)$$

$\therefore S_n = 1 - \frac{1}{n} \Rightarrow S_{20} = 1 - \frac{1}{20} = \frac{19}{20}$

Answer: D

4. Let A and B be two events. Suppose that the probability that neither event occurs in $3/8$. What is the probability that at least one of the events occurs?

A. $1/8$ B. $5/8$ C. $1/4$ D. $3/4$

Telegram: https://t.me/ExtremeSeries

Solution: A and B be two events P(at least one of the events A and B occur) = 1 – P(neither events occur)

P(at least one of the events A and B occur) = $1-\frac{3}{8}=\frac{5}{8}$

Answer: B

5. If the truth value of $(p \wedge \neg p) \Leftrightarrow [(q \vee \neg q) \Rightarrow r]$ is True, then which one of the following must be True?

A. q B. p C. ¬q D. ¬r

Solution: The truth value of

$(p \wedge \neg p) \Leftrightarrow [(q \vee \neg q) \Rightarrow r] \equiv$ True (T)

$p \wedge \neg p \equiv F$ from the definition of negation and conjunction

$q \vee \neg q \equiv T$ from the definition of negation and disjunction

$(q \vee \neg q) \Rightarrow r \equiv$ False because from the given

$(p \vee \neg p) \Leftrightarrow [(q \vee \neg q) \Rightarrow r] \equiv T$ and from the definition of bi implication $r \equiv F$ whereas $\neg r \equiv T$

Answer: D

6. What is the maximum value of the function $f(x) = x^4 - 2x^2$ on $[-2, 1]$?

A. 8 B. 12 C. 24 D. 40

Solution: Here at, we set $f'(x) = 0$

$f(x) = x^4 - 2x^2 \Rightarrow f'(x) = 4x^3 - 4x = 0$

$\Rightarrow 4x(x^2 - 1) = 4x(x-1)(x+1) = 0$

Hence, the critical numbers are –1, 0, and 1.

We evaluate f at the critical points and the end points of the interval:

$f(-2) = (-2)^4 - 2(-2)^2 = 16 - 8 = 8 \rightarrow$ is the maximum value

$f(-1) = (-1)^4 - 2(-1)^2 = 1 - 2 = -1$

$f(0) = 0 - 0 = 0$, and $f(1) = 1^4 - 2 = -1$

Answer: A

7. What are the greatest lower bound and least upper bound of the sequence $\left\{(-1)^n\left(1+\frac{1}{n}\right)\right\}$, respectively?

A. –2 and 2 C. –2 and $\frac{3}{2}$

B. $-\frac{3}{2}$ and 2 D. –2 and $-\frac{3}{2}$

Solution:

i) For even numbers: n = 2, 4, 6, 8, …,∞, then:

$$\left\{(-1)^n\left(1+\frac{1}{n}\right)\right\} = \left\{1+\frac{1}{n}\right\} = \left\{\frac{n+1}{n}\right\} = \left\{\frac{3}{2}, \frac{5}{4}, \frac{7}{6}, \ldots 1\right\}$$

ii) For odd numbers: n = 1, 3, 5, 7, …∞, then:

$$\left\{(-1)^n\left(1+\frac{1}{n}\right)\right\} = -\left\{\left(1+\frac{1}{n}\right)\right\} = \left\{-2, \frac{-3}{2}, \frac{-5}{4}, \frac{-7}{6}, \ldots -1\right\}$$

$$\therefore -2 \le \left\{(-1)^n\left(1+\frac{1}{n}\right)\right\} \le \frac{3}{2}$$

Answer: C

8. Which one of the following is an equation of the circle whose endpoints of a diameter are (0, –2) and (2, 2)?

A. $x^2 + y^2 = 4$ C. $(x-1)^2 + y^2 = 4$

B. $x^2 + y^2 - 2y - 4 = 0$ D. $x^2 + y^2 - 2x - 4 = 0$

Solution:

✓ Diameter: $D = \sqrt{(2-(-2))^2 + (2-0)^2} = \sqrt{20} = 2\sqrt{5}$

✓ Radius $= \frac{D}{2} = \frac{2\sqrt{5}}{2} = \sqrt{5}$ and center $= \left(\frac{0+2}{2}, \frac{-2+2}{2}\right) = (1,0)$

✓ Equation of circle: $(x-1)^2 + (y-0)^2 = \sqrt{5}^2$

$\Rightarrow (x-1)^2 + y^2 = 5 \Leftrightarrow x^2 - 2x + 1 + y^2 = 5$

$\therefore x^2 + y^2 - 2x - 4 = 0$

Answer: D

9. What is the equation of the line that passes through (1, 1) and is parallel to the line $3y - x = 1$?

A. $x - 3y + 2 = 0$ C. $3y - x + 2 = 0$

B. $x + 3y = 4$ D. $3x - y = 2$

Solution: Any line parallel to the line $3y - x = 1$ is of the form $3y - x = k$, where $k \in \mathbb{R}$. If $3y - x = k$, passes through (1, 1).

$\Rightarrow 3(1) - 1 = k \Rightarrow k = 2$

$\therefore$ The required equation $3y - x = 2 \Leftrightarrow x - 3y + 2 = 0$

Answer: A

10. Which one of the following is true about signum, absolute value and greatest integer functions?

A. $\text{sgn}(x) = \pm|x|$, for all $x \in \mathbb{R}$

B. $\text{sgn}(x) \le [x]$, for all $x \le 0$

C. $|x| = x\,\text{sgn}(x)$, for all $x \in \mathbb{R}$

D. $\text{sgn}(x) \le [x]$, for all $x \ge 0$

Telegram: https://t.me/ExtremeSeries

Solution: Properties of signum; absolute value and greatest integer function

A. $\text{sgn}(x)=\begin{cases}1 \text{ for } x>0\\0 \text{ for } x=0\\-1 \text{ for } x<0\end{cases}$ then $\text{sgn}(x)\neq\pm|x|$ for $x\in\mathbb{R}$

B. $\text{sgn}(x)=\begin{cases}-1 \text{ for } x<0\\0 \text{ for } x=0\end{cases}\Rightarrow \text{sgn}(x)\geq[x]$ for all $x\leq 0$

C. $|x|=\begin{cases}x \text{ for } x>0\\0 \text{ for } x=0\\-x \text{ for } x<0\end{cases}$ implies to $|x|=x\,\text{sgn}\,x$ for $x\in\mathbb{R}$

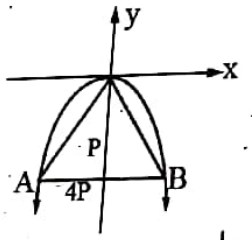
D. Let $x=\frac{1}{2}, \text{sgn}\left(\frac{1}{2}\right)\leq\left[\frac{1}{2}\right]\Rightarrow 1\leq 0$ is false

Answer: C

11. What is the area of the triangle (in sq. units) formed by the lines joining the vertex of the parabola $x^2=-36y$ to the end points of the latus rectum?

A. 126 B. 261 C. 216 D. 162

Solution:



✓ The length of the latus rectum is $AB=|4P|=36$ unit

✓ The distance between the focus and the vertex is $P=\frac{36}{4}=9$

$\therefore \text{Area}=\frac{1}{2}(\text{base})(\text{height})=\frac{1}{2}(36)(9)=162$

Answer: D

12. What is the partial fraction decomposition of $\frac{x^2+x+1}{(x+2)(x^2+1)}$?

A. $\frac{3}{5(x+2)}+\frac{2x+1}{5(x^2+1)}$ C. $\frac{2}{5(x+2)}+\frac{3x+1}{5(x^2+1)}$

B. $\frac{5}{3(x+2)}+\frac{2x+1}{3(x^2+1)}$ D. $\frac{2}{3(x+2)}+\frac{2x+1}{3(x^2+1)}$

Solution: Partial fraction decomposition of $\frac{x^2+x+1}{(x+2)(x^2+1)}$ is

$\frac{x^2+x+1}{(x+2)(x^2+1)}=\frac{A}{x+2}+\frac{Bx+C}{x^2+1}=\frac{Ax^2+A+Bx^2+Cx+2Bx+2C}{(x+2)(x^2+1)}$

$\frac{x^2+x+1}{(x+2)(x^2+1)}=\frac{Ax^2+Bx^2+(C+2B)x+A+2C}{(x+2)(x^2+1)}$

$\Rightarrow A+B=1$ .....*

$C+2B=1\Rightarrow C=1-2B$ .....**

$A+2C=1$ .....***

Substitute ** in to ***

$A+2(1-2B)=1$

$A+2-4B=1$

$-A+4B=1$ .....****

Solve simultaneously * and ****

$\begin{cases}A+B=1\\-A+4B=1\end{cases}$

$5B=2\Rightarrow B=2/5, A=3/5$ and $C=1-\frac{4}{5}=\frac{1}{5}$

$\therefore$decomposition of $\frac{x^2+x+1}{(x+2)(x^2+1)}=\frac{3}{5(x+2)}+\frac{2x+1}{5(x^2+1)}$

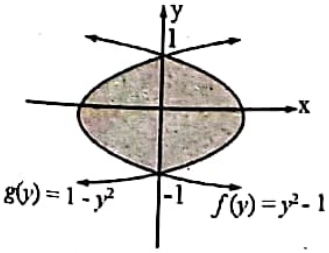
Answer: A

13. What is the area of the region enclosed by the graph of $y^2=x+1$ and $y^2=-x+1$?

A. $\frac{3}{8}$sq.units C. $\frac{8}{3}$sq.units

B. $\frac{4}{3}$sq.units D. $\frac{3}{4}$sq.units

Solution: $y^2=x+1\Rightarrow x=y^2-1$ and $y^2=-x+1\Rightarrow x=1-y^2$



Let $f(y)=y^2-1$ and $g(y)=1-y^2$

The intersection of $f(y)$ and $g(y)$ is

$f(y)=g(y)\Rightarrow y^2-1=1-y^2$

$\Rightarrow 2y^2=2\Rightarrow y=1$ or $y=-1$

$\therefore A=\int_{-1}^{1}[g(y)-f(y)]dy$

Telegram: https://t.me/ExtremeSeries

$$\therefore A=\int_{-1}^{1}[1-y^2-(y^2-1)]dy=\int_{-1}^{1}(-2y^2+2)dy$$

$$\therefore A=\frac{-2}{3}y^3+2y\Big|_{-1}^{1}=\left(\frac{-2}{3}+2\right)-\left(\frac{2}{3}-2\right)=4-\frac{4}{3}=\frac{8}{3}$$

Answer: C

14. What is the value of $\int_1^2 \frac{x+4}{x(x+2)}dx$?

A. 2 B. ln 3 C. ln 2 D. ln 4 − ln 3

Solution: $\int_1^2 \frac{x+4}{x(x+2)}dx$ but partial fraction decomposition of

$$\frac{x+4}{x(x+2)}=\frac{A}{x}+\frac{B}{x+2}=\frac{Ax+2A+Bx}{x(x+2)}$$

$\Rightarrow A+B=1$ and $2A=4$, $A=2$, $B=1-2=-1$

$$\int_1^2 \frac{x+2}{x(x+2)}dx=\int_1^2 \frac{2}{x}dx-\int_1^2 \frac{1}{x+2}dx=2\ln|x|-\ln|x+2|$$

$$\ln\left|\frac{x^2}{x+2}\right|=\ln\frac{x^2}{x+2}\Big|_1^2=\ln\frac{4}{4}-\ln\left(\frac{1}{3}\right)$$

$$=\ln 1-(\ln 1-\ln 3)=\ln 1-\ln 1+\ln 3$$

$$\therefore \int_1^2 \frac{x+4}{x(x+2)}dx=\ln 3$$

Answer: B

15. What is the slope of the tangent line to the graph of $f(x)=3e^x+\sin x+2$ at (0, 5)?

A. 2 B. 3 C. 5 D. 4

Solution:

$f(x)=3e^x+\sin x+2$ at (0, 5) then the slope of the tangent line is

$f'(x)=3e^x+\cos x \Rightarrow f'(0)=3e^0+\cos 0=4$

The slope of tangent line is 4 and equation of the tangent line:

$$y-f(0)=f'(0)(x-0)$$

$$\Rightarrow y-5=4(x-0)\Rightarrow y=4x+5$$

Answer: D

16. The population of certain city is increasing at a rate of 3% per year. If the population was 100,000 in 2010 E.C., then what will be the population in 2020 E.C? (Given $(1.03)^9=1.30$, $(1.03)^{10}=1.34$, $(1.3)^9=10.60$, $(1.3)^{10}=13.78$)

A. 134,000 C. 1,060,000

B. 130,000 D. 1,378,000

Solution: Population in certain country increasing at the rate of 3%

$P_0=100{,}000$ then the population in 2020

$P=P_0(1+r)^{10}\Rightarrow P=100{,}000(1.03)^{10}=134{,}000$

Answer: A

17. If $z=-3+4i$ and $w=1+2i$, then what is the value of $\frac{2z}{w}+\bar{w}$?

A. $2+3i$ B. $3+5i$ C. $3+2i$ D. $3-2i$

Solution: $z=-3+4i$ and $w=1+2i$

$$\frac{2z}{w}+\bar{w}=\frac{2(-3+4i)}{1+2i}+1-2i=\frac{-6+8i}{1+2i}+1-2i$$

$$=\frac{-6+8i+1-2i+2i+4}{1+2i}=\frac{-1+8i}{1+2i}$$

$$\therefore \frac{2z}{w}+\bar{w}=\frac{-1+8i}{1+2i}\times\frac{1-2i}{1-2i}=\frac{-1+2i+8i+16}{5}=3+2i$$

Answer: C

18. A salesman sold items x_1, x_2 and x_3, with different rates of commissions as shown in the table below.

Months	Sales of units			Total Commission (in Birr)
	x_1	x_2	x_3	
February	90	100	20	800
March	130	50	40	900
April	60	100	30	850

What are the rates of commission on items x_1, x_2 and x_3, respectively?

A. 4, 2 and 11 C. 4, 11 and 2

B. 2, 4 and 11 D. 11, 2 and 4

Solution: From the above table $\begin{cases}90x_1+100x_2+20x_3=800\\130x_1+50x_2+40x_3=900\\60x_1+100x_2+30x_3=850\end{cases}$

Solve by using cramer's rule $D=\begin{vmatrix}9&10&2\\13&5&4\\6&10&3\end{vmatrix}=-175$

$$D_x=\begin{vmatrix}80&10&2\\90&5&4\\85&10&3\end{vmatrix}=-350, D_y=\begin{vmatrix}9&80&2\\13&90&4\\6&85&3\end{vmatrix}=-700$$

Telegram: https://t.me/ExtremeSeries

$$D_z = \begin{vmatrix} 9 & 10 & 80 \\ 13 & 5 & 90 \\ 6 & 10 & 85 \end{vmatrix} = -1925 \text{ then } x = \frac{D_x}{D} = \frac{-350}{-175} = 2$$

$$y = \frac{D_y}{D} = \frac{-700}{-175} = 4 \text{ and } z = \frac{-1925}{-175} = 11$$

The rates of common x_1, x_2 and x_3 are 2, 4 and 11 respectively.

Answer: B

19. If A is a square matrix of order 3 and det (A) = 5, then what is the value of det (A . adj (A))?

A. 3 B. 5 C. 125 D. 25

Solution: det (A. adj (A)) = det (A). det (adj. A) = $5 \times 5 = 25$

Answer: D

20. If $f'(x) = e^{x-1} + 3x^2 - \frac{1}{x}$ and $f(1) = 5$, what is $f(x)$?

A. $f(x) = e^{x-1} + 3x^2 + \frac{1}{x^2} + 2$

B. $f(x) = e^{x-1} - x^3 + \ln x + 5$

C. $f(x) = e^{x-1} + 3x^2 - \frac{1}{x} + 5$

D. $f(x) = e^{x-1} + x^3 - \ln x + 3$

Solution: If $f'(x) = e^{x-1} + 3x^2 - \frac{1}{x}$ and $f(1) = 5, f(x) =$

$$f(x) = \int f'(x)dx \Rightarrow f(x) = e^{x-1} + x^3 - \ln x + c$$

$$f(1) = e^{1-1} + 1^3 - \ln 1 + C = 5 \Rightarrow C = 3$$

$$\therefore f(x) = e^{x-1} + x^3 - \ln x + 3$$

Answer: D

21. Suppose the following are the premises of an argument.

He is healthy and he is not angry.
He is angry or his plan fails.
His plan does not fail if he does not travel abroad.

Given that the premises are true, which one of the following can be a conclusion that makes the argument valid?

A. His plan fails and he is angry.
B. His plan does not fail.
C. He travels abroad.
D. His plant fails and he does not travel abroad.

Solution: First translate into symbolic form:

Let P ≡ he is healthy and q ≡ he is angry
S ≡ his plan fails and r ≡ he travel abroad
Then the argument, is of the form: $p \wedge \neg q$, q V s, $\neg r \Rightarrow \neg s$

1. $p \wedge \neg q \equiv T$ if $p \equiv T$ and $\neg q \equiv T$, $q \equiv F$
2. q V s ≡ T, if s ≡ T, because q ≡ F (from 1)
3. $\neg r \Rightarrow \neg s \equiv T$, if $\neg r \equiv F$, because $\neg s \equiv F$... (from 2)

$\therefore p \wedge \neg q$, q V s, $\neg r \Rightarrow \neg s \vdash r$ is valid

Answer: C

22. A man running a race-course noted that the sum of the distance from the two flag posts from him is always 10 meters and the distance between the flag posts is 8 meters. What is the equation of the path traced by the man? (Take flag posts to be on the x-axis with origin at their midpoint)

A. $\frac{x^2}{9} + \frac{y^2}{25} = 1$ C. $\frac{x^2}{100} + \frac{y^2}{64} = 1$

B. $\frac{x^2}{25} + \frac{y^2}{9} = 1$ D. $\frac{x^2}{64} + \frac{y^2}{100} = 1$

Solution: Distance between two foci is 8m

$$2C = 8m \Rightarrow C = 4m$$

$$2b = 10m \Rightarrow b = \frac{10}{2}m = 5m$$

$$a = \sqrt{b^2 - c^2} = \sqrt{25 - 16} = \sqrt{9} = 3m$$

$\therefore$ The equation of the path traced by the man $\frac{x^2}{9} + \frac{y^2}{25} = 1$

Answer: A

23. What is the value of $\int_1^9 \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$?

A. $e^3 - e$ B. $\frac{e^3}{3} - e$ C. $e\left(e^2 - \frac{1}{3}\right)$ D. $2(e^3 - e)$

Solution: The value of $\int_1^9 \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$ let $u = \sqrt{x}$

$$\frac{du}{dx} = \frac{1}{2\sqrt{x}} \Rightarrow dx = 2\sqrt{x}\, du = 2udu$$

$$2\int_1^3 e^u du = 2e^u \Big|_1^3 = 2e^3 - 2e = 2(e^3 - e)$$

Answer: D

Telegram: https://t.me/Extremeseries

24. What is the value of $\int x\sqrt{1-x^2}\,dx$?

A. $\frac{-1}{3}(1-x^2)^{3/2}+C$ C. $(1-x^2)^{3/2}+C$

B. $-2x\sqrt{1-x^2}+C$ D. $\frac{1}{2}\sqrt{1-x^2}+C$

Solution: $\int x\sqrt{1-x^2}\,dx = \int \sqrt{1-x^2}\,x dx$

Let $u = 1-x^2 \Rightarrow du = -2xdx \Rightarrow xdx = \frac{du}{-2}$

$= \frac{-1}{2}\int u^{1/2}du = -\frac{1}{3}u^{3/2}+c$

$\therefore \int x\sqrt{1-x^2}\,dx = \frac{-1}{3}(1-x^2)^{3/2}+c$

Answer: A

25. Which one of the following is **NOT** true about the function $f(x) = 3x^4 - 4x^3$?

A. (0, 0) is point of inflection of f.

B. 0 and 1 are critical number of f.

C. f is concave upward on $\left(0, \frac{2}{3}\right)$ and concave downward on $(-\infty, 0)$ and on $\left(\frac{2}{3}, \infty\right)$.

D. f is decreasing on $(-\infty, 1)$ and increasing on $(1, \infty)$.

Solution: $f(x) = 3x^4 - 4x^3$

$f'(x) = 12x^3 - 12x^2 = 12x^2(x-1) = 0$

$\Rightarrow x = 0$ and $x = 1$ are critical numbers

		0		1	
$12x^2$	− − −	0	+ + +		+ + +
$x - 1$	− −		− −	0	+ + +
$f'(x) = 12x^2(x-1)$	− − −	0	− − −	0	+ + +

$f'(x) > 0$ on $(-\infty, 0)$ and $(1, \infty)$ is increasing

$f'(x) < 0$ on $(0, 1)$ is decreasing

To test concavity

$f''(x) = 36x^2 - 24x = 12x(3x-2) = 0 \Rightarrow x = 0$ or $x = \frac{2}{3}$

$\therefore (0, 0)$ and $\left(\frac{2}{3}, \frac{-16}{27}\right)$ are inflection points

		0		$\frac{2}{3}$	
$12x$	− − −	0	+ + +		+ + ++
$3x - 2$	− −		−	0	+ + +
$f''(x) = 12x(3x-2)$	+ + +	0	− −	0	+ + ++

$f''(x) < 0$ on the intervals $\left(0, \frac{2}{3}\right)$ the graph is concave upward

$f''(x) > 0$ on $(-\infty, 0)$ and $\left(\frac{2}{3}, \infty\right)$ which is concave down ward

Answer: D

26. What is the value of $\lim_{x\to\infty}\frac{1}{x^2}\sin^2\left(\frac{x}{2}\right)$?

A. $\frac{1}{2}$ B. $\frac{1}{4}$ C. 2 D. 4

Solution: $\lim_{x\to 0}\frac{1}{x^2}\sin^2\left(\frac{x}{2}\right)$

Let $u = \frac{x}{2} \Rightarrow x = 2u$ as $x \to 0, u \to 0$

$\lim_{u\to 0}\frac{\sin^2 u}{(2u)^2} = \frac{1}{4}\lim_{u\to 0}\frac{\sin^2 u}{u^2} = \frac{1}{4}$

Answer: B

27. If $f(x) = \frac{|x|}{x}$ and $g(x) = \frac{x+2}{x^3-4x}$ then what is the value of $\lim_{x\to -2}(f(x)+g(x))$?

A. $\frac{-9}{8}$ B. ∞ C. $\frac{9}{8}$ D. $\frac{-7}{8}$

Solution: $f(x) = \frac{|x|}{x}, g(x) = \frac{x+2}{x^3-4x}$

$\lim_{x\to -2}(f(x)+g(x)) = \lim_{x\to -2} f(x) + \lim_{x\to -2} g(x)$

$\lim_{x\to -2}(f(x)+g(x)) = \lim_{x\to -2}\frac{|x|}{x} + \lim_{x\to -2}\frac{x+2}{x(x+2)(x-2)}$

$\therefore \lim_{x\to -2}(f(x)+g(x)) = -1 + \frac{1}{8} = \frac{-7}{8}$

Answer: D

Telegram: https://t.me/ExtremeSeries

28. Let $z=\left(\frac{1-i}{1+i}\right)^{18}$. Then what is the value of z?

A. -1 B. i C. $-i$ D. $1-i$

Solution: Let $z=\left(\frac{1+i}{1+i}\right)^{18}$

$$z=\left(\frac{1-i}{1+i}\times\frac{1-i}{1-i}\right)^{18}=\left(\frac{-2i}{2}\right)^{18}=(-i)^{18}=(-1)^{18}.((i)^2)^9=1.(-1)^9=-1$$

Answer: A

29. Let A be a 3×3 invertible matrix and B be any 3×3 matrix. If $|A|=a$ and, $|B|=b$, then which one of the following is **NOT** true?

A. $|A^TA|=a^2$

B. $|kA|=k^3|A|$ for any $k\in\mathbb{R}$

C. $|A^{-1}B|=ab$

D. If $b=0$, then B is not invertible

Solution: A be non – singular 3×3 matrix and B be any matrix

If $|A|=a$ and $|B|=b$

$|A|=|A^T|=a$

A. $|A^TA|=|A^T||A|=a.a=a^2$

B. $|kA|=k^3|A|$ for any $k\in\mathbb{R}$

C. $|A^{-1}||B|=\frac{1}{a}.b=\frac{b}{a}$

D. If $b=0$ then B is not invertible

Answer: C

30. The time needed to type a sample of 8 business letters in an office is 7, 8, 6, 8, 9, 7, 5, 6 minutes. What are the mean $(\bar{x})$ and the standard deviation (s) of the data in minute?

A. $\bar{x}=7, s=\sqrt{1.5}$ C. $\bar{x}=8, s=\sqrt{2}$

B. $\bar{x}=7, s=\sqrt{2}$ D. $\bar{x}=8, s=\sqrt{1.5}$

Solution:

Sample of 8 business letters in an office is 7, 8, 6, 8, 9, 7, 5, 6 minutes

$$\text{Mean}: \bar{x}=\frac{7+8+6+8+9+7+5+6}{8}=\frac{56}{8}=7$$

Standard deviation:

$$\delta=\sqrt{\frac{(5-7)^2+2(6-7)^2+2(7-7)^2+2(8-7)^2+1(9-7)^2}{8}}$$

$$\delta=\sqrt{\frac{4+2+0+2+4}{8}}=\sqrt{\frac{12}{8}}=\sqrt{1.5}$$

$\therefore$ Mean $(\bar{x})=7$ and standard deviation $(\delta)=\sqrt{1.5}$

Answer: A

31. A private college has 1000 students. 60% of these students are males. 45% of these students pay their payment by credit card including 175 females. What is the probability that the student is a male or a credit card user?

A. 0.675 B. 0.225 C. 0.325 D. 0.775

Solution: Let S – students; M – male students; F – Female students; S – Credit card user students

$n(S)=1000$

$n(M)=60\%$ of $1000=600$

$n(F)=40\%$ of $1000=400$

$P(M)=0.6$, $P(F)=0.4$, P (of credit card user) $=0.45$

P(male or a credit card user) $=0.6+0.45-0.275=0.775$

Answer: D

32. Which one of the following is true about the graph of $f(x)=\frac{2x^3+2x^2+3x}{x^2+x}$?

A. The vertical asymptote of the graph is only $x=-1$ and its oblique asymptote is $y=2x$.

B. The graph has y-intercept at $(0, 3)$.

C. The graph has at least one x-intercept.

D. The vertical asymptotes of the graph are at $x=0$ and $x=-1$ but it has not horizontal asymptote.

Solution: $f(x)=\frac{2x^3+2x^2+3x}{x^2+x}=\frac{x(2x^2+2x+3)}{x(x+1)}$

f has a hole at $x=0$ and vertical asymptote $x=-1$ and oblique asymptote $y=2x$

The graph has no x and y intercept.

Answer: A

33. Let $f(x)=\frac{3x+1}{x-2}$. Then what is the range of $f(x)$?

A. $\mathbb{R}\setminus\{2\}$ B. $\mathbb{R}$ C. $\mathbb{R}\setminus\{3\}$ D. $\mathbb{R}\setminus\left(\frac{1}{3}\right)$

Solution: Let $y=\frac{3x+1}{x-2}$

Telegram: https://t.me/ExtremeSeries

$yx - 2y = 3x + 1$

Solve for x:

$yx - 3x = 2y + 1 \Rightarrow x(y-3) = 2y + 1$

$\therefore x = \dfrac{2y+1}{y-3}$

$\therefore$ Range of $f = \mathbb{R} \setminus \{3\}$

Answer: C

34. What is the value of $\int \frac{1}{x}(\ln x + x^2 e^{-x})dx$?

A. $\frac{1}{2}x^2 \ln x - (x+1)e^{-x} + C$ C. $\frac{1}{2}x^2 \ln x + (2-x)e^{-x} + C$

B. $\frac{1}{2}\ln^2 x - (x+1)e^{-x} + C$ D. $\frac{1}{2}\ln^2 x + (2-x)e^{-x} + C$

Solution: $u = x$; $du/dx = 1$; $dv = e^{-x}dx$; $v = -e^{-x}$

$$\int \frac{1}{x}(\ln x + x^2 e^{-x})dx = \int \frac{\ln x}{x}dx + \int xe^{-x}dx$$

$$u = \ln x \Rightarrow du/dx = \frac{1}{x} \Rightarrow dx = xdu$$

$$\int \frac{1}{x}(\ln x + x^2 e^{-x})dx = \int \frac{ux}{x}du + \int xe^{-x}dx$$

$$= \frac{u^2}{2} + (uv - \int vdu) = \frac{1}{2}(\ln x)^2 + (xe^{-x} - \int -e^{-x}dx)$$

$$= \frac{1}{2}(\ln x)^2 - xe^{-x} - e^{-x} + c$$

$$\therefore \int \frac{1}{x}(\ln x + x^2 e^{-x})dx = \frac{1}{2}(\ln x)^2 - e^{-x}(x+1) + c$$

Answer: B

35. If $f(x) = \frac{1}{3}x^3 + cx^2 + ax + 5$ has a local minimum value at $x = 1$, then which one of the following is true about the possible values of a and c?

A. $a = 3, c = -2$

B. $a = -2c - 1$, any real number

C. $a = -2c - 1, c < -1$

D. $a = -2c - 1, c > -1$

Solution:

$f(x) = \frac{1}{3}x^3 + cx^2 + ax + 5$ has a local minimum value at $x = 1$

$f(1) = \frac{1}{3} + c + a + 5 = a + c + \frac{16}{3}$

$f'(x) = \frac{3}{3}x^2 + 2cx + a \Rightarrow f'(x) = x^2 + 2Cx + a = 0$

$f'(1) = 1 + 2c + a = 0 \Rightarrow a + 2c = -1$

$a = -1 - 2c = -2c - 1$ and $c > -1$

Answer: D

36. Let $\{a_n\}$ be a sequence with $a_1 = a$, $a_2 = f(a_1) = f(a)$, $a_3 = f(a_2) = f(f(a))$, ... $a_n + 1 = f(a_n)$, where f is a continuous function. If $\lim_{n\to\infty} a_n = 5$, what is $f(5)$?

A. 1 B. 5^n C. 5^{n+1} D. 1

Solution: $\{a_n\}$ – be a sequence with

$a_1 = a$, $a_2 = f(a_1) = f(a)$

$a_3 = f(a_2) = f(f(a))$

$a_{n+1} = f(a_n)$ where f is a continuous function

$\lim_{n\to\infty} a_n = 5, f(5) = 5$

Answer: A

37. If the sum of the first three consecutive terms of an arithmetic progression $\{A_n\}$, with $A_n > 0$ for all n, is 9 and the sum of their squares is 35, then what is the sum S_n of the first n terms?

A. $n^2 + 1$ B. $n^2 - 1$ C. n^2 D. $2n^2 + 1$

Solution: $\{A_n\}$ be an arithmetic sequence with $A_n > 0$

$$a_1 + a_2 + a_3 = 9 \Rightarrow s_3 = \frac{3}{2}(a_1 + a_3) = 9 \Rightarrow a_1 + a_3 = 6$$

$$a_1^2 + a_2^2 + a_3^2 = 35 \Rightarrow a_1^2 + a_3^2 = 26$$

$$\Rightarrow a_1 + a_3 + a_2 = 9 \Rightarrow 6 + a_2 = 9 \Rightarrow a_2 = 3$$

$$(a_1 + a_3)^2 = a_1^2 + 2a_1a_3 + a_3^2 = 36$$

$$a_1^2 + a_3^2 = 36 - 2a_1a_3$$

$$26 + 2a_1a_3 = 36 \Rightarrow 2a_1a_3 = 10$$

$$\Rightarrow a_1a_3 = 5 \Rightarrow a_3 = \frac{5}{a_1}$$

$$a_1 + 3 + \frac{5}{a_1} = 9 \Rightarrow a + \frac{5}{a_1} = 6 \Rightarrow a_1^2 - 6a_1 + 5 = 0$$

Telegram: https://t.me/Extremeseries

$a_1 = \frac{6 \pm \sqrt{36-4\times5}}{2} = \frac{6\pm4}{2} = 5$ or 1

$a_3 = \frac{5}{5} = 1$ or $a_3 = \frac{5}{1} = 5$

$\therefore (a_1, a_2, a_3) = (1, 3, 5)$ or $(a_3, a_2, a_1) = (5, 3, 1)$

If $a_1 = 1$, $d = 2$ then $a_n = a_1 + (n-1)d$

$a_n = 1 + (n-1)2 = 2n - 1$

$S_n = \frac{n}{2}(1 + 2n - 1) = n^2$

$\therefore$ The sum of the first n – terms is $S_n = n^2$

Answer: C

38. Let A be a 3 × 3 matrix and |A| = –2. Then what is the value of |adj(A)|?

A. –8 B. –2 C. $-\frac{1}{2}$ D. 4

Solution:

Let A be 3 × 3 matrix and |A| = –2 then |adj A| = _____

Note: $|A| = |A^T| = |\text{adj } A|$

Since |A| = –2 then |adj A| = –2

Answer: B

39. Let A = {1, 2, 3, 4, 5, 6, 7}, B = {7, 8, 9} and C = {8, 9, 10}. If one of the numbers is deleted randomly from each of these sets, what is the probability that all the three deleted numbers are even or all are multiples of 3?

A. $\frac{4}{5}$ B. $\frac{2}{21}$ C. $\frac{1}{9}$ D. $\frac{8}{63}$

Solution:

A = {1, 2, 3, 4, 5, 6, 7}, B = {7, 8, 9}, C = {8, 9, 10}

A ∪ B ∪ C both even and multiplies of 3 = {6}

$$P(\text{even or multiple of } 3) = \frac{4}{10} + \frac{2}{10} + \frac{3}{10} - \frac{1}{10} = \frac{8}{10} = \frac{4}{5}$$

Answer: A

40. The age distribution of students in a certain class is given below:

Age	10 – 14	15 – 19	20 – 24	25 – 29
No. of students	2	10	6	7

What is the modal value of the distribution?

A. 17.83 B. 17.38 C. 18.37 D. 18.73

Solution:

Age (Class boundaries)	9.5–14.5	14.5–19.5	19.5–24.5	24.5–29.5
No. of students	2	10	6	7

Modal class 14.5 – 19.5

$d_1 = 10 - 2 = 8$ and $d_2 = 10 - 6 = 4$ and i = 5

then $M_o = B_L + \left(\frac{d_1}{d_1 + d_2}\right) i$

$M_o = 14.5 + \left(\frac{8}{3+4}\right)5 \Rightarrow M_o = 14.5 + \left(\frac{8}{3+4}\right)5$

$M_o = 14.5 + \frac{40}{12} = 17.83$

Answer: A

41. If $f(x) = k \ln x + e^{\sin x}$ and $f''(\pi) = -1$, then what is the value of k?

A. $2\pi^2$ B. π^2 C. π D. 2π

Solution: $f(x) = k \ln x + e^{\sin x}$, $f''(\pi) = -1$ then $k =$ _____

$f'(x) = \frac{k}{x} + \cos x e^{\sin x}$ and let $u = e^{\sin x}$: $dv = \cos x\, dx$, $v = \sin x$

$f''(x) = \frac{-k}{x^2} - \sin x e^{\sin x} + \cos^2 x e^{\sin x}$

$f''(\pi) = \frac{-k}{\pi^2} - 0 + 1 = -1 \Rightarrow \frac{-k}{\pi^2} = -2 \Rightarrow k = 2\pi^2$

Answer: A

42. Let $f(x) = x - x^2$ and $g(x) = \frac{1}{x}$. Then what is $g\left(f\left(\frac{1}{x}\right)\right)$ equals to

A. $x - x^2$ B. $\frac{x^2}{x-1}$ C. $\frac{1}{x^2 - x}$ D. $\frac{x-1}{x^2}$

Solution: $f(x) = x - x^2$ and $g(x) = \frac{1}{x}$ then

$g\left(f\left(\frac{1}{x}\right)\right) = g\left(\frac{1}{x} - \frac{1}{x^2}\right) = g\left(\frac{x-1}{x^2}\right) = \frac{1}{\frac{x-1}{x^2}} = \frac{x^2}{x-1}$

Answer: B

43. A cylindrical tank whose inner diameter is 2m contains 4π m^3 oil. If the oil is discharged from the tank at the rate of $\frac{2\pi}{3}$ m^3/min, then how long (in min) does it take for the tank to be empty?

Telegram: https://t.me/ExtremeSeries

A. $\frac{4}{3}$ B. 4 C. 12 D. 6

Solution: $V = \pi r^2 h$, radius: $r = \frac{D}{2} = \frac{2}{2} = 1$

$$\frac{dv}{dt} = \pi r^2 \frac{dh}{dt}$$

$$\frac{2\pi}{3} = \pi(1)\frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{2}{3} \Rightarrow dh = \frac{2}{3}dt$$

$$\int dh = \frac{2}{3}dt \Rightarrow h = \frac{2}{3}t \text{ and } h = \frac{4\pi}{\pi(1)} = 4m$$

$$\therefore t = \frac{3}{2}h = \frac{3}{2}(4) = 6 \text{ minute}$$

Answer: D

44. What are the values of a and b so that the function $f(x) = \begin{cases} x+1, x<1 \\ ax+b, 1 \le x < 2 \\ 3x, x \ge 2 \end{cases}$ is continuous everywhere?

A. $a = 4, b = -2$ C. $a = 4, b = 2$

B. $a = -4, b = -2$ D. $a = -4, b = 2$

Solution: f is continuous everywhere so that, limit exists

✓ $\lim_{x\to 1^+} f(x) = \lim_{x\to 1^-} f(x)$ and $\lim_{x\to 2^-} f(x) = \lim_{x\to 2^-} f(x)$

$\Rightarrow \lim_{x\to 1^+}(ax+b) = \lim_{x\to 1^-}(x+1)$ and $\lim_{x\to 2^-} 3x = \lim_{x\to 2^-}(ax+b)$

$\Rightarrow a + b = 2$ and $6 = 2a + b$

Solving together: $\begin{cases} a+b=2 \\ 2a+b=6 \end{cases}$

$-a = -4$

$\therefore a = 4$ and $b = 2 - a = -2$

Answer: A

5. Let U = ℕ (the set of natural numbers) be the universe. Which one of the following proposition is true?

A. $(\exists x)(x + x = x)$ C. $(\forall x)(\exists y)(x \div y = y \div x)$

B. $(\forall x)(\exists y)(y < x)$ D. $(\forall x)(\exists y)(x - y = x)$

Solution: u = the set of natural numbers

A. $(\exists x)(x + x = x)$ is false, because there is no natural number x, which is $x + x = x$

B. $(\forall x)(\exists y)(y < x)$. For every natural number x, there is a natural number y such that $y < x$ is false. For example, let $x=1$ and $y<1$.
There is no natural number less than 1.

C. $(\forall x)(\exists y)(x \div y = y \div x)$ is true
For every natural number x, there is a natural number y such that $x \div y = y \div x$ is True. For example; if $x = 1$, $y = 1$; if $x = 3$, $y = 3$, and so on.

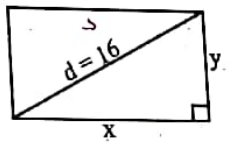
D. For every natural number x, there is a natural number y such that $x - y = x$ is false.

Answer: C

46. What is the maximum possible area of a rectangle in square units with diagonal of length 16 units?

A. 48 B. 128 C. 64 D. 256

Solution: Area $= xy$



- $x^2 + y^2 = 16^2 = 256 \Rightarrow y = \sqrt{256 - x^2}$
- Area: $A(x) = x\sqrt{256 - x^2} = \sqrt{256x^2 - x^4}$

Let $g(x) = 256x^2 - x^4$ and $g'(x) = 512x - 4x^3$

$g'(x) = 0 \Rightarrow 512x - 4x^3 = 0 \Leftrightarrow 4x(128 - x^2) = 0$

$\therefore x = 0$ or $x = \sqrt{128} = 8\sqrt{2}$

$\therefore$ The maximum area $= \left(8\sqrt{2}\right)\left(8\sqrt{2}\right) = 128$

Answer: B

47. Let $f(x) = \ln\left(x\sqrt{x}\right)$. Then what is $f'(x)$ equal to?

A. $\frac{2x}{3}$ B. $\frac{\sqrt{x}}{2}$ C. $\frac{2}{x\sqrt{x}}$ D. $\frac{3}{2x}$

Solution: $f(x) = \ln\left(x\sqrt{x}\right) = \ln\sqrt{x^3} = \ln x^{3/2}$

$$\Rightarrow f(x) = \frac{3}{2}\ln x \text{ and } f'(x) = \frac{3}{2x}$$

Answer: D

48. Which one of the following is equal to $\frac{d}{dx}\log_2^{\sqrt{6x}}$?

A. $\frac{1}{2x\ln(2)}$ C. $\frac{1}{6x\ln(2)}$

Telegram: https://t.me/Extremeseries

B. $\frac{3}{2x\ln(2)}$ D. $\frac{1}{6x\ln(2)}$

Solution: $\log_2\sqrt{6x} = \frac{\ln\sqrt{6x}}{\ln 2} = \frac{\ln(6x)^{1/2}}{\ln 2} = \frac{\ln 6x}{2\ln 2}$

$\frac{d}{dx}\log_2\sqrt{6x} = \frac{d}{dx}\left(\frac{\ln 6x}{2\ln 2}\right) = \frac{6}{(6x)2\ln 2} = \frac{6}{12x\ln 2} = \frac{1}{2x\ln 2}$

Answer: A

49. What is the sum of the series $\sum_{n=1}^{\infty} 3^n 4^{-n}$?

A. ∞ B. $\frac{3}{16}$ C. 3 D. 4

Solution: The sum of the series $\sum_{n=1}^{\infty} 3^n 4^{-n} = \frac{3}{4} + \frac{9}{16} + \ldots$

$G_1 = \frac{3}{4}, r = \frac{3}{4}$

$S_{n\to\infty} = \frac{G_1}{1-r} = \frac{3/4}{1-3/4} = \frac{\frac{3}{4}}{1/4} = 3$

Answer: C

50. If $z = (1+\sqrt{3}i)(1+i)$ then which one of the following is the polar representation of z?

A. $z = 4(\cos(105°) + i\sin(105°))$

B. $z = 2\sqrt{2}(\cos(105°) + i\sin(105°))$

C. $z = 2\sqrt{2}(\cos(15°) + i\sin(15°))$

D. $z = 4((\cos 75°) + i\sin(75°))$

Solution: Let $z_1 = 1+\sqrt{3}i$ and $z_2 = 1 + i$

✓ Let $\theta_1 = Arg(z_1) = Arg(1+\sqrt{3}i) = \tan^{-1}\left(\frac{\sqrt{3}}{1}\right) = 60°$

✓ Let $\theta_2 = Arg(z_2) = Arg(1+i) = \tan^{-1}\left(\frac{1}{1}\right) = 45°$

✓ $|z_1| = \sqrt{1^2+\sqrt{3}^2} = \sqrt{4} = 2$ and $|z_2| = \sqrt{1^2+1^2} = \sqrt{2}$

✓ Polar form of $z = z_1 . z_2 = r_1.r_2(\cos(\theta_1+\theta_2) + i(\sin(\theta_1+\theta_2))$

$\therefore z = 2\sqrt{2}[\cos(60°+45°) + i\sin(60°+45°)]$

$= 2\sqrt{2}(\cos 105° + i\sin 105°)$

Answer: B

51. The variance of 20 observations is 5. If each observation is multiplied by 2, then what is the variance of the resulting observations?

A. 5 B. 10 C. 20 D. 40

Solution: If a data multiplying by constant k:

New variance = k^2 (old variance)

$\therefore$ New variance = $(2)^2(5) = (4)(5) = 20$

Answer: C

52. What is the value of $\lim_{n\to\infty}\sum_{k=1}^{n}\frac{2}{n}\left[\left(\frac{2k}{n}\right)^3 + 5\left(\frac{2k}{n}\right)\right]$?

A. 4 B. 14 C. 10 D. - 18

Solution: $\lim_{n\to\infty}\sum_{k=1}^{n}\frac{2}{n}\left[\left(\frac{2k}{n}\right)^3 + 5\left(\frac{2k}{n}\right)\right]$

$\lim_{n\to\infty}\frac{16}{n^4}\sum_{k=1}^{n}k^3 + \frac{20}{n^2}\sum_{k=1}^{n}k$

$\lim_{n\to\infty}\frac{16}{n^4}\left(\frac{n^2(n+1)^2}{4}\right) + \lim_{n\to\infty}\frac{20(n(n+1))}{2n^2}$

$\frac{16}{4}\lim_{n\to\infty}\frac{n^4+2n^3+n^2}{n^4} + \frac{20}{2}\lim_{n\to\infty}\frac{n^2+n}{n^2} = 4+10 = 14$ **Answer: B**

53. Let A (a, 2, 5) for $a > 0$, be a point on the sphere $x^2 + y^2 + z^2 - 6z = 0$ and C be the center of the sphere. If P(k, 2, 4) is a point in space such that $\overline{PA}$ is perpendicular to $\overline{CA}$, what is the cosine of the angle between $\overline{PA}$ and $\overline{PC}$?

A. $\frac{5}{\sqrt{35}}$ B. $\frac{7}{\sqrt{35}}$ C. $\frac{7}{\sqrt{70}}$ D. $\frac{5}{\sqrt{70}}$

Solution:

A (α, 2, 5) for α > 0 be point on the sphere $x^2 + y^2 + z^2 - 6z = 0$

$\Rightarrow x^2 + y^2 + z^2 - 6z + 9 = 9 \Rightarrow x^2 + y^2 + (z-3)^2 = 9$ and $r = 3$

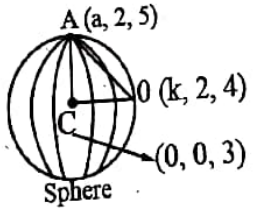
C – be center of the sphere i.e., (0, 0, 3)

P(k, 2, 4) is a point in the sphere

$\overline{PA} \perp \overline{CA}, \overline{CA} = (1,2,2)$

$r = \overline{CA} = \sqrt{\alpha^2+2^2+2^2} = 3$

$\Rightarrow \alpha^2 + 4 + 4 = 9 \Rightarrow \alpha^2 = 1 \Rightarrow \alpha = 1$



Telegram: https://t.me/Extremeseries

$\overrightarrow{PA} = (1-k, 0, 1)$

$\overrightarrow{PA}.\overrightarrow{CA} = 0 \Rightarrow 1-k+0+2=0 \Rightarrow k=3$

$\cos\theta = \dfrac{PA.CP}{|PA||CP|}$

$\overrightarrow{PC} = (-3,-2,-1), |\overrightarrow{CP}| = \sqrt{3^2+2^2+1^2} = \sqrt{14}$

$\overrightarrow{PA} = (-2,0,1), |\overrightarrow{PA}| = \sqrt{4+0+1} = \sqrt{5}$

$\cos\theta = \dfrac{\overrightarrow{PA}.\overrightarrow{CP}}{|\overrightarrow{PA}||\overrightarrow{CP}|} = \dfrac{6+0+(-1)}{\sqrt{5}.\sqrt{14}} = \dfrac{5}{\sqrt{5}\times\sqrt{14}} = \dfrac{5}{\sqrt{70}}$ **Answer: D**

54. What is the period (p) and the range of $f(x) = 5\sin\left(\frac{1}{3}x+2\right)+3$?

A. p = 6π, range = [−2, 8] C. p = 6π, range = [−5, 5]

B. $p = \frac{2\pi}{3}$, range = [−5, 5] D. $p = \frac{2\pi}{3}$, range = [−2, 8]

Solution: $f(x) = 5\sin\left(\frac{1}{3}x+2\right)+3$

$\text{Period} = \dfrac{2\pi}{1/3} = 6\pi$

Range = $[-a+c, a+c] = [-5+3, 5+3] = [-2, 8]$

∴ The period and Range of f is 6π and [−2, 8] respectively.

Answer: A

55. Let ℓ be the line given by the vector equation $(x, y) = (-2, 1) + \lambda(1, 1), \lambda \in \mathbb{R}$. Which one of the following is the equation of the image of ℓ after being translated by the vector u = (2, −1) followed by a rotation through 45° about the origin?

A. $y = \sqrt{2}x$ B. $y = \sqrt{2}$ C. $x = 0$ D. $y = -2\sqrt{x}$

Solution: Vector equation $(x, y) = (-2, 1) + \lambda(1, 1)\ \lambda \in \mathbb{R}$

$x+2 = x = y-1 \Rightarrow y = x+3$

$y = x+3 \xrightarrow{\text{translated by u}(2,-1)} y = x$ take (1, 1) be points on $y = x$ then $(1,1) \xrightarrow{\theta = 45°}$

$x' = x\cos\theta - y\sin\theta$

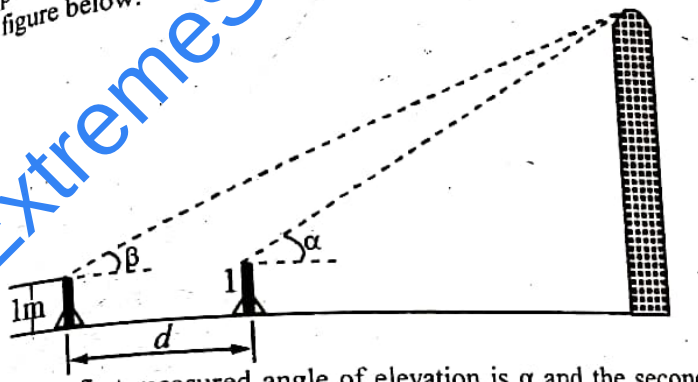
$x' = 1\cos 45° - 1\sin 45° = 0$

$y' = x\sin\theta + y\cos\theta$

$y' = 1\sin 45° + 1\cos 45° = \dfrac{2\sqrt{2}}{2} = \sqrt{2}$

∴ The image is $y = \sqrt{2}$. **Answer: B**

56. In order to measure the height of a tower, suppose a surveyor takes two sightings from a transit 1 meter high which are positioned d meters apart on the same ground level as in the figure below.

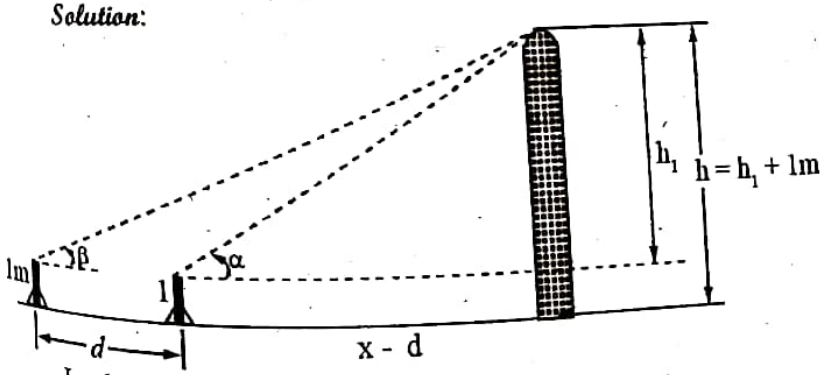


If the first measured angle of elevation is α and the second is β (see, the figure), then what is the height of the tower (in meter) in terms of α, β and d?

A. $\dfrac{\tan\alpha\tan\beta}{(\tan\alpha-\tan\beta)d}+1$ C. $\left(\dfrac{\tan\alpha-\tan\beta}{\tan\alpha\tan\beta}\right)d+1$

B. $\left(\dfrac{\tan\alpha\tan\beta}{\tan\alpha-\tan\beta}\right)d+1$ D. $\left(\dfrac{\tan\alpha\tan\beta}{\tan\alpha+\tan\beta}\right)d+1$

Solution:



Let $h = h_1 + 1m$

h – be the height of tower

Telegram: https://t.me/ExtremeSeries

x – be the distance between the end surveyor sighting

$\tan\alpha = \frac{h_1}{x-d} \Rightarrow h_1 = (x-d)\tan\alpha$ ..... (1)

$\tan\beta = \frac{h_1}{x} \Rightarrow h_1 = x\tan\beta$ ..... (2)

From 1 and 2

$x\tan\beta = x\tan\alpha - d\tan\alpha$

$\Rightarrow x\tan\beta + d\tan\alpha = x\tan\alpha$

$x(\tan\alpha - \tan\beta) = d\tan\alpha \Rightarrow x = \frac{d\tan\alpha}{\tan\alpha - \tan\beta}$

$h = h_1 + 1m = \frac{d\tan\alpha}{\tan\alpha - \tan\beta}\tan\beta + 1$

$h = \left(\frac{\tan\alpha\tan\beta}{\tan\alpha - \tan\beta}\right)d + 1$ **Answer: B**

57. If the dot products of a vector $\vec{A}$ with the vectors $i - j + k$, $2i + j - 3k$ and $i + j + k$ are 4, 0 and 2, respectively, what is $\vec{A}$?

A. $\vec{A} = (2,1,1)$ C. $\vec{A} = (-2,-1,1)$

B. $\vec{A} = (-2,1,-1)$ D. $\vec{A} = (2,-1,1)$

Solution: Let $u = i - j + k$; $v = 2i + j - 3k$; $w = i + 5 + k$

Let $A = x_i + yj + zk$

A.u $= x - y + z = 4$ 1

A. $v = 0 \Rightarrow 2x + y - 3z = 0$ 2

A. $W = 2 \Rightarrow x + y + z = 2$ 3

From 1 and 2 and 1, and 3 respectively

$3x - 2z = 4 \Rightarrow 5x = 10$

$2x + 2z = 6 \Rightarrow x = 2$

$z = 3 - x \Rightarrow z = 3 - 2 = 1$

$y = x + z - 4$

$y = 2 + 1 - 4 = -1$

∴ vector A is (x, y, z) = (2, –1, 1) **Answer: D**

58. What is the image of the circle $x^2 + y^2 - 4x - 6y + 12 = 0$ when it is reflected with respect to the line $y = -x$?

A. $(x+3)^2 + (y+2)^2 = 1$ C. $(x+2)^2 + (y+3)^2 = 1$

B. $(x-2)^2 + (y-3)^2 = 1$ D. $(x-3)^2 + (y-2)^2 = 1$

Solution: The image of circle $x^2 - 4x + y^2 - 6y + 12 = 0$

$x^2 - 4x + 4 + y^2 - 6y + 9 = 4 + 9 + 12$

$(x-2)^2 + (y-3)^2 = 1$

The center of circle is (2, 3)

$(2,3) \xrightarrow{\text{reflected } y=-x} (-3,-2)$

∴ Image $(x+3)^2 + (y+2)^2 = 1$ **Answer: A**

59. Let P(1, α, α) and Q = (α – 1, 1, 1) be two points in space and the distance between P and Q is 3m. Then what is the value(s) of α?

A. $\alpha = -1, \alpha = 9$ C. $\alpha = 3, \alpha = -\frac{1}{3}$

B. $\alpha = 1, \alpha = -9$ D. $\alpha = -3, \alpha = \frac{1}{3}$

Solution: P(1, α, α) and Q = (α – 1, 1, 1) $\overline{PQ} = 3m$

$\overline{PQ} = \sqrt{(\alpha-2)^2 + (1-\alpha)^2 + (1-\alpha)^2} = 3$

$\alpha^2 - 4\alpha + 4 + 1 - 2\alpha + \alpha^2 + 1 - 2\alpha + \alpha^2 = 9$

$3\alpha^2 - 8\alpha + 6 = 9 \Rightarrow 3\alpha^2 - 8\alpha - 3 = 0$

$3\alpha^2 - 9\alpha + \alpha - 3 = 0$

$(\alpha - 3)(3\alpha + 1) = 0$

$\alpha - 3 = 0$ or $3\alpha + 1 = 0 \Rightarrow \alpha = 3$ or $\alpha = \frac{-1}{3}$ **Answer: C**

60. If the image of the line $2x - 3y = 7$ under a translation is $2x - 3y = 0$, which one of the following is a translation vector of the translation line?

A. u = (2, –1) C. u = (–1, 2)

B. u = (–2, 1) D. u = (1, –2)

Solution: The image of the line $2x - 3y = 7$ under a translation is $2x - 3y = 0$ then the translation vector is equal to: $y = \frac{2}{3}x - \frac{7}{3}$

direction vector is parallel to $y = \frac{2}{3}x$

$u = (a,b) = \left(a; \frac{2}{3}a\right) = a(3,2)$ take any point on $y = \frac{2}{3}x - \frac{7}{3}$

if $x = 2, y = -1$ i.e., u = (a, b) = (2, –1) position vector. **Answer: A**

61. Let P(n) be an open proposition on the set of natural numbers (ℕ). Which one of the following is a correct application of the principle of mathematical induction?

Telegram: https://t.me/ExtremeSeries

A. If P(1) is true for n = 1; and if both P(n) and P(n + 1) are true for certain n ∈ ℕ, then P(n) is true for all n ∈ ℕ.

B. If P(10) is true; and if P(n) is true implies that P(n + 1) is true, then P(n) is true for all n ∈ ℕ.

C. If P(1) is true; and P(n) ⇒P(n + 1) is true for any n ∈ ℕ, then P(n) is true for all n ∈ ℕ.

D. If P(10) is true; and assuming P(n) is true for any n > 10 if it follows that P(n + 1) is true, then P(n) is true for all n ≥ 10.

Solution: Let P_n – be an open proposition on the set of natural numbers (N) then correct application of the principle of mathematical induction such that P(1) is true for n = 1 P(n) and P(n + 1) is true for some n ∈ ℕ then P(n) is true for all n ∈ ℕ.

Answer: C

62. If, in ΔABC, AB = 3, BC = 4 and m(∠B) = 60°, then what are the length of AC and the cosine of ∠A, respectively?

A. $\sqrt{13}$ and $\frac{1}{\sqrt{13}}$

B. $\sqrt{13}$ and $\frac{-1}{\sqrt{13}}$

C. $\sqrt{13}$ and $\frac{6}{5\sqrt{13}}$

D. $\sqrt{13}$ and $\frac{-6}{5\sqrt{13}}$

Solution: In triangle ABC, $\overline{AB} = 3$, BC = 4 and m(∠B) = 60° then

$\overline{AC}$ = _____ and cos (∠A) = _____

By using cosine law

$AC^2 = AB^2 + BC^2 - 2AB.BC\cos(\angle B)$

$\overline{AC} = \sqrt{3^2 + 4^2 - 2\times3\times4\cos 60°}$

$\overline{AC} = \sqrt{25-12}$

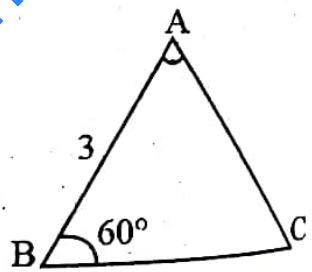
$\overline{AC} = \sqrt{13}$

$BC^2 = AB^2 + AC^2 - 2AB.AC\cos(A)$

$4^2 = 3^2 + \left(\sqrt{3}\right)^2 - 2\times3\times\sqrt{3}\cos(\angle A)$

$16 = 9 + 13 - 6\sqrt{13}\cos A$

$\cos A = \frac{1}{\sqrt{13}}$



$\therefore \overline{AC} = \sqrt{13}$ and $\cos(\hat{A}) = \frac{1}{\sqrt{13}}$

Answer: A

63. What is the standard equation of the line passing through the point (2, 3) and parallel to the line given by $\begin{cases} x = 1 + 2\lambda \\ y = -2 - \lambda \end{cases}, \lambda \in \mathbb{R}$?

A. $\frac{x-2}{-1} = \frac{y-3}{2}$

B. $\frac{x-1}{2} = \frac{y-3}{-2}$

C. $\frac{x-2}{2} = \frac{y-3}{-1}$

D. $\frac{x-1}{1} = \frac{y-3}{-2}$

Solution:

Use vector equation of the line: $r = r_o + \lambda v$.

$\lambda = \frac{x-1}{2} = \frac{y+2}{-1}$, and parallel vector v = (2, −1)

$\therefore r = r_o + \lambda v \Rightarrow r = (2, 3) + \lambda(2, -1)$

$\therefore x = 2 + 2\lambda$ and $y = 3 - x$

$\therefore \lambda = \frac{x-2}{2}$ and $\lambda = \frac{y-3}{-1}$

$\therefore$ The standard equation of the line is $\frac{x-2}{2} = \frac{y-3}{-1}$

Answer: C

64. Consider the following assertion:

$p + 2^n$ is an odd number for any prime p and any n ∈ ℕ.

Which one of the following is correct about a prove or disprove of the assertion?

A. The assertion can be proved by direct method; because p is odd and $2^n = 2(2^{n-1})$ is even imply that $p + 2^n$ is odd since the sum of odd and even is odd.

B. There is a counter example that disproves the assertion.

C. The assertion can be proved by the indirect method; because if n ∉ ℕ, then $2^n \notin \mathbb{N}$ and hence $p + 2^n$ is not odd.

D. The assertion can be disproved by the method of contradiction.

Solution:

$P + 2^n$ is odd number for any prime P and only n ∈ ℕ. The assertion can be proved by indirect method because there is a counter example that disprove the assertion p = 2 is the only even prime number $P + 2^n$ is even.

Answer: B

Telegram: https://t.me/ExtremeSeries

65. If θ is a fourth quadrant angle and $\sec\theta = \sqrt{2}$ then what is $\csc\theta$ equals to?

A. $\frac{-1}{\sqrt{2}}$ B. $-\sqrt{2}$ C. $\frac{1}{\sqrt{2}}$ D. $\sqrt{2}$

Solution: Let θ be fourth quadrant angle and $\sec\theta = \sqrt{2}$ then csc θ is equal to in the fourth quadrant sine, cosec, tangent and cotangent are negative.

$$\therefore \cos\theta = \frac{1}{\sec\theta} \Rightarrow \cos\theta = \frac{1}{\sqrt{2}}$$

$$\sin\theta = \pm\sqrt{1-\cos\theta} = \pm\sqrt{1-\left(\frac{1}{\sqrt{2}}\right)^2} \quad \sin\theta = \pm\sqrt{\frac{2-1}{2}} = \pm\sqrt{\frac{1}{2}}$$

$$\sin\theta = -\frac{1}{\sqrt{2}} \text{ then } \csc\theta = \frac{1}{\sin\theta} = -\sqrt{2}$$

Answer: B

SECTION 2 – CHEMISTRY

1. Nitrogen – fixing bacteria convert:
A. nitrate to nitrogen gas C. nitrate to ammonia
B. nitrogen gas to nitrate D. ammonia to nitrate

Explanation: In nitrogen cycle, nitrogen is taken from atmosphere by nitrogen fixing bacteria (Rhizobium bacteria) in root nodules of leguminous plants. So, the bacteria change atmospheric nitrogen to nitrate. **Answer: B**

2. What is the standard entropy change for the reaction:

$2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$

Given the following data; s° of SO_2 (g) = 248.1 J/mol. K; S° of O_2 (g) = 205.0 J/mol. K; S° of SO_3 (g) = 256.6 J/mol. K

A. -188.0 J/K C. -196.5 J/K
B. -172.0 J/K D. +60.1 J/K

Solution: The entropy change, ΔS, for a chemical reaction is equal to the sum of the entropies of the products of that reaction minus the sum of the entropies of the reactants.

For the reaction,

$aA + bB \longrightarrow cC + dD$, the entropy change is

$$\Delta S = \sum S_{Products} - \sum S_{reactants}$$

$$= [(c \times S_C) + (d \times S_D)] - [(a \times S_A) + (b \times S_B)]$$

Based on the above formula;

$2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$

The change in entropy (ΔS) = 2 (256.6 J/mol. k) – [2 (248.1 J/mol. k) + (205 J/mol. k)]

= (513.2 J/mol. k) – (701.2 J/mol. k)

= -188 J/mol. K **Answer: A**

3. How many moles are there in 159 grams of alanine, $C_3H_7NO_2$?
A. 0.560 B. 0.992 C. 1.78 D. 3.31

Solution: m = 159 grams

M = 89 g/mol

$$\text{Number of moles (n)} = \frac{\text{given mass (m)}}{\text{molar mass (M)}}$$

$$n = \frac{159 \text{ gram}}{89 \text{ gram/mol}} = 1.786 \text{ moles}$$

Answer: C

Telegram: https://t.me/ExtremeSeries

4. Which one of the following is NOT true of metallic bonding?
   A. It gives rise to excellent electrical conductivity.
   B. Electrons are free to move throughout the structure.
   C. The strength of metallic bonds increases down a group.
   D. The strength of metallic bonding affects the boiling point of metals.

Explanation

Experimental results indicate that the strength of metallic bonding first increases with increasing number of electrons, then decreases.

Answer: C

5. Synthetic fibers
   A. naturally produced
   B. manufactured through the use of chemical substance
   C. produced from animal hair fiber
   D. produced from plant substances

Answer: B

6. An amphiprotic species is a molecule or ion that can
   A. accept protons
   B. donate more than one proton
   C. both accept and donate protons
   D. be formed into a double ion

Answer: C

7. Which one is true for a triportic acid, such as phosphoric acid, H_3PO_4?
   A. $Ka_1 > Ka_2 > Ka_3$
   B. $Ka_3 > Ka_2 > Ka_1$
   C. $Ka_1 > Ka_2 = Ka_3$
   D. $Ka_1 = Ka_2 = Ka_3$

Explanation: The dissociation decreases due to the greater difficulty of removing a positively charged proton from a negatively charged ion (such as $H_2PO_4^-$) then from the neutral molecule H_3PO_4. And HPO_4^{2-} than $H_2PO_4^-$.

In short the dissociation decreases in order $HPO_4^{2-} < H_2PO_4^{2-} < H_3PO_4$. Hence $Ka_1 > Ka_2 > Ka_3$.

Answer: A

8. Which group of elements are the most reactive of all the metallic elements?
   A. Alakli metals
   B. Alkali earth metals
   C. Transition metals
   D. Group 2B metals

Answer: A

9. All of these are characteristics of most ionic compounds in the solid phase except
   A. high melting point
   B. solubility in water
   C. high electrical conductivity
   D. insolubility in organic solvents

Explanation: Ionic compounds don't conduct electricity in solid phase, because of lack of mobile ions.

Answer: C

10. Consider the following:

$$2NO\,(g) + Cl_2\,(g) \rightleftharpoons 2NOCl\,(g) \quad \Delta H = -78.38\ kJ$$

Which conditions of temperature and pressure will produce the highest yield of NOCl at equilibrium?
   A. high temperature – High pressure
   B. high temperature – low pressure
   C. low temperature – high pressure
   D. low temperature – low pressure

Explanation

- Increasing pressure will shift the equilibrium towards less number of gaseous molecules. Hence, in the above reaction the rxn will shift towards the product.
- Lowering the temperature favours the exothermic direction so for the above reaction it favours the product because the forward reaction has negative ΔH value.
- Increasing temperature favours the endothermic direction (to the reactant of the above reaction). This is because if the forward reaction is exothermic, the backward reaction is endothermic.

Answer: C

11. In the process known as 'roasting' a (n) _____ is chemically converted to a (an) _____
   A. carbonate/ oxide
   B. oxide/ sulfate
   C. sulfide/ oxide
   D. phosphate/ phosphide

Answer: C

12. Which of the following statement is NOT true according to 1st law of thermodynamics?
   A. Energy can neither be created nor destroyed although it can be changed from one form to another.
   B. The total energy of the universe is constant.

Telegram: https://t.me/Extremeseries

C. The change in the internal energy of a closed system is equal to the sum of heat absorbed by the system and work done by the system.

D. The change in internal energy of a system is zero in all process.

Explanation

According to 1st law of thermodynamics, energy of the universe is conserved. i.e., $\Delta E_{system} + \Delta E_{surrounding} = 0$

Answer: D

13. How many chloride ions are in 1 mole of $CaCl_2$?

A. $3.01 \times 10^{23}\ Cl^-$ ions C. $1.81 \times 10^{24}\ Cl^-$ ions

B. $6.02 \times 10^{23}\ Cl^-$ ions D. $1.20 \times 10^{24}\ Cl^-$ ions

Solution

$CaCl_2 \longrightarrow Ca^{2+} + 2Cl^-$

1 mole 1 mole 2 mole (from the stoichiometry)

1 mole $Cl^- = 6.02 \times 10^{23}$ ions

2 mole $Cl^- = ?x$

Solving for the unknown

$x = 2 \times 6.02 \times 10^{23}$ ions $= 12.04 \times 10^{23}\ Cl^-$ ions $= 1.204 \times 10^{24}$

Answer: D

14. Which of the following types of solutions are possible?

I. solid dissolved in a liquid

II. gas dissolved in a liquid

III. gas dissolved in a gas

IV. solid dissolved in a solid

A. I and II C. I, II and IV

B. I, II, III and IV D. I

Answer: B

15. What is the normality of 1.0 M solution of Na_2CO_3?

A. 1N B. 0.5N C. 2N D. 3N

Solution

Normality = Molarity × charge Z (charge) = no of positive charge

$N = M \times Z$

$N = 1 \times 2 = 2N$

$Na_2CO_3 \longrightarrow 2Na^+ + CO_3$

charge = 2

Answer: C

16. What is the type of monomer unit in a natural protein polymer?

A. Amino acids C. α – amino acids

B. Essential amino acids D. Nucleic acids

Answer: C

17. Which of the following substance could be amphiprotic?

I. NO_3^- II. H_2O III. NH_3 IV. HCO_3^-

A. I, II and III C. II and IV

B. I and II D. II, III and IV

Explanation

Molecules or ions gain or lose a proton, under appropriate conditions. Such species are said to amphiprotic.

Example H_2O, NH_3, CH_3COOH, OH^-, HSO_4^-, $H_2PO_4^-$ . . .

Answer: D

18. Which one of the following does NOT form hydroxide ions when placed in water?

A. Ionic hydrides C. Nonmetal oxides

B. Ionic metal oxides D. Ionic nitrides

Answer: C

19. Which of the following is true for a 0.10 M solution of a weak acid, HB?

A. $[H^+] = 0.10$ M C. $[HB] > [H^+]$

B. $[B^-] = 0.10$ M D. pH = 1.0

Explanation

$HB + H_2O \rightleftharpoons H_3O^+(aq) + B^-(aq)$

Initial conc.	0.1	-	-
change	-x	x	x
Equil.conce	0.1 – x	x	x

From the above table it is clear to observe that $[HB] > [H^+]$

Answer: C

20. What is the bond order of O_2^+ ?

A. 2 B. 2.5 C. 1.5 D. 3

Explanation

The total electrons in O_2^+ (Z = 8) = 15 electrons.

The molecular orbital arrangement will be

$(\delta 1s)^2\ (\delta^* 1s)^2 (\delta 2s)^2 (\delta^* 2s)^2 (\delta 2px)^2 (\pi 2py)^2 (\pi 2pz)^2 (\pi^* 2py)^1$

Telegram: https://t.me/ExtremeSeries

Bond order $= \frac{1}{2}$(number of bonding electrons − number of anti bonding electrons) $= \frac{1}{2}(10-5) = 2.5$

Note: * represents anti bonding molecular orbitals.

21. Which of the following is characteristics of an anti bonding molecular orbital?

Answer: B

A. It is a molecular orbital with a high probability of finding electrons parallel to the reign between bonded atoms.
B. It has no electrons.
C. It is a molecular orbital with high probability of finding electrons away from the region between bonded atoms.
D. It is a molecular orbital with high probability of finding electrons in the region between the bonded atom.

22. Which of the following doesn't affect the rate of chemical reaction?

Answer: C

A. Enthalpy of the reaction
B. Concentration of reactants
C. surface area
D. Temperature

Explanation:

Factors that affect rate of a chemical reaction are nature of reactants, surface area, concentration, temperature and catalyst.

23. What is the molarity of a solution obtained by dissolving 0.01 moles of NaCl in 500 ml of solution?

Answer: A

A. 0.01 M B. 0.05M C. 0.02M D. 0.10M

Solution:

$$\text{Molarity (M)} = \frac{\text{number of moles of solute}}{\text{volume of solution (L)}}, V = 500\text{mL}, V = 0.5\text{L}$$

$$M = \frac{0.01\ \text{mol}}{0.5\ \text{Lit}} = 0.02M$$

Answer: C

24. Which of the following is NOT a valid expression for the rate of the reaction given below?

$$4NH_3 + 7O_2 \longrightarrow 4NO_2 + 6H_2O$$

A. $-\frac{1}{7}\Delta\frac{[O_2]}{\Delta t}$
B. $\frac{1}{7}\Delta\frac{[NO_2]}{\Delta t}$
C. $-\frac{1}{6}\Delta\frac{[H_2O]}{\Delta t}$
D. $-\frac{1}{4}\Delta\frac{[NH_3]}{\Delta t}$

Explanation

$$4NH_3 + 7O_2 \longrightarrow 4NO_2 + 6H_2O$$

$$\text{rate} = -\frac{1}{4}\frac{\Delta[NH_3]}{\Delta t} = -\frac{1}{7}\frac{\Delta[O_2]}{\Delta t} = \frac{1}{4}\frac{\Delta[NO_2]}{\Delta t} = \frac{1}{6}\frac{\Delta[H_2O]}{\Delta t}$$

Answer: C

25. Each of the choices below gives a reaction and the corresponding rate law. Of these choices, which one could be an elementary process or individual step in a chemical reaction?

A. $2A \longrightarrow P$ rate = k [A]
B. $A + B \longrightarrow P$ rate = k [A] [B]
C. $A + 2B \longrightarrow P$ rate = k $[A]^2$
D. $A + B + C \longrightarrow P$ rate = k [A] [C]

Explanation

- Molecularity is the number of reactant molecules taking part in elementary reaction.
- The rate law in an elementary step follows directly from the molecularity of that step.

Answer: B

26. Consider the reaction in which nitric oxide is oxidized to nitrogen dioxide.

$$2NO\ (g) + O_2\ (g) \longrightarrow 2NO_2\ (g)$$

for which of the rate law is rate = k $[NO]^2$ $[O_2]$. If the reaction takes place in a sealed vessel and the partial pressure of nitric oxide is doubled, what effect would this have on the rate of the reaction?

A. The reaction rate would increase by a factor of four.
B. The reaction rate would increase by a factor of three.
C. The reaction rate would increase by a factor of eight.
D. The reaction rate would increase by a factor of two.

Telegram: https://t.me/ExtremeSeries

Explanation
If a given reactantsincreased by a factor, x, then the rate increases by a factor of x^a wherea is the order the reaction with respect to the reactant. So, from the rate law equation rate = k $[NO]^2[O_2]$, the order with respect to NO is 2, and the partial pressure of NO is doubled, then the rate increases by $2^2 = 4$.

Answer: A

27. Which of the following will change the value of an equilibrium constant?
A. Changing the temperature.
B. Adding other substancesthat do not react with any of the species involved in the equilibrium,
C. Varying the initial concentration of reactants.
D. Varying the initial concentration of products.

Answer: A

28. The equilibrium constant for reaction (1) is k. What is the equilibrium constant for reaction (2)?

$$SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g) \ldots (1)$$

$$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g) \ldots (2)$$

A. k^2 B. 2k C. $\frac{1}{2}$ D. $\frac{1}{k^2}$

Explanation
- The equilibrium expression for a reaction written in reverse is the reciprocal of that for the original reaction.
- When the balanced equation for reaction is multiplied by a factor n, the equilibrium expression for the new reaction is the original expression raised to the n^{th} power. Thus

$$k_C^{new} = \left(k_C^{original}\right)^n$$

1) $SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g) \ldots k$

Reversing the first equation and multiplying by 2 we get

$$2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g) \ldots \left(\frac{1}{k}\right)^2 = \frac{1}{k^2}$$

Answer: D

29. Which of the following are carbohydrates?
I. nucleic acids
II. polyhydroxyl ketone
III. disaccharides
IV. aldoses
V. triglycerides
A. I and V
B. II, III and IV
C. I, III and V
D. III and IV

Answer: A

30. Which of the following is true concerning this acid – base reaction?

$$H\bar{S} + CH_3Cl \longrightarrow CH_3SH + Cl^-$$

A. CH_3Cl is a Bronsted – Lowery acid
B. $H\bar{S}$ is a Bronsted – Lowry acid
C. $H\bar{S}$ is a Lewis acid
D. $H\bar{S}$ is a Lewis base

Answer: D

31. For the stepwise dissociation of aqueous, H_3PO_4, which of the following is NOT a conjugate acid-base pair?
A. HPO_4^{2-} and PO_4^{3-} C. $H_2PO_4^-$ and HPO_4^{2-}
B. H_3PO_4 and $H_2PO_4^-$ D. H_2PO_4 and PO_4^{3-}

Explanation

Acid	Conjugate base
H_3PO_4	$H_2PO_4^-$
$H_2PO_4^-$	HPO_4^{-2}
HPO_4^{-2}	PO_4^{3-}

Answer: D

32. Which solution below has the highest concentration of hydroxide ions?
A. pH = 3.21 C. pH = 7.93
B. pH = 12.59 D. pH = 9.82

Explanation
The higher the pH value, the lesser the pOH and the higher will its hydroxide ion concentration be.

Answer: B

Telegram: https://t.me/ExtremeSeries

33. The standard reduction potential of Zn^{2+}/Zn is (-0.76V) and that of Cu^{2+}/Cu is +0.34V. What would be the electromotive force (emf) of the cell constructed between these two electrodes?
A. 1.10V B. 0.42V C. -1.10V D. -0.42V

Explanation
Zinic is a reducing agent and Cu is an oxidizing agent. Thus the reaction is as follows:

$$Zn + Cu^{2+} \rightleftharpoons Zn^{2+} + Cu$$

$$E°_{cell} = E°_{oxi} + E°_{red}$$
$$= -(-0.76V) + (0.34V) = 1.1V$$

Answer: A

34. At constant T and P, which one of the following statements about enthalpy (ΔH and internal energy (Δu)) changes is correct for the reaction?

$$CO(g) + \frac{1}{2}O_2(g) \longrightarrow CO_2(g)$$

A. $\Delta H = \Delta U$ C. $\Delta H > \Delta U$
B. $\Delta H < \Delta U$
D. ΔH and Δu are independent of physical states of the reactant.

Explanation
The enthalpy, H, is defined as
$H = U + PV$
$\Delta H = \Delta u + P\Delta V + V\Delta P$
For temperature constant (ideal gas)
$\Delta H = \Delta u + \Delta(nRT)$, $\Delta n = n_p - n_r$

$$\Delta n = 1 - \left(1 + \frac{1}{2}\right)$$

$$\Delta n = -\frac{1}{2} \Rightarrow \Delta H = \Delta U - \frac{1}{2}RT$$

Therefore $\Delta H < \Delta$

Answer: A

35. Which one of the following conditions regarding a chemical process ensures its spontaneity at all temperatures?
A. $\Delta H < 0, \Delta S < 0$ C. $\Delta H < 0, \Delta S > 0$
B. $\Delta H > 0, \Delta S < 0$ D. $\Delta H > 0, \Delta S > 0$

Explanation
From the Gibbs free energy formula
$\Delta G = \Delta H - T\Delta S$
- If ΔG is negative, the reaction is spontaneous.
- If ΔG is positive, the reaction is non spontaneous.
- If ΔG is zero, the reaction is at equilibrium.

And ΔG will be negative when ΔH is negative and ΔS is positive at any temperature.

Answer: C

36. Why silicate minerals are not commonly used as source of metals.
A. They are rare minerals.
B. They usually don't contain important metals.
C. They are difficult to reduce and concentrate.
D. They are only found at excessive depths in the ocean.

Answer: C

37. Which of the following metals is the best conductor of heat and electricity?
A. Copper C. Gold
B. Silver D. Tungsten

Answer: B

38. HNO_3 is used as
I. a hydrogenating agent
II. an acid
III. an oxidizing agent
IV. an ammoniating agent
A. I and II C. I, II, III and IV
B. I and II D. II, III and IV

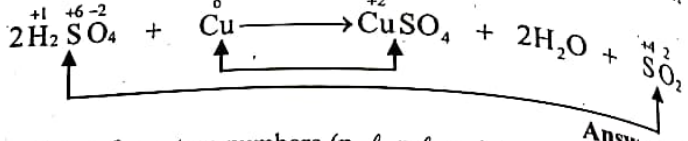
Answer: D

39. Which equation represents an oxidation – reduction reaction?
A. $H_2SO_4 + 2NH_3 \longrightarrow (NH_4)_2SO_4$
B. $H_2SO_4 + Na_2CO_3 \longrightarrow Na_2SO_4 + H_2O + CO_2$
C. $2K_2CrO_4 + H_2SO_4 \longrightarrow K_2Cr_2O_7 + K_2SO_4 + H_2O$
D. $2H_2SO_4 + Cu \longrightarrow CuSO_4 + 2H_2O + SO_2$

Explanation Redox reaction that occurs as a result of electron transfer. The best way to check a reaction is Redox or not is to assign the oxidation number value for each element in the

Telegram: https://t.me/ExtremeSeries

reactant and product side. Hence, if there is a change in oxidation number that reaction is Redox.



Answer: D

40. Which set of quantum numbers (n, ℓ, mℓ, ms) is not permitted by the value of quantum mechanics.

A. $1, 0, 0, \frac{1}{2}$ C. $3, 3, 1, -\frac{1}{2}$

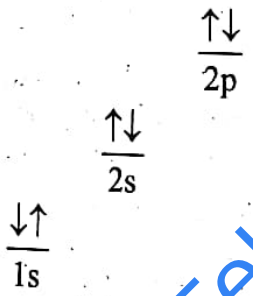
B. $2, 1, -1, -\frac{1}{2}$ D. $4, 3, 2, \frac{1}{2}$

Explanation

- Principal quantum number (n) takes a value of 1, 2, . . .
- Angular momentum quantum number (ℓ) take a value from 0 to n – 1 and is always less than n.
- Magnetic quantum number (mℓ) takes places from $-\ell$ to $+\ell$.
- Magnetic spin quantum number (mℓ) takes values either $+\frac{1}{2}$ or $-\frac{1}{2}$

Answer: C

41. What can you conclude from the figure below?



A. Hund's rule has been violated.
B. The Pauli exclusion principle has been violated.
C. The Autbau principle has been violatd.
D. This is valid orbital diagram.

Explanation

Hund's rule states that, "For any set of orbitals that are equal in energy the configuration with the maximum number of parallel spins results in the lowest electron – electron repulsion." Besides Hnnd's rule can also be explained as:

"No pairing of electrons takes place in degenerate orbitals before orbitals are singly filled."

Pauli exclusion principle states that, "No two electrons can have the same four quantum numbers."

Aufbau principle states that, "During fulfilling of electrons in an atom oribital with lower energy will be filled first that orbitals with higher energy."

Answer: A

42. Which set contains only covalently bonded molecules?

A. BCl_3, $SiCl_4$, PCl_3 C. I_2, H_2S, NaI

B. NH_4Br, N_2H_4, HBr D. Al, O_3, As_4

Answer: A

43. How many grams of $Ca(OH)_2$ are contained in 1500 ml of 0.0250M $Ca(OH)_2$ solution?

A. 1.85 gram C. 3.17 gram

B. 2.78 gram D. 4.25 gram

Solution

$$\text{Molarity} = \frac{\text{mass of } Ca(OH)_2}{\text{Molar mass of } Ca(OH)_2} \times \frac{1}{\text{Volume (L)}}$$

Molar mass = 74 g/mol

Mass of $Ca(OH)_2$ = Molarity × molar mass × Volume (Liter)

$= 0.0250 \times 74 \times 1.5$

$= 2.78$ gram

Answer: B

44. The conventional equilibrium constant expression (k_C) for the system as described by the equation.

$2ICl(s) \rightleftharpoons I_2(s) + Cl_2(g)$

A. $[Cl_2]$ C. $[I_2][Cl_2][ICl]^2$

B. $[Cl_2]/[ICl]^2$ D. $[I_2][Cl_2]/2[ICl]$

Explanation

- The equilibrium is heterogeneous equilibrium because the phase of the reactant and the products are different.

Telegram: https://t.me/Extremeseries

- The k_C value of heterogeneous equilibrium doesn't depend on the amount of pure solids or liquids present. Thus, the k_C value does not include pure solids or liquids.

$$k_C = [Cl_2]$$

45. What is the conjugate base of HSO_4^-?

Answer: A

A. H^+ B. H_2SO_4 C. OH^- D. SO_4^{2-}

46. An electrochemical cell constructed for the reaction.

Answer: D

$Cu^{2+}(aq) + M(s) \longrightarrow Cu(s) + M^{2+}(aq)$, has an $E°_{cell} = 0.75V$. The standard reduction $E°_{cell}$ potential for $Cu^{2+}(aq)$, is 0.34V. What is the standard reduction potential for $M^{2+}(aq)$?

A. -1.09 V C. 0.410 V
B. -0.410V D. 1.09V

Explanation

$E°_{cell} = E°_{ox} + E°_{red}$
$0.75V = E°_{ox} + 0.34\ V$
$\Rightarrow E°_{red} = -E°_{ox} = -0.410V$
$E°_{ox} = +0.41\ V$

Answer: B

47. Which of the following compounds would be expected to have the highest melting point?

A. BaF_2 B. $BaCl_2$ C. $BaBr_2$ D. BaI_2

Explanation: All of the compounds given are ionic. Hence, the strength of an ionic bond affects the melting point of the compounds. Ionic compounds with high lattice energies generally are harder and have higher melting point than compounds with low lattice energies.

Lattice energy increases with increasing ionic charge and decreasing ionic radii. Thus, the lattice energy decreases in the order $BaF_2 > BaCl_2 > BaBr_2 > BaI_2$

Answer: A

48. Which of the following is true about chlorofluoro carbons?

A. React directly with stratospheric ozone to destroy it.
B. Interact with UV energy and becomes free radicals which destroy ozone.
C. Become free radicals that react with oxygen to create ozone.
D. React with free radicals to remove carbondioxide.

Answer: B

49. Which metal can be found as a free element.

A. Cr B. Fe C. Mn D. pt

Answer: D

50. The enthalpy change for which reaction represents the standard enthalpy of formation for hydrogen cyanide, HCN?

A. $H(g) + C\ (graphite) + N\ (g) \longrightarrow HCN\ (g)$

B. $\frac{1}{2}H_2(g) + C(graphite) + \frac{1}{2}N_2(g) \longrightarrow HCN(g)$

C. $HCN\ (g) \longrightarrow \frac{1}{2}H_2(g) + C\ (graphite) + \frac{1}{2}N_2(g)$

D. $H_2\ (g) + 2C\ (graphite) + N_2\ (g) \longrightarrow 2HCN\ (g)$

Explanation: Standard enthalpy of formation ($\Delta H°_f$) is one mole of a compound is formed from the elements in standard states. Since hydrogen and nitrogen are diatomic molecules, they should be written as H_2 and N_2.

Answer: B

51. Which one of the compounds below is most likely to be ionic?

A. CCl_4 B. NO_2 C. $ScCl_3$ D. ClO_2

Answer: C

52. What is the correct order when the substance O_2, H_2O, OF_2 and H_2O_2 are arranged in order of increasing oxidation number for oxygen?

A. O_2, H_2O, OF_2, H_2O_2 C. H_2O_2, O_2, H_2O, OF_2
B. H_2O, H_2O_2, O_2, OF_2 D. OF_2, O_2, H_2O_2, H_2O

Explanation

Molecule oxidation number of oxygen

O_2	0
H_2O	-2
OF_2	+2
H_2O_2	-1

According the increasing order $H_2O < H_2O_2 < O_2 < OF_2$

Answer: B

53. A mono atomic ion that has 20 protons and a+2 charge

A. has 16 protons C. has 18 neutrons
B. has the symbol Ar^{2+} D. is isoelectronic with Ar

Explanation: a+2 charge indicates that 2 electrons are lost. So the number of electrons become 18, which makes a+2 isoelectronic with Ar.

Answer: D

Telegram: https://t.me/Extremeseries

54. What is the molality of a solution that contains 51.2 gram of naphthalene, $C_{10}H_8$, in 500 ml of carbon tetrachloride? (Given the density of CCl = 1.60 g/ml)

A. 0.25 m B. 0.5 m C. 0.75 m D. 0.84 m

Solution

Given:

$m_{C_{10}H_8} = 51.2$ gram

Molar mass = 128 g/mol

V = 500 ml

density (s) = 1.60 g/ ml

$$\text{moles of } m_{C_{10}H_8} = \frac{51.2 \text{ gram}}{128 \text{ g/mol}} = 0.4 \text{ mole}$$

$$\text{density (s)} = \frac{m_{CCl_4}}{V_{CCl_4}}$$

$m_{CCl_4} = 1.60$ g/mol × 500 ml

= 800 gram = 0.8 kg

$$\therefore \text{molality (m)} = \frac{\text{no of moles}}{\text{mass of solvent (kg)}} = \frac{0.4}{0.8} = 0.5 \text{ m}$$

Answer: B

55. A homogeneous liquid reaction mixture is often heated to increase the rate of reaction. This is best explained by the fact that raising the temperature.

A. increases the heat of reaction
B. decreases the energy of activation
C. increases the vapor pressure of the liquid
D. increases the average kinetic energy of the reactants

Answer: D

56. Which of the following titrations will have an equivalence point pH > 7.00?

A. weak acid with strong base
B. strong acid with strong base
C. weak acid with weak base
D. strong acid with a weak base

Explanation

Titration type	pH at equivalence point
Strong acid – strong base	pH = 7
Weak acid – strong base	pH > 7
Strong acid – weak base	pH < 7
Weak acid – weak base	Difficult to determine

Answer: A

57. A 0.10M solution of weak acid, Hx, is 0.059% ionized. What is the dissociation constant for the acid?

A. 3.8×10^{-9} C. 6.5×10^{-7}
B. 3.5×10^{-8} D. 4.2×10^{-6}

Solution

Percentage dissociation (%) = 0.059%

$$\text{percentage dissociation (x\%)} = \frac{\text{amount of dissociated ion (x)}}{\text{initial concentration (C)}} \times 100$$

$$x\% = \frac{x}{C} \times 100$$

$$x = \frac{\% \times C}{100} = \frac{0.059 \times 0.1}{100} = 5.9 \times 10^{-5} \text{ M} = [H_3O^+] = [A^-]$$

$$HA + H_2O \rightleftharpoons H_3O^+(aq) + A^-(aq)$$

	HA	H_2O	H_3O^+	A^-
Initial conc.	0.1	-	-	
change	-x		+x	+x
Equilibrium	0.1 – x		x	x

$$Ka = \frac{[H_3O^+][A^-]}{[HA]} = \frac{x \cdot x}{0.1 - x} \cong \frac{x^2}{0.1} = \frac{(5.9 \times 10^{-5})^2}{0.1}$$

$= 3.5 \times 10^{-8}$

Answer: B

58. Consider the standard voltaic (or galvanic) cell: Fe, Fe^{2+} versus Au, Au^{3+}. Which answer identifies the cathode and gives the $E°_{cell}$ for the cell?

Given:

$E°(Au^{3+}/Au) = 1.50$ V, $E°(Fe^{2+}/Fe) = -0.44$ V

A. Fe, -0.44V C. Au, 1.94 V
B. Au, 1.06V D. Fe, 1.94 V

Telegram: https://t.me/ExtremeSeries

Explanation
The half cell with more positive electrode potential is always placed to the cathode. Therefore, gold will be reduced and iron is oxidized.

$$E^\circ_{cell} = E^\circ_{ox} + E^\circ_{red}$$
$$= 0.44\ V + (1.50V) = 1.94\ V$$

59. What is the enthalpy for the reaction **Answer: C**

$$C_2H_6\,(g) + \frac{7}{2}O_2\ (g) \longrightarrow 2CO_2\,(g) + 3H_2O\,(\ell)?$$

Given the following reaction

1. $2C(s) + 3H_2\,(g) \longrightarrow C_2H(g)$ $\Delta H^\circ_f = -84.7\ kJ.mol^{-1}$
2. $C(s) + O_2\,(g) \longrightarrow CO_2\,(g)$ $\Delta H^\circ f = -393.5\ kJ.mol^{-1}$
3. $H_2(g) + \frac{1}{2}O_2\,(g) \longrightarrow H_2O\,(l)$ $\Delta H^\circ f = -285.8\ kJ.mol^{-1}$

A. -764 kJ B. -1560 kJ C. -1664 kJ D. -312 kJ

Explanation

- Reversing the first equation we get,

A. $C_2H_6\,(g) \longrightarrow 2C\,(s) + 3H_2\,(g)$ $\Delta H^\circ_f = +84.7\ kJ.mol^{-1}$

- Multiplying the second equation by two, we get

B. $2C(s) + 2O_2\,(g) \longrightarrow 2CO_2\,(g)$
$\Delta H^\circ f = -393.5\ kJ.mol^{-1} \times 2 = -787\ kJ/mol$

- Multiplying the third equation by three, we get

C. $3H_2(g) + \frac{3}{2}O_2\,(g) \longrightarrow 3H_2O\,(l)$

$\Delta H^\circ f = -285.8\ kJ.mol^{-1} \times 3 = -857.4\ kJ/mol$

Adding equations A, B, C we get

$$C_2H_6\,(g) + \frac{7}{2}O_2\,(g) \longrightarrow 2CO_2\,(g) + 3H_2O\,(\ell)$$

$\Delta H^\circ_{rxn} = 84.7 + (-787) + (-857.4)$

$\Delta H^\circ_{rxn} = -1559.7\ kJ = -1560\ kJ$

Answer: B

60. How many atoms are present in 64 gram of SO_2?

A. 6.02×10^{23} C. $3 \times 6.02 \times 10^{23}$
B. $2 \times 6.02 \times 10^{23}$ D. $4 \times 6.02 \times 10^{23}$

Explanation
1 mole of SO_2 = Avogadro's Number = Mass of SO_2 (in gram)
1 mole of $SO_2 = 6.02 \times 10^{23}$ molecules = 64 gram
1 molecule of SO_2 is made up of 3 atoms then 6.02×10^{23} molecules of SO_2 are made up of $6.02 \times 10^{23} \times 3 = 12.06 \times 10^{23}$ atoms

Answer: C

61. What is the correct decreasing order of the atomic radius in the elements below?

A. $Mg < Al < Cl$ C. $Mg > Al > Cl$
B. $Al < Mg < Cl$ D. $Cl > Al > Mg$

Explanation: Atomic radius decrease across a period.

Answer: C

62. The outermost electron (s) of which the following experiences, is the greatest effective nuclear charge?

A. K B. Ca C. Br D. P

Answer: C

63. Which of the following is not possible shape of a molecules?

A. Tetrahedral $GeCl_4$
B. A bent HCN
C. A trigonalbipyramidal PCl_5
D. A T – shaped BrF_3

Answer: B

64. Four elements, A, B, C, D have atomic numbers Z – 1, Z, Z + 1 and Z + 2, respectively. If Z is 9, then the bond between which pair of elements will be ionic?

A. A and C C. B and D
B. C and D D. B and C

Explanation: The atomic numbers are 8, 9, 10 and 11 respectively. And the configuration of the elements will be

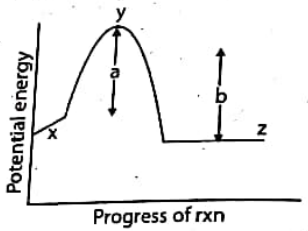
	K	L	M
8^A	2	6	-
9^B	2	7	-
10^C	2	8	-
11^D	2	8	1

From the configuration C is necessarily a noble gas i.e., Ar, which is not expected to form a compound at normal condition. Besides element B is in group VIIA and an element D is a group IA element. Therefore element B and D forms an ionic compound.

Answer: C

Telegram: https://t.me/ExtremeSeries

65. Which of the following statements is INCORRECT for the reaction represented by the figure below?



A. y represents the transition state of the reaction
B. the enthalpy of the reaction is given by b – a
C. the forward reaction is endothermic
D. b is the activation energy of the formed reaction

Explanation

'x' represents the reactant, and 'z' represents the product and y is the transition state

- b isheat content of product
- a is the activation energy
- The reaction is exothermic since heat of product (z) is lower in energy than the reactant (x).

No answer (B, C and D are incorrect)

66. Given the initial rate below the reaction?

$aA + bB \longrightarrow$ products

What is the correct rate law?

Experiment	Concentration of A (M)	Concentration of B (M)	Initial Rate M/s
1	0.100	0.100	4.00×10^{-5}
2	0.200	0.100	1.6×10^{-4}
3	0.100	0.200	$8.0 \times 10+-5$

A. Rate = k [A] [B]
B. Rate = k $[A]^2$ [B]
C. Rate = l $[A]^2$ $[B]^2$
D. Rate = k [A] $[B]^2$

Explanation:

For the reaction $aA + bB \longrightarrow$ product
The rate law expression will be
rate = k $[A]^x$ $[B]^y$ where x and y are order of the rxn
To determine the order with respect to A, take experiments 1 and 2 (where the concentration of B are similar)

$rate_1 = k [A]^x [B]^y = k (0.2)^x (0.1)^y = 4.0 \times 10^{-5}$
$rate_2 = k [A]^x [B]^y = k (0.2)^x (0.1)^y = 1.6 \times 10^{-4}$
Taking the ratio of $rate_1$ and $rate_2$

$$\frac{rate_1}{rate_2} = \frac{k(0.1)^x (0.1)^y}{k(0.2)^x (0.1)^y} = \frac{4.0\times10^{-5}}{1.6\times10^{-4}}$$

$$\left(\frac{1}{2}\right)^x = \frac{1}{4}$$

$$\left(\frac{1}{2}\right)^x = \left(\frac{1}{2}\right)^2$$

x = 2 (order with respect to A)
To determine the order with respect to B, take experiments 1 and 3 (where the concentratin of A are similar).
$rate_1 = k (0.1)^x (0.1)^y = 4.0 \times 10^{-5}$
$rate_3 = k (0.1)^x (0.2)^y = 8.0 \times 10^{-5}$

$$\left(\frac{1}{2}\right)^y = \frac{1}{2}$$

y = 1 (order with respect to B)
Therefore, the rate law expression will be

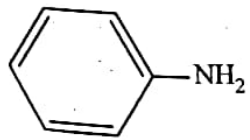
rate = k $[A]^2$ [B]

Answer: B

67. Which of the following compounds is the strongest base?

A. NH_2Cl
B. NH_3
C.

D.



Solution: Electron with drawing groups such as halogens and phenyl decrease basicity.

Answer: B

Telegram: https://t.me/ExtremeSeries

68. For preparing a buffer solution of pH 6.0 by mixing sodium acetate with acetic acid. What would be the ratio of the concentratin of salt and acid (ka = 10^{-5})?

A. 1: 10 B. 10: 1 C. 100: 1 D. 1: 100

Explanation

Using Handresson's Hassel Balch formula

$$pH = p^{Ka} + \log\left(\frac{[CH_3COONa]}{[CH_3COOH]}\right)$$

$$6 = -\log(10^{-5}) + \log\left(\frac{[CH_3COONa]}{[CH_3COOH]}\right)$$

$$6 = 5 + \log\left(\frac{[CH_3COONa]}{[CH_3COOH]}\right)$$

$$1 = \log\left(\frac{[CH_3COONa]}{[CH_3COOH]}\right)$$

$$\frac{[CH_3COONa]}{[CH_3COOH]} = 10^1 = \frac{10}{1}$$

Therefore, the ratio of the salt to the acid is 10: 1.

Answer: B

69. Given the following datea at 25°C.

1. $2NOCl\ (g) \longrightarrow 2NO\ (g) + Cl_2\ (g)$ $\Delta H^\circ = +75kJ$
2. $2NO\ (g) + O_2\ (g) \longrightarrow 2NO_2(g)$ $\Delta H^\circ = -113.05\ kJ$
3. $2NO_2\ (g) \longrightarrow N_2O_4\ (g)$ $\Delta H^\circ = -58.03\ kJ$

CalcualteΔH° at 25°C for the reaction

$N_2O_4\ (g) + Cl_2\ (g) \longrightarrow 2NOCl\ (g) + O_2\ (g)$

A. +96.08 kJ C. 186.8 kJ

B. +299 kJ D. 85.52 kJ

Explanation

- Reversing the first equation, results the following

  1. $2NO\ (g) + Cl_2\ (g) \longrightarrow 2NOCl\ (g)$, $\Delta H^\circ = -75kJ$

- Reversing the second equation, results the following

  2. $2NO_2\ (g) \longrightarrow 2NO\ (g) + O_2\ (g)$, $\Delta H^\circ = 113.0\ kJ$

And reversing also the third equation, gets the following

3. $N_2O_4\ (g) \longrightarrow 2NO_2\ (g)$, $\Delta H^\circ = 58.03\ kJ$

Adding equations 1, 2 and 3 results

$N_2O_4\ (g) + Cl_2\ (g) \longrightarrow 2NOCl\ (g) + O_2(g)$

$\Delta H^\circ_{rxn} = -75\ kJ + 113.0\ kJ + 58.03\ kJ$

$= +96.08\ kJ$

Answer: A

70. How many moles of $KBrO_3$ are required to prepare 0.0700 moles of Br_2 according to the equation.

$KBrO_3 + KBr + HNO_3 \longrightarrow KNO_3 + Br_2 + H_2O$

a. 0.0233 B. 0.0732 C. 0.210 D. 0.220

Explanation

First balance the given equation

$KBrO_3 + 5KBr + 6HNO_3 \longrightarrow 6KNO_3 + 3Br_2 + 3H_2O$

So using the balanced equation

$$\frac{x}{1\ mole} = \frac{0.07\ moles\ of\ Br_2}{3\ mole}$$

$$x = \frac{1 \times 0.07\ moles}{3} = 0.0233\ moles$$

Answer: A

71. A possible mechanism for a reactin $2A + B \longrightarrow C + D$ is

a.	$A + A \longrightarrow A_2$	fast, equilibrium
b.	$A_2 + A \longrightarrow A_3$	slow
c.	$A_3 + B \longrightarrow A + C + D$	fast

According to the mechanism, the rate law will be

A. Rate = k [A] C. Rate = k $[A]^3$

B. Rate = k $[A]^2$ D. Rate = k $[A]^2$

Explanation

Rate determining step: the elementary reaction in a mechanism that is so much slower than the rest that it governs the rate of the over all reaction.

So, the rate determining step is the second step which is the slowest step

$A_2 + A \longrightarrow A_3$ (slow)

Rate = k $[A_2]$ [A]

Telegram: https://t.me/ExtremeSeries

Let us check wheather the mechanism works or not to give the over allbalnced equation.

$A + A \rightleftharpoons A_2$

$A_2 + A \longrightarrow A_3$

$A_3 + B \longrightarrow A + C + D$

$2A + B \longrightarrow C + D$

Here A_2 and A are intermidate.

The rate law for the slowest step is Rate = k $[A_2]$ [A], contains an intermidate species which is A_2. Therefore A_2 should be substituted $A + A \underset{k-1}{\overset{k_1}{\rightleftharpoons}} A_2$

Rate of forward reaction = $k_1 [A]^2$

Rate of the backward reaction = $k_1 [A_2]$

At equilibrium, the rate of forward reaction is equal to the rate of backward reaction.

$$k_1 [A]^2 = k_1 [A_2]$$

Solving fir A_2

$$[A_2] = \frac{k_1[A]^2}{k-1}$$

Now substituting the expression for $[A]_2$ into the rate law expression for the slow step.

$$\text{Rate } k [A_2] [A_2][A] = k\left(\frac{k_1[A]^2}{k-1}\right)[A] = \frac{k.k_1}{k-1}[A]^2[A]$$

$$= k [A]^2 [A] = k [A]^3$$

Note: $k = \frac{k.k_1}{k-1}$

Answer: C

72. What is the $[H_3O^+]$ in a 0.050 M solution of $Ba(OH)_2$?

a. 1.0×10^{-13} M C. 1.0×10^{-5} M

b. 5.0×10^{-10} M D. 5.0×10^{-2} M

Explanation

$Ba(OH)_2 + H_2O \longrightarrow Ba^{2+}(aq) + 2OH^-(aq)$

0.05M 0.05 2×0.05 M = 0.1

$k_w = [H_3O^+][OH^-]$

At 25°C $k_w = 1 \times 10^{-14}$

$$[H_3O^+] = \frac{k_w}{[OH^-]} = \frac{1\times10^{-14}}{0.1} = 1\times10^{-13}\text{M}$$

Answer: A

73. Which species has the largest F – A – F bond angle where A is the central atom?

a. BF_3 B. CF_4 C. NF_4 D. OF_2

Explanation

Moelcu le	Type	Bond Angle	Shape
BF_3	AB_3	120°	Trigonal planar
CF_4	AB_4	109.5°	Tetrahedral
NF_3	AB_3E	107°	Trigonal pyramidal
OF_2	AB_2E_2	104.5°	Angular/V – shape

Note: A – central atom B – surrounding atom E – loneparis

Answer: A

74. When the atoms Li, Be, B, Na are arranged inorder of increasing atomic radius. Which is correct order?

a. Li, Be, B, Na C. Na, Li, Be, B

b. Li, Na, B, Be D. B, Be, Li, Na

Explanation: Except Na, all the other elements are found at period two. Across a period atomic radius decreases. Hence insize Li > Be > B and Na belongs IA element found next Li. Hence, Na > Li because size increases down a group.

Therefore in size Na > Li > Be > B. **Answer: D**

75. One of the steps in the manufacture of nitiric acid is the oxidation of ammonia shown in this equation

$4NH_3(g) + 5O_2(g) \longrightarrow 4NO(g) + 6H_2O(g)$

If gaseous water appears at a rate of 0.025 mol. min^{-1}. At what rate does ammonia dis appear?

A. 0.0040 mol. min^{-1} C. 0.038 mol. min^{-1}

B. 0.017 mol.min^{-1} D. 0.150 mol. min^{-1}

Solution: $4NH_3(g) + 5O_2(g) \longrightarrow 4NO(g) + 6H_2O(g)$

The rate law of expression will be

$$\frac{-\Delta[NH_3]}{4\Delta t} = \frac{-[O_2]}{5\Delta t} = \frac{\Delta[NO]}{4\Delta t} = \frac{\Delta[H_2O]}{6\Delta t}$$

rate of decomposition of ammonia = rate of formation of water

Telegram: https://t.me/ExtremeSeries

$$\frac{1}{4}\frac{\Delta[NH_3]}{\Delta t} = \frac{1\Delta[H_2O]}{6\Delta t}$$

$$\frac{1}{4}\left(\frac{\Delta[NH_3]}{\Delta t}\right) = \frac{1}{6}(0.025\ \text{mol/min})$$

$$\frac{\Delta[NH_3]}{\Delta t} = \frac{4(0.025\ \text{mol/min})}{6} = 0.017\ \text{mol.min}^{-1}$$

Answer: B

76. What is the half life of the irreversible first order reaction A $\longrightarrow$ B, if 75% of A is converted into B in 60 minutes?

a. 30 minutes C. 60 minutes
b. 45 minutes D. 80 minutes

Explanation: If the reaction is 75% complete, 75% of the reactant has been consumed. This leaves 25% in the original form.

For the 1st order reaction $\log\frac{[A]_o}{[A]_t} = \frac{kt}{2.303}$

$$\log\left(\frac{[A]_o}{\frac{1}{2}[A]_o}\right) = \frac{k.60}{2.303}$$

$k = 0.023\ \text{min}^{-1}$

$$t\frac{1}{2}\text{ for } 1^{st}\text{ order} = \frac{0.693}{k} = \frac{0.693}{0.023\,\text{min}^{-1}} = 30\ \text{min}$$

Answer: A

77. At a certain temperature for the reaction:

$2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$

the equilibrium constrations are $[NO] = 0.52$ M; $[O_2] = 0.24$ M $[NO_2] = 0.18$ M. What is the value of k_C at this temperature?

A. 0.0693 B. 0.50 C. 1.4 D. 2.0

Solution: $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$

$$k_c = \frac{[NO_2]^2}{[NO]^2[O_2]} = \frac{(0.18)^2}{(0.52)^2(0.24)} = \frac{0.0324}{0.065} = 0.5$$

Answer: B

78. A 25.00 ml sample of 0.1050 M H_2SO_4 is titrated with a NaOH solution of unknown concentration. The phenolphthalein end point was reached when 17.23 ml of the NaOH solution had been added. What is the cncentration of the NaOH?

a. 0.07617 M C. 0.1524 M
b. 0.1447 M D. 0.3047 M

Solution

At the end point, number of equivalents of the acid and the base are equal. That is

Normality of the acid × Volume of the acid = Normality of the base × volume of the base

$N_{Acid} \cdot V_{Acid} = N_{Base} \cdot V_{Base}$

Normality of H_2SO_4 = charge × Molarity

$= 2 \times (0.1050\ \text{M}) = 0.21\ \text{N}$

$0.21\ \text{N} \cdot 25\ \text{ml} = N_{Base}(17.23\ \text{ml})$

$N_{Base} = 0.3047\ \text{N}$

$$\text{Molarity of the base} = \frac{0.3047}{\text{charge}} = \frac{0.3047}{1} = 0.3047\text{M}$$

Answer: D

79. During electrolysis of an aquesous solution of $AgNO_3$. What would happen to the mass of siliver metal deposited if the current is doubled and the electrolysis time is decreased to $\frac{1}{2}$ of its intial value?

a. It would stay the same.
b. It would increase twice the initial value.
c. It would decrease to $\frac{1}{4}$ of its initial value.
d. It would decrease to $\frac{1}{2}$ of its value.

Explanation

Let m_1 – be the mass of Ag – metal produced by the current (I_1)
m_2 – be the mass of Ag – metal produced by the current (I_2)
given that

Telegram: https://t.me/ExtremeSeries

The current is increased by two. That is $I_2 = 2I$, the time is reduced by $\frac{1}{2}$. That is $t_2 = \frac{1}{2}t$.

Applying Faraday's 1st law, for I_1 and I_2

$$\frac{m_1}{M} = \frac{I_1.t_1}{nF} \ldots\ldots\ldots (1) \qquad m - \text{mass}$$

$$\frac{m_2}{M} = \frac{2I_1.t\frac{1}{2}}{nF} \ldots\ldots (2) \qquad M - \text{molar mass}$$

Taking the ratio of (1) and (2), we get

$$\frac{m_1}{M}.\frac{M}{m_2} = \frac{I_1.t_1}{nF}.\frac{nF}{2I_1.t\,1/2}$$

$$\frac{m_1}{m_2} = 1; \qquad m_1 = m_2$$

Answer: A

80. What is the balanced equation for the cell notation of the following gavalnci cell?

$Pb(s) / Pb^{2+}(aq) // Cl_2(g) / Cl^-(aq) / C(s)$

a. $Pb(s) + 2Cl^-(aq) \longrightarrow Pb^{2+}(aq) + Cl_2(g)$

b. $Pb(s) + Cl_2(g) \longrightarrow Pb^{2+}(aq) + 2Cl^-(aq)$

c. $Pb^{2+}(aq) + 2Cl^-(aq) \longrightarrow Pb(s) + Cl_2(g)$

d. $Pb^{2+}(aq) + 2Cl^-(aq) \longrightarrow PbCl_2(s)$

Explanation

- The left side of the double arrow (II) represents the anode reaction and the right side indicate the cathode reaction.
- Carbon (graphite) is an electrode.
- The double arrow (II) indicates the salt bridge.

Answer: B

1. Which of the following is NOT a state function?
   A. Work C. Internal Energy
   B. Enthalpy D. Pressure

Explanation

A thermodynamic function that depends on the initial and final state of a system, regardless of how the conditions are achieved is called a state function.

Examples: Energy, volume, pressure, temperature, internal energy, Enthalpy, Entropy etc.

Functions that depend on the path are path functions. Examples, Heat, Work

Answer: A

2. Which of the following metals has the highest positive oxidation state?
   A. Ti C. Mn
   B. Cr D. Cu

Explanation

Element	Oxidation State
Ti	+2, +3, +4
Cr	+2, +3, +4, +5, +6
Mn	+2, +3, +4, +5, +6, +7
Cu	+1, +2, +3

Answer: C

3. Which of the following is NOT a natural polymer?
   A. Proteins C. Cellulose
   B. Starch D. Teflon

Answer: D

4. The coefficients required to correctly balance the reaction below are

$$?Al(NO_3)_3 + Na_2S \longrightarrow Al_2S_3 + NaNO_3$$

   A. 2, 1, 3, 2 C. 1, 1, 1, 1
   B. 2, 3, 1, 6 D. 4, 6, 3, 2

Answer: B

5. One mole of $(NH_4)_2HPO_4$ contains _____ moles of hydrogen atom.
   A. one C. six
   B. five D. nine

Solution:

1 mol $(NH_4)_2HPO_4 = 2 \times 4 + 1 = 9$ moles H

Answer: D

Telegram: https://t.me/ExtremeSeries

6. A 500 grams tooth paste sample has 0.2 g of flouride concentration. What is the concentration of fluoride in terms of ppm?

A. 200 B. 250 C. 400 D. 1000

Solution

Given mass of solute = 0.2 gram

Mass of sample (soln) = 500 gram

Required ppm (parts per million)

$$ppm = \frac{\text{mass of solute}}{\text{mass of solution}} \times 10^6 = \frac{0.2}{500} \times 10^6 = 400 \text{ ppm}$$

Answer: C

7. Which solution below has the highest concentration of hydroxide ion?

A. pH = 3.21 C. pH = 9.82

B. pH = 7.93 D. pH = 12.59

Explanation

The pH of the solution is inversely proportional to $[H^+]$ and directly proportional to $[OH^-]$.

Answer: D

8. Which substance is reduced in the following reaction?

$$Cr_2O_7^{2-} + 6S_2O_3^{2-} + 14H^+ \longrightarrow 2Cr^{3+} + 3S_4O_6^{2-} + 7H_2O$$

A. $Cr_2O_7^{2-}$ C. H^+

B. $6S_2O_3^{2-}$ D. Cr^{3+}

Explanation

$\overset{+6}{Cr_2}O_7^{2-} \longrightarrow 2Cr^{3+}$, the oxidation state of Cr is reduced from +6 to +3.

Answer: A

9. What percent is gold in 18 karat gold?

A. 18 B. 84 C. 25 D. 75

Solution

Pure gold is 24 karat

$$\%18 \text{ karat} = \frac{18}{24} \times 100 = 75\%$$

Answer: D

10. Which of the following is a colligative property of a solvent in a solution?

A. Lowering of vapor pressure

B. Osmotic pressure

C. Boiling point

D. Change in entropy

Explanation

The lowering of vapor pressure of a solvent when non – volatile solute is added is colligative property of a solvent.

Answer: A

11. Which one of the following is the criterion for a spontaneous change using thermodynamic parameter entropy(s)?

A. $\Delta S_{system} < 0$ C. $\Delta S_{universe} \leq 0$

B. $\Delta S_{system} \leq \Delta S_{surrounding}$ D. $\Delta S_{universe} > 0$

Explanation

2nd law of thermodynamic states that, "For a spontaneous process the total entropy of a system and its surrounding process always increases."

$$\Delta S_{universe} = \Delta S_{system} + \Delta S_{surrounding} > 0$$

Answer: D

12. Which element corresponds to the electron configuration $[Ne]3s^2 3p^1$?

A. Mg B. Al C. Si D. Ga

Answer: B

13. Which bond between the following groups of elements will be most polar?

A. S – F C. S – Br

B. S – Cl D. S – I

Explanation

- Electro negativity is the power of an atom to attract shared pair of electrons towards itself.
- Electro negativity decreases down a group. Hence the one with high electro negative element forms the most polar bond.

Answer: A

14. What is the unit for the rate of a chemical reaction if the concentration is expressed in moles per liter and the time in second?

A. $L\ mol^1 S^{-1}$ C. S^{-1}

B. $mol\ L^{-1} S^{-1}$ D. $mol\ L^{-1} S^{-2}$

Telegram: https://t.me/ExtremeSeries

Solution:

$$\text{rate} = \frac{\Delta C}{\Delta t} = \frac{\text{mol/L}}{\text{s}} = \text{molL}^{-1}\text{s}^{-1}$$

Answer: B

15. What is the most important type of solute-solvent interaction in a solution of $CaCl_2$ in water?

A. Dipole – dipole
B. Dispersion force
C. Hydrogen bonding
D. Ion – dipole

Explanation

- Dipole – dipole interaction exists between polar solute and polar solvents.
- Dispersion (London force) exists between non polar solutes and non polar solvents.
- Ion – dipole exists between ionic compounds and polar solvents.
- Hydrogen bonding exists between molecules containing hydrogen together with most electronegative elements (F, O, N). Example H_2O, HF, NH_3

Answer: D

16. A bleaching solution contains 5.2 mass percent sodium hypochlorite, NaOCl. What is the mass of NaOCl in a bottle containing 2500 gram of bleaching solution?

A. 90
B. 130
C. 480
D. 520

Solution

Given:

Percentage by mass (%m/m) = 5.2%

Mass of solution = 2500 gram

Required:

Mass of NaOCl =?

$$\%\text{m/m} = \frac{\text{mass of NaOCl}}{\text{mass of soln}} \times 100$$

$$\text{mass of NaCl} = \frac{\%\text{ m/m} \times \text{mass of solution}}{100}$$

$$= \frac{5.2 \times 2500}{100} = 130 \text{ gram}$$

Answer: B

17. What is the correct order of increasing acid strength in the following species: $HBrO_3$, $HBrO_2$, $HBrO_3$ and $HClO_3$?

A. $HBrO < HBrO_2 < HBrO_3 < HClO_3$
B. $HBrO < HBrO_2 < HClO_3 < HBrO_3$
C. $HClO_3 < HBrO_3 < HBrO_2 < HBrO$
D. $HBrO_3 < HBrO_2 < HBrO < HClO_3$

Explanation: In strength $HBrO_3 > HBrO_2 > HBrO$ and $HClO_3 > HBrO_3$ because Cl is more electronegative than Br

Answer: A

18. A 50.0 mL sample of 0.50 M acetic acid (CH_3COOH), Ka = 1.8×10^{-5} is titrated with 0.15M NaOH solution. What is the pH of the solution when no NaOH is added?

A. 6
B. 2.25
C. 2.58
D. 3.17

Solution: Because there is no added base, the pH of the solution will be simply calculated as the following.

$$CH_3COOH \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq)$$

Initial conc	0.5	-	-
change	-x	+x	+x
Equil. conc	0.5 – x	x	x

$$Ka = \frac{x^2}{0.5 - x}$$

$$\frac{x^2}{0.5} = 1.8 \times 10^{-5}$$

$$x = \sqrt{(0.5)(1.8 \times 10^{-5})} = 0.003 = [H_3O^+]$$

$$\text{pH} = -\log [H_3O^+] = -\log^{0.003} = 2.58$$

Answer: C

19. The enthalpy change for the conversion of white phosphorous to red phosphorus is -17.7 $kJ.mol^{-1}$. Which of the following statement is true?

A. One mole of red phosphorus will give off more heat when born in air than one mole of white phosphorus.
B. One mole of white phosphorus will give off more heat when burned in air than one mole of red phosphorus.
C. The heat generated by both types is the same.
D. The enthalpy change for the conversion of red phosphorus to white will be -35.4 kJ/mol.

Answer: B

Telegram: https://t.me/Extremeseries

20. Which of the following is NOT a spontaneous process?
A. The rusting of iron.
B. The dissolution of salt in water.
C. A clear solution becomes in uniform color after a few drops of dye are added.
D. Cleaning a room.

Explanation: A spontaneous process is a process that occurs without an external influence.

Answer: D

21. What are the expected products of the electrolysis of an aqueous solution of $CuCl_2$?
A. Cu and H_2 at the cathode and Cl_2 and O_2 at the anode.
B. Cu at the cathode and Cl_2 at the anode.
C. Cu and H_2 at the cathode and only Cl_2 at the anode.
D. Cu at the cathode and Cl_2 and O_2 at the anode.

Solution:

Anode rxn: $2Cl^- \longrightarrow Cl_2 + 2e^-$

Cathode rxn: $Cu^{2+} + 2e^- \longrightarrow Cu$

Cell rxn: $Cu^{2+} + 2Cl^- \longrightarrow Cu + Cl_2$

Answer: B

22. Which is the most abundant element in the earth's crust?
A. Aluminum
B. Silicon
C. Oxygen
D. Iron

Explanation

The twelve most abundant elements in earth's crust are oxygen, silicon, aluminum, iron, calcium, magnesium, sodium, potassium, titanium, hydrogen, phosphorous and manganese.

Answer: C

23. The reaction of a dialcohol with diprotic carboxylic acid is an example of
A. a condensation polymerization
B. a copolymer
C. an addition polymerization
D. an thermoses polymer

Explanation

- Condensation polymerization is a combination of monomers with the elimination of simple molecules like water. Monomers are usually bifunctional.

- Addition polymerization: involves unsaturated compounds containing double or triple bonds.
- Copolymer: are prepared by polymerizing more than one kind of monomer unit.
- Thermo set polymers are polymers that can be moulded into different shapes by heating and become hard when cooled. However, once solidified and set they cannot softened or remoulded by heating. Example Bekelite.

Note: Polyester (polyethylene terephthalate or Dacron) is a consation polymer made from 1, 4 – Benzene dicarboxylic acid (terephthalic acid) and 1, 2 – ethanediol (ethylene glycol).

Answer: A

24. Which of the following is NOT a carbohydrate?
A. Glucose
B. Cellulose
C. Glycine
D. Starch

Answer: C

25. The exothermic step in the dissolution of a salt in water is
A. Separation of the salt into ions in gaseous state.
B. Separation of the water molecules.
C. Hydration (solvation) of the ions from the salt.
D. Separation of the salt into ions in gaseous state and separation of the water.

Answer: C

26. Which of the following is NOT correct?
A. Addition of a strongly solvated solute decreases the solubility of a gas is liquid.
B. For solid dissolved in liquid that is exothermic, increase in temperature increase the solubility.
C. The solubility of a substance is its concentration in the saturated solution.
D. For solution to occur solvent – solute attractions must overcome solute – solute and solvent – solvent attractions.

Explanation: For exothermic mixing, increasing temperature, decreases solubility.

Answer: B

27. How many milligrams are in 0.2 decigrams?
A. 20 milligrams
B. 2000 milligrams
C. 0.002 milligrams
D. 0.00002 milligrams

Solution 1 mg = 10^{-2} decigram

?x = 0.2 decigram

x = 20 mg

Answer: A

Telegram: https://t.me/Extreme Series

28. How many grams of $CaCO_3$ will be formed by passing excess CO_2 gas through 100 ml of 0.25 M $Ca(OH)_2$ solution?

A. 10 grams C. 2.5 grams
B. 2 grams D. 5 grams

Solution

$$CO_2 + Ca(OH)_2 \longrightarrow CaCO_3 + H_2O$$

Number fo moles of $Ca(OH)_2$ = Molarity × volume (Lit)
= 0.25 × 0.1
= 0.025 moles

$$\underset{0.025}{CO_2} + \underset{x}{Ca(OH)_2} \longrightarrow CaCO_3 + H_2O$$

$$\frac{0.025 \text{ mole}}{1 \text{ mole}} = \frac{x}{1 \text{ mole}}$$

x = 0.025 mole of $CaCO_3$
Mass of $CaCO_3$ = number of mole × molar mass
= 0.025 × 100 = 2.5 grams

Answer: C

29. Which one of the following electromagnetic radiation has the longest wave length?

A. Gamma ray C. radio waves
B. Micro wave D. x-rays

Explanation:
In wave length cosmio ray < gamma ray < x – ray < UV < visible < infrared < microwave < radio wave

Answer: C

30. What are the possible values of azimuthal quantum number for an electron in the 4f subshell?

A. 0, 1, 2, 3 C. 3
B. 1, 2, 3 D. 4

Explanation

The azimuthal quantum number (ℓ) takes a value from 0 to n – 1.

In 4f n = 4, hence the values of ℓ for s – sub shell, p – sub shell, d – sub shell and f – sub shell are 0,1, 2 and 3 respectively.

Answer: C

31. Which of the following is the correct electronic configuration of Ni?

A. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^2$
B. $1s^2 2s^2 2p^6 3s^2 3s^2 3p^6 3d^9 4s^1$
C. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 4p^6 5s^2$
D. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 4p^2$

Answer: A

32. What type of hybridization is used by carbon in C_2H_2 (acetylene)?

A. sp B. sp^3 C. sp^2 D. sp^2d

Answer: A

33. Which of the following rate relationships is valid for the reaction?

$$2NO_2(g) \longrightarrow 2NO(g) + O_2(g)$$

A. $\frac{\Delta[NO_2]}{\Delta t} = \frac{\Delta[NO]}{\Delta t}$ C. $\frac{-1}{2}\frac{\Delta[NO_2]}{\Delta t} = \frac{\Delta[O_2]}{\Delta t}$

B. $\frac{-\Delta[NO_2]}{\Delta t} = \frac{\Delta[O_2]}{\Delta t}$ D. $\frac{-\Delta[O_2]}{\Delta t} = \frac{\Delta[NO]}{\Delta t}$

Answer: C

34. Which of the following statement is NOT correct regarding equilibrium constant? The value of the equilibrium constant does not depend upon the _____

A. initial concentration of the reactant
B. initial concentration of the products
C. nature of the reactants
D. presence of a catalyst

Answer: D

35. Consider the reaction,

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

in a closed container at equilibrium at a constant temperature. What will be the effect of adding more PCl_5 on the equilibrium concentration of Cl_2 (g)? It

A. decreases
B. cannot be predicted without the value of k_P
C. remains unaffected
D. increases

Explanation: Increasing the concentration of the reactant shifts the equilibrium to the right.

Answer: D

Telegram: https://t.me/ExtremeSeries

36. What is the mole fraction of a solute in one molal aqueous solution?
A. 0.009 B. 0.018 C. 0.027 D. 0.036

Solution:
Molality and mole fraction of solute (X_B) are related as follows:
Where
X_B = mole fraction of solute
m = molality
M_A = molecular mass of the solvent

$$\frac{1}{X_B} = 1 + \frac{100}{m \times M_A}$$

$$\frac{1}{X_B} = 1 + \frac{100}{1 \text{ molal} \times 18 \text{ g/mol}}, \quad \frac{1}{X_B} = 56.5556$$

$X_B = 0.018$

Answer: B

37. Which of the following aqueous solution will have the lowest freezing point? 0.1 molal solution of _____
A. Urea C. sodium chloride
B. acetic acid D. calcium chloride

Explanation

Compound	Van't Hoff factor (i)
Urea	1
Acetic acid	2
Sodium chloride	2
Calcium chloride	3

Theoretically,
Note: i = number of cations + number of anions
$T_f = T°f - ikfm$
$T°f$ for H_2O is 0°C

A. $T_f = 0°C - (1)\, k_f (0.1)$
$= 0°C - 0.1\, k_f$
$= -0.1\, k_f$

B. $T_f = 0°C - (2)\, k_f (0.1)$
$= 0°C - 0.2\, k_f$
$= 0.2\, k_f$

C. $T_f = 0°C - (2) k_f (0.1)$
$= -0.2 k_f$

D. $T_f = 0°C - (3)\, k_f (0.1)$
$= 0°C - 0.3\, k_f$
$= -0.3\, k_f$

Answer: D

38. Which one of the following pair CANNOT be mixed together to form a buffer solution?
A. NH_3 and NH_4Cl C. RbOH and HBr
B. $NaC_2H_3O_2$ and $HC_2H_3O_2$ D. H_3PO_4 and KH_2PO_4

Explanation
- Buffer solution is made from
  - A weak acid with its conjugate base.
  - A weak base with its conjugate acid.
- A buffer solution will not be formed between RbOH and HBr.

Answer: C

39. Which of the following aqueous solutions will be basic?
A. NaCl C. Na_2CO_3
B. Na_2SO_4 D. $FeCl_3$

Explanation
- NaCl is made from NaOH and HCl (strong acid and strong base).
- Na_2SO_4 is made from NaOH and H_2SO_4 (strong base and strong acid).
- Na_2CO_3 is made from NaOH and H_2CO_3 (strong base and weak acid).
- $FeCl_3$ is made from $Fe(OH)_3$ and HCl (weak base and strong acid).

A basic salt is formed between strong base and weak acid.

Answer: C

40. Consider the reaction:

$CH_4(g) + 4Cl_2(g) \longrightarrow CCl_4(g) + 4HCl(g), \Delta H = -434$ kJ

What energy change occurs when 1.2 moles of methane reacts?
A. 5.2×10^5 J are released
B. 5.2×10^5 are absorbed
C. 3.4×10^5 J are released
D. 3.6×10^5 J are absorbed

Solution
From the reaction
1 mole of CH_4 releases -434 kJ
1.2 mole of CH_4 releases ?
Solving for the unknown
1.2 mole of CH_4 release 520.8 kJ
5.208×10^5 J (released)

Answer: A

Telegram: https://t.me/Extremeseries

41. When electricity is passed through aqueous solution of aluminum chloride, 13.5 gram of Al are deposited. What must be the number of Faraday's of electricity passed?

A. 2.0 B. 1.5 C. 1.0 D. 0.5

Explanation

Given mass of Al = 13.5 gram

Charge of Al in $AlCl_3 = 3$ $\quad Al^{3+} + 3e^- \longrightarrow Al$

Molar mass of Al = 27 g/mol

$$\text{number moles of Al} = \frac{\text{mass of Al}}{\text{molar mass}} = \frac{13.5}{27} = 0.5 \text{ moles}$$

$Al^{3+} + 3e^- \longrightarrow Al$

1 mole of Al^{3+} needs 3 moles of electrons. Hence 3 Faraday's of electricity are required to produce 1 mole of Al

1 mole = 3 Faraday

0.5 mole of Al =?

x = 1.5 Faraday

Answer: B

42. Which of the following is the anodic reaction in the discharge of lead – acid storage battery?

A. $PbSO_4(s) + 2H_2O(l) \longrightarrow PbO_2(s) + 4H^+(aq) + SO_4^{2-}(aq) + 2e^-$

B. $Pb(s) + SO_4^{-2}(aq) \longrightarrow PbSO_4(s) + 2e^-$

C. $PbSO_4(s) + 2e^- \longrightarrow Pb(s) + SO_4^{2-}(aq)$

D. $PbO_2(s) + SO_4^{2-}(aq) + 4H^+ + 2e^- \longrightarrow PbSO_4(s) + 2H_2O(\ell)$

Explanation

The anode and cathode reaction for lead – acid storage battery are

Anode reaction $\quad Pb(s) + SO_4^{2-}(aq) \rightleftharpoons PbSO_4(s) + 2e^-$

Cathode rxn

$PbO_2(s) + 4H^+(aq) + SO_4^{2-}(aq) + 2e^- \rightleftharpoons PbSO_4(s) + 2H_2O(\ell)$

Answer: A

43. With which one of the following is gold complex to recover it from its ore?

A. NO_3^- B. CO_3^{2-} C. SO_4^{2-} D. CN^-

Answer: D

44. For the reaction, $3O_2(g) \longrightarrow 2O_3(g)$; $\Delta H = 284$ kJ. What are the signs of ΔG and ΔS respectively?

A. positive and negative
B. positive and positive
C. negative and negative
D. negative and positive

Explanation

$3O_2(g) \longrightarrow 2O_3(g)$

The entropy is negative because the number of moles of reactant is greater than number of moles of product.

From the formula $\Delta G = \Delta H - T\Delta S$

ΔH = positive

ΔS = negative

$\therefore \Delta G$ is always positive.

Answer: A

45. Which of the following is NOT true about anode of a voltaic cell?

A. Reduction occurs at it.
B. It is assigned as the negative pole.
C. Electrons flow from it.
D. Cations in solution move away from the surface of the electrode as the redox reaction occurs.

Answer: A

46. What is the orbital hybridization of the oxygen atom in water?

A. sp B. sp^2 C. sp^3 D. sp^3d

Answer: C

47. Within a specified period, an increase in atomic number's usually accompanied by:

A. an increase in atomic radius and an increase in electronegativity
B. a decrease in atomic radius and an increase in electronegativity
C. an increase in atomic radius and a decrease in electronegativity
D. a decrease in atomic radius and a decrease in electronegativity

Answer: B

Telegram: https://t.me/ExtremeSeries

48. Which of the following sets of quantum numbers is allowed in the hydrogen atom?

A. $n = 2, \ell = 1, m\ell = -1$ C. $n = 8, \ell = 1, m\ell = -6$

B. $n = 1, \ell = 0, m\ell = 0$ D. $n = 2, \ell = 0, m\ell = 2$

Explanation

H (atomic no 1) = $1s^1$

$n = 1, \ell = 0$ (s-sub shell), $m\ell = 0$

Answer: B

49. NF_3, BF_3 and ClF_3 all have molecular formulae of the type XF_3. Which molecule(s) has (have) trigonal pyramidal molecular geometry?

A. NF_3 and BF_3 C. NF_3 and ClF_3

B. BF_3 and ClF_3 D. NF_3

Explanation

Molecule	Molecular geometry
NF_3	Trigonal pyramidal
BF_3	Trigonal planar
ClF_3	T-shaped

Answer: D

50. Which one of the following molecules cannot be formed?

A. NCl_3 B. OF_6 C. ONF_3 D. PCl_3

Answer: B

51. Of Br_2, Ne, HCl and N_2 which is likely to have the largest molecular dispersion forces?

A. Br_2 B. Ne C. HCl D. N_2

Explanation

Molecules with higher molecular mass have large dispersion force.

Answer: A

52. Thiosolfate ion is oxidized by iodine according to the following reaction:

$$2S_2O_3^{2-}(aq) + I_2(aq) \longrightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$$

If a certain experiment, 0.006 mol of $S_2O_3^{2-}$ is consumed in 1.0 L of solution each second, at what rates are $S_4O_6^{2-}$ and I^- produced in this solution?

A. 0.012 mol L^{-1} s^{-1} ($S_4O_6^{2-}$) and 0.006 mol L^{-1} s^{-1} (I^-)

B. 0.006 mol L^{-1} s^{-1} ($S_4O_6^{2-}$) and 0.006 mol L^{-1} s^{-1} (I^-)

C. 0.003 mol L^{-1} s^{-1} ($S_4O_6^{2-}$) and 0.006 mol L^{-1} s^{-1} (I^-)

D. 0.012 mol L^{-1} s^{-1} ($S_4O_6^{2-}$) and 0.012 mol L^{-1} s^{-1} (I^-)

Explanation

$$2S_2O_3^{2-}(aq) + I_2(aq) \longrightarrow S_4O_6^{2-}(aq) + 2I(aq)$$

$$\text{rate} = \frac{\Delta[S_2O_3^{2-}]}{2\Delta t} = \frac{-\Delta[I_2]}{\Delta t} = \frac{\Delta[S_4O_6^{2-}]}{\Delta t} = \frac{\Delta[I^-]}{2\Delta t}$$

$$\Rightarrow \frac{-\Delta[S_2O_3^{2-}]}{2\Delta t} = \frac{\Delta[S_4O_6^{2-}]}{\Delta t}$$

$$\frac{-1}{2}(0.006) = \Delta[S_4O_6^{2-}]$$

$$[S_4O_6^{2-}] = \frac{0.006}{2} = 0.003 \text{ mol/L.s}$$

and

$$\frac{-\Delta[S_2O_3^{2-}]}{2\Delta t} = \frac{\Delta[I^-]}{2\Delta t}$$

$\Delta[I^-] = 0.006$ mol/L. s

Answer: C

53. A particular reaction has found to depend on the concentration of hydrogen ion, $[H^+]$. The initial rate change as a function of $[H^+]$ as follows.

$[H^+]$ (M)	0.005	0.01	0.02
Initial rate (M/s)	3.4×10^{-7}	1.7×10^{-7}	0.85×10^{-7}

What is the order of the reaction in $[H^+]$?

A. -1 B. 1 C. 2 D. 0

Explanation

$$\text{rate} = K[H^+]^x$$

Take any of the two rates with their corresponding concentrations of $[H^+]$.

Telegram: https://t.me/ExtremeSeries

$rate_1 = k(0.005)^x = 3.4 \times 10^{-7}$

$rate_2 = k(0.01)^x = 1.7 \times 10^{-7}$

Taking the ratio of the two rates

$$\frac{rate_1}{rate_2} = \frac{k(0.005)^x}{k(0.01)^x} = \frac{3.4\times10^{-7}}{1.7\times10^{-7}}$$

$$\left(\frac{1}{2}\right)^x = 2$$

$(2^{-1})^x = 2$

$2^{-x} = 2$

$-x = 1$

$x = -1$

Answer: A

54. The conversion of ozone into O_2 proceeds in two elementary steps as follows.

$$O_3(g) \longrightarrow O_2(g) + O(g)$$

$$O_3(g) + O(g) \longrightarrow 2O_2(g)$$

What is the equation for the overall reaction?

A. $2O_3(g) \longrightarrow 3O_2(g)$

B. $2O_3(g) + O(g) \longrightarrow 3O_2(g) + O(g)$

C. $O_3(g) + O(g) \longrightarrow O_2(g) + 2O(g)$

D. $2O_3(g) + O(g) \longrightarrow 3O_2(g)$

Explanation

O is an intermediate so it should not appear in the overall reaction.

$$O_3(g) \longrightarrow O_2(g) + O(g)$$

$$O_3(g) + O(g) \longrightarrow 2O_2(g)$$

$$2O_3(g) \longrightarrow 3O_2(g)$$

Answer: A

55. Which of the following factors does not affect the rate of a chemical reaction that is of zero order?

A. concentration of reactants

B. surface area of reactants

C. temperature

D. presence of catalyst

Answer: A

56. How would you prepare 0.90 L of 1.2×10^{-2} M NaCl aqueous solution?

A. weigh 0.012 g of NaCl, dissolve in a small amount of water and dilute to 0.90L.

B. weigh 0.63 gram of NaCl, dissolve in a small amount of water and dilute to 0.90L.

C. weigh 1.2 gram of NaCl, dissolve in a small amount of water and dilute to 0.90L.

D. weigh 1.4 gram of NaCl, dissolve in a small amount of water and dilute to 0.90L.

Explanation

$$\text{Molarity} = \frac{\text{mass}}{\text{molar mass}} \times \frac{1}{V(L)} \Rightarrow \text{mass} = \text{molarity} \times \text{molar} \times V$$

$= 1.2 \times 10^{-2} \times 58.5 \times 0.9 = 0.631$ gram

Answer: B

57. Which of the following solutes will produce the largest increase in boiling point upon addition to 1 kg of water?

A. 1 mole of $Co(NO_3)_2$

B. 2 mole of KCl

C. 3 mol of ethylene glycol ($C_2H_6O_2$)

D. 1 mol of $CaCl_2$

Explanation

$T_b = T_b^\circ + ik_b\, m$

where T_b: B. p of the solution

T_b°: B. p of pure solvent

i : Vant Hoff factor

k_b: boiling point elevation constant

m: molality

A. 1 mol of $CO(NO_3)_2$

$i = 2$

$$T_b = T_b^\circ + ik_b\, m = 100°C + 2k_b\left(\frac{1\text{ mol}}{1\text{ kg}}\right) = 100°C + 2k_b$$

B. 2 mole of KCl

$i = 2$

Telegram: https://t.me/ExtremeSeries

$$T_b = T_b^o + 1\,kbm \Rightarrow 100°C + 2k_b\left(\frac{2\text{ mol}}{1\text{ kg}}\right) = 100°C + 4k_b$$

C. 3 mole of $C_2H_6O_2$
i = 1 (blse it is non electrolyte)

$$T_b = T_b^o + i\,k_b m$$

$$T_b = 100°C + 1\,k_b\left(\frac{3}{1}\right) = 100°C + 3k_b$$

D. 1 mole of $CaCl_2$
i = 3

$$T_b = T_b^o + ik_b m \Rightarrow 100°C + 3k_b\left(\frac{1}{1}\right) = 100°C + 3k_b$$ Answer: B

58. What is the percentage of HF molecules ionized in a 0.10 M HF solution ($Ka = 6.8 \times 10^{-4}$)?
A. 2 B. 5 C. 8 D. 10

Solution

$$HF + H_2O \rightleftharpoons H_3O^+(aq) + F^-(aq)$$

Initial conc.	0.1	–	–
Change	-x	x	+x
Equili. Conc	0.1 – x	X	x

$$Ka = \frac{x^2}{0.1-x} = 6.8\times10^{-4}$$

Using approximation method $\frac{x^2}{0.1-x} = 6.8\times10^{-4}$

$$\frac{x^2}{0.1} = 6.8\times10^{-4}$$

$$x^2 = 6.8 \times 10^{-5}$$

$$x = \sqrt{6.8\times10^{-5}} = 8.24\times10^{-3}$$

$$\text{Percentage dissociation (x\%)} = \frac{\text{amount of dissiciation (x)}}{\text{initial concentration (c)}}\times100$$

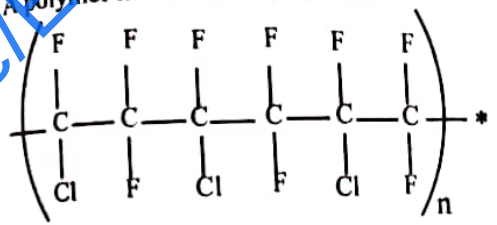
$$= \frac{8.24\times10^{-3}}{0.1}\times100 = 8.24\%$$ Answer: C

59. Which of the following reaction is a redox reaction?
A. $3Se(s) + 4HNO_3(aq) \longrightarrow 3SeO_2(s) + 4NO(g) + H_2O(\ell)$
B. $CaCl_2(aq) + 2AgNO_3(aq) \longrightarrow Ca(NO_3)_2(aq) + 2AgCl(s)$
C. $BaCO_3(s) \longrightarrow BaO(s) + CO_2(g)$
D. $3Mg(OH)_2 + 2NH_3(g) \longrightarrow Mg_3N_2(s) + 6H_2O(\ell)$

Explanation
The way to check whether a given reaction is redox or not is by assigning oxidation number value for reactants and products. As a result, if there is a change in oxidation number the reaction is redox otherwise it is non redox. All acid base and precipitation reactions are non redox.

Answer: A

60. A polymer has the following structure:



What is the monomer of this polymer?
A. $CFCl = CF_2$ C. CF_2Cl_2
B. CF_3Cl D. CF_4

Answer: A

61. How many significant figures are there in 0.0500?
A. one B. three C. two D. Four

Answer: B

62. If 16 grams of O_2 react with excess C_2H_6, how many grams of CO_2 will be formed? The unbalanced equation is

$$C_2H_6(g) + O_2(g) \longrightarrow CO_2(g) + H_2O(g)$$

A. 22 grams C. 13 grams
B. 9 grams D. 12.6 grams

Solution

$$C_2H_6(g) + O_2(g) \longrightarrow CO_2(g) + H_2O(g)$$

When it is balanced

$$2C_2H_6(g) + 7O_2(g) \longrightarrow 4CO_2(g) + 6H_2O(g)$$

Telegram: https://t.me/ExtremeSeries

Using mass – mass relationship

$$\frac{\text{mass of } O_2}{7\times \text{ molar mass}} = \frac{x}{4 \times \text{molar mass of } CO_2}$$

$$\frac{16}{7\times 32} = \frac{x}{4\times 44}$$

$$\frac{16}{224} = \frac{x}{176}$$

x = 12.57 grams

Answer: D

63. How does the titration of a strong acid with a strong base differ from the titration of a weak acid with a strong base with respect to the quantity of base required to reach the equivalence point?
   A. The quantity of base required to reach the equivalence point is the same in both titrations.
   B. The quantity of base required to reach the equivalence point for the titration of a strong acid with a strong base is higher than for the titration of a weak acid with a strong base.
   C. The quantity of base required to reach the equivalence point for the titration of a strong acid with a strong base is lower than the titration of weak acid with a strong base.
   D. The quantity of base required to reach the equivalence point for the titration of a strong acid with a strong base is slightly higher than for the titration of a weak acid with a strong base.

Answer: A

64. Hydrogen atoms are excited to the stationary state designated by the principal quantum number, n = 4. What will be the number of lines (frequencies) emitted by the hydrogen atoms?
   A. 1 B. 4 C. 6 D. 16

Answer: B

65. According to the molecular orbital theory, which one of the following is paramagnetic
   A. F_2 B. H_2 C. NO^+ D. NO

Explanation

Species with odd number of electrons are paramagnetic:

$F_2 = 2 \times 9 = 18e^-$

$H_2 = 2 \times 1 = 2e^-$

$NO^+ = 7 + 8 - 1 = 14e^-$

$NO = 7 + 8 = 15e^-$

Answer: D

66. For the reaction

$$2NO\,(g) + O_2\,(g) \longrightarrow 2NO_2\,(g)$$

the following data were obtained

$[NO]_0$ (M):	0.10	0.10	0.40
$[O_2]_0$ (M):	0.10	0.20	0.20
Rate (M/s):	2.5×10^{-4}	5.0×10^{-4}	8.0×10^{-3}

What is the rate law for the reaction?

A. rate = $k[NO][O_2]$ C. rate = $k[NO]^2[O_2]$

B. rate = $k[NO][O_2]^2$ D. rate = $k[NO]^2[O_2]^2$

Solution: $2NO\,(g) + O_2\,(g) \longrightarrow 2NO_2\,(g)$

rate = $k[NO]^x[O_2]^y$

To determine y order with respect to O_2, take trial 1 and 2 where the concentration of O_2 are the same

$rate_2 = k\,(0.1)^x\,(0.1)^y = 2.5 \times 10^{-4}$

$rate_3 = k\,(0.1)^x\,(0.2)^y = 5.0 \times 10^{-4}$

$$\frac{rate_1}{rate_2} = \frac{k\,(0.1)^x\,(0.1)^y}{k(0.1)^x\,(0.2)^y} = \frac{2.5\times10^{-4}}{5.0\times10^{-4}}$$

$$\left(\frac{1}{2}\right)^y = \left(\frac{1}{2}\right)^1$$

y = 1

∴ order with respect to O_2 is 1 (1^{st} order)

To determine x (order with respect to NO), take trials 2 and 3 where the concentrations of O_2 are the same.

$rate_1 = k\,(0.1)\,(0.2)^y = 5.0 \times 10^{-4}$

$rate_2 = k\,(0.4)\,(0.2)^y = 8.0 \times 10^{-3}$

Taking the ratio of $rate_3$ to $rate_2$

$$\frac{rate_3}{rate_2} = \frac{k(0.4)^x(0.2)^y}{k(0.1)^x(0.2)^y} = \frac{8.0\times10^{-3}}{5.0\times10^{-4}} = 4^x = 16$$

$4^x = 4^2$

x = 2

∴ order with respect to NO is 2 (second order)

Hence the complete rate expression would be

rate = $k[NO]^2[O_2]$

Answer: C

Telegram: https://t.me/ExtremeSeries

67. The value of k_C for the reaction:

$$COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$$

is 2.5×10^{-10} at 300k. Starting with pure $COCl_2$ (g) in a closed vessel, if the concentration of CO (g) is 1×10^{-5} M at equilibrium. What would be the equilibrium concentration of $COCl_2$ (g)?

A. 2.5M B. 2.5×10^{-5}M C. 2.5×10^{-5} M D. 0.4 M

Explanation

$$COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$$

$$k_C = \frac{[CO][Cl_2]}{[COCl_2]}$$

At equilibrium $[CO] = [Cl_2] = 1 \times 10^{-5}$ M

$$2.5 \times 10^{-5} = \frac{(1\times10^{-5})(1\times10^{-5})}{[COCl_2]}$$

$[COCl_2] = 0.4$ M **Answer: D**

68. An aqueous solution freezes at -0.186°C (k_f = 1.86°C/m; k_b = 0.512°C/m). What is the elevation in boiling point (in °C)?

A. 0.186 B. 0.512 C. 0.5.12/1.86 D. 0.0512

Solution

$\Delta T_f = T_f^\circ - T_f$

$= 0°C - (-0.186°C)$

$= 0.186°C$

Where

T_f° = freezing point of pure solvent

T_f = freezing point of the solution

T_b = boiling point of the solution

$\Delta T_f = kfm$

m = molality

$$m = \frac{\Delta T_f}{k_f} = \frac{0.186}{1.86}$$

$$\Delta T_b = k_b m = (0.512)\left(\frac{0.186}{1.86}\right) = 0.0512°C$$ **Answer: D**

69. The P^{ka} of acetyl salicylic acid (aspirin) is 3.5. The pH of gastric juice in human stomach is about 2 to 3 and the pH in the small intestine is about 8. Which of the following statement is correct? Aspirin is

A. unionized in the small intestine and in the stomach

B. completely ionized in the small intestine and in the stomach

C. ionized in the stomach and almost unionized in the small intestine

D. ionized in the small intestine and almost unionized in the stomach

Explanation: The stomach medium is acidic and that of the small intestine is basic, so, when aspirin is added to a gastric juice with acidic pH, it lowers the dissociation of the acid. And the acid ionization increase when it inserted in to a small intestine.

Answer: D

70. A 10 ml sample of tartaric acid (a diprotic acid) is titrated with 20 mL of 1.0 M NaOH. What is the molarity of the acid?

A. 2.0 B. 1.0 C. 4.0 D. 0.5

Solution

Number of equivalent of the acid = number of equivalent of the base

Normality of NaOH = Z × M = 1 × 1 = 1

$N_1V_1 = N_2V_2$

$$N_1 = \frac{N_2V_2}{V_1} = \frac{1\times20}{10} = 2N$$

$$\text{Molarity} = \frac{N}{\text{charge (Z)}} = \frac{2N}{2} = 1M$$

i.e., Z = no of replaceable hydrogen **Answer: B**

71. When the equation for the following reaction in basic solution is balanced, what is the sum of the coefficients?

$$MnO_2 + HO_2^- \longrightarrow MnO_4^-$$

A. 11 B. 31 C. 14 D. 9

Explanation: $MnO_2 + HO_2^- \longrightarrow MnO_4^-$

1. Identify the oxidation and reduction half reaction.

oxidation half reaction reduction half reaction

$$\overset{+4}{Mn}O_2 \longrightarrow \overset{+7}{Mn}O_4^- + 3e^- \qquad HO_2^- + 2e^- \longrightarrow OH^-$$

2. Balance atom other than oxygen and hydrogen.

$$\overset{+4}{Mn}O_2 \longrightarrow MnO_4^- + 3e^- \text{ (already balanced)}$$

$$HO_2^- + 2e^- \longrightarrow OH^- \text{ (already balanced)}$$

Telegram: https://t.me/ExtremeSeries

3. Balance oxygen atoms by adding water to side where oxygen is deficient.

$2H_2O + MnO_2 \longrightarrow MnO_4^- + 3e^-$ $HO_2^- + 2e^- \longrightarrow OH^-$

4. Balance hydrogen atoms by adding H^+ to side where hydrogen is deficient.

$2H_2O + MnO_2 \longrightarrow MnO_4^- + 4H^+ + 3e^-$

$2H^+ + HO_2^- + 2e^- \longrightarrow OH^- + H_2O$

5. Balance number of electrons lost and gained.
   - Multiply the oxidation half reaction by two.
   - Multiply the reduction half reaction by three.

$4H_2O + 2MnO_2 \longrightarrow 2MnO_4^- + 8H^+ + 6e^-$

$6H^+ + 3HO_2^- + 6e^- \longrightarrow 3OH^- + 3H_2O$

6. Write the oxidation and reduction half reactions together.

$4H_2O + 2MnO_2 + 6H^+ + 3HO_2 + 6e^- \longrightarrow$

$2MnO_4^- + 8H^+ + 3OH^- + 3H_2O + 6e^-$

$\Rightarrow$ $H_2O + 2MnO_2 + 3HO_2^- \longrightarrow 2MnO_4^- + 2H^+ + 3OH^-$

add 2 moles of OH^- to the reactant and product side $2OH^-$

$+ H_2O + 2MnO_2 + 3HO_2^- \longrightarrow 2MnO_4^- +$

$2H^+ + 2OH^- + 3OH^-$

$\Rightarrow$ $2OH^- + H_2O + 2MnO_2 + 3HO_2^- \longrightarrow$

$2MnO_4^- + 2H_2O + 3OH^-$ canceling the H_2O molecule

$OH^- + 2MnO_2 + 3HO_2^- \longrightarrow 2MnO_4^- + H_2O + 3OH^-$

$\Rightarrow$ $2MnO_2 + 3HO_2^- \longrightarrow 2MnO_4^- + H_2O + OH^-$

(in basic medium)

Number of atoms are conserved!

$\therefore$ The sum of the coefficients will be 9

Answer: D

72. The standard emf for the all reaction, $Zn + Cu^{2+} \longrightarrow Cu + Zn^{2+}$, is 1.1V. What would the emf for the all reaction be when 0.1 M Cu^{2+} and 0.1 M Zn^{2+} solutions are used at 298 K?

A. 1.10V B. 0.10V C. -1.10V D. -0.110V

Solution

Anode rxn: $Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$

Cathode rxn: $Cu^{2+} + 2e^- \longrightarrow Cu(s)$

$E°cell = 1.1$ V

Using Nernest equation

$$E_{cell} = E°_{cell} - \frac{0.0592}{2}\log\left(\frac{[Zn^{2+}]}{[Cu^{2+}]}\right)$$

$$= 1.1 - \frac{0.0592}{2}\log\left(\frac{0.1}{0.1}\right) = 1.1 - \frac{0.0592}{2}\log 1 = 1.1$$

Answer: A

73. Which one of the following is a polyamide molecule?

A. Terylene C. Nylon

B. Rayon D. Polystyrene

Answer: C

74. Which metal atom can form ionic bonds by losing electrons from both the outermost and next to outermost principal energy levels?

A. Fe B. Mg C. Pb D. Ca

Answer: A

75. The energy required to remove an electron from an iron atom amounts to 7.21×10^{-19} J. What is the maximum wave length of light that can perform this?

A. 250 nm C. 289 nm

B. 276 nm D. 300 nm

Solution

$$\Delta E = \frac{hC}{\lambda}$$ Where E = energy

$$\lambda = \frac{hC}{\Delta E}$$ h = plank's content

Telegram: https://t.me/ExtremeSeries

$$= \frac{(6.626\times10^{-34})(3\times10^{8}\ m/s)}{7.21\times10^{-19}}$$ C = speed of light

$= 2.757 \times 10^{-7}$ m $= 276$ nm

76. What is the molecular geometry of O_3? **Answer: B**

A. Trigonal planar C. Trigonal pyramidal
B. Bent D. Tetrahedral

Explanation

O_3 is AB_2E type so its molecular geometry is Bent/Angular / V-shape and its electron pair geometry is trigonal planar.

77. At 295 K, a 3.0L flask contains of mol HI, 4 mol H_2, and 0.6 mol I_2 in equilibrium. What is the value of k at 295k? **Answer: B**

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

A. 10.2 B. 20.4 C. 30.4 D. 61.2

Solution

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

$$[H_2] = \frac{\text{no of mole}}{\text{volume (L)}} = \frac{4\ mol}{3} = 1.3\ M$$

$$[I_2] = \frac{0.6}{3} = 0.2M$$

$$[HI] = \frac{1}{3} = 2.3M$$

$$k_C = \frac{[HI]^2}{[H_2][I_2]} = \frac{(2.3)^2}{(1.3)(0.2)} = \frac{5.4}{0.26} = 20.769$$ **Answer: B**

78. The equilibrium constant is 0.09 at 295 K for the reaction

$$H_2O(g) + Cl_2O(g) \rightleftharpoons 2HOCl(g)$$

For which of the following sets of conditions is the system of equilibrium?

A. A 1.0L flask contains 1.0 mol of HOCl, 0.10 mol of Cl_2O and 0.1 mol of H_2O.

B. A 2.0 L flask contains 0.084 mol of HOCl, 0.080 mol of Cl_2O, and 0.98 mol of H_2O.

C. A 3.0 L flask contains 0.25 mol of HOCl, 0.000 mol of Cl_2O, and 0.56 mol of H_2O.

D. A 4.0 L flask contains 0.25mol HOCl, 0.001 mol of Cl_2O and 0.56 mol of H_2O.

Explanation: For choice

$$[HOCl] = \frac{0.084\ mol}{2\ Lit} = 0.04M$$

$$[H_2O] = \frac{0.98\ mol}{2} = 0.49M$$

$$[Cl_2O] = \frac{0.080\ mol}{2\ Lit} = 0.040\,M$$

$$H_2O(g) + Cl_2O(g) \rightleftharpoons 2HOCl(g)$$

$$k_C = \frac{[HOCl]^2}{[H_2O][Cl_2O]} = \frac{(0.042)^2}{(0.040)(0.40)} = 0.09$$

$\therefore$ The $K'_C = K_C$ **Answer: B**

79. 1000 grams of a 55% (mass percentage) nitric acid solution is to be diluted to 20% (mass percentage) nitric acid solution. How many liters of water should be added to the starting solution?

A. 0.45 B. 1.55 C. 2.30 D. 2.75

Explanation

1 gram = 1 ml

1000 gram = 1000 ml = 1 L

Applying dilution formula

$$\%\ m/m \cdot V_1 = \%\ m/m \cdot V_2$$

$$\frac{55}{100}.1L = \frac{20}{100}.V_2$$

$$\frac{0.55}{0.2} = .V_2 = 2.75\ L$$

V_{H_2O} added $= V_2 - V_1$

$= 2.75L - 1L$

$= 1.75L$

Answer: D

80. When pure methanol is mixed with water, the solution becomes warm. Is this an ideal solution?

A. yes, for an ideal solution $\Delta H_{soln} = 0$
B. No, for ideal solution $\Delta H_{soln} = 0$
C. yes, for ideal solution $\Delta H_{soln} > 0$
D. No, for ideal solution $\Delta H_{soln} < 0$

Answer: B

Telegram: https://t.me/ExtremeSeries

1. Which of the following is the right order of the steps of a scientific method?
   A. Performing experiments – formulating hypothesis – marking observation
   B. Formulating hypothesis – making observation – performing experiment
   C. Making observation – formulating hypothesis – performing experiments
   D. Making observation – performing experiments – formulating hypothesis

Explanation:
Scientific method is a careful, logical and systematic approach towards answering scientific questions. The steps are stated under choice C.

Answer: C

2. The compound CuCl emits blue light having a wave length of 450 nm. When heated about 120°C. What is the increment in energy (quantum) that is emitted at 450 am?
   A. 2.25×10^{-19} J
   B. 4.41×10^{-19} J
   C. 8.20×10^{-19} J
   D. 16.20×10^{-19} J

Solution:

$$\Delta E = \frac{hC}{\lambda}$$ where h = plank's constant (6.626×10^{-34} J.s)

C = speed of light (2.9979×10^{8} m/s^{-1})

λ = wave length

$$\Delta E = \frac{(6.626\times10^{-34}\,\text{J.s})(2.9979\times10^{8}\,\text{m/s})}{450\times10^{-9}\,\text{m}} = 4.41\times10^{-19}\,\text{J}$$

Answer: B

3. What is the difference between chlorine -35 and chlorine 37?
   A. Chlorine – 37 has two more protons than chlorine – 35
   B. Chlorine – 37 has two more neutrons than chlorine – 35
   C. Chlorine – 37 has two more electrons than chlorine – 35
   D. Chlorine – 37 has one more proton and one more neutron than chlorine – 35

Explanation
Isotopes are atoms of the same elements that have the same atomic number but different atomic mass. In a neutral atom atomic number, number of electron = number of proton

$A = Z + n$ where A = mass number
Z = atomic number
n = neutron number

For chlorine 35, A = 35, Z = 17, n = A – Z = 35 – 17 = 18
For chlorine 37, A = 37, Z = 17, n = A – Z = 37 – 17 = 20

Answer: B

4. Which one of the following electromagnetic radiation has the shortest wavelength?
   A. x-ray
   B. uv-ray
   C. gamma ray
   D. microwave

Explanation
- The following order of electromagnetic radiation is based on increasing order of energy y is cosmic ray > gamma ray > x-ray > uv-ray > visible > infrared > micro wave > TV > Radio (in energy).
- The energy of the wave is:
  (1) Directly proportional to the frequency
  (2) Inversely proportional to the wave length

Answer: C

5. Which group of elements is characterized with ns^2np^2 outer-electron configuration?
   A. Group 2A
   B. Group 4A
   C. Group 4B
   D. Group 3B

Explanation
ns^2np^2 indicates the two sub shells exists at the same energy level because they have the same n-value. Hence to find the group number one should add the number of electrons s and p sub shells (2 + 2 = 4 electrons). Therefore, ns^2np^2 belongs to group IVA = 4A.

Answer: B

6. Which quantum number/s determine the energy of an electron in hydrogen atom?
   A. n
   B. n and ℓ
   C. n, ℓ and m
   D. n, ℓ, m and s

Explanation
Hydrogen has 1 electron, so its configuration is $1s^1$. Hence n = 1, ℓ = 0 (s- sub shell). Therefore, n value determines the energy of an electron in hydrogen atom.

Answer: A

Telegram: https://t.me/ExtremeSeries

7. For elements in the left-most column of the periodic table, properties that have increasing values as the atomic number increases include which of the following?
I. Ionization II. Atomic radius III. Atomic mass
A. III only C. I and II only
B. I, II and III D. II and III only

Explanation:
Atomic size, atomic mass and metallic property increase down a group.

Answer: D

8. For an electron that has quantum numbers n = 4 and ml = 0, which of the following is true?
A. It must have the quantum number n = 0.
B. It must have the quantum number $\ell = 0$.
C. It must have the quantum number ms = $+\frac{1}{2}$.
D. It may have the quantum number $\ell = 0, 1, 2, 3$.

n	ℓ (0 to n – 1)	mℓ (– ℓ to+ ℓ)	Ms
4	0	0	+1/2
	1	-1, 0, +1	+1/2
	2	-2,-1,0,1,+2	+1/2
	3	-3,-2,-1,0,1,2,3	+1/2
	.		+1/2

If n = 4 and mℓ = 0 implies the ℓ – value should be 0, 1, 2 or 3.

Answer: D

9. Which of the following electron transition requires the smallest energy to be absorbed by the hydrogen atom?
A. from n = 1 to n = 2 C. from n = 3 to n = 4
B. from n = 2 to n = 3 D. from n = 4 to n = 5

Explanation
Using the following formula:

$$\Delta E = 2.18\times10^{-18}\left(\frac{1}{n_i 2} - \frac{1}{n_f 2}\right)$$ where,

n_i=initial quantum number
n_f = final quantum number

- For electron transition from n = 1 to n = 2.

$$\Delta E = 2.18\times10^{-18}\left(\frac{1}{1^2} - \frac{1}{2^2}\right) = 2.18\times10^{-18}\left(1 - \frac{1}{4}\right) = 1.64\times10^{-18}$$

- For an electron transition from n = 2 to n = 3.

$$\Delta E = 2.18\times10^{-18}\left(\frac{1}{2^2} - \frac{1}{3^2}\right) = 2.18\times10^{-18}\left(\frac{1}{4} - \frac{1}{9}\right) =$$

3.03×10^{-19} J

- For an electron transition from n = 3 to n = 4.

$$\Delta E = 2.18\times10^{-18}\left(\frac{1}{3^2} - \frac{1}{4^2}\right) = 2.18\times10^{-18}\left(\frac{1}{9} - \frac{1}{16}\right)$$

$= 1.06 \times 10^{-19}$

- For an electron transition from n = 4 to n = 5.

$$\Delta E = 2.18\times10^{-18}\left(\frac{1}{4^2} - \frac{1}{5^2}\right) = 2.18\times10^{-18}\left(\frac{1}{10} - \frac{1}{25}\right)$$

$= 4.91 \times 10^{-20}$

Answer: D

10. Which of the following below best explains why atoms react chemically with each other?
A. When atoms react, they gain protons and are more stable.
B. When atoms react they lose all their electrons and become more stable.
C. When atoms react, they lose, gain or share electrons and are then less stable.
D. When atoms react they lose gain or share electrons to attain for outer energy level and are then more stable.

Answer: D

11. What did Rutherford's alpha particle experiment show?
A. Electrons have a negative charge.
B. A proton is a hydrogen atom without electron.
C. Electrons circle the nucleus of an atom in orbit.
D. Most of the mass and all of the positive charge of an atom is found in a tiny nucleus.

Answer: D

12. For which of the following elements is Hund's rule used in writing the electron configuration?
A. C B. B C. Be D. Li

Telegram: https://t.me/ExtremeSeries

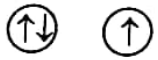
Explanation:
Hund's rule states that, "For any set of orbitals that are equal in energy (degenerate orbitals), the configuration with maximum number of parallel signs results in the lowest electron-electron repulsion."

$_6C = 1s^2 2s^2 2p^2$

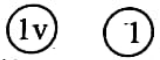
1s 2s 2px 2py 2pz

$_6B = 1s^2 2p^2 2p^1$

$_4Be = 1s^2 \quad 2s^2$



$_3Li = 1s^2\ 2s^1$



Since only carbon contains degenerate orbitals, it need Hund's rule to write its electronic configuration. **Answer: A**

13. What is the total number of valence shell electrons in BrO_3^-?

A. 20 B. 26 C. 32 D. 36

Explanation

- Valence shell is the outermost shell of an atom.
- Valence shell electrons are electrons that are found in the outer most shell.

For main group elements (Representative elements), the valence electrons determines the group to which that element belongs to i.e.,

Valence electrons = Group number

So, Br is group VIIA element hence its valence electrons are seven. Likewise, oxygen is group VIA element hence its valence electrons are six. Besides the negative ion on the species (BrO_3^-) indicates one electron is added. Therefore, the total number of valence electrons will be 7 electrons + 6 electrons + 1 electrons = 26 electrons. **Answer: B**

14. Which term describes the units that make up compounds with covalent bonds?

A. Ions C. Salts
B. Acids D. Molecules

Answer: D

15. There is a strong covalent bond between the N atoms in nitrogen gas, N_2. Why, then, does nitrogen have such a low boiling point of –196°C?

A. The bond between the N-atom is triple.
B. N is very electronegative, only next ot F and O.
C. The strong bond an intra molecular one, determines the boiling point of the substance.
D. Boiling point is determined by intermolecular force, which is this case as the molecule is non-polar.

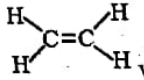
Explanation:
The type of inter-molecular force that holds nitrogen molecule together is London force (weakest force). **Answer: D**

16. Which molecule listed below has two sigma (σ) and two pi (π) bonds?

A. N_2 B. C_2H_4 C. N_2F_2 D. HCN

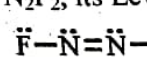
Explanation
To identify the type and number of bonds, the Lewis structure of each compound should be known.

- For N_2, it is Lewis structure is $N \equiv N$. The existence of triple bond shows 2π bonds and one sigma bond.
- For C_2H_4, it Lewis structure is



which implies are 5σ bonds and 2π bond

- For N_2F_2, its Lewis structure is



which implies there are 1π and 3σ bonds.

- For HCN

$H - C \equiv N$ which implies there are 2σ bonds and 2π bonds

Note that:

Bond type	Number of sigma and π-bonds
Single bond (–)	has only one sigma bond
Double bond (=)	has one sigma and one pi bond
Triple bond (≡)	has one sigma and two pi bonds

Answer: D

Telegram: https://t.me/ExtremeSeries

17. Which of the following explains why, at room temperature, I_2 is a solid, Br_2 is a liquid and Cl_2 is a gas?
A. Ionic bonding C. Hydrogen bonding
B. Hybridization D. London dispersion forces

Answer: D

Explanation:
As size increases, the strength of London force also increases. That is why, I_2 is solid. Br_2 is liquid as Cl_2 is gas.

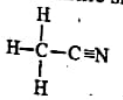
18. Which molecule has a Lewis structure that does not obey the octet rule?
A. NO B. CS_2 C. PF_3 D. HCN

Answer: D

Explanation
Molecules with an odd number of valence electrons does not obey octet rule.

Molecule	Total no of valence electrons
NO	5 + 6 = 11 electrons
CS_2	4 +2 (6) = 16 electrons
PF_3	5 + 3(7) = 26 electrons
HCN	1 + 4 + 5 = 10 electrons

19. What is the hybridization of carbon atom attached to nitrogen in acetonitile show below?



A. sp B. sp^2 C. sp^3 D. sp^4

Answer: A

Explanation
The number of sigma bonds around carbon which is attached to nitrogen atom are two sigma bonds. Hence the type of hybridization is sp.

20. What is the half life, $t_{\frac{1}{2}}$ for a zero order reaction $A \longrightarrow B$, (k is the rate constant)?
A. ln 2k B. [A]0/2k C. ln k$[A]_o$ D. ln 2$[A]_o$k

Answer: A

Explanation
Consider A reaction $A \longrightarrow B$

Order	Rate reaction	Half life $\left(t_{\frac{1}{2}}\right)$
0	$[A]_t = -kt + [A]_o$	$[A]o/2\,k$
1	$\ln [A]_t = -k + \ln [A]_o$	$\frac{0.693}{k}$
2	$\frac{1}{[A]_t} = kt + \frac{1}{[A]_o}$	$\frac{1}{k[A]_o}$

$[A]_o$ = initial concentration, $[A]_t$ = concentration of reactant A at a certain time.

21. Consider the following reaction:

Answer: B

$$2S_2O_3^{2-}(aq) + I_2(aq) \longrightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$$

If, in an experiment, 0.05 mol $S_2O_3^{2-}$ is consumed in 1.0L of solution each second, at what rates are $S_4O_6^{-2}$ and I^- produced in this solution?

A. $S_4O_6^{2-} = 0.025, I^- = 0.025$ C. $S_4O_6^{2-} = 0.025; I^- = 0.05$
B. $S_4O_6^{2-} = 0.05; I^- = 0.05$ D. $S_4O_6^{2-} = 0.05; I^- = 0.025$

Explanation
For the general hypothetical reaction
$aA + bB \longrightarrow cC + dD$ where a, b, c, d are coefficients.
The rate expression in terms of the concentration of each reactant and each product is related as:

$$\text{Rate of reaction} = \frac{\Delta[A]}{a\Delta t} = \frac{-\Delta[A]}{b\Delta t} = \frac{\Delta[C]}{c\Delta t} = \frac{\Delta[D]}{d\Delta t}$$

For the given reaction

$$2S_2O_2^{2-}(aq) + I_2(aq) \longrightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$$

$$\text{Rate} = \frac{-\Delta[S_2O_3^{2-}]}{2\Delta t} = \frac{-\Delta[I_2]}{\Delta t} = \frac{\Delta[S_4O_6^{2-}]}{\Delta t} = \frac{\Delta[I^-]}{2\Delta t}$$

rate of $S_2O_3^{2-} = 0.05\ mol.L^{-1}.s^{-1}$

$$-\frac{1}{2}(0.05) = \frac{\Delta[S_4O_6^{2-}]}{\Delta t}$$

Telegram: https://t.me/ExtremeSeries

$\therefore$ rate of $S_4O_6^{2-} = \frac{\Delta[S_4O_6^{2-}]}{\Delta t} = 0.025$

rate of $I^- = \frac{\Delta[I^-]}{2\Delta t} = \frac{2}{2}(0.05) = 0.05$ **Answer: C**

22. The reaction between NO and I_2 is second order in NO and first order in I_2. What change occurs in the rate of the reaction if the concentration of NO is doubled and I_2 left unchanged?

A. double C. eight times
B. quadruple D. three times

Explanation

If the concentration of a given reactants is increased by a factor x, the rate increases by a factor x^a where "a" is the order of the reaction with respect to that reactant rate = $k[NO]^2[I]$, so if the concentration NO is doubled, the rate becomes quadrupled.

Answer: B

23. A reaction is 50% complete in 2 hours and 75% complete in 4 hours. What is the ratio of this reaction?

A. 0 B. 1 C. 2 D. 3

Explanation

At t = 2hr, 50% of the reaction is complete implies $50\%[A]_o$. At t = 4 hr, 75% of the reaction is complete shows if 75% of the starting material is reacted, the amount left is (100% − 75%) which is 25% i.e., $\frac{25}{100}[A]_o$. Taking the two values ratio

$$\frac{\frac{5}{100}[A]_o}{\frac{25}{100}[A]_o} = \frac{50}{100} \times \frac{100}{25} = 2$$

Answer: C

24. Consider the following reaction:

$$HOCl + H_2O \rightleftharpoons H_3O^+ + OCl^-$$

What will happen if NaOCl is added to the above reaction at equilibrium?

A. The concentration of both HOCl and H_3O^+ would increase.
B. The concentrations of both HOCl and H_3O^+ would decrease.

C. The concentration of HOCl would increase and the concentration of H_3O^+ would decrease.
D. The concentration of HOCl would decrease and the concentration of H_3O^+ would increase.

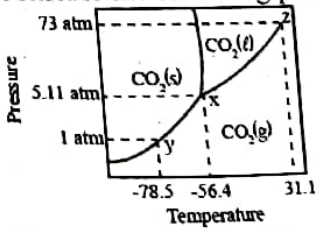
Explanation

Lechatelier's principle states that, "If an external stree is applied to a system at equilibrium the system will adjust itself in such away that it reduces the stress to reach a new state of equilibrium." The work stress means a change in temperature, pressure and concentration.

- Increasing concentration of the reactant shifts the equilibrium to be the right (product).
- Pressure affects only equilibrium system that involve gases.
- Increasing pressure affects the equilibrium to shift towards the least number of moles.
- A decrease in pressure affects the equilibrium to shift to greater no of gaseous molecules.
- Increasing pressure favors the endothermic reaction.
- Decreasing pressure favors the exothermic reaction.

$HOCl + H_2O \rightleftharpoons H_3O^+ + OCl^-$, hence adding NaOCl in this reaction increases the concentration of OCl^- because $NaOCl + H_2O \rightleftharpoons Na^+ + OCl^-$. Therefore, according to Lechatlier's principle if one of the product concentration increases, the reaction shifts to the reactant. **Answer: C**

25. Consider the following phase diagram for CO_2.



What happens in a CO_2 sample intialling at 1 atom and -70°C, the temperature increases from -70°C to −10°C at a constant pressure of 60 atm?

A. $CO_2(g) \longrightarrow CO_2(s)$ C. $CO_2(g) \longrightarrow CO_2(g)$
B. $CO_2(s) \longrightarrow CO_2(\ell)$ D. $CO_2(g) \longrightarrow CO_2(\ell)$

Telegram: https://t.me/Extremeseries

Explanation

- A phase diagram is defined as a graphical representation the condition when two or more phases (such as vapor-liquid, liquid-solid) of a system can exist in equilibrium.
- A substance exists in solid state at low temperature and high pressure.
- A substance exists in vapor phase at high temperature and low pressures.
- The remaining area between two regions represents liquid phase.

Hence keeping the pressure constant (p = 60 atom) and increasing temperature from –70°C to –10°C changes (CO_2 (s) to CO_2 (ℓ)).

Answer: B

26. The dye bromothymol blue (HBb) is a weak acid whose ionization can be represented as follows:

$$HBb\ (aq) \rightleftharpoons H^+\ (aq) + Bb^-\ (aq)$$

Which way will the equilibrium shift when NaOH is added?

A. to the left
B. to the right
C. initially to the left and after a while to the right
D. initially to the right and after a while to the left

Explanation

The NaOH will dissociate as follows:

$$NaOH \longrightarrow Na^+\ (aq) + OH^-\ (aq)$$

Hence, the OH^- obtained by the dissociation of NaOH reacts with H^+ obtained by HBb as a result its concentration decreases. As a result, decreasing one of the product concentration favors the reaction to move towards the product.

Answer: B

27. Consider the following equilibrium:

$$CaCO_3\ (s) \rightleftharpoons CaO\ (s) + CO_2\ (g)$$

Which of the following mixtures, each placed in a closed container and allowed to stand is not capable of reaching the equilibrium given above?

A. pure $CaCO_3$
B. $CaCO_3$ and CaO

C. some CaO and a pressure of CO_2 greater than the value of K_P
D. some $CaCO_3$ and a pressure of CO_2 than the value of k_P

28. If the following reaction is at equilibrium, which one of the following changes will shift the equilibrium to the left?

$$N_2 + 3H_2 \longleftrightarrow 2NH_3 + Heat$$

A. increasing pressure
B. decreasing temperature
C. adding more N_2 and H_2
D. increasing volume of the reaction container

Explanation

Factors affecting chemical equilibrium are change in concentration, temperature and pressure.

Equilibrium factor	Position of an equilibrium
• increasing reactant concentration	• shifts the equilibrium to the left (product)
• decreasing reactant concentration	• shifts the equilibrium to the right (reactants)
• increasing pressure	• shifts the equilibrium to the less no of gas molecules
• decreasing pressure	• shifts the equilibrium to the higher no of gas molecules
• increasing temperature	• shifts the rxn towards endothermic side
• decreasing temperature	• shifts the equilibrium towards the exothermic side
• increasing product concentration	• shifts the equilibrium to the reactant
• decreasing product concentration	• shifts the equilibrium to the product
• increasing volume	• shifts the rxn to higher no of gas molecules
• decreasing volume	• shifts the rxn to the less no of gas molecules

Answer: D

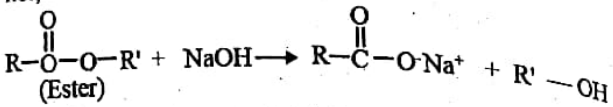
29. What is the name of a base-promoted ester hydrolysis reaction?

A. Acylation
B. Esterification
C. condensation
D. saponification

Telegram: https://t.me/ExtremeSeries

Explanation

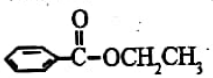
- Condensation is a reaction in which two molecule combine to form a larger one and a small molecule is eliminated.
- Esterification is a kind of condensation reaction in which an ester is formed from carboxylic acid and alcohol. i.e., carboxylic acid of alcohol → ester + water
- Saponificatioin is a base catalyzed ester hydrolysis. i.e.,



- Acylation is a kind of substitution in which aromatic rings are substituted with acyl group (RCO–).

30. What is the following compound?

Answer: D



A. Benzoate ester
B. Ethyl benzoate
C. Phenyl butyrate
D. Ethyl benzyl ketone

Explanation:

The esters are named by converting acid to oate and alcohol portion to alkyl group. The name for the given ester will be Ethylbenzoate.

Answer: B

31. Which of the following is a chemical formula that represents an amino acid?

A. CH_4
B. CH_3NH_3
C. CH_3COOH
D. NH_2CH_2COOH

Explanation

Amino acids are basic units of proteins.

- The general formula of amino acids is:

- Amino acids have both carboxylic acid (–COOH) part and basic part ($–NH_2$).

Answer: D

32. Which of the following statements is true about esters?

A. Esters can form intermolecular hydrogen bonds.
B. Ester molecules can form intermolecular hydrogen bonds.

C. Ester molecules cannot form intermolecular hydrogen bonds.
D. Esters have higher boiling points than alcohols of comparable molecular weight.

Answer: C

33. The organic compound $CH_3C(O)CH_3$ is:

A. aldehyde
B. ester
C. carbonyl
D. ketone

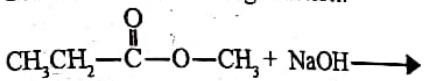
Explanation:

The general formula for ketone is:

$CH_3C(O)CH_3 = CH_3—C—CH_3$ - propanone

Answer: D

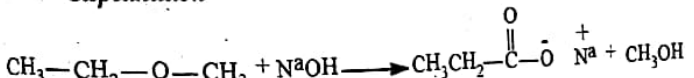
34. Consider the following reaction:



What are the products of this reaction?

A. Sodium acetate and ethanol.
B. Sodium acetate and methanol.
C. Sodium propionate and methanol.
D. Methyl prop ionic acid and methanol.

Explanation



Answer: C

35. Which catalyst is used in the hydrogenation of vegetable oils?

A. Iron
B. Nickel
C. Platnium
D. Molybednum

Explanation

- Hydrogenation is addition of hydrogen.
- The catalytic hydrogenation of oils (liquids) gives fats (solids). The process of converting oils or soft fats to hard fats is called hardening. The catalyst used is nickel.

Answer: B

Telegram: https://t.me/ExtremeSeries

36. Which of the following is **NOT** a solution?
A. Milk C. Whisky
B. Brass D. Coca Cola drink

Explanation Milk: Heterogeneous mixture
Brass: Homogeneous mixture and it is an alloy.
Whisky: Homogeneous mixture
Coca Cola: Homogeneous mixture
A solution is a homogeneous mixture of solutes and solvents.

Answer: A

37. How many ml of water is required to dilute 50 ml of 3.5 M H_2SO_4 to 2.00 M H_2SO_4?
A. 37.5 B. 45 C. 75 D. 87.5

Solution: Applying dilution formula.
Number of moles before solution = Number of moles after dilution $(\text{Molarity} \times \text{Volume})_{\text{before dilution}} = (\text{Molarity} \times \text{volume})_{\text{after dilution}}$

$$M_1V_1 = M_2V_2$$

Given: $M_1 = 3.5\ M$ $V_1 = 50\ ml$
$M_2 = 2.00\ M$ $V_2 = ?$
$M_1V_1 = M_2V_2$

V_{H_2O} added $= V_2 - V_1 = 87.50 - 50 = 37.5\ mL$

$$V_2 = \frac{M_1V_1}{M_2} = \frac{3.5M \times 50ml}{2M} = 87.5ml$$

Answer: A

38. How much water has to be evaporated from 250 ml of 1 M $Ca(OH)_2$ to make it 3M?
A. 100 ml B. 150 ml C. 167 ml D. 200 ml

Solution: Applying dilution formula
$M_1V_1 = M_2V_2$
Given:

$V_1 = 250\ ml$ $V_2 = \frac{M_1V_1}{M_2} = \frac{250}{3} \times 1 = 83.3mL$

$M_1 = 1\ M$
$M_2 = 3\ M$
$V_2 = ?$

The amount of H_2O to be evaporated is: 250 ml – 83.3mL = 166.7 ml.

Answer: C

39. The solubility of sodium selenate, Na_2SiO_4, is 84 g/ 100 g of water at 35°C. If a solution is obtained by dissolving 92 gram of Na_2SiO_4 in 200 gram of water at 35°C, what do you call this solution?
A. Dilute C. Unsaturated
B. Saturated D. Supersaturated

Explanation
Dilute: the amount of solvent is higher than the amount of solute.
Saturated solution: a solution in which dissolved and undissolved solute are in dynamic equilibrium.

undissolved solid $\underset{\text{crystallization}}{\overset{\text{dissolution}}{\rightleftharpoons}}$ dissolved ions

Rate of crystallization = Rate of dissolution
Supersaturated solution: A solution containing more solute than a saturated solution.
Solubility is the quantity of a substance dissolved in a given quantity (volume, mass) of the solvent at a given temperature to form a saturated solution. Having all these things, at 35°C 84 grams of Na_2SiO_4. Hence at the same temperature $2 \times 84 = 168$ gram of Na_2SiO_4 should be dissolved in 200 gram of water to form saturated solution. Taking amount of Na_2SiO_4 less than 168 gram in 200 gram of H_2O makes the solution unsaturated.

Answer: C

40. Which of the following is most likely deviate from ideal gas behavior?
A. He B. Ar C. Cl_2 D. CCl_2F_2

Explanation
- The hypothetical gas that obeys the ideal gas law exactly at all temperature and pressures is called ideal gas.
- All gases are found to obey the law of an ideal gas more and more closely as the reduced to very low values.
- Real gases don't behave exactly as predicted by the ideal gas law. This deviation arises from intermolecular force. And this force is important at high temperature.
- Relatively speaking CCl_2F_2 will have a dipole-dipole force.

Answer: D

Telegram: https://t.me/ExtremeSeries

41. Which of the following compounds would give the lowest freezing point depression when 100 g of each are dissolved in 1 kg of water (k_f for water = 1.86°C/m)? Assume complete dissociation.

A. NaCl
B. NH_4NO_3
C. $(NH_4)_2SO_4$
D. glucose, $C_6H_{12}O_6$

Explanation

Compound	Theoretical Vant Hoff factor
NaCl	2
NH_4NO_3	2
$(NH_4)_2SO_4$	3
Glucose, $C_6H_{12}O_6$	1

Note: The theoretical Van't Hoff's factor is calculated by summing up he cations and anions.

$\boxed{T_f = T°_f - ik\,fm}$ ΔT_f – freezing point depression

$T°_f$ – freezing point of pure solvent

m – molality

k_f – freezing point depression constant

i – vant hoff's factor

$$\text{molality} = \frac{\text{no of moles}}{\text{mass of solvent in kg}}$$

For NaCl:

Number of moles of NaCl

$$= \frac{\text{mass}}{\text{molar mass}} = \frac{100 \text{ gram}}{58.5 \text{ gram/mol}} = 1.7 \text{ mol}$$

$$\text{molality} = \frac{\text{number of moles}}{\text{mass of solvent (kg)}} = \frac{1.7 \text{ mol}}{1 \text{ kg}} = 1.7 \text{ mol}$$

$T_f = T°_f - ik_fm = 0°C - (2)(1.86)(1.7) = -6.324°C$

For NH_4NO_3:

$$\text{Number of moles} = \frac{\text{mass}}{\text{Molar mass}} = \frac{100 \text{ gram}}{80 \text{ g/mol}} = 1.25 \text{ mol}$$

$$\text{molality} = \frac{\text{number of moles}}{\text{mass of the solvent in kg}} \frac{1.25}{1} = 1.25 \text{ molal}$$

$T_f = T°_f - ik_fm = 0°C - (2)(1.84)(0.76) = -4.6°C$

For $(NH_4)_2SO_4$:

$$\text{Number of moles} = \frac{\text{mass}}{\text{molar mass}} = \frac{100 \text{ gram}}{132 \text{ gram/mol}} = 0.76 \text{ moles}$$

$$\text{molality} = \frac{\text{no of moles}}{\text{mass of the solvent (kg)}^2} = \frac{0.76}{1} = 0.76 \text{ molal}$$

$\Delta T_f = T°_f - ikfm = 0°C - 3(1.86)(0.76) = -4.22°C$

For glucose, $C_6H_{12}O_6$:

$$\text{Number of moles} = \frac{100}{180} = 0.56$$

$$\text{Molality} = \frac{\text{number of moles}}{\text{mass of solvent (kg)}} = \frac{0.56}{1} = 0.56$$

$T_f = T°_f - ik_fm = 0\,°C - (1)(1.86)(0.56) = -1.03°C$

Answer: A

42. Consider the following balanced equation.

$3Ba(NO_3)_2(aq) + Fe_2(SO_4)_3(aq) \longrightarrow 3BaSO_4(s) + 2Fe(NO_3)_3(aq)$

the net ionic equation to describe this balance equation is?

A. $3Ba^+(aq) + SO_4^{2-}(aq) \longrightarrow 3BaSO_4(s)$

B. $3Ba^{2+}(aq) + 3SO_4^{2-}(aq) \longrightarrow 3BaSO_4(s)$

C. $6NO_3^-(aq) + 2Fe^{2+}(aq) \longrightarrow 2Fe(NO_3)_3(aq)$

D. $3Ba^{2+}(aq) + 2NO_3^-(aq) + 2Fe^{3+}(aq) + 3SO_4^{2-}(aq) \longrightarrow 3BaSO_4(s) + 2Fe^{3+}(aq) + 6NO_3^-(aq)$

Explanation

- By looking at the state of the reactant and product tells whether the compound is written in the form of ion or not.
- If the compound exists in aqueous (aq) form, write that compound in the form of ion otherwise leave it as it is (i.e., in its molecular form). For the reaction given below:

$3Ba(NO_3)_2(aq) + Fe_2(SO_4)_3(aq) \longrightarrow 3BaSO_4(s) + 2Fe(NO_3)_3(aq)$ the ionic equation becomes

$3Ba^{2+}(aq) + 6NO_3^-(aq) + 2Fe^{2+}(aq) + 3SO_4^{2-}(aq) \longrightarrow 3BaSO_4(s) + 2Fe^{3+}(aq) + 6NO_3^-(aq)$.

Telegram: https://t.me/ExtremeSeries

And to find the net ionic equation, the spectator ions should be cancelled. The net ionic equation is:

$3Ba^{2+}(aq) + 3SO_4^{2-}(aq) \longrightarrow 3BaSO_4(s)$

Note: The spectator ions are ions that exists at the left and right side of the reaction. For the above reaction the spectator ions are NO_3^-, and Fe^{3+}.

Answer: B

43. Which one of the following organic molecules has the highest water solubility?

A. $HOCH_2CH_2CH_2OH$
B. $CH_3CH_2CH_2CH_2OH$
C. $HOCH_2CH_2CH_2CH_2OH$
D. $CH_3CH_2CH_2CH_2OH$

Explanation

- All the choices given are alcohols. The solubility of alcohols can be affected by:

A. Number of OH (hydroxyl) groups.
- Solubility of alcohol increases with increasing number of hydroxyl groups.

B. Carbon number
- Solubility of alcohols decreases with increasing carbon number.

Answer: A

44. A solution is labeled 0.500 M HCl. What is $[H^+]$?

A. 0.5 B. 1.5 C. 1.6 D. 1.69

Explanation

Strong acids dissociates almost completely. Hence

$HCl(aq) \longrightarrow H^+(aq) + Cl^-(aq)$

0.5 M 0.5M 0.5 M

For strong mono protic acid, [Acid] = $[H^+]$ **Answer: A**

45. How is a buffer solution prepared?

A. By mixing a weak acid and a strong acid.
B. By mixing a weak acid and its conjugate base.
C. By mixing a strong acid and its conjugate base.
D. By mixing a strong base and its conjugate acid.

Explanation

- Buffer solution is a homogenous mixture that resists the pH change up on small addition of acids or bases.
- Buffer solution contains an acid and a base parts together that should not consume each other but react with externally added acid or base.

- Buffer solutions are made by conjugate acid-base pair. There are two kinds of buffer solutions.

A. Acidic buffer: A buffer solution made form a weak acid and its conjugate base. Example CH_3COOH/CH_3COO^-.

B. Basic buffer: A buffer solution made from a weak base and its conjugate acid. Example: NH_3/ NH_4^+

Answer: B

46. Which of the following compounds would be the most basic?

A. 0.1 M acetic acid
B. 0.1 M sodium acetate
C. 0.1 M hydrochloric acid
D. 0.1 M ammonium chloride

Explanation

- 0.1 M CH_3COOH is an acid with pH < 7
- 0.1 M CH_3COONa is a salt obtained from a weak acetic acid (CH_3COOH) and strong base is always basic (pH>7)
- 0.1 M HCl is an acid with pH < 7
- 0.1 M NH_4Cl is a salt drived from weak base ammonia (NH_3) and strong acid HCl. Hence, a salt obtained from weak base and strong acid is acidic solution i.e., pH < 7.

Answer: B

47. Which one of the following statement is **NOT** true about acids?

A. an acid is a proton donor
B. an acid is an electron pair donor
C. an acid is an electron pair acceptor
D. an acid is a substance that is ionized in water to produce H^+ (aq).

Explanation: According to:

1. Arrhenius Definition "An acid is a substance that is ionized in water to give H^+ (aq)"
2. Bronsted – Lowry definition "An acid is a proton donor"
3. Lewis "An acid is an electron pair acceptor"

Answer: B

48. Consider the following acids;

I. CH_3COOH, $Ka = 1.8 \times 10^{-5}$
II. HCO_2H, $Ka = 1.8 \times 10^{-4}$
III. HOBr, $Ka = 2.4 \times 10^{-9}$
IV. C_6H_5OH, $Ka = 1.0 \times 10^{-10}$

Telegram: https://t.me/ExtremeSeries

Which one of the following aqueous solutions will have the highest pH?

A. 0.1 M NaOBr
B. 0.1 M HCO_2Na
C. 0.1 M C_6H_5ONa
D. 0.1 M CH_3COONa

Solution:

- The Ka value of the acid is directly proportional to the strength of the acid. That is $\boxed{Ka \ \alpha \ \text{Acid strength}}$
- The pH value is inversely proportional to the strength of the acid.

That is, $\boxed{pH \ \alpha \ \frac{1}{\text{Acid strength}}}$

So, having highest pH value implies the weakest acid. And the weakest acid is the one having lowest Ka value:

(A) $NaOBr \longrightarrow Na^+ + OBr^-$
0.1M 0.1 M 0.1 M

Na^+ has no reaction with water.

$OBr^- + H_2O \rightleftharpoons HBr + OH^-$

Initial conc.	0.1	-	-
change	-x	+x	+x
equilibrium	0.1 – x	x	x

$$K_b = \frac{K_w}{K_a} = \frac{x^2}{0.1-x}$$

At 25°C $K_w = 1 \times 10^{-14}$

$$\frac{1\times10^{-14}}{2.4\times10^{-9}} = \frac{x^2}{0.1-x}$$, using approximation method.

$$\frac{x^2}{0.1-x} \cong \frac{x^2}{0.1}$$

$$\frac{1\times10^{-14}}{2.4\times10^{-9}} = \frac{x^2}{0.1}, \ x = 6.45 \times 10^{-4}$$

$[OH^-] = x = 6.45 \times 10^{-4}$
pOH = 3.91
pH = 14 – pOH = 10.8

(B) $HCOONa \longrightarrow HCOO^-(aq) + Na^+(aq)$

0.1 M 0.1 M 0.1 M

$HCOO^- + H_2O \rightleftharpoons HCOOH + OH^-$

initial conc.	0.1	-	-
change	-x	+x	+x
equil.	0.1 – x	x	x

$$Kb = \frac{Kw}{Ka} = \frac{x^2}{0.1-x} = \frac{1\times10^{-14}}{1.8\times10^{-4}} = \frac{x^2}{0.1}$$ . . . using approximation method

Solving for $x = 2.35 \times 10^{-6}$
$[OH^-] = x = 2.35 \times 10^{-6}$
$pOH = -\log(2.35 \times 10^{-6}) = 5.62$
pH = 8.37

(C) $C_6H_5ONa \longrightarrow C_6H_5O^-(aq) + Na^+(aq)$
0.1 M 0.1M 0.1 M

$C_6H_5O^- + H_2O \rightleftharpoons C_6H_5OH + OH^-$

initial conc.	0.1	-	-
change	-x	+x	+x
equil.	0.1 – x	x	-x

$$Kb = \frac{Kw}{Ka} = \frac{x^2}{0.1-x} = \frac{1\times10^{-14}}{1\times10^{-10}} = \frac{x^2}{0.1-x}$$

Using approximation and solving for x.
$x = 3.16 \times 10^{-3}$
$[OH^-] = x = 3.16 \times 10^{-3}$
pOH = 2.5, pH = 11.5

(D) 0.1 M CH_3COONa

$CH_3COONa \longrightarrow CH_3COO^-(aq) + Na^+$
0.1M 0.1M 0.1M

$CH_3COO^- + H_2O \rightleftharpoons CH_3COO^- + OH^-$

Initial conc.	0.1	-	-
change	-x	+x	+x
equil. conc.	0.1 – x	x	x

$$Kb = \frac{Kw}{Ka} = \frac{x^2}{0.1-x} = \frac{1\times10^{-14}}{1.8\times10^{-5}} = \frac{x^2}{0.1-x}$$

Telegram: https://t.me/ExtremeSeries

Using approximation and solving for x

$x = 7.45 \times 10^{-6}$

$[OH^-] = x = 7.45 \times 10^{-6}$

pOH = 5.12, pH = 8.87

Answer: C

49. Which of the following titrations will have an equivalence point at a pH < 7?

A. strong acid with weak base
B. weak acid with weak base
C. strong acid with strong base
D. weak acid with strong base

Explanation

Titration type	Equivalence point region
Strong acid – weak base	pH < 7
Weak acid – weak base	Difficult to predict
Strong acid – strong base	pH = 7
Weak acid – strong base	pH > 7

Answer: A

50. Which of the following is a conjugate acid/base pair?

A. HCl/OCl^- C. NH_4^+ / NH_3

B. H_3O^+/OH^- D. H_2SO_4 / SO_4^{2-}

Explanation

Conjugate acid: is Bronsted acid formed when the Bronsted base has accepted a proton. Example NH_4^+ is the conjugate acid of NH_3.

Conjugate base: The Bronsted base formed when a Bronsted acid has donated a proton. Example NH_3 is he conjugate base of NH_4^+. Therefore, the correct conjugate acid/base pair of the choices are: HCl/Cl^-, H_3O^+/H_2O, NH_4^+/NH_3, H_2SO_4/ HSO_4^-.

Answer: C

51. All of the following can act as Bronsted-Lowery bases except:

A. I^- B. NH_3 C. HCO_3^- D. H_2CO_3

Explanation: Bronsted-Lowery base is a proton acceptor. For a given species to be Bronsted-Lowery base, it should contain negative charge or should be compound that undergo self ionization (example H_2O, NH_3, CH_3COOH).

Answer: D

52. Consider the following oxidation – reduction equation.

$2H_2O\,(l) + Al\,(s) + MnO_4^-\,(aq) \longrightarrow Al(OH)_4^-\,(aq) + MnO_2\,(s)$

What are the reducing and oxidizing agents in this reaction?

A. Al (s) is the reducing agent and H_2O is the oxidizing agent.
B. Al (s) is oxidizing agent and H_2O is the reducing agent.
C. Al (s) is an oxidizing agent and MnO_4^- (aq) is the reducing agent.
D. Al (s) is the reducing agent and MnO_4^- (aq) is the oxidizing agent.

Explanation

- Oxidizing agent is a substance that causes oxidation (and itself is reduced in the redox reaction).
- Reducing agent is a substance that caused reduction (and is itself oxidized) in a redox reaction.
- Reduction is an electron gain.
- Oxidation is an electron loss.
- A substance that under goes reduction is said to be reduced.
- A substance that under goes oxidation is said to be oxidized.

In order to identify the oxidizing and reducing agents in a given redox reaction, assign the oxidation number values for the reactant and product based on the rue of assuming oxidation number.

$$\overset{+1\ -2}{2H_2O}(\ell) + \overset{0}{Al}(s) + \overset{+7}{MnO_4^-} \longrightarrow \overset{+3}{Al(OH)_4^-} + \overset{+4}{MnO_2}$$

Oxidation | Reducing agent | Reduction oxidizing agent

Answer: D

53. Consider the following reaction:

$I_2O_5\,(s) + 5CO\,(g) \longrightarrow I_2(s) + 5CO_2\,(g)$

What is the magnitude of the change in the oxidation number of the elements?

A. I, +5 to O, C, +2 to +4 C. I, +0 to 0, C, +2, to +4
B. I, +5 to 0, C, +4 to + 4 D. I, +4 to 0, C, +2 to +4

Explanation By applying the rule of assigning an oxidation number assign the oxidation number values for the reactants and products.

Telegram: https://t.me/ExtremeSeries

$$\overset{+5}{I_2}O(s) + 5\overset{+2}{C}O \longrightarrow I_2(s) + 5\overset{+4}{C}O_2$$

54. Which of the following is true? **Answer: A**
A. A battery is rechargeable.
B. A battery produces electricity spontaneously.
C. A battery has generally no liquid components.
D. A battery produces the same amount of electricity, regardless of composition.

Explanation: A battery is a galvanic cell or a series of combined galvanic cells that can be used as a direct current source. It produces electricity spontaneously. Based on rechargebility batteries can be classified as primary and secondary batteries.

55. All of the following when added to water will produce an electrolytic solution **EXCEPT**. **Answer: B**
A. N_2 (g) B. NaI (s) C. HCl D. KOH (s)

Explanation
Electrolytic solution: A solution that conducts electricity.
Electrolytes: A substance that dissolves to give ions (cations and anions) in a solution.
Electrolytic substance include Acids, Bases and Salts.

56. Which one of the following substance is a non-conductor of electricity? **Answer: A**
A. Graphite
B. $MgCl_2$ (s)
C. Silver
D. H_2SO_4 (aq)

Explanation
There are two kinds of conductivity:
(1) Metallic conduction: is conduction through metallic elements by free mobile electrons.
(2) Ionic conduction: is conduction of electricity by the mobility of cations and anions in the solution.
- Ionic compounds do not coduct electricity in soldi state. Salts, acids and bases are electrolytes.

Note: Carbon has two crystalline alltropes.
- Graphite: conducts electricity
- Diamond: do not conduct electricity

Answer: B

57. A solution of 25°C contains the metal ions Ni^{2+}, Pt^{2+}, and Pd^{2+}, all at 1.0M concentrations. Consider the following standard reduction potentials:

$Ni^{2+} + 2e^- \longrightarrow Ni$ $E^\circ = -0.23V$
$Pt^{2+} + 2e^- \longrightarrow Pt$ $E^\circ = 1.20V$
$Pd^{2+} + 2e^- \longrightarrow Pd$ $E^\circ = 0.99V$

Which metal (s) will plate out first when the solution is electrolyzed?
A. Ni B. Pd C. Pt D. Ni and Pd

Explanation
The more positive the reduction potential (E°_{red}), the greater tendency for the substance to be reduced.

58. How can silver be plated onto Nickel? **Answer: C**
A. Electricity to a nickel anode in a solution of silver ions.
B. Electricity to a silver anode in a solution of nickel ions.
C. A solution of nickel ions to react with a piece of silver.
D. Electricity to a nickel cathode in a solution of silver ions.

Explanation
One application of electrolysis is Electroplating. During electroplating one should keep the plating material at the anode and the material to be plated should be kept at cathode. Besides the electrolytic solution be a solution of the plating material ion.

Answer: D

59. For the reaction $A + B \longrightarrow C + D$, $\Delta H^\circ = +40$ kJ and $\Delta S^\circ = +50$J/k. Therefore, the reaction under standard condition is:
A. Spontaneous at all temperatures.
B. Non spontaneous at all temperatures.
C. Spontaneous at temperatures greater than 800k.
D. Spontaneous only at temperatures between 10k and 800k.

Explanation
Finding the Gibbs free energy
$\Delta G = \Delta H - T\Delta S$
$\Delta G = 40 \times 10^3$ J $-$ T(50 J/k), which implies the value of ΔG depends on the value of temperature.
1. If $\Delta H > T\Delta S$, $\Delta G > 0$ and the process will be nonspontaneous.

Telegram: https://t.me/ExtremeSeries

2. At a certain temperature when $\Delta H = T\Delta S$, ΔG becomes zero hence the process is at equilibrium.
3. At a temperature when $\Delta H = T\Delta S$, ΔG becomes negative so the process is spontaneous.

When T = 800 k

$\Delta G = \Delta H - T\Delta S = (40 \times 10^3\ J) - 800k\ (50\ J/k) = 0$ (equilibrium)

When T= 10 k

$\Delta G = 39500$ (nonspontaneous)

∴ The process becomes spontaneous at temperature > 800k.

Answer: C

60. For the conversion of C (diamond) ⟶ C (Gra phite), $\Delta H = -3kJ$. What does this mean?
   A. Both are equally stable.
   B. Graphite is more stable than diamond.
   C. Diamond is more stable than graphite:
   D. Graphite has more stable energy than diamond.

Explanation

The conversion of diamond to graphite is spontaneous, this implies graphite is more stable than diamond.

Answer: B

61. What is the change in internal energy, ΔE of a system if it absorbs 300 kJ heat from the surroundings and does 500 kJ of work on the surroundings?
   A. 100kJ B. −200kJ C. 400kJ D. 500kJ

Solution:

q = +300 kJ (the system absorbs heat energy)

w = −500 kJ (the system does work on the surrounding)

$\Delta E = q + w = 300\ kJ - 500\ kJ = -200\ kJ$ **Answer: B**

62. Which of the following element has the highest melting point?
   A. Iodine C. Mercury
   B. Tungsten D. Bromine

Explanation: Metals in general have high melting and boiling points. Exceptionally, tungsten has the highest melting point above 3000°C.

Answer: B

63. Which of the following compounds is unlikely to contain ionic bonds?
   A. CO B. NaF C. LiCl D. $MgBr_2$

Explanation:

Ionic bond is formed by complete transfer of one or more electrons. It is formed between metals and non – metals and between ammonium ion (NH_4^+) and anions.

Answer: A

64. What is the number of moles of atoms and the number of atoms in a 10.0 g sample of copper?
   A. 0.08 mol Cu atoms and 2.16×10^{23} atoms
   B. 0.16 mol Cu atoms and 9.63×10^{22} atoms
   C. 0.16 mol Cu atoms and 9.63×10^{23} atoms
   D. 0.31 mol Cu atoms and 4.16×10^{23} atoms

Solution:

$$n = \frac{m}{M} = \frac{10\ g}{64\ g/mol} = 0.16\ mol\ Cu\ atoms$$

$1\ mol\ Cu = 6.02 \times 10^{23}$ atoms

$0.16\ mol\ Cu = ?$

$$\Rightarrow ? = \frac{0.16\ mol\ Cu \times 6.02 \times 10^{23}\ atoms}{1\ mol\ Cu}$$

$? = 9.63 \times 10^{22}$ atoms

$\Rightarrow 10\ g\ Cu = 0.16\ mol\ Cu = 9.63 \times 10^{22}$ atoms of Cu

Answer: B

65. Which of the following statements correctly describes a chemical reaction at equilibrium?
   A. The concentrations of the products and reactants are equal.
   B. The change in the concentrations of the products and reactants is constant.
   C. The rate of the forward reaction is less than the rate of the reverse reaction.
   D. The rate of the forward reaction is greater than the rate of the reverse reaction.

Explanation:

At equilibrium:

- Rate of forward reaction is equal to rate of reverse reaction.
- The change in the concentrations of the products and reactants is constant.

Answer: B

Telegram: https://t.me/ExtremeSeries

66. Under what conditions will the enthalpy change of a process equals the amount of heat transferred into or out of the system?
A. Under constant volume.
B. Under constant pressure.
C. Under constant pressure and volume.
D. Under constant pressure and temperature.

Explanation:
Enthalpy is defined as the sum of the internal energy and the pressure – volume product of a system.
$H = E + PV$
$\Delta H = \Delta E + P\Delta V$ (at constant pressure)
But $\Delta E = q - P\Delta V$
$\Rightarrow \Delta H = (q - P\Delta V) + P\Delta V$
$\Delta H = q_P$
q_P – heat transferred at constant pressure.
Enthalpy change is heat transferred at constant pressure. However, change in internal energy is heat transferred at constant volume.

Answer: B

67. What is the hydroxide ion concentration for a solution with a pH of 10 at 25°C?
A. 10^{-14}M B. 10^{-10}M C. 10^{-7}M D. 10^{-4}M

Solution:
$pH = 14 - \log [OH^-]$
$pH - 14 = \log [OH^-]$
$[OH^-] = 10^{pH-14} = 10^{10-14} = 10^{-4}$M

Answer: D

68. Which law relates the concentration of a dissolved gas, C_g, to its partial pressure?
A. Henry's law C. Boyle's law
B. Raoul's law D. Ideal gas law

Explanation:
The quantitative relationship between gas solubility (concentration) and pressure was given by Henry's law. This law states that at a given temperature the solubility of a gas in a liquid is directly proportional to the partial pressure of the gas over the solution.
$C_g = kH.P_g$
C_g – molar concentration of the gas in mol/L

K_H – Henry's law constant in $\frac{mol}{atm.L}$
P_g – partial pressure of the gas

Answer: A

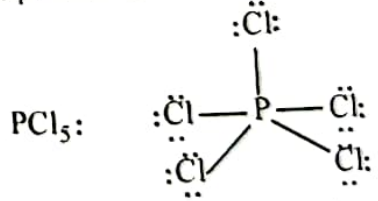
69. How many atoms are present in 22g of CO_2?
A. 3.01×10^{23} C. $2 \times 6.02 \times 10^{23}$
B. 6.02×10^{23} D. $1.5 \times 6.02 \times 10^{23}$

Solution:
$n = \frac{m}{M} = \frac{22g}{44g/mol} = 0.5$ mol
Number of atoms in 22 g of CO_2 = n_{CO2} = (subscript of C + subscript of O) $\times 6.02 \times 10^{23}$ atoms = 0.5 mol $\times$ (1 + 2) $\times 6.02 \times 10^{23}$ atoms = $1.5 \times 6.02 \times 10^{23}$ atoms = 9.03×10^{23} atoms

Answer: D

70. What is the hybridization of phosphorous atom in PCl_5?
A. sp^3d B. sp^3d^2 C. sp^3 D. sp^2

Explanation:
Draw the Lewis structure for PCl_5. Number of hybrid orbitals equal to number of sets of electrons on the central atoms.

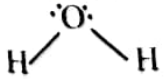


$\Rightarrow$There are five sets of electrons on P. Therefore, the hybridization of P in PCl_5 will be sp^3d.

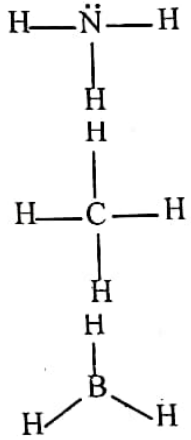
Answer: A

71. Which of the following species has the smallest H – X – H bond angle where X is the central atom?
A. OH_2 B. NH_3 C. CH_4 D. BH_3

Explanation:

Lewis structure	Shape	Bond angle
	Bent (V – shape)	104.5°

Telegram: https://t.me/ExtremeSeries



Trgonal pyramid 107°

Tetrahedral 109°

Trigonal planar 120°

Answer: A

72. The reaction, $2X + Y \longrightarrow Z$, was studied and the following data were obtained.

Expt	[X]	[Y]	Rate (mol. L^{-1}, S^{-1})
1	3.0	3.0	1.8
2	3.0	1.5	0.45
3	1.5	1.5	0.45

What is the proper rate expression?

A. rate = k [X] C. rate = K $[Y]^2$

B. rate = K [Y] D. rate = K [X][Y]

Solution:

Let the rate law be: rate = k $[X]^a[Y]^b$. Find a by using expt 2 and 3 and b by using expt 1 and 2.

$$\frac{rate_2}{rate_3} = \frac{k[x]_2^a[y]_2^b}{k[x]_3^a[y]_3^b}$$

$$\frac{0.45}{0.45} = \frac{(3.0)^a(1.5)^b}{(1.5)^a(1.5)^b}$$

$1 = 2^a$

$2^0 = 2^a$, $a = 0$

$$\frac{rate_2}{rate_3} = \frac{[y]_1^b}{[y]_2^b}$$

$$\frac{1.8}{0.45} = \left(\frac{3.0}{1.5}\right)^b$$

$4 = 2^b$

$2^2 = 2^b$

$b = 2$

$\Rightarrow$ rate = k $[x]^0[y]^2$ = k $[y]^2$

Answer: C

73. What mass of aluminum is produced in one hour by the electrolysis of molten $AlCl_3$ with a current of 10A?

A. 1.5g B. 2.5g C. 3.4g D. 4.3g

Solution:

$Al^{3+} + 3e^- \longrightarrow Al \Rightarrow Z = 3$

$M_{Al} = 27$ g/mol

I = 10A

t = 1 hr = 3600 s

$$\frac{m}{M} = \frac{It}{ZF} = 3600s$$

$$m = \frac{10A \times 3600s \times 27g/mol}{3 \times 96500C/mol} = 3.36\ g$$

Answer: C

74. Suppose reactions A $\longrightarrow$ B and B $\longrightarrow$ A are both elementary of $8 \times 10^2\ s^{-1}$ and $4 \times 10^4\ s^{-1}$, respectively. What is the value of the equilibrium constant for the equilibrium,

$A \rightleftharpoons B$?

A. 2×10^{-2} C. 4×10^{2}

B. 0.5×10^{2} D. 4×10^{-2}

Solution: $A \longrightarrow B$, $r_f = k_f[A]$

$B \longrightarrow A$, $r_r = k_r[B]$

$A \rightleftharpoons B$, At equilibrium, $r_f = r_r \Rightarrow k_f[A] = k_r[B]$

$$\frac{k_f}{k_r} = \frac{[B]}{[A]} = K_C$$

$$K_C = \frac{k_f}{k_r} = \frac{8 \times 10^2 s^{-1}}{4 \times 10^4 s^{-1}} = 2 \times 10^{-2}$$

Answer: A

Telegram: https://t.me/ExtremeSeries

75. Which of the following ion is the most abundant in sea water?
A. Na^+ B. Ca^{2+} C. Cl^- D. HCO_3^-

Explanation:
The most abundant ion in sea water is the sodium ion because the water is salty due to the presence of excess sodium chloride.

Answer: A

76. Which element is added to natural rubber to make it harder and reduce its susceptibility to oxidation and chemical attackers?
A. Sulfur C. Carbon
B. Silicon D. Nitrogen

Explanation:
The properties of natural rubber are improved by heating it with sulphur. This process is called vulcanization.

Answer: A

77. Which of the following is NOT a carbohydrate?
A. Starch C. Glycine
B. Glucose D. Cellulose

Explanation:
Starch – is a poly saccharide
Glucose – is a mono saccharide
Glycine – is an amino acid
Cellulose – is a polysaccharide

Answer: C

78. Why does a pressure cooker reduce cooking time?
A. The heat is more evenly distributed.
B. A stronger flame is used for cooking.
C. The high pressure tenderizes the food.
D. The boiling point of water inside the cooker is increased.

Answer: B

79. The pH at room temperature of a 0.1M solution of formic acid ($HCHO_2$) was measured to be 4.50. What is the hydrogen ion concentration?
A. 3.16×10^{-5}M C. 6.32×10^{-3}M
B. 3.16×10^{-12}M D. 6.32×10^{-4}M

Solution:
$pH = -\log [H^+]$
$[H^+] = 10^{-pH} = 10^{-4.5} = 3.16 \times 10^{-5}$M

Answer: A

80. What is the molarity of a solution containing 10 g o sulfuric acid in 500 mL of solution?
A. 0.02 B. 0.03 C. 0.12 D. 0.2

Solution:

$$M = \frac{m_{H_2SO_4}}{M_{H_2SO_4}}$$

$$M = \frac{10g}{98g/mol \times 0.5L} = 0.2M$$

Answer: D

Telegram: https://t.me/ExtremeSeries

1. Which of the following is NOT a basic SI unit?
   A. Candela C. Mole
   B. Gram D. Second

Explanation:
There are seven basic quantities with seven basic units.

Basic quantity	Basic SI units
Temperature	Kelvin
Mass	Kilogram
Time	Second
Length	Meter
Current	Ampere
Amount of substance	Mole
Luminuous intensity	Candela

Answer: B

2. What is the number of significant figures in 0.0030050?
   A. 4 B. 5 C. 7 D. 8

Explanation: Zeros before non-zero digits are not significant. But between and after non – zero digits, they are significant. As a result, the number of significant figures in 0.0030050 are 5.

Answer: B

3. Which of the following properties of a substance does NOT represent an intensive physical property?
   A. Boiling point C. Density
   B. Color D. Volume

Explanation: Intensive properties are properties which do not depend on the amount of matter present. Example: color, melting point, boiling point, density, hardness, conductivity, odour, specific gravity, freezing point, etc. Extensive physical properties are properties that depend on the amount of substance.
Example: Volume, length thickness, area, weight of substance, diameter etc.

Answer: D

4. What is the kelvin scale (K) corresponding to the temperature readings when the degree Celsius (°C) is identical to the degree Fahrenheit (°F)?
   A. 0°K B. 37K C. 233K D. 273K

Solution: $^{\circ}C = \frac{5}{9}(^{\circ}F - 32)$ $\quad \frac{9}{5}{}^{\circ}C - {}^{\circ}C = -32$

$^{\circ}F = \frac{9}{5}\cdot{}^{\circ}C + 32$ $\quad \frac{4}{5}{}^{\circ}C = 32$

If $^{\circ}C = {}^{\circ}F$ $\quad {}^{\circ}C = -32 \times \frac{5}{4}$

$\Rightarrow {}^{\circ}C = \frac{5}{9}(^{\circ}C - 32)$ $\quad {}^{\circ}C = -40^{\circ}C$

$\frac{9}{5}{}^{\circ}C = {}^{\circ}C - 32$

Therefore, °C = °F at -40
K = °C + 273 = -40 + 273 = 233°K

Answer: C

5. If a piece of aluminum (Al) foil measuring 24 cm by 31 cm has a mass of 10.35 g, (density of Al = 2.70 g/cm^3). What is the thickness of the foil in millimeters?
   A. 5.15×10^{-3} C. 3.833
   B. 5.15×10^{-2} D. 744

Solution:

$\rho = \frac{m}{v}$ $\quad$ ρ – density; m – mass; v – volume

$v = \frac{m}{\rho} = \frac{10.35g}{2.70g/cm^3} = 3.833cm^3$

v = length × width × thickness

$\text{thickness} = \frac{3.833cm^3}{24cm \times 31cm} = 5.15 \times 10^{-3} cm$

1 cm = 10 mm
5.15×10^{-3} cm =?

$? = \frac{5.15 \times 10^{-3} cm \times 10mm}{1\ cm} = 5.15 \times 10^{-2} mm$

⇒ Thickness of the Al foil = 5.15×10^{-2} mm

Answer: B

6. What sizes of particles and velocities can one consider quantum effect?
   A. Particles with very large mass and large velocities.
   B. Particles with large mass and large velocities.
   C. Particles with small mass and large velocities.
   D. Particles with small mass and small velocities.

Telegram: https://t.me/ExtremeSeries

Explanation: The size of particles and velocities of the particles are inversely related.

$v = \frac{h}{m\lambda}$ v – velocity of a particle

λ – wave length
m – mass of the particle
h – plank's constant

Answer: C

The quantum numbers listed below are meant for four different electrons in an atom.

I. $n = 4, \ell = 0, m_\ell = 0, m_s = +½$

II. $n = 3, \ell = 1, m_\ell = 1, m_s = +½$

III. $n = 4, \ell = 2, m_\ell = 0, m_s = +½$

IV. $n = 4, \ell = 1, m_\ell = 0, m_s = -½$

When these sets of quantum numbers are arranged in order to increasing energy, one may set:

A. I < III < II < IV C. II < I < III < IV
B. I < II < III < IV D. IV < III < II < I

Explanation: The atomic sub levels or sub shell are given by the values of n and ℓ.

Sub shell	Value of n	Value of ℓ	Value of m_ℓ	Value of m_s
1s	1	0	0	± ½
2s	2	0	0	± ½
2p	3	1	-1, 0, 1	± ½
3s	3	0	0	± ½
3p	4	1	-1, 0, 1	± ½
4s	3	0	0	± ½
3d	4	2	-2, -1, 0, 1, 2	± ½
4p	5	1	-1, 0, 1	± ½
5s	4	0	0	± ½
4d	5	2	-2, -1, 0, 1, 2	± ½
5p	6	1	-1, 0, 1	± ½
6s	6	0	0	± ½
4f	4	3	-3, -1, -1, 0, 1, 2, 3	± ½
.	.	.	.	.
.	.	.	.	.
.	.	.	.	.

⇒ I. 4s, II. 3p (one of the three p – orbitals)
III. 4d (one of the five d – orbitals)
Iv. 4p (one of the three p – orbitals)
⇒ In increasing order of energy, according to building up principle:
3p < 4s < 4p < 4d i.e., II < I < IV < III **Answer: ___**

(8) What is the wave length associated with an electron of mass, $m = 9.11 \times 10^{-28}$ g, travelling at 40% of the velocity of light?

A. 6.06×10^{-13} m C. 6.06×10^{-12} m
B. 2.42×10^{-13} m D. 2.42×10^{-11} m

Solution

$V_e = \frac{h}{M_e h}$, but, $V_e = 40\%C$

$\lambda = \frac{h}{M_e v_e}$ $V_e = \frac{40}{100} \times 3 \times 10^8 \text{ m/s}$

M_e – mass of an electron $V_e = 1.2 \times 10^8$ m/s
C – speed of light
V_e – velocity of an electron

$$\Rightarrow \lambda = \frac{6.626 \times 10^{-34} \text{ J.s}}{9.11 \times 10^{-31} \text{kg} \times 1.2 \times 10^8 \text{ m/s}}$$

$1J = 1 \text{ kg m}^2/\text{s}^2$

$$\Rightarrow \lambda = \frac{6.626 \times 10^{-34} \text{ kg.m}^2/\text{s}^2}{9.11 \times 10^{-31} \text{kg} \times 1.2 \times 10^8 \text{ m/s}}$$

$\lambda = 6.06 \times 10^{-12}$ m **Answer: C**

(9) An electron has a spin quantum number, $m_s = +½$ and a magnetic quantum number $m_\ell = +1$. In which of the following orbital will it NOT be present?

A. s – orbital C. d – orbital
B. p – orbital D. f – orbital

Explanation:

Each sub level is designated by the azimuual quantum number, ℓ :

ℓ = 0, is an s sub level ℓ = 2, is a d – orbital

ℓ = 1, is a p – orbital ℓ = 3, is an f – orbital

The value of magnetic quantum number, m_ℓ, is from $-\ell$ to $+\ell$.

Telegram: https://t.me/Extremeseries

For $\ell = 0, m_\ell = 0$

For $\ell = 1, m_\ell = -1, 0, 1$

For $\ell = 2, m_\ell = -2, -1, 0, 1, 2$

For $\ell = 3, m_\ell = -3, -2, -1, 0, 1, 2, 3$

$\Rightarrow m_\ell = +1$ is never found in an s – orbital since the value of ℓ is 0.

Answer: A

10. Which of the following represents the general configuration of the transition elements?

A. ns^2np^6 C. $ns\,(n-2)\,f$

B. $ns\,(n-1)\,d$ D. $ns^2np^6\,(n-1)\,d^{10}$

Explanation:

Transition elements are sub – group (IB – VIIIB) or d – block elements that have variable valence. Their general electron configuration is represented by $ns\,(n-1)\,d$. For example:

IB – $ns^1\,(n-1)\,d^{10}$
IIB – $ns^2\,(n-1)\,d^{10}$
IIIB – $ns^2\,(n-1)\,d^1$
IVB – $ns^2\,(n-1)\,d^2$
VB – $ns^2\,(n-1)\,d^3$
VIB – $ns^1\,(n-1)\,d^5$
VIIB – $ns^2\,(n-1)\,d^5$

VIIIB $\begin{cases} ns^2(n-1)d^6 \\ ns^2(n-1)d^7 \\ ns^2(n-1)d^8 \end{cases}$

Answer: B

11. Which of the following quantum number (s) is (are) related to the size and energy of an electron in a hydrogen atom?

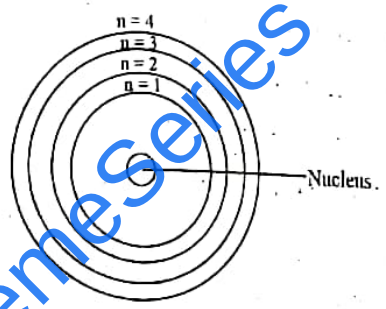
A. n B. n, ℓ C. n, ℓ, m D. n, ℓ, m, s

Explanation:

The size and energy of an electron in a hydrogen atom is affected by the principal quantum number, n. As n increases, the energy and size of an electron increases.

$E_{e^-} = \dfrac{-2.18\times10^{-18}\,J}{n^2}$; E_{e^-} – energy of an electron

The negative sign (–) implies, as n increases the energy also increases.



Answer: A

12. Which of the following statements is TRUE?

A. Ultra violet light has longer wave length than visible light.

B. The energy of radiation decreases as the wave length decreases.

C. The frequency of radiation increases as the wave length decreases.

D. Wave length of an electromagnetic radiation increases as wave length increases.

Explanation:

Gamma rays x-rays UV visible IR →

- Wave length increases
- Energy and frequency decreases

$E = h\nu$ E – energy

$E = \dfrac{hc}{\lambda}$ ν – frequency

c – speed of light

$\Rightarrow$ Energy is directly related with frequency but inversely related with wave length

$c = \lambda\,\nu$ $\nu = \dfrac{c}{\lambda}$

$\Rightarrow$ Frequency is inversely related with wave length.

Answer: C

13. Which of the following ionic compounds is formed from the reaction between magnesium and nitrogen?

A. MgN_2 C. Mg_3N_2

B. Mg_2N_2 D. Mg_2N_3

Telegram: https://t.me/Extremeseries

Explanation:
Magnesium loses two electrons and nitrogen gains three electrons to achieve the nearest noble gas configuration.

$$Mg \longrightarrow \overset{2+}{Mg} + 2e^-$$

$$N + 3e^- \longrightarrow N^{3-}$$

We use a simple criss – cross method to write the formula of ionic compounds by omitting the charge.

$Mg^{2+} \rightleftharpoons N^{3-} = Mg_3N_2$

Answer: C

14. Which of the following molecules represents a non –polar covalent bond?

A. B – Cl C. Cl – Cl
B. C – Cl D. Mg – Cl

Explanation:
Non – polar covalent bond is formed when electrons are shared equally because of similar electronegativity of the bonded atoms.
Example:

H – H O = O F – F I – I etc
N ≡ N Cl – Cl Br – Br

Answer: C

15. Which one of the following groups in the periodic table has paramagnetic atoms?

A. Group zero C. Group IIB
B. Group IIA D. Group IVA

Explanation:
Group IVA elements have unpaired electrons.
Example: ${}_6C: 1s^2 2s^2 2p_x^1 2p_y^1 2p_z^0$

Answer: D

16. Which of the following molecules has a trigonal bi pyramidal structure?

A. SF_4 B. IF_5 C. ICl_4^- D. BrF_5

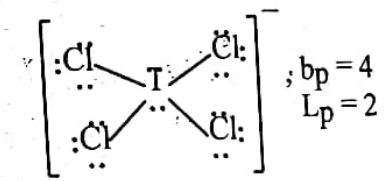
Explanation:
Draw the Lewis structure to determine the structure of the given molecules.

SF_4 : , $L_p = 1$, $b_p = 4$

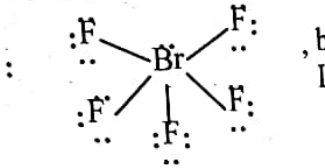
If there are 4 bonding pairs (bp) and 1 lone pair on the central atom, the shape of the molecule is see – saw and trigonal bipyramidal electron arrangement.

IF_5 : , $b_p = 5$, $L_p = 1$

If there are 5 bonding pairs and 1 lone pair on the central atom, the shape (structure) of the molecules is square pyramidal.

ICl_4^- :  ; $b_p = 4$, $L_p = 2$

If there are 4 bonding pairs and 2 lone pairs on the centeral atom, the structure of the molecule is square planar.

BrF_5 :  , $b_p = 5$, $L_p = 1$

There are 5 bonding pairs and one lone on the centeral atom. As a result, the shape of the molecule is square pyramidal.

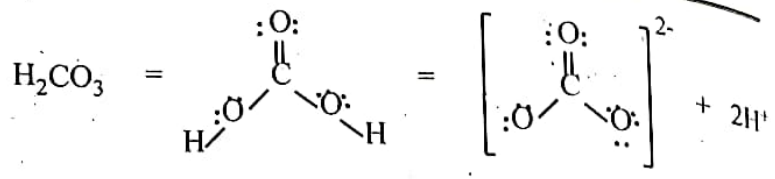
Answer: _____

17. The total number of electrons participating in the bond formation of carbonate ion CO_3^{2-}, in the molecule of carbonic acid are:

A. 16 B. 10 C. 8 D. 5

Explanation:
Draw the Lewis structure of carbonic acid, H_2CO_3:

Telegram: https://t.me/Extremeseries



Four bounding pairs (8 electrons) participate in bond formation of the carbonate ion.

Answer: C

18. Which of the following crystals possess high electrical and thermal conductivities?

A. Ionic crystals C. Molecular crystals
B. Metallic crystals D. Covalent network

Explanation:
Metallic crystals transmit heat and electricity due to free electrons that carry charge.

Answer: B

19. Which of the following hybrid orbitals is favoring the formation of trignal bipyramidal?

A. sp^3d B. sp^3 C. sp^3d^2 D. sp^3d^3

Explanation:

Electron arrangement	Hybridization
Linear	sp
Trigonal planar	sp^2
Tetrahedral	sp^3
Trigonal bipyramidal	sp^3d
Octahedral	sp^3d^2

Answer: A

20. Which of the following molecules/molecular ions is paramagnetic according to the molecular orbital theory?

A. O_2^{2-} B. O_2 C. F_2 D. O_2^{2+}

Explanation: A species with unpaired electron exhibits paramagnetic property. Such species is attracted by an external magnetic field. A species in which all the electrons are paired, exhibits diamagnetism. All species with odd number of electrons are paramagnetic species. For species with even number of electrons, write electron configuration by using molecular orbitals. O_2^{2-}, total electrons $= 2 \times 8 + 2 = 18$

$(\delta 1s)^2(\delta 1s^*)^2(\delta 2s)^2(\delta^* 2s)^2(\delta 2px)^2(\pi 2py)^2(\pi 2py^*)^2(\pi 2p2)^2(\pi^* 2p2)^2$

$\Rightarrow O_2^{2-}$ is diamagnetic species.

O_2, total electrons $= 2 \times 8 = 16$

$\Rightarrow (\delta 1s)^2(\delta^* 2s)^2(\delta 2s)^2(\delta^* 2s)^2(\delta 2px)^2(\pi 2py)^2(\pi^* py)^1(\pi 2p2)^2(\pi 2pz)^1$

O_2 is paramagnetic species because it has two unpaired electrons.

F_2 is similar with O_2^{2-}

O_2^{2+}, total electrons $= 2 \times 8 - 2 = 14$

$\Rightarrow (\delta 1s)^2(\delta 1^* s)^2(\delta 2s)^2(\delta^* 2s)^2(\pi 2py)^2(\pi 2py)^2(\delta 2px)^2(\pi 2p2)^2$

O_2^{2+} is diamagnetic species.

Answer: B

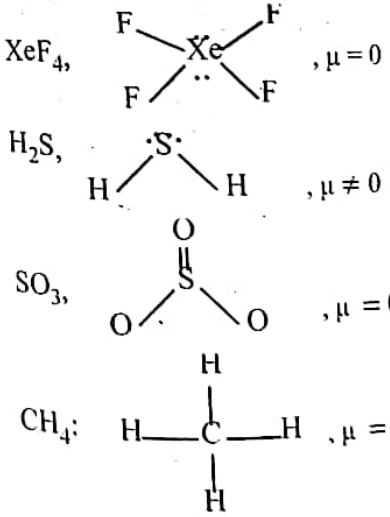
21. Which of the following molecules has a dipole moment?

A. XeF_4 B. H_2S C. SO_3 D. CH_4

Explanation:
Dipole moment, μ, is defined as the magnitude of the separated charges g multiplied by the distance, d, between their centers,

$$\mu = q \times d$$

A molecular dipole is represented by a crossed arrow directed toward the more negative atom.



Answer: B

Telegram: https://t.me/ExtremeSeries

22. What is a valid rate expression for the following reaction?

$2NO + 2H_2 \longrightarrow N_2 + 2H_2O$

A. $\frac{1}{2}\frac{\Delta[H_2]}{\Delta t}$ C. $-\frac{1}{2}\frac{\Delta[NO]}{\Delta t}$

B. $-\frac{1}{2}\frac{\Delta[H_2O]}{\Delta t}$ D. $-\frac{\Delta[N_2]}{\Delta t}$

Explanation:

Rate is the speed at which reactants disappear and products appear. For a general reaction:

$aA + bB \longrightarrow cC + dD,$

$$\text{Rate} = -\frac{1}{a}\frac{\Delta[A]}{\Delta t} = -\frac{1}{b}\frac{\Delta[B]}{\Delta t} = \frac{1}{c}\frac{\Delta[C]}{\Delta t} = \frac{1}{d}\frac{\Delta D}{\Delta t}$$

$$\Rightarrow \text{Rate} = -\frac{1}{2}\frac{[NO]}{\Delta t} = -\frac{1}{2}\frac{\Delta[H_2]}{\Delta t} = \frac{\Delta[N_2]}{\Delta t} = \frac{1}{2}\frac{\Delta[H_2O]}{\Delta t}$$

Answer: C

23. For the reaction;

$2A + B \longrightarrow C$

The following experimental results were obtained.

Experiment	[A]	[B]	Rate [mol $L^{-1}s^{-1}$]
1	0.50	0.50	0.300
2	0.50	0.25	0.075
3	0.25	0.25	0.075

What is the value of the rate constant?

A. 0.6 mol $L^{-1}s^{-1}$ C. 1.2 L mol^{-1} s^{-1}

B. 0.6 mol^{-1} Ls^{-1} D. 2.4 mol L^{-1} s^{-1}

Solution:

Determine the rate law by using initial method. Find the order with respect to A by using experiment 2 and 3. Determine the order with respect to B by using experiment 1 and 2.

Rate = $k[A]^x[B]^y$, k – rate constant

$$\frac{\text{Rate 2}}{\text{Rate 3}} = \frac{k[A]_2^x[B]_2^y}{k[A]_3^x[B]_3^y}$$

$$\frac{0.075}{0.075} = \frac{k(0.50)^x(0.25)^y}{k(0.25)^x(0.25)^y}$$

$$1 = \left(\frac{1}{2}\right)^x$$

$$\left(\frac{1}{2}\right)^0 = \left(\frac{1}{2}\right)^x$$

$x = 0 \Rightarrow$ The order of a reaction with respect to A is zero.

$$\frac{\text{Rate 1}}{\text{Rate 2}} = \frac{k[A]_1^x[B]_1^y}{k[A]_2^x[B]_2^y}$$

$$\frac{0.300}{0.075} = \frac{k(0.50)^x(0.50)^y}{k(0.50)^x(0.25)^y}$$

$4 = 2^y$

$2^2 = 2^y \Rightarrow y = 2$

The reaction is second order with respect to reactant B.

$\Rightarrow \text{Rate} = k[B]^2$

Use either of the experiments to find the value of rate constant, k.

$\text{Rate}_1 = k[B]^2$

$$k = \frac{\text{Rate}_1}{[B]_1^2} = \frac{0.300\text{mol } L^{-1}s^{-1}}{(0.50\text{mol/L})^2} = 1.2\text{Lmol}^{-1}\text{s}^{-1}$$ **Answer: C**

24. Increase in temperature of rate of a given reaction is due to the increase in the:

A. Extent of molecular dissociation.

B. Activation energy of the reaction.

C. Frequency of collisions of the reacting species.

D. Numerical value of the rate constant of the reaction.

Explanation: Temperature is the measure of the average kinetic energy of particles. As temperature increases, the number of collision between reacting species also increases. As a result, the rate of a reaction also increases. **Answer: C**

25. The reaction for the formation of nirosyl chloride:

$2NO\ (g) + Cl_2\ (g) \rightleftharpoons 2NOCl\ (g)$ was studied at 25°C. The value of K_P for this reaction at 25°C is 1.9×10^3 atm^{-1}. What is the value of K_C at 25°C?

A. 1.9×10^{-3} L/mol C. 4.6×10^4 L/mol

B. 3.8×10^{-3} L/mol D. 4.6×10^5 L/mol

Telegram: https://t.me/ExtremeSeries

Solution:

$$K_P = \frac{P_{NOCl}^2}{P_{NO}^2 P_{Cl_2}}$$

$$K_C = \frac{[NOCl]^2}{[NO]^2[Cl_2]}$$

From ideal gas equation, we know that pv = nRT, R – universal gas constant, T – kelvin temperature, P – pressure and v – volume.

$P = \frac{n}{v}RT$, but $\frac{n}{v}$ is concentration

$P = cRT$

$$K_P = \frac{([NOCl]RT)^2}{([NO]RT)^2[Cl_2](RT)^1}$$

$$K_P = \frac{[NOCl]}{[NO)]^2} \cdot (RT)^{2-(2+1)}$$

$K_P = K_C (RT)^{-1}$

$R = 0.082$ atm. L/ mol. K, $T = 25°C = 25 + 273 = 298$ k

$\Rightarrow K_C = 1.9 \times 10^3$ atm^{-1} $\times$ 0.082 atm.L/mol. K $\times$ 298 k

$K_C = 4.6 \times 10^4$ L/mol **Answer: C**

26. Three gases are in equilibrium in a closed chamber sealed with a piston. The following equilibrium is established.
$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$. What will happen if the piston is pushed into the champer?

A. The mole fraction of N_2 is increases.
B. The mole fraction of N_2 decreases.
C. The mole fraction of N_2 remains same.
D. The mole fraction of N_2 increases and then decreases.

Explanation:

Increasing pressure shifts the equilibrium to the smaller number of molecules and decreasing pressure shifts to the larger number of molecules. In the above chamber, the equilibrium shifts to the left. That is, formation of ammonia (NH_3) is favoured. The amount of N_2 and H_2 decreases.

Answer: B

27. Which of the following statements is TRUE about equilibrium reaction?

A. No more reactants are transformed into products.
B. There are equal amounts of reactants and products.
C. The rate constant for the forward reactions equals that of the reverse reaction.
D. The rate for the forward reaction equals that of the reverse reaction.

Explanation:

Characteristics of chemical equilibrium:

- The rate of the forward reaction is equal to the rate of reverse reaction.
- The equilibrium is dynamic in nature.
- All reactants and products are present at equilibrium.
- The concentration of reactants and products are generally not equal.
- The free energy change of the reaction is zero at equilibrium.

Answer: D

28. What is the process that converts liquid vegetable oils to solid fats?

A. Hydration
B. Hydrogenation
C. Hydrolysis
D. Saponification

Explanation:

Oils can be converted to fats by the addition of hydrogen (hydrogenation) at high pressure in the presence of nickel or palladium as catalyst.

Answer: B

29. What would be the solubility of $HOCH_2(CH_2)_6CH_2OH$ compared to $CH_3(CH_2)_6CH_2OH$?

A. Less soluble in water.
B. More soluble in water.
C. The same solubility in water.
D. More soluble in a non – polar solvent such as dichloro ethane.

Explanation:

Polyhydric alcohol (alcohols with more than one hydroxyl groups) are more soluble than monohydric alcohols in water.

Answer: B

Telegram: https://t.me/Extremeseries

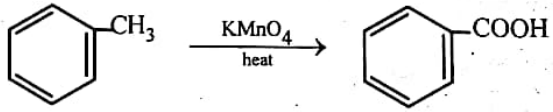
30. Which of the following reactions will produce an alkyl carboxylic acid?
A. Heating methyl ketone with an acid and iodine.
B. Reacting an alkyl halide with hydrogen gas and platinum.
C. Reacting an alcohol with ozone.
D. Oxidation of a primary alcohol with hot permanganate or chromate.

Explanation:
Carboxylic acids are organic compounds that contain at least one carboxyl group in their structure. Carboxylic acids are prepared by oxidation reaction. Alkyl carboxylic acids are manufacture by oxidation of primary alcohols, while aromatic carboxylic acids are prepared by oxidation of alkyl benzene.

$R-CH_2-OH \xrightarrow{KMnO_4} RCOOH$
1° alcohol — Carboxylic acid

Example

$CH_3-CH_2OH \xrightarrow{KMnO_4} CH_3-COOH$
Ethanol — Ethanoic acid



Methyl benzene — Benzoic acid

Answer: D

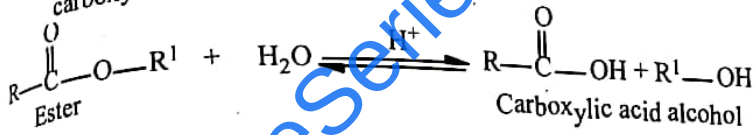
31. Which of the following statements is NOT true?
A. Naturally derived soaps consist of a soluble salt of a long chain fatty acid.
B. Triacyl glycerols are esters of glycerol and long chain carboxylic acids.
C. Long chain caroboxylic acids also known as fatty acids.
D. The major acidic component of vinegar is formic acid.

Explanation:
Formic acid is found in ants and bees. The major acidic components of vinegar is acetic acid.

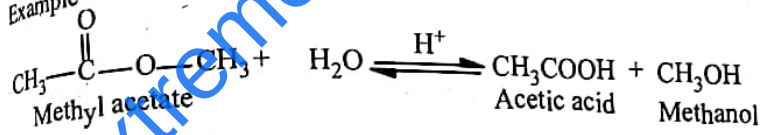
Answer: D

32. What is the product of the hydrolysis of esters in the presence of a mineral acid catalyst?
A. Alcohol
B. Carbon dioxide
C. Ether
D. Ketones

Explanation:
Hydrolysis of esters yields the corresponding alcohols and carboxylic acids.



Example



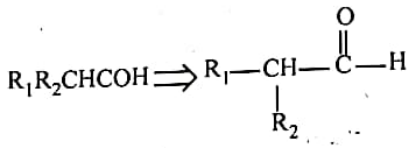
Answer: A

To which organic functional group does the following molecular representation, i.e., R_1R_2CHCOH belong? (R_1 and R_2 represent different alkyl chains).
A. Amides
B. Aldehydes
C. Ethers
D. organic acids

Explanation:
Functional group is part of the compound that determines its property.

Organic compound	General formula	Functional group
Carboxylic acid	R—C(=O)—OH	—COOH
Ester	R—C(=O)—O—R¹	—C(=O)—O—C
Ether	R—O—R¹	—C—O—C—
Aldehyde	R—C(=O)—H	—C(=O)—H
Ketone	R—C(=O)—R¹	—C(=O)—
Alcohol	R—OH	—OH

Telegram: https://t.me/ExtremeSeries



Answer: B

34. What is the name of triacylglycerol (triglycerides) that is solid at room temperature?
A. Lecithin B. Fat C. Wax D. Oil

Explanation:
Fats and oils belong to a class of biomolecules called lipids. They are triesters of glycerol which are collectively known as triglycerides or triacylglycerols. Fat is solid at room temperature while oil is liquid.

Answer: B

35. Which of the following reactions will convert carboxylic acids to primary amines?
A. Decarboxylation with HBr and peroxide, then reaction of the alkyl bromide with ammonia.
B. Reduction of the acid to the alcohol with NaOH formaldehyde, then reaction.
C. A two – step conversion, first to the amide with ammonia and heat and then by reduction with lithium aluminum hydride or hydrogen plus catalyst.
D. Lactase fermentation in the presence of ammonia atmosphere.

Explanation:
Carboxylic acids are weak organic acids that possess all the properties of acids.

Answer: C

36. A solution was prepared by dissolving 3.75 g pure hydrocarbon in 95 g of cylclohexane. The boiling point of pure cyclohexane was observed to be 80.70°C. What is the approximate molecular weight of the hydrocarbon? (k_b for cyclohexane = 2.79 °C/m).
A. 71.0 g/mol C. 147 g/mol
B. 105 g/mol D. 312 g/mol

Solution:

$\Delta T_b = mk_b = T_b - T_b^o$

ΔT_b – boiling point elevation
T_b – boiling point of solution
T_b^o – boiling point of pure solvent

m – molality
k_b – boiling point elevation constant for the solvent
$\Delta T_b = 81.45°C - 80.70°C = 0.75°C$

$$m = \frac{\Delta T}{kb} = \frac{0.75°C}{2.7g°C/m} = 0.2\,m$$

$$m = \frac{n}{\text{kg of solvent}}$$

n = m × kg of solvent
n = 0.27 mol/kg × 0.095 kg = 0.0255 mol

$$n = \frac{m}{M}$$

$$M = \frac{m}{n} = \frac{3.75g}{0.0255mol} = 147g/mol$$

Answer: C

37. A solution was prepared by adding 48 g of methanol into 81 g of water (H_2O). What is the mole fraction of methanol in this solution?
A. 0.25 B. 0.75 C. 1.5 D. 4.5

Solution:

$$n_{CH_3OH} = \frac{m}{M} = \frac{48g}{32g/mol} = 1.5mol$$

$$n_{H_2O} = \frac{m}{M} = \frac{81\,g}{18\,g/mol} = 4.5\,mol$$

$$X_{CH_3OH} = \frac{n_{CH_3OH}}{n_{CH_3OH} + n_{H_2O}}$$

$$X_{CH_3OH} = \frac{1.5\,mol}{(1.5+4.5)mol} = 0.25$$

Answer: A

38. What is the number of chloride ions (Cl^-) present in 1.0×10^{-5} mol of $AlCl_3$?
A. 1.80×10^{19} C. 6.02×10^{23}
B. 6.02×10^{18} D. 6.02×10^{28}

Solution:
Number of Cl^- ions=number of mole of $AlCl_3$ × subscript of Cl×6.02×10^{23}
$= 1.0 \times 10^{-5} \times 3 \times 6.02 \times 10^{23} = 1.80 \times 10^{19}$

Answer: A

Telegram: https://t.me/ExtremeSeries

39. What is the quantity of water in mL, required to prepare 0.5M of HCl from a concentrated solution of 3.5M in 50mL?
A. 50 mL B. 100mL C. 300mL D. 350mL

Solution:
$M_1 = 3.5M$ $M_2 = 0.5M$
$V_1 = 50\ mL$ $V_2 = ?$
$M_1V_1 = M_2V_2$

$$V_2 = \frac{M_1V_1}{M_2} = \frac{3.5M \times 50mL}{0.5M} = 350mL$$

V_{H_2O} added $= V_2 - V_1 = 350\ mL - 50\ mL = 300\ mL$ **Answer: C**

40. Which of the following is the most important type of solute – solvent interaction in a solution of n – butanol in water?
A. Dispersion C. Dipole – dipole
B. Ion – dipole D. Hydrogen bonding

Explanation: Alcohols possess hydrogen bond, that is why they are soluble in water. However, as carbon number increases, solubility of alcohols in water decreases. Ionic compounds dissolve in water due to ion – dipole interaction. Non – polar solutes dissolve in non –polar solvents due to dispersion or London force. Polar compounds are soluble in water due to dipole – dipole interactions. **Answer: D**

41. What is the mass of one molecule of water?
A. 3.0×10^{-23} g C. 0.0003 g
B. 1.8×10^{-22} g D. 18.0 g

Solution: 1 mole = molecular mass $= 6.02 \times 10^{23}$ molecules
1 mole of H_2O = 18 g $H_2O = 6.02 \times 10^{23}$ molecules H_2O
$\Rightarrow$ 18 g $H_2O = 6.02 \times 10^{23}$ molecules of H_2O
? = 1 molecule H_2O

$$? = \frac{18\ g\ H_2O \times 1\ molecule\ H_2O}{6.02 \times 10^{23}\ molecules\ H_2O}$$

$? = 2.99 \times 10^{-23}$ g H_2O
$\Rightarrow$ 1 molecule of water weighs 2.99×10^{-23}g **Answer: A**

42. Which one of the mixture of the following pairs will NOT give a buffer solution?
A. HCN and NaCN C. H_3PO_4 and KH_2PO_4
B. NH_3 and NH_4Cl D. HNO_3 and $NaNO_3$

Explanation:
Buffer solution is a solution that resists pH change upon the addition of small amount of acid or base. Buffer solution is formed from weak acid and its conjugate base or from weak base and its conjugate acid.
HA/A^-, acidic buffer
B/BH^+, basic buffer
A buffer cannot be formed from strong acid and strong base because there is no unionized form in the solution.

Answer: D

43. Which of the following is TRUE for salts formed from strong acids and strong bases?
A. No hydrolysis takes place.
B. Produces ions which are proton acceptors.
C. Produces ions that are proton donors.
D. Depends on the P_{ka} and P_{kb} of the parent acids and bases.

Explanation: Salts formed from strong acids and strong bases do not undergo hydrolysis reaction. Because the cations and anions are very weak acids and bases respectively.
Example

$$NaOH + HCl \longrightarrow NaCl + H_2O$$
Strong base strong acid Natural salt

$NaCl \longrightarrow Na^+ + Cl^-$
$Na^+ + H_2O \longrightarrow$ no reaction
$Cl^- + H_2O \longrightarrow$ no reaction
Adding such types of salts to water does not affect the pH of water. **Answer: A**

44. Given the following equilibira and equilibrium constants:
(I) $HC_2H_3O_2 + H_2O \rightleftharpoons H_3O^+ + C_2H_3O_2^-$; $Ka = 1.8 \times 10^{-5}$
(II) $H_2CO_3 + H_2O \rightleftharpoons H_3O^+ + HCO_3^-$; $Ka = 4.2 \times 10^{-7}$
(III) $NH_4^+ + H_2O \rightleftharpoons H_3O^+ + NH_3$; $Ka = 5.6 \times 10^{-10}$
(IV) $HCOOH + H_2O \rightleftharpoons H_3O^+ + HCOO^-$; $Ka = 1.80 \times 10^{-4}$
What is the strength of the acids in decreasing order?
A. I, IV, II and III C. III, II, I and IV
B. II, III, IV and I D. IV, I, II and III

Telegram: https://t.me/Extremeseries

Explanation: The strength of an acid is directly related with the acid dissociation constant, K_a. The larger the value of Ka, the stronger is the acid. $\Rightarrow HCOOH > HC_2H_3O_2 > H_2CO_3 > NH_4^+$

Answer: D

45. Given the reaction:

$$H_2PO_4^- + H_2O \longrightarrow H_3O^+ + HPO_4^{2-}$$

Which of the following represents a conjugate acid – base pair?

A. $H_2PO_4^-$ and H_2O C. $H_2PO_4^-$ and H_3O^+

B. $H_2PO_4^-$ and HPO_4^{2-} D. H_2O and HPO_4^{2-}

Explanation:

Conjugate acid is formed when a base accepts a proton.

Conjugate acid = base + hydrogen ion

Conjugate base is formed when an acid donates a proton.

Conjugate base = acid – hydrogen ion

The conjugate base of HPO_4^- is $HPO_4^- - H^+ = HPO_4^{2-}$

The conjugate acid of H_2O is $H_2O + H^+ = H_3O^+$.

Answer: B

46. Which of the following statements is TRUE about colligative properties?

A. Both vapour pressure and freezing point increase when a non – volatile solute is added to a solvent.

B. Both freezing point and boiling point increase when a non-volatile solute is added to a solvent.

C. Both freezing point and boiling point increase when a non – volatile solute is added to a solvent.

D. Colligative properties depend only upon number of solute properties in a solution and not upon their identity.

Explanation:

- Colligative properties depend on the concentration of the solute particles that are present in a given amount of solvent and not on their chemical identity. These are: vapor pressure lowering, boiling point elevation, freezing point depression and osmotic pressure.
- When a non – volatile solute is added to a solvent, vapor pressure and freezing point decrease while boiling point and osmotic pressure increase.

Answer: D

47. What is the pH of 0.005M solution of $Ca(OH)_2$?

A. 2.3 B. 10 C. 12 D. 14

Solution:

$$Ca(OH)_2 \longrightarrow 2OH^- + Ca^{2+}$$

1M 2M

0.005M ?

$$\Rightarrow [OH^-] = \frac{2M \times 0.005M}{1M} = 0.01M = 10^{-2}$$

$pH = 14 + \log [OH^-]$

$pH = 14 + \log 10^{-2} = 12$

Answer: C

48. The indicator Bromothymol blue (HBb) is a weak acid with Ka = 1.0×10^{-7} ionizes as follows:

$$HBb\ (aq.\ yellow) \rightleftharpoons H^+\ (aq.\ colorless) + Bb^-\ (aq.\ blue)$$

Which way will the equilibrium shift when NaOH is added and what will the color of NaOH solution be containing this indicator?

A. Equilibrium will shift to the right and the color of NaOH solution will turn green.

B. Equilibrium will shift to the right and the color of NaOH solution will turn blue.

C. Equilibrium will shift to the left and the color of NaOH solution will turn yellow.

D. Equilibrium will shift to the left and the color of NaOH solution will turn blue.

Explanation:

Acid – base indicators are weak organic acids or organic bases that indicate whether a solution is acidic or basic.

$$HBb\ (aq,\ yellow) \rightleftharpoons H^+\ (aq,\ colorless) + Bb^-\ (aq.\ blue)$$

If the indicator is in sufficiently acidic medium, the equilibrium shifts to the left and the predominant color is that of HBb, on the other hand, in basic medium, the equilibrium shifts to the right and the color of the solution will be that of Bb^-.

Answer: B

Telegram: https://t.me/EXtremeSeries

49. To 0.2M solution of a weak monoprotic acid, HA, enough quantity of its sodium salt, NaA, was dissolved to give a concentration of 0.2M of the salt. What will be the acid concentration, $[H_3O^+]$ in the final solutions? (Ka of HA = 1.80×10^{-5})

A. 3.60×10^{-6} M C. 1.80×10^{-5} M
B. 1.00×10^{-5} M D. 1.90×10^{-3} M

Solution:

The common ion effect is caused by the addition of a compound having an ion in common with the dissolved substance that shift the equilibrium. This is very important to adjust the concentration of hydronium ion.

$NaA \longrightarrow Na^+ + A^-$
0.2M 0.2 M 0.2M

$\Rightarrow$ Initially the $[A^-]$ is 0.2M.

$HA + H_2O \rightleftharpoons H_3O^+ + A^-$

Initial	0.20	0	0.2
Change	-x	+x	+x
Equilibrium	0.2 – x	x	0.2 + x

$$Ka = \frac{[H_3O^+][A]^-}{[HA]}$$

$$1.8\times10^{-5} = \frac{x(0.2+x)}{0.2-x}, \text{ approximation}$$

$0.2 + x \approx 0.2, 0.2 - x \approx 0.2$

$$1.8 \times 10^{-5} = \frac{x(0.2)}{0.2}$$

$x = [H_3O^+] = 1.8 \times 10^{-5}$ M **Answer: C**

50. A 50 mL solution of H_2SO_4 0.205 M is titrated with NaOH solution of unknown concentration. The end point against phenolphthalein indicator was known after 41 mL of NaOH added. What is the concentration of NaOH solution?

A. 0.10MB. 0.25 C. 0.41M D. 0.50M

Solution:

Write the neutralization reaction and determine the number of mole of NaOH from the given number of moles of H_2SO_4

$n_{H_2SO_4} = M \times V = 0.205\ M \times 50\ mL = 10.25\ m\ mol$

10.25 m mol x

$H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2H_2O$

1 2

$$\frac{0.25mol}{1} = \frac{x}{2}, x = n_{NaOH} = 20.5\ mol \Rightarrow M_{NaOH} = \frac{n_{NaOH}}{V_{NaOH}}$$

$$M_{NaOH} = \frac{20.5\ mmol}{41.0\ mL} = 0.50M$$ **Answer: D**

51. What is the pH of an aqueous solution prepared to contain 1.3×10^{-3} M sodium nitrite ($NaNO_2$) if the acid dissociation equilibrium constant, Ka, for nitrous acid (HNO_2) is 5.1×10^{-4}? (Kw = 1.0×10^{-14})

A. 3.1 B. 5.1 C. 6.0 D. 7.3

Solution:

$NaNO_2 \longrightarrow Na^+ + NO_2^-$

$\Rightarrow [NaNO_2] = [NO_2^-] = 1.3 \times 10^{-3}$ M

$NO_2^- + H_2O \rightleftharpoons HNO_2 + OH^-$

Initial	1.3×10^{-3}	0	0
Change	-x	+x	+x
Equilibrium	1.3×10^{-3}	x	x

$$K_b \text{ for } NO_2^- = \frac{K_w}{Ka \text{ of } HNO_2} = \frac{[HNO_2][OH^-]}{[NO_2^-]}$$

$$\frac{1.0\times10^{-14}}{5.1\times10^{-4}} = \frac{x.x}{1.3\times10^{-3}-x}$$

$$1.96 \times 10^{-11} = \frac{x^2}{1.3\times10^{-3}-x}, 1.3 \times 10^{-3} - x \approx 1.3 \times 10^{-3}$$

$2.548 \times 10^{-14} = x^2$

$x = [OH^-] = 1.6 \times 10^{-7}$ M

$pH = 14 + \log [OH^-]$

$pH = 14 + \log^{1.6\times10^{-7}} = 7.3$ **Answer: D**

Telegram: https://t.me/Extremeseries